

# Characterization and Convergence of Markov Processes via Martingale Problems

A blueprint for formalization

August 30, 2026

## Abstract

This manuscript collects, in a self-contained and formalization-oriented form, four results on martingale problems: (i) existence of a càdlàg modification of a solution (Ethier–Kurtz [7], Theorem 4.3.6); (ii) the fact that uniqueness of the one-dimensional distributions already implies uniqueness of the finite-dimensional distributions and the (strong) Markov property ([7], Theorem 4.4.2); (iii) duality as a tool for proving uniqueness ([7], Lemma 4.4.10, Theorem 4.4.11 and its corollaries); (iv) existence: five routes, of which two produce a solution out of nothing — an explicit construction of jump processes, and the correspondence with stochastic differential equations — one constructs it from a *dual* process (Depperschmidt–Greven–Pfaffelhuber [3]), and one obtains it as a weak limit of solutions of approximating martingale problems ([7], Lemma 4.5.1 and Remark 4.5.2), together with the strictly more general convergence theorem of Criens–Pfaffelhuber–Schmidt [2]. Each of (i)–(iii) is first proved in the abstract setting of [2], without any operator, and [7]’s statement is then recovered as a corollary; the abstract versions are new, since [2] prove no characterization results. A Markov structure is introduced only where a result needs one: uniqueness of the finite-dimensional distributions does not, and is proved for an arbitrary family of solution measures, with the Markov property falling out afterwards from a shift hypothesis. Two non-Markovian examples of [2] are used as test objects. Throughout, the time index is an abstract ordered set carrying a *clock*, the state space is graded from “measurable space” to “Polish”, and local martingale problems are treated wherever the statement permits it. Every statement is tagged with the weakest hypotheses on the index and on the state space that it uses, and Section 2.9 tabulates the result: the partially ordered index survives in (ii) for the Markov property but provably not for uniqueness, and in (iii) for an atomic clock; a merely measurable state space suffices for (ii) and (iii) throughout. Section 1 explains, and justifies, the choice of the underlying setting; Section 9 records the design decisions that are relevant for the Lean formalization; and Section 10 audits what is here that is not in the sources, including four apparent generalizations that turned out to be empty.

## Contents

|          |                                                                               |          |
|----------|-------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Scope, and the choice of the setting</b>                                   | <b>3</b> |
| 1.1      | The two candidate settings . . . . .                                          | 3        |
| 1.2      | Recommendation . . . . .                                                      | 4        |
| 1.3      | Generality: time index, Polish state space, local martingale problems . . . . | 5        |
| 1.4      | Generalizations not made, and why . . . . .                                   | 7        |

|          |                                                                        |           |
|----------|------------------------------------------------------------------------|-----------|
| <b>2</b> | <b>Preliminaries</b>                                                   | <b>8</b>  |
| 2.1      | Notation . . . . .                                                     | 8         |
| 2.2      | The time index and the clock . . . . .                                 | 10        |
| 2.3      | The state space . . . . .                                              | 13        |
| 2.4      | From Chapter 1: operator semigroups . . . . .                          | 15        |
| 2.5      | From Chapter 2: stochastic processes and martingales . . . . .         | 16        |
| 2.6      | From Chapter 3: weak convergence and the Skorokhod space . . . . .     | 18        |
| 2.7      | From the Appendixes of [7] . . . . .                                   | 20        |
| 2.8      | Where the prerequisites are used . . . . .                             | 22        |
| 2.9      | Which result needs which bundle . . . . .                              | 22        |
| <b>3</b> | <b>The Skorokhod space</b>                                             | <b>25</b> |
| 3.1      | The setting . . . . .                                                  | 25        |
| 3.2      | Time changes . . . . .                                                 | 26        |
| 3.3      | The metric . . . . .                                                   | 27        |
| 3.4      | The Borel field . . . . .                                              | 27        |
| 3.5      | Compactness . . . . .                                                  | 28        |
| 3.6      | Where the graduation stops . . . . .                                   | 28        |
| 3.7      | Notes for the formalization . . . . .                                  | 29        |
| <b>4</b> | <b>Martingale problems</b>                                             | <b>30</b> |
| 4.1      | The abstract (local) martingale problem . . . . .                      | 30        |
| 4.2      | The Markovian martingale problem . . . . .                             | 31        |
| <b>5</b> | <b>Sample path regularity: existence of a càdlàg modification</b>      | <b>37</b> |
| 5.1      | An abstract regularization theorem . . . . .                           | 37        |
| 5.2      | The Markovian case . . . . .                                           | 41        |
| <b>6</b> | <b>Uniqueness, the Markov property, and the strong Markov property</b> | <b>41</b> |
| 6.1      | Uniqueness without a Markov structure . . . . .                        | 42        |
| 6.2      | The Markov layer: shift systems . . . . .                              | 46        |
| 6.3      | The local martingale problem . . . . .                                 | 54        |
| 6.4      | Local uniqueness . . . . .                                             | 59        |
| 6.5      | Non-Markovian test objects . . . . .                                   | 61        |
| 6.6      | The Markovian case . . . . .                                           | 63        |
| <b>7</b> | <b>Uniqueness via duality</b>                                          | <b>65</b> |
| 7.1      | The anti-diagonal lemma . . . . .                                      | 65        |
| 7.2      | The continuous-time theory . . . . .                                   | 70        |
| <b>8</b> | <b>Existence</b>                                                       | <b>75</b> |
| 8.1      | From a transition semigroup . . . . .                                  | 76        |
| 8.2      | From a dual process . . . . .                                          | 76        |
| 8.3      | Jump processes . . . . .                                               | 86        |
| 8.4      | From stochastic differential equations . . . . .                       | 90        |
| 8.5      | By convergence: the Ethier–Kurtz version . . . . .                     | 91        |
| 8.6      | By convergence: the abstract theorem . . . . .                         | 93        |
| 8.7      | By convergence: the general version . . . . .                          | 95        |
| 8.8      | Approximation on different clocks . . . . .                            | 97        |
| 8.9      | Relaxing the continuity hypothesis . . . . .                           | 100       |

|           |                                                                            |            |
|-----------|----------------------------------------------------------------------------|------------|
| 8.10      | Convergence in measure . . . . .                                           | 104        |
| 8.11      | Hawkes processes and their Volterra limit . . . . .                        | 107        |
| <b>9</b>  | <b>Notes for the formalization</b>                                         | <b>110</b> |
| 9.1       | What is available and what has to be built . . . . .                       | 110        |
| 9.2       | Design decisions . . . . .                                                 | 113        |
| <b>10</b> | <b>What this manuscript adds to its sources</b>                            | <b>115</b> |
| 10.1      | Results that are not in the sources . . . . .                              | 115        |
| 10.2      | The split that organizes everything: uniqueness is not Markovian . . . . . | 116        |
| 10.3      | Generalizations that cost nothing . . . . .                                | 117        |
| 10.4      | Gaps and slips in the sources . . . . .                                    | 119        |
| 10.5      | Generalizations that turned out to be empty . . . . .                      | 119        |

# 1 Scope, and the choice of the setting

## 1.1 The two candidate settings

There are two natural frameworks in which the results of this manuscript can be phrased.

**(EK) The Ethier–Kurtz setting [7], Chapter 4.** A martingale problem is attached to an *operator*  $A \subset M(E) \times M(E)$  on a metric space  $E$ : a measurable  $E$ -valued process  $X$  solves the martingale problem for  $A$  if

$$f(X(t)) - \int_0^t g(X(s)) \, ds, \quad t \geq 0, \quad (1)$$

is a martingale for every  $(f, g) \in A$ . This is the setting in which *characterization* results are traditionally stated: uniqueness, the Markov property, duality, and the correspondence with semigroups and generators.

**(A) The abstract setting.** A martingale problem is attached to a *family*  $\mathcal{X}$  of  $\mathbb{K}$ -valued *test processes* on a filtered space carrying, as yet, no measure: a probability measure  $P$  solves the martingale problem  $\text{MP}(\mathcal{X})$  if every  $Y \in \mathcal{X}$  is a  $P$ -martingale, and the local martingale problem if every  $Y \in \mathcal{X}$  is a  $P$ -local martingale. The state space is Polish and the path space is an arbitrary Polish subset of  $E^{\mathbb{T}}$ ; no operator and no Markov structure are assumed.

It is worth being accurate about where this second setting comes from, since this manuscript leans on it heavily. It is not new with [2]. In essentially the form used here it is [10], Definition III.1.3, which attaches a martingale problem to a family of optional processes and does so with *local* martingales as the default rather than as a variant, and with the initial condition given on an initial  $\sigma$ -field rather than by an initial distribution. [2] rediscover it in the shape that suits weak convergence — canonical test processes, determining sets,  $P$ -continuity — and it is their formulation that Section 4 follows, but Remark 4.7 records what [10] already had and what is worth importing from them. [16], Chapter 32 works in the same spirit for diffusions, again with local martingales throughout.

**What each setting is good for.** The division of labour is not the one the first draft of this manuscript assumed, and the correction is the main structural finding here. Setting (A) is where *all four* target results are naturally proved — including the three that [2] do

not state. Setting (EK) is where they are naturally *applied*: in practice one is handed an operator, not a family of test processes, and the hypotheses of Section 6.6 are the ones that can be checked on  $A$  directly. Section 1.2 sets out the resulting three-layer architecture.

## 1.2 Recommendation

We use a *hybrid*, organised in three layers. The reader should be warned that the balance between them shifted while this manuscript was written, and that the outcome is not the one the first draft anticipated.

- (1) **Definitional layer: (A).** “Solution of a martingale problem” is defined as in [2], Definition 3.2, relative to a family  $\mathcal{X}$  of test processes, with a *local* variant available from the outset (Definition 4.2). This costs one line and is what makes local martingale problems and non-càdlàg path spaces expressible at all.
- (2) **Abstract working layer: also (A).** Each of the three characterization results is stated and proved in this setting, with no operator anywhere: Theorem 5.4 for the càdlàg modification, Proposition 6.13 and Theorems 6.23, 6.27 for uniqueness and the (strong) Markov property, and Lemma 7.1 with Proposition 7.3 for duality. This layer is itself graded: a *Markov structure* is introduced only where it is needed, namely in Section 6.2, and Remark 6.50 tabulates what lies on either side of that line. Uniqueness, in particular, lies on the non-Markovian side.
- (3) **Markovian layer: (EK).** The martingale problem for an operator  $A$  is the special case  $\mathcal{X} = \mathcal{X}_A(X)$  of (1) (Definition 4.6), and the four target results are recovered on it: Theorem 5.6, Theorem 6.51, Theorem 7.12 and Lemma 8.34. Each is a corollary of layer (2), and each proof is a verification of hypotheses rather than an argument. Section 8 sits partly outside this scheme: the existence results there are not characterizations and have no abstract counterpart to be corollaries of, except Theorem 8.40, which is itself the abstract one.

Layer (2) is the substance of this manuscript, and it is the point on which the first draft was wrong. It assumed that uniqueness, the Markov property and duality are *intrinsically* statements about an operator  $A$  and its domain, and that (A) could therefore serve only as packaging. That is not so. Each of the three proofs, read carefully, uses of the operator only that it produces test processes of a particular shape, and each hypothesis it really needs can be stated directly:

- the càdlàg theorem needs a class  $\Phi$  for which  $f(X)$  splits into a martingale plus a time-regular remainder — Definition 5.2;
- uniqueness and the Markov property need the test family to be stable under the shift — Definition 6.14 — and nothing else, the whole proof reducing to one lemma applied four times (Lemma 6.17);
- duality needs only two increment representations of a function of two time variables (Lemma 7.1), and no probability at all.

Remarks 5.5, 6.29 and 7.10 record what each abstraction buys. Briefly: path-dependent and atomic compensators, which cannot be written with an operator  $A \subset C_b(E) \times B(E)$  at all; the disappearance of the auxiliary filtration (4) from everything except Proposition 4.9; and a precise account of which time indices and which clocks each result admits (Section 2.9).

Two caveats keep this from being an argument for (A) alone.

- *[2] contains none of the three.* That paper is concerned exclusively with identifying weak limits; it proves no uniqueness statement, no Markov property and no path regularity statement. For the càdlàg theorem the omission is structural rather than accidental: their Setting 4.1 *presupposes* a Polish path space  $F$  and right continuous test processes, which is exactly what a regularization theorem has to produce. Layer (2) is therefore new here, not imported.
- *Layer (3) is not decoration.* The abstract hypotheses are the ones a proof needs; the operator is the one a user has. In applications one is handed  $A$ , and the value of Theorem 6.51 over Theorem 6.23 is that its hypotheses are checkable on  $A$  without first reconstructing  $\mathcal{X}_A$ ,  $\theta_r$  and a determining set.

For the fourth target result — existence by convergence, [7], Lemma 4.5.1 — the [2] formulation is a strict generalization (Theorem 8.44): it drops boundedness of  $f$  and  $g$ , replaces exact solutions of approximating martingale problems by *approximate* martingales, and replaces uniform boundedness by uniform integrability. Here [2] is used directly, and Theorem 8.40 isolates the abstract statement behind it, which needs neither càdlàg paths nor the Skorokhod topology and is proved here in full.

### 1.3 Generality: time index, Polish state space, local martingale problems

Four deliberate strengthenings relative to [7] are made.

**Abstract time index.** [7] and [2] index time by  $[0, \infty)$ . Here the index is an abstract set  $\mathbb{T}$ , and every statement is tagged with the weakest of the bundles (T0)–(T4) of Definition 2.1 that it needs; the compensator becomes a separate datum, a *clock* in the sense of Definition 2.3. Section 2.9 tabulates the result. There are two reasons for the effort. The first is that the classical formulation conflates a topological hypothesis on  $\mathbb{T}$  with an algebraic one: Section 5 needs the former, Sections 6 and 7 the latter, and  $[0, \infty)$  happens to supply both. The second is that Mathlib indexes `Filtration`, `Martingale` and `IsStoppingTime` by an arbitrary `[Preorder ι]`, so fixing  $\mathbb{T} = \mathbb{R}_+$  at the outset would mean working against the library rather than with it.

The gain is not merely cosmetic. Discrete time ( $\mathbb{T} = \mathbb{N}_0$ ,  $q$  counting measure) and processes with fixed times of discontinuity ( $q$  with atoms) become special cases rather than analogues, and the partially ordered case survives in Section 4, in the Markov half of Section 6, in the abstract convergence theorem and in the kernel layer of Section 8.2. It fails in exactly three places, and for reasons worth knowing: the uniqueness induction needs the times to form a chain (Remark 6.25); the Kolmogorov step of Theorem 8.5 needs finite subsets to be chains (Remark 8.8(ii)); and Corollary 7.4 is a time change, so it lives on (T3) and needs the clock to be atomless (Remark 7.5) — a limitation of that particular device, since for an atomic clock Proposition 7.8 covers a partially ordered index, needing only that the atoms below a point form a chain.

**State space: measurable, not Polish, where possible.** [7] and [2] take  $E$  Polish throughout. Here  $E$  is graded exactly as the time index is: (E0) a measurable space, (E1) standard Borel, (E2) separable metrizable, (E3) Polish (Definition 2.7). The grading is not decoration. Sections 4 and 7 and almost all of Section 6 run under (E0); regular

conditional distributions and the Borel structure, and hence (E1), are needed at five places listed in Remark 2.8; the topology of  $E$  is used only in Sections 5 and 8. The payoff is concrete: martingale problems for distribution-valued processes, where  $E = \mathcal{S}'(\mathbb{R}^d)$  is standard Borel but not metrizable, are covered by the first three sections with no new proof. See Remarks 2.8 and 2.9.

Where a topology is used,  $E$  is *not* assumed locally compact. On a general Polish space there is no space  $C_c(E)$  and no one-point compactification argument, and the rôle of local compactness is taken over throughout by the *compact containment condition* (27) resp. (12). This is the essential mechanism, and it is worth isolating it as a definition (Definition 4.21) in the formalization.

**Local martingale problems.** Whether “martingale” may be weakened to “local martingale” differs from result to result, and it is important not to overclaim. The situation is:

- *Theorem 5.6 (càdlàg modification)*: must be stated for genuine martingales. Localization requires stopping times, stopping times require a right-continuous filtration and a progressively measurable process, and path regularity is precisely what is being proved. The argument is circular otherwise. See Remark 5.8.
- *Theorem 6.51 (uniqueness and Markov property)*: the proof conditions on events of positive probability and uses the martingale property under the conditioned measures; boundedness of  $f$  and  $g$  enters through the integrability needed for (19). A local version is available, but only after localization along stopping times that are *common* to all solutions; see Remark 6.53.
- *Theorem 7.12 (duality)*: is already stated for unbounded  $f, g, h \in M(E_1 \times E_2)$ , controlled by the integrability hypotheses (58)–(59). Nothing has to be changed.
- *Section 8.3 (jump processes)*: the local version is not an afterthought but the general case — with unbounded rates the process explodes, and the jump times are a localizing sequence by construction; see Remark 8.26.
- *Section 8.11 (path-dependent rates)*: here the local version is not merely the general case but the *primary* one. With a path-dependent rate the compensator need not be integrable even when the process does not explode, so “is a martingale” has no content before stopping; see Remark 8.72. This is the cleanest instance in the manuscript of why Definition 4.2 carries a local variant from the outset.
- *Theorem 8.44 (convergence)*: concludes that the limit solves the *global* martingale problem. A local martingale problem is obtained in the limit by applying the theorem to a localized family of test processes; see Remark 8.45.

Section 6.3 carries out the local theory in the abstract setting, following [16], Theorems 32.10 and 32.11. Three hypotheses are needed and are isolated there as (L1)–(L3): a localizing sequence that is a *strict* stopping time and common to all  $P$ ; covariance of that sequence under the shift; and integrability of the increments after a restart. The point of isolating them is that the local martingale property is *not* linear in  $P$ , so that the two workhorse lemmas of Section 6.1 — mixture stability and the restart lemma — do not survive localization for free.

## 1.4 Generalizations not made, and why

Four axes have been pushed (index, clock, state space, local versus global) and one has been decided against on principle (locally compact state spaces). It is worth going through the remaining candidates systematically, because for a blueprint the question “what did you deliberately not do” is as informative as the list of what was done, and because two of the answers are “nothing, it is already there”.

| Candidate                                      | Verdict                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Complex-valued test processes                  | <b>carried out:</b> the scalar field is $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ throughout, and Remark 4.5 lists the four places needing $\mathbb{K} = \mathbb{R}$                                                                                                                                                |
| Submartingale problems (reflecting boundaries) | <i>transverse</i> , not wider: it forces $\mathbb{K} = \mathbb{R}$ , so it excludes the row above rather than adding to it. Over $\mathbb{R}$ it is a genuine generalization, by (37); the mixture and restart lemmas transfer, the determining set needs Lemma 6.20, and uniqueness fails. See Remarks 6.19 and 6.21 |
| Quasimartingale decompositions in Section 5    | <i>already there, and weaker:</i> Definition 5.2 asks of $C^f$ only one-sided limits and $L^1$ right continuity, not finite variation                                                                                                                                                                                 |
| Banach-valued test processes                   | rejected: measure- and distribution-valued processes are handled by pairing with a test function, so the gain is nil, and Section 5 needs an order                                                                                                                                                                    |
| Path-dependent (random) clock                  | <b>the real one;</b> see below                                                                                                                                                                                                                                                                                        |
| Nonlinear dependence on $P$ (McKean–Vlasov)    | a genuine boundary of Section 6; see below                                                                                                                                                                                                                                                                            |
| Control variables, relaxed solutions           | available in [2] and deliberately dropped; see below                                                                                                                                                                                                                                                                  |
| Signed or complex solution measures            | rejected: disintegration fails and nothing wants it                                                                                                                                                                                                                                                                   |

**A path-dependent clock.** The clock of Definition 2.3 is a *deterministic* measure on  $\mathbb{T}$ . Replacing it by a kernel  $\mathbf{q}$  from the path space to  $\mathbb{T}$ , so that the compensator is  $\int g(\pi_u) \mathbf{q}(\omega, du)$  — equivalently, by a predictable increasing process — is the step from a martingale problem to the semimartingale characteristics of [10], Chapter II. We write  $\mathbf{q}$  and not  $A$ , which is the operator throughout.

Two observations. First, *the abstract layer already contains it:* Definition 4.2 asks only that  $\mathcal{X}^\circ$  be a family of adapted path functionals, and  $f(\pi_t) - \int g(\pi_u) \mathbf{q}(\omega, du)$  is one. It is only Definition 4.6 that fixes a deterministic  $q$ , so the cost of the generalization is confined to Section 4.2 and to the results that quantify over  $q$ . Second, an *absolutely continuous* random clock buys nothing:  $\int g(\pi_u) a_u(\omega) du$  is the path-dependent integrand of Setting 8.69, which is already treated. What is genuinely new is a *singular*  $\mathbf{q}$  — a local time, or one with random jumps.

And that is where the two halves of Section 8.9 separate. A deterministic clock has its atoms at deterministic times, and Theorem 8.55 then removes the discontinuity of the compensator by enlarging the path space with those finitely or countably many coordinates. A random clock has its atoms at *random* times, no fixed family of coordinates captures

them, and the control variables and weak-strong convergence of [2] become necessary — exactly as Remark 8.58 predicts. So the material of Section 8.9 that this manuscript does not need is precisely the material that the path-dependent clock would need, which is a good reason for having kept it.

**Nonlinear dependence on  $P$ .** In a McKean–Vlasov or nonlinear martingale problem the operator depends on the law of the solution,  $A = A[\mathcal{L}(X_t)]$ , so that  $\mathcal{X}$  depends on  $P$ . Every lemma in Section 6.1 then fails at once, and for a reason the manuscript already knows: Lemma 6.2 and Lemma 6.17 rest on the martingale property being an identity *linear* in  $P$ . This is the same failure that Section 6.3 confronts — “ $Y$  is a local martingale” is not linear in  $P$  either — and it is worth recording that the two have the same shape and different repairs: localization repairs it by a hypothesis (L1) that restores linearity, whereas the nonlinear problem cannot be repaired that way and needs a fixed point instead. Section 8 is less affected: Theorem 8.40 concludes membership in a *fixed*  $\mathcal{M}(\mathcal{X})$ , so a nonlinear version needs the limit law to be identified before the theorem is applied, which is again a fixed point. This is a different subject and is not attempted.

**Control variables and relaxed solutions.** [2] carry a control space  $U$  throughout, and this manuscript takes  $U$  to be a singleton everywhere except Section 8.9. The general version would allow the approximating and limiting objects to be measures on  $U \times F$ , which is the setting of relaxed controls and of martingale problems with control. It is a genuine generalization, it is available in the source, and it was dropped for a reason worth stating: with  $U$  a singleton every statement in Sections 4–7 is unchanged, so the control space would be an index carried through five sections and used in one. Remark 8.58 shows what it is for when it is needed.

## 2 Preliminaries

Following the project plan, all material of [7], Chapters 1–3, may be cited. This section makes that interface explicit: it collects, in the form of numbered *Facts*, precisely those definitions and results of [7], Chapters 1–3 (together with a few items from the Appendixes of [7]) that are used later on. Nothing here is proved; the rôle of the section is to fix statements and to serve as the list of prerequisites for the formalization. Section 2.8 records where each item is used.

### 2.1 Notation

Let  $(E, \mathcal{E})$  be a measurable space; the topological assumptions on  $E$  are made bundle by bundle in Definition 2.7, and  $\mathcal{B}(E)$  is written for  $\mathcal{E}$  whenever  $E$  carries a topology. We write

- $\mathbb{K}$  for the scalar field, either  $\mathbb{R}$  or  $\mathbb{C}$ , fixed once and for all — in Lean this is [RCLike  $\mathbb{K}$ ], and the test functions and test processes below are the RCLike-valued ones;
- $M(E)$  for the  $\mathbb{K}$ -valued Borel measurable functions on  $E$ ;
- $B(E) \subset M(E)$  for the bounded ones, with  $\|f\| = \sup_{x \in E} |f(x)|$ ;
- $C(E)$  for the continuous functions,  $C_b(E) = C(E) \cap B(E)$ , and, for  $E$  locally compact,  $\hat{C}(E)$  for the continuous functions vanishing at infinity;
- $\mathcal{P}(E)$  for the Borel probability measures on  $E$  and  $V(E)$  for the finite signed Borel measures on  $E$ ;

- $D_E[0, \infty)$  for the space of càdlàg functions  $[0, \infty) \rightarrow E$  and  $C_E[0, \infty)$  for the continuous ones.

$L$  always denotes a real Banach space with norm  $\|\cdot\|$ .

Every statement is to be read over  $\mathbb{K}$ ; the places where  $\mathbb{K} = \mathbb{R}$  is *needed* are few, and Remark 4.5 lists all of them. The reader who wants only the real theory may take  $\mathbb{K} = \mathbb{R}$  throughout and lose nothing.

An *operator* is a subset  $A \subset M(E) \times M(E)$ ; it need not be single-valued nor linear. Its domain is  $\mathcal{D}(A) = \{f : (f, g) \in A \text{ for some } g\}$ , its range is  $\mathcal{R}(A) = \{g : (f, g) \in A \text{ for some } f\}$ , and  $A_S$  denotes its linear span inside  $M(E) \times M(E)$ . For  $\lambda \in \mathbb{R}$  we write  $\lambda - A = \{(f, \lambda f - g) : (f, g) \in A\}$ .

Throughout, references of the form “[7], Theorem 3.9.1” mean Chapter 3, Section 9, item 1 of [7].

### Typography, and symbols that carry two meanings

Blackboard bold is reserved for the two things that do not vary with the problem: the number systems  $\mathbb{R}, \mathbb{N}, \mathbb{Q}, \mathbb{C}, \mathbb{K}, \mathbb{R}_+$ , and the index  $\mathbb{T}$ , which is fixed once for the whole manuscript. Families that *do* vary — the test processes  $\mathcal{X}$ , the determining sets  $\mathcal{Z}$ , the solution sets  $\mathcal{M}$  — are calligraphic. Filtrations are the one exception:  $\mathbb{F}$  and  $\mathbb{G}$  are blackboard bold because [10] made that standard for  $(\mathcal{F}_t)$  and  $(\mathcal{G}_t)$ , and departing from it would cost more than it saves. An earlier version wrote  $\mathbb{X}$  and  $\mathbb{Z}$  for the two families, which is doubly unfortunate: they are not canonical objects, and  $\mathbb{Z}$  is the integers.

Some letters are nevertheless used twice, and the following are deliberate. Each pair is separated by enough of the manuscript that no single statement contains both; the table is here so that the reader who meets the second use first is not misled.

| Symbol          | First meaning                          | Second meaning                                  |
|-----------------|----------------------------------------|-------------------------------------------------|
| $\Phi$          | regularizing class, Definition 5.2     | the two-variable function of Section 7          |
| $\Theta$        | mixing parameter space, Lemma 6.2      | the dual flow of Lemma 8.14                     |
| $\gamma$        | the density in (50)                    | the augmenting map of Theorem 8.55              |
| $\Lambda$       | the time changes of Definition 3.4     | the path-dependent rate of Setting 8.69         |
| $\lambda$       | the resolvent parameter, Lemma 4.17    | the rate of Setting 8.20; also Lebesgue measure |
| $T$             | a time horizon                         | a strict stopping time in Section 6.4           |
| $H$             | a dense subset of $C_b(E)$ , Fact 2.43 | the duality function of Section 8.2             |
| $\alpha, \beta$ | paths in $F$ , Definition 6.43         | the Feynman–Kac potentials of Theorem 7.12      |
| $M$             | the measurable functions $M(E)$        | a separating class, Definition 2.37             |

Three collisions that would have been harmful were removed rather than declared, and are recorded here because the original notation is that of the sources: [7] write  $q$  for the truncated metric  $r \wedge 1$  in the Skorokhod metric, which is the clock here, so the truncation is written inside Definition 3.9 and  $\varrho$  is free for the metric on the index  $\mathbb{T}$ , which [7] do not have; [7] write  $F$  for a dense subset of the index in their regularization lemmas, which is the path space here, so we write  $D$ ; and the generalized inverse of the clock in Corollary 7.4 is written  $Q^{\leftarrow}$  rather than  $\tau$ , which is a stopping time everywhere else.

## 2.2 The time index and the clock

[7] and [2] both index time by  $\mathbb{R}_+$ . Almost nothing in this manuscript needs that, and the parts that do need it need it for visibly different reasons: Section 5 needs a *topological* property of the index, Sections 6 and 7 an *algebraic* one. We therefore fix an abstract index  $\mathbb{T}$  and record, for each statement, the weakest bundle of assumptions it uses. Two motives: it separates the topological from the algebraic hypotheses, which the classical formulation conflates; and it matches Mathlib, which indexes `Filtration`, `Martingale` and `IsStoppingTime` by an arbitrary `[Preorder ι]`, so that committing to  $\mathbb{R}_+$  at the outset would mean working against the library.

### Structure on the index

**Definition 2.1.** We use the following bundles of assumptions on  $\mathbb{T}$ . They are displayed as tagged equations so that every later occurrence of a bundle name is a link back to this definition.

$\mathbb{T}$  is a preordered set with a least element 0. (T0)

(T0), and  $\mathbb{T}$  is a directed lattice:  $s \wedge t$  and  $s \vee t$  exist. (T1)

(T1) in the concrete form of [7], Section 2.8: a metric lattice whose order intervals are separable from above. (T1')

(T0), and  $\mathbb{T}$  is linearly ordered. (T2a)

(T2a),  $\mathbb{T}$  carries the order topology and admits a countable dense subset  $D$ , and every non-maximal  $t \in \mathbb{T}$  is a limit from the right:  $D \cap (t, u) \neq \emptyset$  whenever  $t < u$ . A maximal element is allowed —  $\mathbb{T} = [0, T]$  satisfies it — and is the one point at which a right limit is not available. (T2b)

$\mathbb{T} = [0, \infty)$  or  $\mathbb{T} = [0, T]$ . (T3)

(T0), and  $\mathbb{T}$  is a cancellative ordered commutative monoid:  $r + \cdot$  is order preserving, and  $t - s$  exists for  $s \leq t$ . (T4)

Three remarks on the design of Definition 2.1.

(T4) is independent of (T2a)–(T3). The monoid structure is what the shift  $X(r + \cdot)$  needs; it presupposes no linear order. Keeping it separate makes visible that Sections 6 and 7 rest on an algebraic property of  $\mathbb{T}$  whereas Sections 5 and 8 rest on a topological one.

Why (T2a) and (T2b) are separate. The uniqueness half of Theorem 6.23 and the whole of Section 7 need the order to be linear but use no topology at all; only Section 5 and the concrete convergence results need the topology. Bundling the two, as one is tempted to do, hides that.

(T1') is not used in this manuscript. It is listed because it is the only place where [7] themselves treat a general index set, and because it is easy to misremember what it is for; Remark 2.6 sets that out. But every statement here that involves a stopping time — the strong Markov property of Theorem 6.27, and the localized statements of Remarks 6.53, 5.8 and 8.45 — already assumes (T2b), hence a linear order, for which the lattice structure is automatic. (T1') would become relevant only for a strong Markov property on a genuinely directed index, and Remark 6.25 shows that such a statement is blocked earlier, on the uniqueness side.

**Lemma 2.2** (Shifts are order embeddings; (T0) + (T4)). *Assume (T4). Then for every  $r \in \mathbb{T}$  the map  $r + \cdot$  is an order embedding:  $a \leq b \iff r + a \leq r + b$ . If moreover (T2a) holds, then  $r + (a \wedge b) = (r + a) \wedge (r + b)$ .*

*Proof.* “ $\Rightarrow$ ” is order preservation. For “ $\Leftarrow$ ” let  $r + a \leq r + b$  and put  $c = (r + b) - (r + a)$ , which exists by (T4). Then  $(r + a) + c = r + b$ , and cancelling  $r$  gives  $a + c = b$ . Since  $0 \leq c$  and  $a + \cdot$  is order preserving,  $a = a + 0 \leq a + c = b$ . The second assertion is immediate: under (T2a) the meet is the minimum, and an injective order preserving map between linear orders carries minima to minima.  $\square$

## The clock

The index set alone does not carry a martingale problem. The test processes of Definition 4.6 have the form  $Y_t = f(X_t) - C_t^{f,g}$ , and what distinguishes the compensator of a *generator* from an arbitrary adapted process is that it is additive in time and generated pointwise from  $g$ . That is what carries Proposition 4.9, and without a finite-dimensional characterization the phrase “martingale problem for the operator  $A$ ” has no content, since every process solves the martingale problem for *some* family of test processes, namely the family of its own martingales.

**Definition 2.3** (Clock). A *clock* on  $\mathbb{T}$  is a  $\sigma$ -field  $\mathcal{T}$  on  $\mathbb{T}$  for which the down-sets

$$\mathbb{T}_{\leq t} = \{u \in \mathbb{T} : u \leq t\}, \quad \mathbb{T}_{< t} = \mathbb{T}_{\leq t} \setminus \{u \in \mathbb{T} : t \leq u\}$$

are measurable for every  $t$ , together with a measure  $q$  on  $(\mathbb{T}, \mathcal{T})$  such that  $q(\mathbb{T}_{\leq t}) < \infty$  for every  $t$ . For  $s \leq t$  we write

$$(s, t] := \mathbb{T}_{\leq t} \setminus \mathbb{T}_{\leq s}, \quad [s, t) := \mathbb{T}_{< t} \setminus \mathbb{T}_{< s}, \quad (2)$$

and call these the *optional* and the *predictable* interval. Both  $t \mapsto \mathbb{T}_{\leq t}$  and  $t \mapsto \mathbb{T}_{< t}$  are monotone, so both intervals are additive:

$$(s, u] = (s, t] \uplus (t, u], \quad [s, u) = [s, t) \uplus [t, u), \quad (s \leq t \leq u). \quad (3)$$

We write  $\iota \in \{o, p\}$  for the choice of convention and  $\langle s, t \rangle_\iota$  for  $(s, t]$  resp.  $[s, t)$ .

The clock is *shift invariant* if  $\mathbb{T}$  satisfies (T4) and  $q(r + B) = q(B)$  for all  $r \in \mathbb{T}$  and  $B \in \mathcal{T}$ ; it is *reflective* if in addition  $\mathbb{T}$  satisfies (T2a); and it is *atomless* if  $q(\{u : t \leq u \leq t\}) = 0$  for every  $t$ , which is exactly the condition under which the two conventions coincide.

The additivity in (2) is not an assumption: it follows from transitivity of  $\leq$ , which is why a preorder suffices. This is the point at which a measure is genuinely needed rather than an additive interval function  $q_{s,u} = q_{s,t} + q_{t,u}$ : the proof of Proposition 4.9 uses Fubini in the form

$$E \left[ \int_{(s,t]} g(X_u) q(du) \cdot Z \right] = \int_{(s,t]} E[g(X_u)Z] q(du),$$

and additivity along chains does not give that. On  $\mathbb{T} = \mathbb{R}_+^d$ , for instance, chain additivity is strictly weaker than  $d$ -monotonicity, and only the latter produces a measure on boxes.

**Example 2.4.** (i)  $\mathbb{T} = \mathbb{R}_+$ ,  $q$  Lebesgue measure:  $\mathbb{T}_{\leq t} = [0, t]$  and  $C_t = \int_0^t g(X_u) du$ .

This is [7] and [2] throughout.

(ii)  $\mathbb{T} = \mathbb{N}_0$ ,  $q$  counting measure:  $C_n = \sum_{0 < k \leq n} g(X_k)$ , discrete time.

- (iii)  $\mathbb{T} = \mathbb{R}_+$ ,  $q$  with atoms: processes with fixed times of discontinuity, the subject of [2], Section 5.3.
- (iv)  $\mathbb{T} = \mathbb{R}_+^d$ ,  $q$  Lebesgue measure:  $\mathbb{T}_{\leq t}$  is the box  $[0, t]$  and  $C_t = \int_{[0, t]} g(X_u) du$ , the multiparameter martingale problem.

All four are clocks in the sense of Definition 2.3; (i), (ii) and (iv) are shift invariant, and only (i) and (ii) are reflective.

*Remark 2.5* (Optional or predictable: both). The compensator of Definition 4.6 is  $C_t = \int_{(0, t]_\iota} g(X_u) q(du)$ , and the two choices of  $\iota$  genuinely differ. For  $\iota = o$  the compensator is right continuous; for  $\iota = p$  it is predictable and left continuous. They agree exactly when the clock is atomless, in particular in Example 2.4(i) — which is why the choice is invisible in [7] and [2], and why neither source has to make it.

We carry both. The reason is that neither serves the whole manuscript: Theorem 5.4 holds only for  $\iota = o$ , because an atom of  $q$  at  $t$  must lie inside the compensator up to  $t$  for (R3) to hold, and it is that limit which makes the càdlàg process a *modification*; Proposition 7.3 is stated for  $\iota = p$ , because with  $\iota = o$  the chain identity telescopes to  $\gamma(t, 0) - \gamma(0, t)$  instead of to 0; but that quantity vanishes as well (Proposition 7.8), so the *conclusion* of Section 7 is convention-agnostic, as are Sections 4 and 6 and Lemma 7.1 itself, all of which use only the additivity (3). Section 2.9 records which  $\iota$  each result is most naturally stated in.

Two special cases are worth keeping in view. For  $\mathbb{T} = \mathbb{N}_0$  with counting measure,  $\iota = p$  gives  $C_n = \sum_{k \leq n} g(X_k)$ , the Doob decomposition, with  $A = P - I$ ;  $\iota = o$  gives  $C_n = \sum_{k < n} g(X_k)$  and the condition  $Pf - f = Pg$ . For  $\mathbb{T} = \mathbb{R}_+$  with Lebesgue measure the distinction is empty.

*Remark 2.6* (Comparison with [7], Section 2.8). [7], Section 2.8 is the only part of [7] that works with a general index set, and it is worth recording exactly what it does, because three things about it are easy to get wrong.

(i) *What [7] assume.* Their index  $\mathcal{J}$  is a *partially* ordered set — their (8.1)–(8.3) include antisymmetry — which is moreover a *metric lattice*: a metric space in which  $u \wedge v$  and  $u \vee v$  exist, are unique, and depend *continuously* on  $(u, v)$ . Separability from above of the intervals is a further hypothesis, imposed where it is needed (their Propositions 8.4, 8.5(c) and Theorem 8.7), not globally. Our (T0) is weaker on one point: we do not assume antisymmetry, because nothing below uses it and Mathlib’s `Preorder` does not have it.

(ii) *The lattice is needed for  $\vee$ , not for  $\wedge$ .* It is tempting to say that stopping times require  $\tau_1 \wedge \tau_2$ . The opposite is true. [7], Proposition 8.1(a) shows that  $\max_{k \leq n} \tau_k$  is a stopping time —  $\{\max_k \tau_k \leq u\} = \bigcap_k \{\tau_k \leq u\}$  — whereas their Remark 8.3 warns that  $\tau \wedge a$  need not be one. The reason is that in a lattice  $x \wedge y \leq u$  is strictly weaker than “ $x \leq u$  or  $y \leq u$ ”: in  $\mathbb{R}_+^2$  one has  $(1, 0) \wedge (0, 1) = (0, 0) \leq (0, 0)$  with neither factor below  $(0, 0)$ . So  $\{\tau \wedge a \leq u\}$  is genuinely larger than  $\{\tau \leq u\} \cup \{a \leq u\}$  and there is no reason for it to lie in  $\mathcal{F}_u$ . [7] therefore replace  $\tau \wedge a$  by the truncation  $\tau^a = \tau$  on  $\{\tau \leq a\}$  and  $\tau^a = a$  otherwise, their (8.10). Nor does  $\mathcal{F}_\tau$  require infima: their (8.6) is the usual  $\{A : A \cap \{\tau \leq u\} \in \mathcal{F}_u\}$ , verbatim.

(iii) *What it is for.* [7] build Section 2.8 for their Chapter 6, and it is worth being precise about how it is used there. In their Theorem 6.3.4 the multiparameter index carries the *filtration*  $\{\mathcal{F}_u : u \in \mathbb{R}_+^k\}$  and a nondecreasing family of  $\mathbb{R}_+^k$ -valued stopping times  $\tau(t)$ ; the processes themselves live in  $\prod_k D_{E_k}[0, \infty)$ , and so does the time-changed process  $Z(t) = Y(\tau(t))$ . That is, [7] never index a *process* by  $\mathbb{R}_+^d$ : the multiparameter index is a device inside a one-parameter theory, and what Section 2.8 delivers for it is optional sampling and nothing else. This is consistent with — indeed is indirect evidence

for — Remark 6.25: there is no multiparameter uniqueness or Markov theorem in [7] to measure ours against, because there is none to be had.

(iv) *A conflation we avoid.* [7] declare “we assume throughout this section that  $\mathcal{J}$  is a metric lattice” before defining filtration, stopping time,  $\mathcal{F}_\tau$ , adaptedness and, in their (8.15), the martingale property itself — none of which uses the lattice, and all of which are correct on a bare preorder. The lattice is first used in their Proposition 8.1. This is exactly the conflation that Definition 2.1 is meant to remove, and it is reassuring rather than surprising: the entire martingale layer of this manuscript rests on [7] (8.15) alone, which is stated for a partially ordered index.

## Notation for processes

For an  $E$ -valued measurable process  $X$  on  $(\Omega, \mathcal{F}, P)$ , indexed by  $\mathbb{T}$ , we write  $\mathcal{F}_t^X = \sigma(X(s) : s \leq t)$  and, following [7], (4.3.2),

$$*\mathcal{F}_t^X = \mathcal{F}_t^X \vee \sigma\left(\int_{\langle 0, s \rangle_t} h(X(u)) q(du) : s \leq t, h \in B(E)\right). \quad (4)$$

If  $X$  is progressively measurable — in particular if  $\mathbb{T}$  satisfies (T2b) and  $X$  is right continuous — then  $*\mathcal{F}_t^X = \mathcal{F}_t^X$ . Only (T0) and a clock are needed for (4) to make sense.

## 2.3 The state space

[7] and [2] take the state space  $E$  to be Polish throughout. As with the time index, almost nothing needs that, and the parts that do need it need it for visibly different reasons. We therefore separate them.

**Definition 2.7.** We use the following bundles of assumptions on the state space, again as tagged equations.

$$(E, \mathcal{E}) \text{ is a measurable space.} \quad (\text{E0})$$

$$(\text{E0}), \text{ and } (E, \mathcal{E}) \text{ is standard Borel: } \mathcal{E} \text{ is the Borel } \sigma\text{-field of some Polish topology on } E, \text{ equivalently } (E, \mathcal{E}) \text{ is Borel isomorphic to a Borel subset of a Polish space.} \quad (\text{E1})$$

$$(\text{E1}), \text{ and } E \text{ carries a separable metrizable topology with } \mathcal{E} = \mathcal{B}(E). \quad (\text{E2})$$

$$E \text{ is Polish.} \quad (\text{E3})$$

Thus  $(\text{E3}) \Rightarrow (\text{E2}) \Rightarrow (\text{E1}) \Rightarrow (\text{E0})$ , and both gaps are inhabited: a Borel non- $G_\delta$  subset of  $\mathbb{R}$  satisfies (E2) but not (E3), and the space  $\mathcal{S}'(\mathbb{R}^d)$  of tempered distributions satisfies (E1) but not (E2), being a Lusin space and hence standard Borel while not metrizable.

*Remark 2.8* (What each bundle is for). The four bundles correspond to four different uses, and Section 2.9 records which result makes which.

- (i) (E0): *nothing but measurability.* Sections 4 and 7, and the whole of Section 6.1 apart from Lemma 6.3, mention  $E$  only through  $B(E)$  and  $\mathcal{E}$ . [2] already state their duality theorem for measurable  $E_1, E_2$ ; the observation here is that uniqueness and the Markov property are of the same kind.
- (ii) (E1): *regular conditional distributions, and the Borel structure.* Five items need it: Lemma 6.3 and its local counterpart Lemma 6.32(c); Fact 2.48; the identification of a law on  $F$  by its finite-dimensional distributions; the measurability of the event

$\{a_T\beta = a_T\alpha\}$  in Definition 6.43, which needs  $\mathcal{E}$  countably generated and separating; and the Kolmogorov extension of Fact 2.47, used in Theorem 8.5. Each is a statement about the Borel structure, not about the topology, and standard Borel is what makes all five available.

- (iii) (E2): *separating classes and compact sets*. Section 5 needs a topology: the compact containment condition (27) refers to compact subsets of  $E$ , and the two hypotheses on  $\mathcal{D}(A)$  refer to  $C_b(E)$  separating points and separating measures.
- (iv) (E3): *weak convergence*. Only Section 8, and there only because  $D_E[0, \infty)$  has to be Polish for Prohorov's theorem and the Skorokhod representation.

*Remark 2.9* (The point of (E1)). The gap between (E1) and (E2) is not academic. Martingale problems for distribution-valued processes — the martingale problem formulation of an SPDE, with  $E = \mathcal{S}'(\mathbb{R}^d)$  or  $E = \mathcal{D}'(\mathbb{R}^d)$  — live on a state space that is standard Borel but not metrizable. Under (E0) and (E1), Sections 4, 6 and 7 apply to them verbatim, with no new proof. Sections 5 and 8 do not, and the repair is a genuine piece of work rather than a change of hypotheses: one needs a substitute for  $D_E[0, \infty)$  and for tightness in it, which for nuclear duals is Mitoma's theorem. That is out of scope here, and is flagged rather than attempted.

### A state space that varies with time

There is one further grading, orthogonal to (E0)–(E3), which costs almost nothing to admit and which the formalization will force anyway: the state space may depend on  $t$ .

**Definition 2.10** (Fibred state space). A *fibred state space* over  $\mathbb{T}$  is a family  $E = (E_t, \mathcal{E}_t)_{t \in \mathbb{T}}$  of measurable spaces. The path space is then  $F \subseteq \prod_{t \in \mathbb{T}} E_t$ , and  $\pi_t : F \rightarrow E_t$  is the coordinate. A bundle (E0)–(E3) is read *fibrewise*:  $E$  satisfies (E1) if every  $E_t$  does, and so on. The *constant* case  $E_t \equiv E$  is Definition 2.7.

Where a statement needs to compare fibres — and only two do — we write  $\Sigma E = \coprod_{t \in \mathbb{T}} E_t$  for the total space, with the  $\sigma$ -field generated by  $\mathcal{T} \otimes$  the fibrewise ones, and require  $\Sigma E$  to satisfy the bundle in question.

A fibred state space forces one change of shape, and it is a change the manuscript has already made for other reasons. The pairs  $(f, g)$  of Definition 4.6 become *sections*:  $f = (f_t)_{t \in \mathbb{T}}$  with  $f_t \in B(E_t)$ ,  $g = (g_u)_{u \in \mathbb{T}}$  with  $g_u \in B(E_u)$ , and

$$Y_t^{f,g} = f_t(\pi_t) - \int_{\langle 0, t \rangle_t} g_u(\pi_u) q(du). \quad (5)$$

For a constant fibre and a  $t$ -independent  $f$  this is Definition 4.6; the time-dependent  $g$  is already there (Setting 8.47). Allowing  $f$  to depend on  $t$  as well costs nothing and buys the *space-time* martingale problem, in which  $g$  absorbs  $\partial_t f + Af$  — the standard device for time-inhomogeneous problems, which the manuscript would otherwise have to add separately.

*Remark 2.11* (Which results are indifferent to the fibration). The audit is short, and its outcome is that the split runs exactly along the line already drawn in Remark 2.8: what mentions  $E$  only through  $B(E_t)$  and  $\mathcal{E}_t$  is indifferent, what needs a topology on  $E$  is not.

| Result                                     | Fibred state space                                                                                                                                                                          |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Section 4.1 (abstract martingale problem)  | <b>free:</b> $\mathcal{X}$ is a family of $\mathbb{K}$ -valued processes and $E$ is never mentioned                                                                                         |
| Section 4.2, Proposition 4.9               | free, with $(f, g)$ read as sections (5); the test variables $h_k \in B(E_{t_k})$                                                                                                           |
| Section 5 (càdlàg modification)            | <b>needs a constant fibre</b> , or a topological bundle: “càdlàg” compares $\omega(s)$ with $\omega(t)$                                                                                     |
| Section 6.1 (uniqueness, Markov)           | free; $h \in B(E_s)$ and $f \in B(E_t)$ in Definition 6.11                                                                                                                                  |
| Section 6.2 (shift systems)                | free, and more natural: $\theta_r$ maps $\prod_t E_t$ to $\prod_t E_{r+t}$ , so $\mathcal{X}_r^\circ \neq \mathcal{X}^\circ$ is forced and homogeneity is impossible unless $E_{r+t} = E_t$ |
| Theorem 6.27 (strong Markov)               | needs $\Sigma E$ : the value $X(\tau)$ lives in $E_{\tau(\omega)}$ , so $\mu_{F_0} = P\{X(\tau) \in \cdot\}$ is a measure on the total space                                                |
| Sections 6.3, 6.4 (local theory)           | free; the localizing times are path functionals and $a_T$ acts fibrewise                                                                                                                    |
| Section 7                                  | free: $H_{s,t} : E_s^{(1)} \times E_t^{(2)} \rightarrow \mathbb{R}$ , and Lemma 7.1 and Corollary 7.4 do not see $E$ at all                                                                 |
| Section 8.2 (existence from a dual)        | free, and this is where it <i>pays</i> : $\mu_{s,t}$ is a kernel from $E_{1,s}$ to $E_{1,t}$ , and Fact 2.47 is stated for a dependent family in Mathlib already                            |
| Sections 8.3–8.11 (existence, convergence) | <b>needs a constant fibre:</b> $D_E[0, \infty)$ , the $J_1$ topology and tightness all presuppose one                                                                                       |

*Remark 2.12* (Should the fibration be built in from the start?). Yes, and for three reasons that have nothing to do with generality for its own sake.

(i) *It is free where it matters and unavailable where it does not.* Remark 2.11 shows that every result proved without a topology on  $E$  is already indifferent, because those proofs mention  $E$  only through  $B(E_t)$ . Conversely, no amount of care will make Section 5 or Section 8 fibred, since both compare values at different times. So the decision is not “how much generality” but “in which of two already-separated halves does one work”, and the answer is fixed by the audit rather than chosen.

(ii) *Mathlib is fibred.* The projective-limit API of Fact 2.47 — `IsProjectiveLimit`, `projectiveFamilyContent`, `Kernel.traj` — is stated for a family  $E : \iota \rightarrow \text{Type}^*$  and a measure on  $\text{Measure } (\Pi i, E i)$ , not for a fixed state space. Fixing  $E_t \equiv E$  at the outset would mean writing `fun _ => E` everywhere and re-deriving the dependent statements by hand. This is the same argument that Section 2.2 makes for `[Preorder  $\iota$ ]`, and it has the same force.

(iii) *Task 19 needs it.* The one identified route from Section 8.2 to the path-dependent theory is to dualize the *historical* process (Remark 8.8), whose state space at time  $t$  is the space of paths on  $\mathbb{T}_{\leq t}$  — a fibred state space and not a constant one. Genealogy-valued processes are the same phenomenon.

What should *not* be done is to rewrite Sections 5 and 8 in fibred notation. They need a constant fibre, they say so, and the bundle table records it; carrying an index that is then immediately frozen would be notation without content.

## 2.4 From Chapter 1: operator semigroups

**Definition 2.13** ([7], Section 1.1). A one-parameter family  $\{T(t) : t \geq 0\}$  of bounded linear operators on  $L$  is a *semigroup* if  $T(0) = I$  and  $T(s+t) = T(s)T(t)$  for all  $s, t \geq 0$ .

It is *strongly continuous* if  $\lim_{t \rightarrow 0} T(t)f = f$  for every  $f \in L$ , and a *contraction semigroup* if  $\|T(t)\| \leq 1$  for all  $t \geq 0$ .

A (possibly unbounded) *linear operator*  $A$  on  $L$  is a linear map whose domain  $\mathcal{D}(A)$  is a subspace of  $L$  and whose range  $\mathcal{R}(A)$  lies in  $L$ . The (*infinitesimal*) *generator* of a semigroup  $\{T(t)\}$  on  $L$  is the linear operator  $A$  defined by

$$Af = \lim_{t \rightarrow 0} \frac{1}{t}(T(t)f - f), \quad (6)$$

with  $\mathcal{D}(A)$  the set of  $f \in L$  for which the limit exists.

**Definition 2.14** ([7], Section 1.2). A linear operator  $A$  on  $L$  is *dissipative* if  $\|\lambda f - Af\| \geq \lambda\|f\|$  for every  $f \in \mathcal{D}(A)$  and  $\lambda > 0$ . For  $A \subset L \times L$  linear,  $A$  is dissipative if  $\|\lambda f - g\| \geq \lambda\|f\|$  for all  $(f, g) \in A$  and  $\lambda > 0$ .

**Definition 2.15** ([7], Section 1.5). Let  $L$  be a function space of the type considered in [7], Section 1.5 (for instance  $L = B(E)$  with the sup norm). A semigroup  $\{T(t)\}$  on  $L$  is *measurable* if  $t \mapsto T(t)f$  is measurable for each  $f \in L$ . The *full generator* of a measurable contraction semigroup  $\{T(t)\}$  on  $L$  is the multivalued operator

$$\hat{A} = \left\{ (f, g) \in L \times L : T(t)f - f = \int_0^t T(s)g \, ds \text{ for all } t \geq 0 \right\}. \quad (7)$$

*Fact 2.16* ([7], Proposition 1.5.1). The full generator  $\hat{A}$  of a measurable contraction semigroup  $\{T(t)\}$  on  $L$  is linear and dissipative, and  $(\lambda - \hat{A})^{-1}h = \int_0^\infty e^{-\lambda t} T(t)h \, dt$  for all  $h \in \mathcal{R}(\lambda - \hat{A})$  and  $\lambda > 0$ .

*Remark 2.17.* Nothing further from [7], Chapter 1 is used below. In particular the Hille–Yosida theorem, cores, multivalued dissipative operators and the exponential formula ([7], Theorems 1.2.6 and 1.4.3, Propositions 1.3.1 and 1.3.3, Corollary 1.6.8) are *not* needed. They enter only through the semigroup-based uniqueness criteria [7], Theorem 4.4.1 and Corollary 4.4.4, which are deliberately not part of this manuscript: uniqueness is obtained here from Theorem 6.51 together with duality (Corollary 7.20), which requires no semigroup theory at all. Should those criteria be wanted later, this subsection is the place to extend.

## 2.5 From Chapter 2: stochastic processes and martingales

**Definition 2.18** ([7], Section 2.1; (T0), except where noted). Let  $\mathbb{T}$  carry a clock (Definition 2.3), with  $\sigma$ -field  $\mathcal{T}$ . An  $E$ -valued process  $X$  on  $(\Omega, \mathcal{F}, P)$  is *measurable* if  $X : \mathbb{T} \times \Omega \rightarrow E$  is  $\mathcal{T} \otimes \mathcal{F}$ -measurable. A *filtration* is an increasing family  $(\mathcal{F}_t)_{t \in \mathbb{T}}$  of sub- $\sigma$ -fields of  $\mathcal{F}$ .  $X$  is  $(\mathcal{F}_t)$ -*adapted* if  $X(t)$  is  $\mathcal{F}_t$ -measurable for each  $t$ , and  $(\mathcal{F}_t)$ -*progressive* if for each  $t \in \mathbb{T}$  the restriction of  $X$  to  $\mathbb{T}_{\leq t} \times \Omega$  is  $(\mathcal{T} \cap \mathbb{T}_{\leq t}) \otimes \mathcal{F}_t$ -measurable.

$Y$  is a *version* of  $X$  if the two have the same finite-dimensional distributions; a *modification* of  $X$  if  $P\{X(t) = Y(t)\} = 1$  for each  $t \in \mathbb{T}$ ; and *indistinguishable* from  $X$  if  $P\{X(t) = Y(t) \text{ for all } t \in \mathbb{T}\} = 1$ .

A  $\mathbb{T} \cup \{\infty\}$ -valued random variable  $\tau$  is an  $(\mathcal{F}_t)$ -*stopping time* if  $\{\tau \leq t\} \in \mathcal{F}_t$  for every  $t \in \mathbb{T}$ , and then  $\mathcal{F}_\tau = \{F \in \mathcal{F} : F \cap \{\tau \leq t\} \in \mathcal{F}_t \, \forall t \in \mathbb{T}\}$ .

Under (T2b) one says in addition that  $(\mathcal{F}_t)$  is *right continuous* if  $\mathcal{F}_t = \mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$  for all  $t$ , and that  $X$  is *right continuous* if its paths are; every right continuous adapted process is then progressive. All of the above except this last paragraph needs only (T0), and is exactly [7] (8.1)–(8.6) of their Section 2.8; see Remark 2.6.

*Fact 2.19* ([7], Propositions 2.1.2 and 2.1.4; (T2b)). Assume (T2b). If  $\tau, \tau_1, \tau_2, \dots$  are  $(\mathcal{F}_t)$ -stopping times and  $c \in \mathbb{T}$ , then  $\tau_1 + c$ ,  $\tau_1 \wedge c$ ,  $\sup_n \tau_n$  and  $\min_{k \leq n} \tau_k$  are  $(\mathcal{F}_t)$ -stopping times; if  $(\mathcal{F}_t)$  is right continuous, so are  $\inf_n \tau_n$ ,  $\liminf_n \tau_n$  and  $\limsup_n \tau_n$ . If  $X$  is  $(\mathcal{F}_t)$ -progressive and  $\tau$  is an  $(\mathcal{F}_t)$ -stopping time, then  $X(\tau)$  is  $\mathcal{F}_\tau$ -measurable on  $\{\tau < \infty\}$  and  $X(\cdot \wedge \tau)$  is  $(\mathcal{F}_t)$ -progressive. The linear order is used throughout: on a merely directed index  $\tau_1 \wedge \tau_2$  need not be a stopping time at all (Remark 2.6(ii)).

**Definition 2.20** ([7], Section 2.2 and (8.15); (T0)). A  $\mathbb{K}$ -valued  $(\mathcal{F}_t)$ -adapted process  $X$  with  $E[|X(t)|] < \infty$  for all  $t$  — the modulus, if  $\mathbb{K} = \mathbb{C}$  — is an  $(\mathcal{F}_t)$ -martingale if  $E[X(t) \mid \mathcal{F}_s] = X(s)$  a.s. for all  $s \leq t$  in  $\mathbb{T}$ . For  $\mathbb{K} = \mathbb{R}$  it is an  $(\mathcal{F}_t)$ -submartingale if  $E[X(t) \mid \mathcal{F}_s] \geq X(s)$  a.s. This is [7] (8.15) verbatim, and it is the whole of what the martingale layer of this manuscript uses; it is also `MeasureTheory.Martingale` of Mathlib, which is stated for `[Preorder ι]` and for values in an arbitrary real Banach space, hence covers both choices of  $\mathbb{K}$  without a word.

*Fact 2.21* (Regularization along a dense set; [7], Lemma 2.2.8; (T2b)). Let  $(E, r)$  be a metric space,  $D \subset \mathbb{T}$  dense, and  $x : D \rightarrow E$ . If  $y(t) := \lim_{s \rightarrow t+, s \in D} x(s)$  exists for every  $t \in \mathbb{T}$ , then  $y$  is right continuous; if  $y^-(t) := \lim_{s \rightarrow t-, s \in D} x(s)$  exists for every  $t > 0$ , then  $y^-$  is left continuous on  $\mathbb{T} \setminus \{0\}$ ; and if both exist for every  $t > 0$ , then  $y^-(t) = y(t-)$  for all  $t > 0$ . In particular  $y$  is càdlàg, and  $y \in D_E[0, \infty)$  when  $\mathbb{T}$  satisfies (T3).

*Fact 2.22* (Submartingale regularization; [7], Proposition 2.2.9; (T2b)). Let  $X$  be a real-valued  $(\mathcal{F}_t)$ -submartingale — here, and in Fact 2.46,  $\mathbb{K} = \mathbb{R}$  is genuinely needed, an order being involved — and let  $D \subset \mathbb{T}$  be countable and dense. Then there is  $\Omega_0$  with  $P(\Omega_0) = 1$  such that for every  $\omega \in \Omega_0$  the limits

$$Y(t, \omega) = \lim_{s \rightarrow t+, s \in D} X(s, \omega) \quad (t \in \mathbb{T}), \quad Y^-(t, \omega) = \lim_{s \rightarrow t-, s \in D} X(s, \omega) \quad (t > 0) \quad (8)$$

exist. Moreover  $\Gamma = \{t : P\{Y(t) \neq Y(t-)\} > 0\}$  is countable,  $P\{X(t) = Y(t)\} = 1$  for  $t \notin \Gamma$ , and setting  $\tilde{X}(t) = Y(t)$  for  $t \notin \Gamma$  and  $\tilde{X}(t) = X(t)$  for  $t \in \Gamma$  defines a modification of  $X$  almost all of whose sample paths have right and left limits at all  $t \in \mathbb{T}$  and are right continuous at all  $t \notin \Gamma$ .

**Corollary 2.23** (Regularization of a  $\mathbb{K}$ -valued martingale; (T2b)). *Let  $Y$  be a  $\mathbb{K}$ -valued  $(\mathcal{F}_t)$ -martingale and let  $D \subset \mathbb{T}$  be countable and dense. Then there is  $\Omega_0$  with  $P(\Omega_0) = 1$  such that for every  $\omega \in \Omega_0$  the limits*

$$\lim_{s \rightarrow t+, s \in D} Y(s, \omega) \quad (t \in \mathbb{T}), \quad \lim_{s \rightarrow t-, s \in D} Y(s, \omega) \quad (t > 0)$$

*exist in  $\mathbb{K}$ .*

*Proof.* Conditional expectation commutes with  $\Re$  and  $\Im$ , so  $\Re Y$  and  $\Im Y$  are real  $(\mathcal{F}_t)$ -martingales, hence submartingales. Apply Fact 2.22 to each and intersect the two sets of full measure; a limit in  $\mathbb{K}$  exists iff the limits of both parts do.  $\square$

*Remark 2.24* (Where the scalar field is felt, and where it is not). Corollary 2.23 is the *only* place in this manuscript where the passage from  $\mathbb{K}$  to  $\mathbb{R}$  is performed. It is worth isolating it, rather than repeating “apply this to  $\Re Y$  and  $\Im Y$ ” in every proof that needs it: a  $\mathbb{K}$ -valued submartingale does not exist, so the retreat is not a convenience but the only route, and it should happen once, at the point where the order is actually used. Everything downstream — Theorem 5.4 and with it the whole of Section 5 — is then stated and proved for  $\mathbb{K}$ -valued test processes with no further mention of the field.

*Fact 2.25* (Optional sampling; [7], Theorem 2.2.13, Remark 2.2.14, and Chapter 2, Problem 11; (T2b)). Let  $X$  be a right continuous  $(\mathcal{F}_t)$ -submartingale and let  $\tau_1, \tau_2$  be  $(\mathcal{F}_t)$ -stopping times. Then for each  $T > 0$ ,

$$E[X(\tau_2 \wedge T) \mid \mathcal{F}_{\tau_1}] \geq X(\tau_1 \wedge \tau_2 \wedge T). \quad (9)$$

If in addition  $\tau_2$  is a.s. finite,  $E[|X(\tau_2)|] < \infty$  and  $\lim_{T \rightarrow \infty} E[|X(T)|\mathbf{1}_{\{\tau_2 > T\}}] = 0$ , then  $E[X(\tau_2) \mid \mathcal{F}_{\tau_1}] \geq X(\tau_1 \wedge \tau_2)$ . Equality holds throughout if  $X$  is a martingale. In particular, a right continuous martingale with bounded increments satisfies the optional sampling identity at arbitrary a.s. finite stopping times.

*Fact 2.26* (Doob's inequalities; [7], Corollary 2.2.17; (T2b)). Let  $X$  be a right continuous martingale. Then for  $x > 0$ ,  $T > 0$  and  $\alpha > 1$ ,

$$P\left\{\sup_{t \leq T} |X(t)| \geq x\right\} \leq x^{-1} E[|X(T)|], \quad E\left[\sup_{t \leq T} |X(t)|^\alpha\right] \leq \left(\frac{\alpha}{\alpha-1}\right)^\alpha E[|X(T)|^\alpha]. \quad (10)$$

**Definition 2.27** ([7], Section 2.3). A  $\mathbb{K}$ -valued process  $X$  is an  $(\mathcal{F}_t)$ -local martingale if there exist  $(\mathcal{F}_t)$ -stopping times  $\tau_1 \leq \tau_2 \leq \dots$  with  $\tau_n \rightarrow \infty$  a.s. such that  $X^{\tau_n} = X(\cdot \wedge \tau_n)$  is an  $(\mathcal{F}_t)$ -martingale for each  $n$ . The sequence  $(\tau_n)$  is a *localizing sequence*. Local submartingales are defined analogously.

*Fact 2.28* ([7], Proposition 2.3.1). If  $X$  is a right continuous  $(\mathcal{F}_t)$ -local martingale and  $\tau$  is an  $(\mathcal{F}_t)$ -stopping time, then  $X^\tau = X(\cdot \wedge \tau)$  is an  $(\mathcal{F}_t)$ -local martingale.

## 2.6 From Chapter 3: weak convergence and the Skorokhod space

Everything in this subsection assumes (T3),  $\mathbb{T} = [0, \infty)$ , and is stated for it without further comment. The Skorokhod space itself is not here: it is the one prerequisite this manuscript builds rather than cites, and Section 3 builds it — under the hypothesis (T3'), which is weaker than (T3) and is the weakest under which the  $J_1$  topology exists at all (Theorem 3.17). Section 2.9 records which results inherit the restriction.

Throughout this subsection  $(S, d)$  is a metric space.

**Definition 2.29** ([7], Section 3.3).  $P_n \Rightarrow P$  in  $\mathcal{P}(S)$  (*weak convergence*) if  $\int f dP_n \rightarrow \int f dP$  for every  $f \in C_b(S)$ .  $S$ -valued random variables  $X_n$  converge in distribution to  $X$ , written  $X_n \Rightarrow X$ , if  $PX_n^{-1} \Rightarrow PX^{-1}$ .

*Fact 2.30* (Portmanteau; [7], Theorem 3.3.1). For  $\{P_n\} \subset \mathcal{P}(S)$  and  $P \in \mathcal{P}(S)$ , conditions (b)–(f) below are equivalent and are implied by (a); if  $S$  is separable, all six are equivalent.

- (a)  $\rho(P_n, P) \rightarrow 0$ , where  $\rho$  is the Prohorov metric;
- (b)  $P_n \Rightarrow P$ ;
- (c)  $\int f dP_n \rightarrow \int f dP$  for all uniformly continuous  $f \in C_b(S)$ ;
- (d)  $\liminf_n P_n(F) \leq P(F)$  for all closed  $F \subset S$ ;
- (e)  $\limsup_n P_n(G) \leq P(G)$  for all open  $G \subset S$ ;
- (f)  $P_n(A) \rightarrow P(A)$  for all  $P$ -continuity sets  $A \subset S$ , i.e. all  $A \in \mathcal{B}(S)$  with  $P(\partial A) = 0$ .

*Fact 2.31* (Continuous mapping theorem; [7], Corollary 3.1.9 and Corollary 3.3.3). Let  $(S, d)$  and  $(S', d')$  be separable metric spaces and  $h : S \rightarrow S'$  Borel measurable. Let  $C_h$  be the set of points at which  $h$  is continuous. If  $P_n \Rightarrow P$  and  $P(C_h) = 1$ , then  $P_n h^{-1} \Rightarrow P h^{-1}$ . Moreover, if  $X_n \Rightarrow X$  and  $d(X_n, Y_n) \rightarrow 0$  in probability, then  $Y_n \Rightarrow X$ .

*Fact 2.32* ([7], Theorems 3.1.7 and 3.1.8). If  $S$  is separable, then  $\mathcal{P}(S)$  is separable; if  $(S, d)$  is complete and separable, then so is  $\mathcal{P}(S)$  with the Prohorov metric. Moreover (*Skorokhod representation*), if  $S$  is separable and  $P_n \Rightarrow P$ , there are  $S$ -valued random variables  $X_n, X$  on a common probability space with laws  $P_n, P$  and  $X_n \rightarrow X$  a.s.

*Fact 2.33* (Prohorov; [7], Lemma 3.2.1 and Theorem 3.2.2). Call  $\mathcal{M} \subset \mathcal{P}(S)$  *tight* if for every  $\varepsilon > 0$  there is a compact  $K \subset S$  with  $\inf_{P \in \mathcal{M}} P(K) \geq 1 - \varepsilon$ . If  $(S, d)$  is complete and separable, then every  $P \in \mathcal{P}(S)$  is tight, and  $\mathcal{M} \subset \mathcal{P}(S)$  is tight if and only if it is relatively compact.

**Definition 2.34** ([7], Section 3.4 and Appendix 3). A sequence  $\{f_n\} \subset B(S)$  *converges boundedly and pointwise* to  $f \in B(S)$ , written  $\text{bp-lim}_n f_n = f$ , if  $\sup_n \|f_n\| < \infty$  and  $f_n(x) \rightarrow f(x)$  for every  $x \in S$ . A set  $M \subset B(S)$  is *bp-closed* if it is closed under bp-limits of sequences, and the *bp-closure* of  $M$  is the smallest bp-closed set containing it. The same terminology is used for subsets of  $B(S) \times B(S)$ , applying bp-lim in each coordinate.

*Fact 2.35* ([7], Lemma 3.4.1, Proposition 3.4.2, and Appendix 3, Proposition 3.1). The bp-closure of a subspace of  $B(S)$  is a subspace.

*Remark 2.36.* [7] prove more:  $C_b(S)$  is bp-dense in  $B(S)$ ; if  $S$  is separable there is a sequence of nonnegative functions in  $C_b(S)$  whose linear span is bp-dense in  $B(S)$ ; and on bounded sequences bp-convergence coincides with convergence in the weak\* topology induced by  $V(S)$ . None of this is used below. Fact 2.35 is invoked at exactly one place, Corollary 4.14, and there only in the form stated; see Remark 4.15 for why even that is optional.

**Definition 2.37** ([7], Section 3.4).  $M \subset C_b(S)$  is *separating* if, for  $P, Q \in \mathcal{P}(S)$ ,  $\int f dP = \int f dQ$  for all  $f \in M$  implies  $P = Q$ ; it is *convergence determining* if, for  $\{P_n\} \subset \mathcal{P}(S)$  and  $P \in \mathcal{P}(S)$ ,  $\int f dP_n \rightarrow \int f dP$  for all  $f \in M$  implies  $P_n \Rightarrow P$ .  $M$  *separates points* if for  $x \neq y$  there is  $f \in M$  with  $f(x) \neq f(y)$ , and *strongly separates points* if for every  $x \in S$  and  $\delta > 0$  there are  $h_1, \dots, h_k \in M$  with  $\inf_{y: d(y, x) \geq \delta} \max_{i \leq k} |h_i(y) - h_i(x)| > 0$ . Every convergence determining set is separating.

*Fact 2.38* ([7], Proposition 3.4.4). If  $(S, d)$  is separable, then the set of uniformly continuous  $f \in C_b(S)$  with bounded support is convergence determining. If  $S$  is in addition locally compact, then  $C_c(S)$  is convergence determining.

*Fact 2.39* (Stone–Weierstrass for separating classes; [7], Theorem 3.4.5). Let  $(S, d)$  be complete and separable and let  $M \subset C_b(S)$  be an algebra. If  $M$  separates points, then  $M$  is separating. If  $M$  strongly separates points, then  $M$  is convergence determining.

For  $\mathbb{K} = \mathbb{C}$  the algebra must in addition be closed under conjugation; this is the one place in the manuscript where the complex case is not automatic, and it is why Remark 4.5 lists it.

*Fact 2.40* ([7], Proposition 3.4.6 and Proposition 3.7.1). Let  $(S_k, d_k)$ ,  $k \geq 1$ , be separable and  $S = \prod_{k \geq 1} S_k$ . If  $M_k \subset C_b(S_k)$  are separating, then

$$M = \left\{ x \mapsto \prod_{k=1}^n f_k(x_k) : n \geq 1, f_k \in M_k \cup \{1\} \right\} \quad (11)$$

is separating on  $S$ ; if the  $(S_k, d_k)$  are complete and separable and the  $M_k$  are convergence determining, so is  $M$ . In particular the finite-dimensional distributions of a process determine its law. That  $\mathcal{B}(D(\mathbb{T}, E)) = \sigma(\pi_t)$ , so that the law on the path space is determined as well, is Theorem 3.13.

*Fact 2.41* ([7], Lemma 3.7.7). If  $X$  has sample paths in  $D_E[0, \infty)$ , then  $D(X) := \{t \geq 0 : P\{X(t) = X(t-)\} = 1\}$  has at most countable complement in  $[0, \infty)$ . In particular  $D(X)$  is dense.

*Fact 2.42* ([7], Theorem 3.7.8). Let  $E$  be separable and let  $X_n, X$  have sample paths in  $D_E[0, \infty)$ .

- (a) If  $X_n \Rightarrow X$  in  $D_E[0, \infty)$ , then  $(X_n(t_1), \dots, X_n(t_k)) \Rightarrow (X(t_1), \dots, X(t_k))$  for every finite  $\{t_1, \dots, t_k\} \subset D(X)$ .
- (b) If  $\{X_n\}$  is relatively compact and there is a dense  $D \subset [0, \infty)$  such that the convergence in (a) holds for every finite subset of  $D$ , then  $X_n \Rightarrow X$  in  $D_E[0, \infty)$ .

*Fact 2.43* (Relative compactness, I; [7], Theorem 3.9.1). Let  $(E, r)$  be complete and separable and let  $\{X_n\}$  be a family of processes with sample paths in  $D_E[0, \infty)$  satisfying the compact containment condition: for every  $\eta, T > 0$  there is a compact  $\Gamma_{\eta, T} \subset E$  with

$$\inf_n P\{X_n(t) \in \Gamma_{\eta, T} \text{ for } 0 \leq t \leq T\} \geq 1 - \eta. \quad (12)$$

Let  $H \subset C_b(E)$  be dense in the topology of uniform convergence on compact sets. Then  $\{X_n\}$  is relatively compact if and only if  $\{f \circ X_n\}$  is relatively compact in  $D_{\mathbb{R}}[0, \infty)$  for each  $f \in H$ .

*Fact 2.44* (Relative compactness, II; [7], Theorem 3.9.4). Let  $(E, r)$  be arbitrary and let  $X_n$  have sample paths in  $D_E[0, \infty)$ , adapted to filtrations  $(\mathcal{F}_t^n)$ . Let  $\mathcal{L}_n$  be the Banach space of real-valued  $(\mathcal{F}_t^n)$ -progressive processes — for  $\mathbb{K} = \mathbb{C}$  one applies the Fact to  $\Re f$  and  $\Im f$  and uses the  $\mathbb{R}^k$ -valued conclusion below — with  $\|Y\| = \sup_{t \geq 0} E[|Y(t)|] < \infty$  and put

$$\mathcal{A}_n = \left\{ (Y, Z) \in \mathcal{L}_n \times \mathcal{L}_n : Y(t) - \int_0^t Z(s) ds \text{ is an } (\mathcal{F}_t^n)\text{-martingale} \right\}. \quad (13)$$

Let  $C_a \subset C_b(E)$  be a subalgebra and let  $\mathcal{D}_a$  be the collection of  $f \in C_b(E)$  such that for every  $\varepsilon, T > 0$  there is  $(Y_n, Z_n) \in \mathcal{A}_n$  with

$$\sup_n E \left[ \sup_{t \in [0, T] \cap \mathbb{Q}} |Y_n(t) - f(X_n(t))| \right] < \varepsilon, \quad \sup_n E[\|Z_n\|_{p, T}] < \infty \text{ for some } p \in (1, \infty]. \quad (14)$$

If  $C_a$  is contained in the sup-norm closure of  $\mathcal{D}_a$ , then  $\{f \circ X_n\}$  is relatively compact for each  $f \in C_a$ ; more generally  $\{(f_1, \dots, f_k) \circ X_n\}$  is relatively compact in  $D_{\mathbb{R}^k}[0, \infty)$  for all  $f_1, \dots, f_k \in C_a$ .

## 2.7 From the Appendixes of [7]

*Fact 2.45* (Uniform integrability; [7], Appendix 2). A family  $\{X_i\}$  of  $\mathbb{K}$ -valued random variables is *uniformly integrable* if  $\sup_i E[|X_i|] < \infty$  and for every  $\varepsilon > 0$  there is  $\delta > 0$  such that  $P(A) < \delta$  implies  $|E[X_i \mathbf{1}_A]| < \varepsilon$  for all  $i$ . Equivalently,  $\lim_{N \rightarrow \infty} \sup_i E[|X_i| - |X_i| \wedge N] = 0$ ; equivalently, there is an increasing convex  $\varphi$  on  $[0, \infty)$  with  $\varphi(x)/x \rightarrow \infty$  and  $\sup_i E[\varphi(|X_i|)] < \infty$ . If  $X_n \Rightarrow X$  and  $\{X_n\}$  is uniformly integrable, then  $E[X_n] \rightarrow E[X]$ .

*Fact 2.46* (Monotone class theorem; [7], Appendix 4). Let  $H$  be a linear space of bounded real-valued functions on  $\Omega$  containing the constants and closed under bounded monotone limits, and let  $K \subset H$  be closed under multiplication. Then  $H$  contains every bounded  $\sigma(K)$ -measurable function.

*Fact 2.47* (Kolmogorov extension; [7], Theorem 4.1.1; (T0) + (E1)). Assume (E1), let  $\nu \in \mathcal{P}(E)$  and let  $(\mu_t)_{t \in \mathbb{T}}$  be a family of probability kernels on  $E$  with  $\mu_0(x, \cdot) = \delta_x$ , with  $(t, x) \mapsto \mu_t(x, \Gamma)$  jointly measurable for every  $\Gamma \in \mathcal{E}$ , and with the Chapman–Kolmogorov property  $\mu_{s+t}(x, \cdot) = \int \mu_s(x, dx') \mu_t(x', \cdot)$  (for (T4)); more generally, let the finite-dimensional distributions built from  $\nu$  and  $(\mu_t)$  be a projective family. Then there is a probability measure on  $E^{\mathbb{T}}$  under which the coordinate process is Markov with initial distribution  $\nu$  and transition kernels  $(\mu_t)$ .

Standard Borel is what the theorem needs: on a general measurable space the projective limit may be empty.

The Mathlib situation as of v4.33.1 is worth recording precisely, because it is easy to assume more than is there. Present are

- the predicate `IsProjectiveLimit`, with its uniqueness statement;
- the additive content `projectiveFamilyContent` built from a projective family;
- the cylinders with closed compact bases, on which the classical proof runs;
- the Ionescu–Tulcea theorem `Kernel.traj`, which needs no topological hypothesis but only a *sequential* index  $\mathbb{N}$ .

Absent is the Kolmogorov extension theorem itself, for an arbitrary index set under topological assumptions — the source file says so in as many words. So Fact 2.47 is a genuine prerequisite to be built, unless  $\mathbb{T}$  is countable and can be enumerated, in which case Ionescu–Tulcea suffices.

*Fact 2.48* (Conditional determination by separating sets; [7], Chapter 3, Problem 7). Assume (E1) and (E2), let  $\mathcal{G} \subset \mathcal{F}$  be a sub- $\sigma$ -field, let  $M \subset C_b(E)$  be separating, and let  $U, V$  be  $E$ -valued random variables with  $V$   $\mathcal{G}$ -measurable. If  $E[f(U) \mid \mathcal{G}] = f(V)$  a.s. for every  $f \in M$ , then  $P\{U = V\} = 1$ .

*Remark 2.49* ([7], Chapter 3, Problem 7 — a proof). This is stated as a *problem* in [7] and is used in the last step of Theorem 5.4, so we prove it. The point worth making is that the countability the argument needs comes from the *state space*, not from  $M$ : no countable subfamily of  $M$  is assumed, and none is available in general.

*Step 1:  $M$  separating gives conditional equality in law.* Let  $G \in \mathcal{G}$  with  $P(G) > 0$ . For  $f \in M$ , the hypothesis gives  $E[\mathbf{1}_G f(U)] = E[\mathbf{1}_G E[f(U) \mid \mathcal{G}]] = E[\mathbf{1}_G f(V)]$ . The two probability measures

$$\nu_1(B) = \frac{P(\{U \in B\} \cap G)}{P(G)}, \quad \nu_2(B) = \frac{P(\{V \in B\} \cap G)}{P(G)}, \quad B \in \mathcal{E},$$

therefore satisfy  $\int f d\nu_1 = \int f d\nu_2$  for every  $f \in M$ , and since  $M$  is separating,  $\nu_1 = \nu_2$ . Hence

$$P(\{U \in B\} \cap G) = P(\{V \in B\} \cap G) \quad \text{for all } B \in \mathcal{E}, G \in \mathcal{G}, \quad (15)$$

i.e.  $P(U \in B \mid \mathcal{G}) = \mathbf{1}_B(V)$  a.s. for each fixed  $B$ .

*Step 2: one null set, using (E1).* By (E1) the  $\sigma$ -field  $\mathcal{E}$  is countably generated; let  $\mathcal{E}_0$  be a countable algebra generating it. Again by (E1) there is a regular conditional distribution  $\mu(\omega, \cdot)$  of  $U$  given  $\mathcal{G}$ . Applying (15) to the countably many  $B \in \mathcal{E}_0$  yields a single set of full measure off which  $\mu(\omega, B) = \mathbf{1}_B(V(\omega))$  for all  $B \in \mathcal{E}_0$  simultaneously. Two probability measures agreeing on a generating algebra agree, so  $\mu(\omega, \cdot) = \delta_{V(\omega)}$  a.s.

*Step 3.* The diagonal  $\{(x, \omega) : x = V(\omega)\}$  is  $\mathcal{E} \otimes \mathcal{G}$ -measurable, again because  $\mathcal{E}$  is countably generated and separates points, so  $P\{U = V\} = E[\mu(\cdot, \{V(\cdot)\})] = E[\delta_V(\{V\})] = 1$ .

Only (E2) is used, in Step 1, to make sense of “ $M \subset C_b(E)$  separating”; all the measure theory is (E1).

## 2.8 Where the prerequisites are used

| Prerequisite    | Used in                                   |
|-----------------|-------------------------------------------|
| Definition 2.14 | Proposition 4.19, Fact 2.16               |
| Fact 2.16       | Remark 4.20                               |
| Fact 2.19       | Theorem 6.51(b), Corollary 7.18           |
| Fact 2.21       | Theorem 5.4                               |
| Fact 2.22       | Corollary 2.23 only                       |
| Corollary 2.23  | Theorem 5.4                               |
| Fact 2.25       | Theorem 6.27                              |
| Fact 2.26       | Remark 8.38 (via Fact 2.44)               |
| Fact 2.28       | Remarks 6.53, 8.45                        |
| Fact 2.30, 2.31 | Lemma 8.34, Theorem 8.44                  |
| Fact 2.32, 2.33 | Remark 8.38                               |
| Fact 2.35       | Corollary 4.14 only                       |
| Fact 2.38       | Remark 5.9                                |
| Fact 2.39       | Remark 8.38, Proposition 8.7              |
| Fact 2.47       | Theorem 8.5                               |
| Theorem 3.13    | Theorem 6.23, Corollary 6.52, Example 4.4 |
| Theorem 3.12    | Setting 4.1, Theorem 8.44                 |
| Theorem 3.15    | Fact 2.43                                 |
| Fact 2.41       | Lemma 8.34, Theorem 8.44                  |
| Fact 2.42       | Remark 8.38                               |
| Fact 2.43, 2.44 | Remark 8.38                               |
| Fact 2.45       | Theorem 8.44                              |
| Fact 2.46       | Proposition 4.9, Example 4.4              |
| Fact 2.48       | Theorem 5.4                               |

## 2.9 Which result needs which bundle

The counterpart of Section 2.8 for Definitions 2.1, 2.3 and 2.7. “Fibred” in the state column means that the result is indifferent to a time-dependent state space in the sense of Definition 2.10; Remark 2.11 is the full audit, and the entries here only mark the two places where the fibration is not merely allowed but does work. “Clock” means Definition 2.3 with no further property; “reflective” is as defined there, and  $\iota$  is the convention of Definition 2.3, shown only where it matters. Bold entries mark the places where the weakest bundle in its column does *not* suffice, for a mathematical reason rather than for convenience. An  $\iota$ -entry means that the result is stated in that convention; only Section 5 *needs* its one — there an atom at  $t$  must lie inside the compensator up to  $t$  — whereas the  $\iota = \mathbf{p}$  entries of the duality table mark the convention in which the proof is shortest, the conclusion being convention-agnostic by Proposition 7.8. Example 8.50 needs  $\iota = \mathbf{o}$  for a third reason, explained in Remark 8.51.

| Result                                           | Index | State | Clock |
|--------------------------------------------------|-------|-------|-------|
| Definition 2.18, Definition 2.20                 | (T0)  | (E0)  | clock |
| Facts 2.19, 2.25, 2.26                           | (T2b) | (E0)  | —     |
| Facts 2.21, 2.22 (regularization)                | (T2b) | (E2)  | —     |
| Section 2.6 entire (Skorokhod, weak convergence) | (T3)  | (E3)  | —     |

| Result                                                                                              | Index        | State              | Clock                       |
|-----------------------------------------------------------------------------------------------------|--------------|--------------------|-----------------------------|
| Setting 4.1, Definitions 4.2, 4.3                                                                   | (T0)         | (E0)               | —                           |
| Definition 4.6, Proposition 4.9,<br>Definition 4.16, (4)                                            | (T0)         | (E0), fibred       | clock                       |
| Lemma 4.17, Proposition 4.19                                                                        | (T3)         | (E0)               | Lebesgue                    |
| Definition 4.21, Example 4.4                                                                        | (T2b)        | (E2)               | —                           |
| Definition 5.2, Theorem 5.4<br>(càdlàg)                                                             | (T2b)        | (E2)               | $\iota = \mathbf{o}$        |
| Theorem 5.6 ([7] 4.3.6)                                                                             | (T2b)        | (E2)               | clock, $\iota = \mathbf{o}$ |
| Lemma 6.2 (mixtures)                                                                                | (T0)         | (E0)               | —                           |
| Lemma 6.3 (disintegration)                                                                          | (T0)         | (E1)               | —                           |
| Lemma 2.2 ( $r + \cdot$ is an order em-<br>bedding)                                                 | (T0) + (T4)  | —                  | —                           |
| Definition 6.14, Lemma 6.17<br>(restart)                                                            | (T0) + (T4)  | (E0)               | clock                       |
| Theorem 6.23(a) (Markov)                                                                            | (T0) + (T4)  | (E0)               | clock                       |
| Definition 6.11, Proposition 6.13<br>(uniqueness)                                                   | (T2a)        | (E0)               | —                           |
| Lemma 6.22 (one-dimensional<br>laws suffice)                                                        | (T0) + (T4)  | (E0)               | clock                       |
| Theorem 6.23(b) (uniqueness, via<br>both)                                                           | (T2a) + (T4) | (E0)               | clock                       |
| Theorem 6.27 (strong Markov)                                                                        | (T2b) + (T4) | (E2) on $\Sigma E$ | clock                       |
| Definition 6.30, Lemmas 6.32,<br>6.34, 6.37, Theorem 6.38 (local<br>theory; (E1) for Lemma 6.32(c)) | (T2b) + (T4) | (E0)               | clock                       |
| Definition 6.42, Definition 6.43,<br>Lemma 6.44, Theorem 6.45 (local<br>uniqueness)                 | (T2b)        | (E0)               | clock                       |
| Corollary 6.46 (its Markovian<br>case)                                                              | (T2b) + (T4) | (E0)               | clock                       |

| Result                                                                  | Index                                        | State               | Clock                              |
|-------------------------------------------------------------------------|----------------------------------------------|---------------------|------------------------------------|
| Lemma 7.1 (chain identity)                                              | (T0)                                         | (E0)                | any                                |
| Proposition 7.3, Corollary 7.16                                         | (T2a) + (T4)                                 | (E0)                | Haar, $\iota = p$                  |
| Corollary 7.4 (atomless clocks)                                         | (T3)                                         | (E0)                | <b>atomless</b> , $\iota = p$      |
| Proposition 7.8 (atomic duality)                                        | (T0)                                         | (E0)                | atomic, atoms a chain, any $\iota$ |
| Lemma 7.11, Theorem 7.12                                                | (T3)                                         | (E0)                | Lebesgue                           |
| Corollary 7.20 (uniqueness via duality)                                 | (T3)                                         | (E2)                | Lebesgue                           |
| Lemma 8.3, Proposition 8.4 (the kernel layer)                           | (T0) + (T4)                                  | (E1), fibred        | shift invariant                    |
| Theorem 8.5 (existence from a dual)                                     | (T2a) + (T4)                                 | (E1), fibred        | shift invariant                    |
| Proposition 8.7 (checking (D3))                                         | (T2a) + (T4)                                 | (E3), $E_1$ compact | shift invariant                    |
| Setting 8.20, Theorem 8.23 (jump processes)                             | (T3)                                         | (E0)                | Lebesgue                           |
| Lemma 6.8 (path lift; vacuous, see Remark 6.9)                          | (T0)                                         | (E0), fibred        | —                                  |
| Setting 8.69, Theorem 8.71, Example 8.74 (path-dependent rates, Hawkes) | (T3)                                         | (E0)                | Lebesgue                           |
| Theorem 8.76 (Hawkes to Volterra)                                       | (T3)                                         | (E3)                | Lebesgue                           |
| Proposition 8.24, Remark 8.26                                           | (T3)                                         | (E2)                | Lebesgue                           |
| Corollary 8.32 (from SDEs)                                              | (T3)                                         | $E = \mathbb{R}^d$  | Lebesgue                           |
| Theorem 8.40 (abstract convergence)                                     | (T0), or (T2b) for $D \subsetneq \mathbb{T}$ | (E3)                | —                                  |
| Lemma 8.54, Theorem 8.55 (augmented form)                               | as Theorem 8.40                              | (E3)                | —                                  |
| Proposition 8.56 (atomic clocks)                                        | (T3)                                         | (E2)                | any, atoms allowed                 |
| Theorem 8.48 (clock change)                                             | (T3)                                         | (E3)                | $q^n$ and $q$                      |
| Lemma 8.34, Theorem 8.44                                                | (T3)                                         | (E3)                | Lebesgue                           |

Two readings of this table are worth recording. First, the partially ordered case — and with it the multiparameter index  $\mathbb{T} = \mathbb{R}_+^d$  — survives in Section 4, in the Markov half of Section 6, in the abstract convergence theorem and in the *kernel layer* of Section 8.2, and dies everywhere else; the bold index entries are where it dies for a *mathematical* reason rather than for want of a Skorokhod space, and Remarks 6.25, 7.10 and 8.8(ii) say which — the middle one only for want of a proof, not because of a counterexample. No *existence* result survives a partial order in full: the duality route comes closest, its obstruction sitting in the very last step, the Kolmogorov extension over antichains. Second, (T1') — the metric lattice of [7], Section 2.8, which the first draft of this manuscript treated as the base assumption — does not occur at all: every statement involving a stopping time already assumes (T2b), under which the lattice structure is automatic. See Remark 2.6.

### 3 The Skorokhod space

Every result of Sections 4, 6.1 and 7 is stated without a path space, and Theorem 8.40 is stated with an abstract Polish one. The Skorokhod space enters at exactly one place, the convergence theory of Section 8.5, and this section builds it.

Two things are decided here that the literature does not decide, because it does not have to. The first is the class of time indices: Billingsley and [7] treat  $[0, 1]$  and  $[0, \infty)$ , in that order and as two constructions, and neither asks what the construction actually needs. The answer, Theorem 3.17, is a hypothesis rather than a list, it covers the two classical cases and the lattices of Section 8.8 at once, and it is where the graduation of Definition 2.1 stops. The second is that  $[0, T]$  and  $[0, \infty)$  are not separate: properness supplies a compact exhaustion, the metric is localized along it, and a bounded index is the case in which the localization is eventually constant.

#### 3.1 The setting

**Definition 3.1** (The time index for  $J_1$ ). We add one bundle to Definition 2.1:

(T2a), and  $\mathbb{T}$  carries a metric  $\varrho$  which induces the order topology, (T3')  
which is *additive along the order*,

$$\varrho(s, u) = \varrho(s, t) + \varrho(t, u) \quad (s \leq t \leq u),$$

and for which closed balls are compact.

Fix once and for all a base point  $t_0 \in \mathbb{T}$  and write  $B_m := \{t : \varrho(t_0, t) \leq m\}$  for  $m \in \mathbb{N}$ , so that  $B_1 \subset B_2 \subset \dots$  and  $\bigcup_m B_m = \mathbb{T}$ . Each  $B_m$  is compact and linearly ordered, hence has a least and a greatest element, and we write

$$\tau_m : \mathbb{T} \rightarrow B_m, \quad \tau_m(t) := \min(\max(t, \min B_m), \max B_m)$$

for the clamp onto it.

*Remark 3.2* (What (T3') is). Additivity along the order makes  $t \mapsto \varrho(t_0, t)$ , signed by the order, an order isomorphism and an isometry onto its image, so (T3') says precisely that  $\mathbb{T}$  is a closed subset of  $\mathbb{R}$  with the induced metric, up to the canonical identification. It is stated as a hypothesis and not as that sentence because the development below never uses the identification, and because as a hypothesis it is a typeclass: the four instances

$$\mathbb{R}, \quad \mathbb{R}_+, \quad [0, T], \quad h\mathbb{Z} \quad (h > 0)$$

and every closed subset of any of them — the closed range of a subordinator, say — satisfy it, and no one of them is the definition. Note in particular that (T3') asks for no least element: the two-sided index  $\mathbb{T} = \mathbb{R}$  and the shifted index  $[-1, \infty)$  are admitted, which (T0) is not. A formalization declares

```
variable {ι : Type*} [LinearOrder ι] [MetricSpace ι]
[OrderTopology ι] [AdditiveDist ι] [ProperSpace ι]
```

of which only `AdditiveDist` has to be introduced; see Remark 3.19.

The base point is immaterial: two base points at distance  $c$  give exhaustions interleaving after  $\lceil c \rceil$  steps, hence uniformly equivalent metrics in Definition 3.9. Nothing below depends on the choice, and no statement mentions it.

**Definition 3.3** (Paths). Assume (T3') and let  $(E, r)$  be Polish. Write

$$D(\mathbb{T}, E) := \{f : \mathbb{T} \rightarrow E : f \text{ is càdlàg in the sense of Definition 2.18}\}.$$

For  $m \in \mathbb{N}$  put  $f^{(m)} := f \circ \tau_m$ , a path which is constant outside  $B_m$  and takes the same values as  $f$  on it.

### 3.2 Time changes

**Definition 3.4** (Time changes and their norm). As in Section 2.6,  $\Lambda$  denotes the set of time changes, here the bi-Lipschitz order isomorphisms  $\lambda : \mathbb{T} \rightarrow \mathbb{T}$ . Their norm *norm* is

$$\gamma(\lambda) := \log \max(\text{Lip}(\lambda), \text{Lip}(\lambda^{-1})) \in [0, \infty], \quad (16)$$

and  $\gamma_m(\lambda)$  denotes the same quantity computed with the Lipschitz constants of  $\lambda$  and  $\lambda^{-1}$  restricted to  $B_m$ .

*Remark 3.5* (Why this and not Billingsley's). For  $\mathbb{T} = [0, 1]$  and  $\lambda$  fixing the endpoints,

$$\sup_{s < t} \left| \log \frac{\lambda t - \lambda s}{t - s} \right| = \max(\log \text{Lip}(\lambda), \log \text{Lip}(\lambda^{-1})) = \gamma(\lambda),$$

because for monotone  $\lambda$  the supremum of the difference quotients is  $\text{Lip}(\lambda)$ , the substitution  $u = \lambda s$ ,  $v = \lambda t$  turns their infimum into  $\text{Lip}(\lambda^{-1})^{-1}$ , and  $\lambda(1) - \lambda(0) = 1 - 0$  forces the infimum below and the supremum above 1. So Definition 3.4 is Billingsley's  $\gamma^\circ$  written without differences and quotients, and Lemma 3.6 is what the identity buys: two facts that are proved by hand in the one-dimensional account become the two standard properties of Lipschitz constants.

**Lemma 3.6** (The norm is a length function). For  $\lambda, \mu \in \Lambda$  and every  $m$ ,

$$\gamma(\text{id}) = 0, \quad \gamma(\lambda^{-1}) = \gamma(\lambda), \quad \gamma(\lambda \circ \mu) \leq \gamma(\lambda) + \gamma(\mu),$$

and the same three with  $\gamma_m$  throughout.

*Proof.* The first is  $\log 1$ , the second holds by construction, and the third is  $\text{Lip}(\lambda \circ \mu) \leq \text{Lip}(\lambda) \text{Lip}(\mu)$  together with the same bound for the inverse. No property of  $\mathbb{T}$  beyond being a metric space is used.  $\square$

**Lemma 3.7** (The norm controls the displacement). Assume (T3'). For  $\lambda \in \Lambda$  and  $t \in B_m$ ,

$$\varrho(\lambda t, t) \leq (e^{\gamma_m(\lambda)} - 1) m' \quad \text{where } m' := \sup_{t \in B_m} \varrho(t_0, t) \leq m.$$

In particular  $\sup_{t \in B_m} \varrho(\lambda t, t) \rightarrow 0$  as  $\gamma_m(\lambda) \rightarrow 0$ , for each fixed  $m$ .

*Proof.* Time changes fix  $t_0$  — an order isomorphism of  $\mathbb{T}$  fixes any point determined by the order, and if  $\mathbb{T}$  has no such point one replaces  $t_0$  by its image throughout, which changes the exhaustion by a bounded amount. With  $\ell(t) := \varrho(t_0, t)$ , additivity gives  $\varrho(s, t) = |\ell(s) - \ell(t)|$  for all  $s, t$  on the same side of  $t_0$ , and  $\ell(\lambda t) = \varrho(\lambda t_0, \lambda t) \leq \text{Lip}(\lambda) \ell(t) = e^{\gamma_m(\lambda)} \ell(t)$  on  $B_m$ ; the same bound for  $\lambda^{-1}$  gives the reverse inequality, and subtracting  $\ell(t)$  yields the claim.  $\square$

*Remark 3.8* (Where additivity is used, and only there). Lemma 3.7 is the only place in this section at which additivity along the order enters, and Theorem 3.17 shows it cannot be dispensed with. Everything before it holds for an arbitrary metric on  $\mathbb{T}$ , and everything after it uses only the conclusion of the lemma.

### 3.3 The metric

**Definition 3.9** (The localized Skorokhod metric). For  $f, g \in D(\mathbb{T}, E)$  and  $m \in \mathbb{N}$  set

$$d_m(f, g) := \inf_{\lambda \in \Lambda, \gamma_m(\lambda) < \infty} \max \left( \gamma_m(\lambda), \sup_{t \in \mathbb{T}} r(f^{(m)}(\lambda t), g^{(m)}(t)) \right),$$

$$d(f, g) := \sum_{m \geq 1} 2^{-m} \min(1, d_m(f, g)).$$

The supremum in  $d_m$  is effectively over the compact set  $B_m$ , since  $f^{(m)}$  and  $g^{(m)}$  are constant outside it, and is therefore finite.

*Remark 3.10* (The two classical cases). For  $\mathbb{T} = [0, 1]$  the exhaustion is eventually constant,  $\tau_m = \text{id}$  for  $m \geq 1$ , and  $d$  is a positive multiple of Billingsley's  $d^o$ . For  $\mathbb{T} = \mathbb{R}_+$  the clamp is  $\tau_m(t) = t \wedge m$  and  $d$  is the metric of [7], Section 3.5, with the integral over the truncation level replaced by a series; in particular no truncation of the *values* is used, so  $E$  need not be a vector space. The unification is what the clamp buys, and the clamp exists because properness makes  $B_m$  compact and a compact linearly ordered space has a greatest element.

**Theorem 3.11** ( $D(\mathbb{T}, E)$  is a metric space). Assume (T3') and let  $(E, r)$  be Polish. Then  $d$  is a metric on  $D(\mathbb{T}, E)$ .

The one point that is not routine is the identity of indiscernibles. If  $d(f, g) = 0$  then  $d_m(f, g) = 0$  for every  $m$ , so by Lemma 3.7 the time changes realizing the infimum converge to the identity uniformly on  $B_m$ , and  $f = g$  on  $B_m$  follows by the argument of [10, Ch. 3, §12] at the continuity points and by right continuity elsewhere. Since the  $B_m$  exhaust  $\mathbb{T}$ ,  $f = g$ . No quotient is taken and no flat extension of the paths is needed: on a bounded index the latter is a way of compensating for a fixed horizon, and the exhaustion already does that work.

**Theorem 3.12** (Polish). Assume (T3') and let  $(E, r)$  be Polish. Then  $(D(\mathbb{T}, E), d)$  is Polish.

Separability is witnessed by the countable family of step paths whose jump times lie in a countable dense subset  $D \subset \mathbb{T}$  — available by (T2b), which (T3') implies — and whose values lie in a countable dense subset of  $E$ . Completeness is the infinite-composition argument of Billingsley's Theorem 12.2, run on each  $B_m$  using Lemmas 3.6 and 3.7 and assembled along the exhaustion; it is the most expensive proof in this section, and Remark 3.19 records where it exists already.

### 3.4 The Borel field

**Theorem 3.13** (Borel equals cylinder). Assume (T3'), let  $(E, r)$  be Polish and let  $D \subset \mathbb{T}$  be countable and dense. Then  $\pi_D : D(\mathbb{T}, E) \rightarrow E^D$ ,  $f \mapsto (f(t))_{t \in D}$ , is a measurable embedding, and consequently

$$\mathcal{B}(D(\mathbb{T}, E)) = \sigma(\pi_t : t \in \mathbb{T}) = \sigma(\pi_t : t \in D).$$

A probability measure on  $D(\mathbb{T}, E)$  is therefore determined by its finite-dimensional distributions along  $D$ , and a map  $X : \Omega \rightarrow D(\mathbb{T}, E)$  is measurable if and only if  $\pi_t \circ X$  is measurable for every  $t \in D$ .

The content is that  $\pi_D$  is measurable, and it is not the triviality it looks like:  $\pi_t$  is *discontinuous* at every path having a jump at  $t$ , so measurability of the coordinates is a theorem and not a remark. It is proved by exhibiting  $\pi_t$  as a pointwise limit of  $d$ -continuous functionals, for which two routes are available; the one that does not require  $E$  to carry a linear structure is to use  $t \in D$  only and to invoke right continuity, which is what the statement is arranged to allow. Injectivity of  $\pi_D$  is right continuity again, and a measurable injection between standard Borel spaces is a measurable embedding.

Theorem 3.13 is what Proposition 4.9 consumes, and it is the one result of this section that the abstract theory of Sections 4–7 actually uses.

### 3.5 Compactness

**Definition 3.14** (The càdlàg modulus). For  $f \in D(\mathbb{T}, E)$ ,  $m \in \mathbb{N}$  and  $\delta > 0$  let  $w'_m(f, \delta)$  be the infimum of those  $\varepsilon \geq 0$  for which there is a finite subdivision  $\min B_m = t_0 < t_1 < \dots < t_n = \max B_m$  of  $B_m$  with  $\varrho(t_{i-1}, t_i) > \delta$  for every  $i$  and  $r(f(s), f(t_{i-1})) \leq \varepsilon$  for all  $s \in [t_{i-1}, t_i)$  and every  $i$ .

**Theorem 3.15** (Relative compactness). Assume (T3') and let  $(E, r)$  be Polish. A set  $A \subset D(\mathbb{T}, E)$  is relatively compact if and only if for every  $m$

- (i)  $\{f(t) : f \in A, t \in B_m\}$  is relatively compact in  $E$ , and
- (ii)  $\lim_{\delta \downarrow 0} \sup_{f \in A} w'_m(f, \delta) = 0$ .

*Remark 3.16* (Is the modulus Borel? An open point). [7], Lemma 3.6.2 asserts, and this manuscript cited until Theorem 3.15 replaced it, that  $w'(\cdot, \delta, T)$  is Borel measurable, and Billingsley says the same. The Lean formalization of [24] does *not* prove it and records why: for a fixed subdivision the functional is a countable supremum, by right continuity, hence Borel; but  $w'$  is an infimum over an uncountable family of subdivisions, and the reduction to subdivisions with rational nodes fails, because a subdivision has to place its nodes at the jumps of  $f$  and those need not be rational. The sublevel sets are then projections of a jointly Borel condition along an uncountable Polish parameter, i.e. analytic and universally measurable, and no Borel witness is produced by that route.

The two statements are not formally in conflict — a route can fail without the conclusion failing, and monotonicity in  $\delta$  leaves room for a limit argument that the naive reduction does not see — but the point should be settled rather than inherited. It is recorded here because it is the only place in this section where a classical assertion and a machine-checked development disagree about what has been proved.

Nothing below depends on the answer. Every use made of Theorem 3.15 bounds the *outer* measure of  $\{f : w'_m(f, \delta) \geq \varepsilon\}$  by that of a cylinder-measurable superset of large increments, and outer measures are monotone and countably subadditive, which is all that is used. A formalization should take that route and leave the measurability question aside.

### 3.6 Where the graduation stops

**Theorem 3.17** ((T3') is sharp). Let  $\mathbb{T}$  satisfy (T2a) and carry a metric  $\varrho$  inducing the order topology, and let  $\Lambda$  and  $\gamma$  be as in Definition 3.4.

- (a) If  $\varrho$  is additive along the order, then  $\mathbb{T}$  is order isomorphic and isometric to a subset of  $\mathbb{R}$ , closed if in addition closed balls are compact.
- (b) Without additivity the conclusion of Lemma 3.7 may fail, and then  $d$  of Definition 3.9 is not a metric: there are  $\mathbb{T}$  and  $f \neq g$  with  $d(f, g) = 0$ .

For (a), additivity says exactly that  $\ell(t) = \pm \varrho(t_0, t)$ , signed by the order, is additive on order intervals, hence an order embedding and an isometry; properness makes the image closed. For (b) it is enough to leave the linear order: on a rooted tree with the tree metric — additive along the order, by the computation  $\varrho(s, u) = \ell(u) - \ell(s) = \varrho(s, t) + \varrho(t, u)$  for  $s \leq t \leq u$  — a root-fixing isometry exchanging two isomorphic branches has  $\gamma(\lambda) = 0$  while  $\sup_t \varrho(\lambda t, t)$  is of the order of the diameter, and two paths differing by that exchange are at distance zero. Restricting  $\Lambda$  to time changes keeping each point on its own branch restores Lemma 3.7, at the price of a group depending on  $\mathbb{T}$ .

*Remark 3.18* (The three walls around this one). Theorem 3.17 is one of four places where the graduation of Definition 2.1 ends, and it is the last of them; Remark 10.1 tabulates all four. Briefly: the martingale property needs no more than (T0); stopping times need  $\vee$  and hence (T1), by Remark 2.6(ii); càdlàg paths need  $\{s : s \leq t\}$  to be a chain, which a tree order supplies and a multiparameter index does not; and  $J_1$  needs (T3'). None of this constrains Section 8.10, whose pseudo-paths need a measure on  $\mathbb{T}$  and a metric on  $E$  and no order at all.

### 3.7 Notes for the formalization

*Remark 3.19* (What exists). Mathlib has none of this section as of August 2026; the string “Skorokhod” occurs in the library only in `docs/1000.yaml`, where the representation and embedding theorems are listed as open and the space is not listed at all. Three developments downstream of it are relevant.

*Càdlàg paths.* RemyDegenne/`brownian-motion` defines `IsCadlag` on `[TopologicalSpace  $\iota$ ] [Preorder  $\iota$ ]`, which is (T2b), together with jump sets and the usual closure properties. Definition 3.3 should reuse it rather than restate it.

*The one-dimensional case, complete.* The repository accompanying [24] constructs  $D([0, 1], \mathbb{R})$  with Billingsley’s metric, the `MetricSpace`, `CompleteSpace` and `PolishSpace` instances, the modulus and the compactness criterion, Aldous’ tightness criterion by the original two-scale argument, and the measurable embedding of Theorem 3.13. It fixes the horizon to  $[0, 1]$  and the state space to  $\mathbb{R}$ , carries boundedness as a structure field, and encodes paths as functions on  $\mathbb{R}$  which are flat outside  $[0, 1]$  — three decisions that Definition 3.9 does not need, the exhaustion doing their work. It also records, and leaves open, the failure of measurability of the modulus, by Remark 3.16. Two repositories accompany [24] and they are licensed differently: `type-D-asymptotics`, which holds the paper sources and eight Lean files, carries the Apache 2.0 licence, whereas `type-D-asymptotics-lean`, which holds the six Skorokhod files described here, carries none. So this material may not be reused without the author’s agreement, and the natural request is that the licence of the first repository be extended to the second.

*The new typeclass.* Of the five hypotheses of (T3') in the form of Remark 3.2, four are Mathlib’s already — `LinearOrder`, `MetricSpace`, `OrderTopology`, `ProperSpace` — and only additivity has to be added:

```
class AdditiveDist (α : Type*) [LinearOrder α]
  [PseudoMetricSpace α] : Prop where
  dist_add : ∀ {s t u : α}, s ≤ t → t ≤ u →
    dist s u = dist s t + dist t u
```

Its instances for  $\mathbb{R}$  and  $\mathbb{Z}$  are four lines each, and the instance for an arbitrary subset of an `AdditiveDist` type is definitional, so the intervals, the lattices and the closed range of a subordinator come for free. `ProperSpace` is inherited by closed subsets through the existing

`ProperSpace.of_isClosed`, and `OrderTopology` through the existing instance for order-connected subsets — but *not* for a lattice embedded in  $\mathbb{R}$ , which is not order connected: there the missing instance has to be proved, or the lattice used as a type of its own, where it is present. That gap is the concrete reason to state (T3') as a typeclass rather than to define  $\mathbb{T}$  as a subset of  $\mathbb{R}$ .

## 4 Martingale problems

### 4.1 The abstract (local) martingale problem

**Setting 4.1** (after [2], Section 3.1; (T0) + (E0)). Let  $(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{t \in \mathbb{T}})$  be a filtered space carrying a family  $\mathcal{X}$  of  $\mathbb{K}$ -valued adapted processes (the *test processes*), right continuous if  $\mathbb{T}$  satisfies (T2b). Let  $(E, \mathcal{E})$  be a measurable space and let  $F \subset E^{\mathbb{T}}$  be a *path space*: a set of maps  $\mathbb{T} \rightarrow E$  equipped with a  $\sigma$ -field  $\mathcal{S}$  for which the coordinates  $\pi_t : F \rightarrow E$  are measurable and which they generate,

$$\mathcal{S} = \sigma(\pi_t : t \in \mathbb{T}). \quad (17)$$

Let  $X = (X_t)_{t \in \mathbb{T}}$  be an  $E$ -valued process with  $X(\omega) \in F$  for every  $\omega$  and  $\omega \mapsto X(\omega)$  measurable  $\mathcal{F}/\mathcal{S}$ .

Under (E3) we take  $F$  to carry a Polish topology and  $\mathcal{S} = \mathcal{B}(F)$ ; (17) is then a theorem rather than an assumption for  $F = D_E[0, \infty)$  (Theorem 3.13), and it is what makes the finite-dimensional distributions determine the law.

**Definition 4.2** ([2], Definition 3.2; (T0) + (E0)). A probability measure  $P$  on  $(\Omega, \mathcal{F})$  is a *solution of the martingale problem*  $\text{MP}(\mathcal{X})$  if every  $Y \in \mathcal{X}$  is an  $(\mathbb{F}, P)$ -martingale, and a *solution of the local martingale problem* if every  $Y \in \mathcal{X}$  is an  $(\mathbb{F}, P)$ -local martingale. The corresponding sets of solutions are denoted  $\mathcal{M}(\mathcal{X})$  and  $\mathcal{M}_{\text{loc}}(\mathcal{X})$ .

**Definition 4.3** ([2], Definitions 3.3 and 3.5; (T0) + (E0)).  $\mathcal{X}$  is *canonical* for  $X$  if for every  $t \in \mathbb{T}$  there is a set  $\mathcal{X}_t^\circ$  of  $\mathcal{B}(F)/\mathcal{B}(\mathbb{K})$ -measurable maps  $Y_t^\circ : F \rightarrow \mathbb{K}$  such that every  $Y \in \mathcal{X}$  satisfies  $Y_t = Y_t^\circ(X)$  for some  $Y_t^\circ \in \mathcal{X}_t^\circ$ . A family  $\mathcal{Z}^\circ = \{Z_t^\circ : t \in \mathbb{T}\}$  of sets of bounded measurable maps  $F \rightarrow \mathbb{R}$  — *real-valued, whatever  $\mathbb{K}$  is* — is *determining* for  $\mathcal{X}$  if for every probability measure  $P$ , every  $Y \in \mathcal{X}$  and all  $s \leq t$  with  $Y_s, Y_t \in L^1(P)$ ,

$$E^P[Y_t Z_s^\circ(X)] = E^P[Y_s Z_s^\circ(X)] \quad \text{for all } Z_s^\circ \in \mathcal{Z}_s^\circ \quad \implies \quad E^P[Y_t | \mathcal{F}_s] = Y_s \quad P\text{-a.s.}$$

**Example 4.4** ([2], Example 3.6(i); (T2b) + (E2)). If  $\mathcal{F}_t = \sigma(X_s : s \leq t)$ ,  $F = D_E[0, \infty)$  or  $F = C_E[0, \infty)$ , and  $D \subset \mathbb{T}$  is dense, then

$$\mathcal{Z}_t^\circ = \left\{ \prod_{i=1}^n h_i(X_{t_i}) : n \in \mathbb{N}, t_1, \dots, t_n \in D \cap [0, t], h_i \in C_b(E) \right\}$$

is determining. This uses  $\mathcal{B}(F) = \sigma(X_t : t \in \mathbb{T})$  (Theorem 3.13) and the monotone class theorem.

*Remark 4.5* (The complex case, carried out). The scalar field  $\mathbb{K}$  of Section 2.1 is  $\mathbb{R}$  or  $\mathbb{C}$ , and every statement of this manuscript is to be read over  $\mathbb{K}$ . This costs almost nothing, and it is worth being precise about the “almost”.

*Why it is nearly free.* A  $\mathbb{K}$ -valued process is a martingale iff its real and imaginary parts are, and every argument in Sections 4, 6 and 7 is  $\mathbb{K}$ -linear in the test process: the mixture, disintegration and restart lemmas integrate a linear identity, the chain identity of Lemma 7.1 is an exact telescoping, and the convergence arguments of Section 8 use only

$|Y_r^\circ|$  and Vitali's theorem, which hold over  $\mathbb{K}$ . Note in Definition 4.3 that the determining sets  $Z_s^\circ$  stay *real-valued* whatever  $\mathbb{K}$  is: testing  $E[(Y_t - Y_s)Z_s^\circ] = 0$  against real  $Z_s^\circ$  already separates the real and imaginary parts of the increment, so nothing is gained by complexifying them and one bookkeeping nuisance is avoided.

*The four places that need  $\mathbb{K} = \mathbb{R}$ .*

- (i) *Order.* There is no such thing as a  $\mathbb{K}$ -valued submartingale, so the *prerequisites* whose content is an inequality stay over  $\mathbb{R}$ : Fact 2.22, and Fact 2.46, whose closure under *bounded monotone limits* presupposes an order. This does *not* propagate to the statements that use them. The submartingale regularization is used only through Corollary 2.23, which is  $\mathbb{K}$ -valued and performs the retreat to  $\Re Y$  and  $\Im Y$  once and for all (Remark 2.24); and the monotone class theorem is used only on the *test variables*, which are real by design — see Definition 4.3 and the  $W$  of Proposition 4.9 — because they multiply a  $\mathbb{K}$ -valued increment, and real ones already separate its two parts. Fact 2.21, by contrast, needs no order at all, being a statement about limits in a metric space, and Fact 2.44 is stated over  $\mathbb{R}$  only because [7] state it so.

The one place where an inequality is the *subject* rather than a tool is Remark 6.19, and there  $\mathbb{K} = \mathbb{R}$  is not a retreat but the hypothesis.

- (ii) *Stone–Weierstrass.* Fact 2.39 needs a complex algebra to be closed under conjugation. This is used in Remark 8.38 and Proposition 8.7.
- (iii) *Positivity.* Proposition 8.7 obtains the kernels of (D3) from a *positive* linear functional, which presupposes an order on the values. Section 8.2 may take  $H$  complex, but the route to (D3) then has to be a transform characterization rather than Riesz–Markov — which is exactly what Remark 8.12 says anyway.
- (iv) *Section 2.4.* Dissipative operators and the Hille–Yosida theory are stated over  $\mathbb{R}$ ; the complex versions differ and are not needed, this section being optional context.

*Why it is worth having.* The transform literature works with  $e^{i\langle u, X_t \rangle}$  rather than with real test functions, and the affine and affine Volterra theory of [14] is stated that way throughout. Theorem 7.12 in particular needs no change at all: its hypotheses (58)–(59) are already phrased with moduli, so complex  $f, g, h, \alpha, \beta$  are covered as they stand, and the complex form is the one in which duality is usually applied.

## 4.2 The Markovian martingale problem

**Definition 4.6** ((T0) + (E0) + clock). Fix a clock  $q$  on  $\mathbb{T}$  and a convention  $\iota \in \{o, p\}$  (Definition 2.3). Let  $A \subset M(E) \times M(E)$  and let  $X$  be a jointly measurable  $E$ -valued process on  $(\Omega, \mathcal{F}, P)$ . Put

$$\mathcal{X}_A^\iota(X) = \left\{ (f(X(t)) - \int_{\langle 0, t \rangle_\iota} g(X(s)) q(ds))_{t \in \mathbb{T}} : (f, g) \in A \right\}, \quad (18)$$

and write  $\mathcal{X}_A(X)$  when the convention is fixed by the context or immaterial.  $X$  is a *solution of the martingale problem for  $A$*  if every element of  $\mathcal{X}_A(X)$  is a martingale with respect to  $(\mathcal{F}_t^X)_{t \in \mathbb{T}}$  of (4). If  $(\mathcal{G}_t)$  is a filtration with  $\mathcal{G}_t \supset \mathcal{F}_t^X$  for all  $t$  and every element of  $\mathcal{X}_A(X)$  is a  $(\mathcal{G}_t)$ -martingale,  $X$  is a *solution of the martingale problem for  $A$  with respect to  $(\mathcal{G}_t)$* . Given  $\mu \in \mathcal{P}(E)$ ,  $X$  solves the martingale problem for  $(A, \mu)$  if in addition  $PX(0)^{-1} = \mu$ . Replacing “martingale” by “local martingale” throughout defines the *local martingale problem for  $A$* .

A probability measure  $P \in \mathcal{P}(D_E[0, \infty))$  is called a solution of the  $D_E[0, \infty)$  *martingale problem* for  $A$ , respectively for  $(A, \mu)$ , if the coordinate process  $X(t, \omega) = \omega(t)$  on the space  $(D_E[0, \infty), \mathcal{B}(D_E[0, \infty)), P)$  is one.

Definition 4.6 is the special case  $\mathcal{X} = \mathcal{X}_A(X)$  of Definition 4.2.

*Remark 4.7* (Provenance of Definition 4.2). The abstract formulation is older than [2]. [10], Definition III.1.3 already attaches a martingale problem to a family of optional processes on a filtered space carrying no measure, and does so with *local* martingales as the default rather than as a variant. Two features of their formulation are worth importing. First, the initial condition is a measure  $P_{\mathcal{H}}$  on an *initial*  $\sigma$ -field  $\mathcal{H} \subset \mathcal{F}_0$ , not merely an initial distribution  $\mu = P\pi_0^{-1}$ ; everything below goes through with  $\sigma(\pi_0)$  replaced by  $\mathcal{H}$ , and Lemma 6.3 in particular. Second, [10], Remark III.1.6 notes explicitly that the elements of that family need be neither càdlàg nor even adapted — which is the observation that Setting 5.1 needs and that [2]’s standing assumptions exclude.

Before the characterization we record two measure-theoretic points that the proof uses and that are easy to pass over. Both concern the compensator alone.

**Lemma 4.8** (The compensator is adapted, and Fubini is available; (T0) + (E0) + clock). *Let  $X$  be jointly measurable,  $g \in M(E)$ ,  $\iota \in \{\text{o}, \text{p}\}$  and  $u \in \mathbb{T}$ , and write  $I_u^g = \int_{\langle 0, u \rangle_\iota} g(X(v)) q(dv)$  on the set  $N_u^g = \{\int_{\langle 0, u \rangle_\iota} |g(X(v))| q(dv) < \infty\}$ .*

- (a)  $N_u^g \in \mathcal{F}_u^X$  and  $I_u^g$  is  $^*\mathcal{F}_u^X$ -measurable on  $N_u^g$ . If  $g \in B(E)$  then  $N_u^g = \Omega$ ,  $I_u^g$  is one of the generators in (4), and  $|I_u^g| \leq \|g\| q(\mathbb{T}_{\leq u})$ .
- (b) The restriction of  $q$  to  $\mathbb{T}_{\leq u}$  is a finite measure. Consequently finite products of such restrictions are well defined and Fubini’s theorem is available on them, without any  $\sigma$ -finiteness assumption on  $q$  itself.

*Proof.* (a) For  $g \in B(E)$  the bound is immediate from  $q(\langle 0, u \rangle_\iota) \leq q(\mathbb{T}_{\leq u}) < \infty$  (Definition 2.3), and  $I_u^g$  appears in (4) by construction. For general  $g$  put  $g_n = (g \wedge n) \vee (-n)$ . Then  $I_u^{g_n} \uparrow \int_{\langle 0, u \rangle_\iota} |g(X(v))| q(dv)$  pointwise, so  $N_u^g$  is the set where that increasing limit is finite, hence lies in  $\sigma(I_u^{g_n} : n) \subset ^*\mathcal{F}_u^X$ ; and on  $N_u^g$ ,  $I_u^{g_n} \rightarrow I_u^g$  pointwise by dominated convergence, so  $I_u^g$  is a pointwise limit of  $^*\mathcal{F}_u^X$ -measurable functions.

(b) Immediate from  $q(\mathbb{T}_{\leq u}) < \infty$ . Note that  $q$  itself need *not* be  $\sigma$ -finite —  $\mathbb{T}$  need not be a countable union of down-sets — so the restriction is what makes the product measure legitimate.  $\square$

**Proposition 4.9** ([7], (3.4); (T0) + (E0) + clock, and (E2) for the variant with  $C_b(E)$ ). *Let  $X$  be jointly measurable, fix  $\iota \in \{\text{o}, \text{p}\}$ , and let  $(f, g) \in A$  be such that  $Y_t^{f,g} = f(X(t)) - \int_{\langle 0, t \rangle_\iota} g(X(s)) q(ds)$  is defined and integrable for every  $t \in \mathbb{T}$ ; by Lemma 4.8(a) it is then automatically  $^*\mathcal{F}_t^X$ -adapted. Then  $Y^{f,g}$  is a  $(^*\mathcal{F}_t^X)$ -martingale if and only if*

$$E \left[ \left( f(X(t)) - f(X(s)) - \int_{\langle s, t \rangle_\iota} g(X(u)) q(du) \right) \prod_{k=1}^n h_k(X(t_k)) \right] = 0 \quad (19)$$

for all  $s \leq t$  in  $\mathbb{T}$ , all  $n \geq 0$ , all  $t_1, \dots, t_n \in \mathbb{T}_{\leq s}$  and all  $h_1, \dots, h_n \in B(E)$  (equivalently,  $h_k \in C_b(E)$ ). Consequently  $X$  solves the martingale problem for  $A$  if and only if (19) holds for every  $(f, g) \in A$ .

*Remark 4.10* (The times need not form a chain). Earlier versions of this manuscript quantified (19) over *chains*  $t_1 \leq \dots \leq t_n \leq s$ , following [7], (3.4), where  $\mathbb{T} = \mathbb{R}_+$  is linearly ordered and the two formulations coincide. On a preordered index they do not, and the chain version is too weak: the proof below tests against the generators of  $\mathcal{F}_s^X = \sigma(X_u : u \in \mathbb{T}_{\leq s})$ , and these are products over *arbitrary* finite subsets of  $\mathbb{T}_{\leq s}$ . On  $\mathbb{T} = \mathbb{R}_+^2$  the times  $(1, 0)$  and  $(0, 1)$  lie in  $\mathbb{T}_{\leq (1,1)}$  without being comparable, and no chain reaches the product  $h_1(X_{(1,0)})h_2(X_{(0,1)})$ . This is the same phenomenon as in Remark 6.25, but with the opposite moral: here it is the *hypothesis* that has to be strengthened, and once it is, nothing is lost.

*Proof.* Throughout,  $s \leq t$  are fixed,  $Y = Y^{f,g}$ , and we abbreviate

$$I_u^h = \int_{\langle 0, u \rangle_\iota} h(X(v)) q(dv), \quad u \in \mathbb{T}, \quad h \in B(E),$$

so that  ${}^*\mathcal{F}_s^X = \sigma(X(u), I_u^h : u \in \mathbb{T}_{\leq s}, h \in B(E))$  by (4). Note  $|I_u^h| \leq \|h\| q(\mathbb{T}_{\leq u}) < \infty$ , so each  $I_u^h$  is *bounded*, by Definition 2.3. By (3),

$$Y_t - Y_s = f(X(t)) - f(X(s)) - \int_{\langle s, t \rangle_\iota} g(X(u)) q(du), \quad (20)$$

which is the bracket in (19); it lies in  $L^1$  by hypothesis.

*Necessity.* Each  $\prod_k h_k(X(t_k))$  with  $t_k \leq s$  is bounded and  $\mathcal{F}_s^X \subset {}^*\mathcal{F}_s^X$ -measurable, so the martingale property gives  $E[(Y_t - Y_s) \prod_k h_k(X(t_k))] = 0$ , which by (20) is (19).

*Sufficiency.* Assume (19) and put

$$\mathcal{H} = \{W : \Omega \rightarrow \mathbb{R} \text{ bounded, } {}^*\mathcal{F}_s^X\text{-measurable, } E[(Y_t - Y_s)W] = 0\}.$$

We must show that  $\mathcal{H}$  contains *all* such  $W$ , for then  $E[Y_t | {}^*\mathcal{F}_s^X] = Y_s$ .

*Step 1:  $\mathcal{H}$  is a monotone vector space.* It is a vector space because  $W \mapsto E[(Y_t - Y_s)W]$  is linear, and it contains the constants by (19) with  $n = 0$ . If  $W_m \in \mathcal{H}$ ,  $0 \leq W_m \uparrow W$  with  $W$  bounded, then  $(Y_t - Y_s)W_m \rightarrow (Y_t - Y_s)W$  pointwise and  $|(Y_t - Y_s)W_m| \leq \|W\| |Y_t - Y_s| \in L^1$ , so dominated convergence gives  $W \in \mathcal{H}$ .

*Step 2: a multiplicative generating class.* Let  $\mathcal{K}$  be the set of finite products

$$Z = \prod_{k=1}^n h_k(X(t_k)) \cdot \prod_{j=1}^m I_{u_j}^{g_j}, \quad n, m \geq 0, \quad t_k, u_j \in \mathbb{T}_{\leq s}, \quad h_k, g_j \in B(E). \quad (21)$$

Every such  $Z$  is bounded, by Lemma 4.8(a), and  $\mathcal{K}$  is closed under multiplication, being a set of finite products of the two kinds of factor. It generates  ${}^*\mathcal{F}_s^X$ : the inclusion  $\sigma(\mathcal{K}) \subset {}^*\mathcal{F}_s^X$  holds factor by factor, and the reverse holds because the cases  $(n, m) = (1, 0)$  and  $(0, 1)$  of (21) are the generators  $h(X(t_1))$  and  $I_{u_1}^{g_1}$  of (4) themselves.

*Step 3:  $\mathcal{K} \subset \mathcal{H}$ .* This is the only step with content, and it is Fubini rather than an approximation. Fix  $Z$  as in (21) and write the product of the  $m$  integrals as one integral over the product:

$$\prod_{j=1}^m I_{u_j}^{g_j} = \int_R \prod_{j=1}^m g_j(X(v_j)) q^{\otimes m}(dv), \quad R = \langle 0, u_1 \rangle_\iota \times \dots \times \langle 0, u_m \rangle_\iota, \quad v = (v_1, \dots, v_m),$$

The map  $(v, \omega) \mapsto (Y_t - Y_s)(\omega) \prod_k h_k(X(t_k, \omega)) \prod_j g_j(X(v_j, \omega))$  is  $\mathcal{T}^{\otimes m} \otimes \mathcal{F}$ -measurable, because  $X$  is jointly measurable, and it is dominated by  $\|h\| \|g\| |Y_t - Y_s|$ , which is integrable

for  $P \otimes q^{\otimes m}$  since  $Y_t - Y_s \in L^1(P)$  and  $q^{\otimes m}(R) \leq \prod_j q(\mathbb{T}_{\leq u_j}) < \infty$ . The product measure is legitimate and Fubini applies by Lemma 4.8(b), giving

$$E[(Y_t - Y_s)Z] = \int_R E\left[(Y_t - Y_s) \prod_{k=1}^n h_k(X(t_k)) \prod_{j=1}^m g_j(X(v_j))\right] q^{\otimes m}(dv).$$

For each  $v \in R$  the times  $t_1, \dots, t_n$  and  $v_1, \dots, v_m$  all lie in  $\mathbb{T}_{\leq s}$ , and the functions  $h_1, \dots, h_n$  and  $g_1, \dots, g_m$  are bounded; so the inner expectation vanishes by (19), applied with  $n+m$  in place of  $n$ . Hence  $E[(Y_t - Y_s)Z] = 0$ .

*This is the step for which the times in (19) must be allowed to form an arbitrary finite subset of  $\mathbb{T}_{\leq s}$ : the  $v_j$  are produced by the integration and cannot be ordered into a chain with the  $t_k$ .*

*Step 4: conclusion.* By Steps 1–3 and the functional monotone class theorem (Fact 2.46),  $\mathcal{H}$  contains every bounded  $\sigma(\mathcal{K})$ -measurable function, i.e. every bounded  ${}^*\mathcal{F}_s^X$ -measurable function. Since  $s \leq t$  were arbitrary,  $Y$  is a  $({}^*\mathcal{F}_t^X)$ -martingale.

*The variant with  $h_k \in C_b(E)$ , and it alone needs (E2).* On a separable metrizable  $E$  the class  $C_b(E)$  generates  $\mathcal{B}(E)$  — for a closed  $C$  the functions  $x \mapsto (1 - m d(x, C))^+$  increase boundedly to  $\mathbf{1}_C$  — so the class (21) built from  $C_b(E)$  has the same generated  $\sigma$ -field, and Steps 1–4 are unchanged. Passing from  $h_k \in C_b(E)$  back to  $h_k \in B(E)$  in (19) is again Step 1 applied one coordinate at a time.

*What was used.* Only the preorder, through  $\mathbb{T}_{\leq u} \subset \mathbb{T}_{\leq s}$  for  $u \leq s$  and through the additivity (3); the finiteness  $q(\mathbb{T}_{\leq u}) < \infty$ , twice, for the boundedness of  $I_u^h$  and for Fubini; and the joint measurability of  $X$ . In particular no topology, no linear order, no property of  $q$  beyond Definition 2.3, and no choice of  $\iota$ : the argument uses only that  $\langle \cdot, \cdot \rangle_\iota$  is additive, which (3) gives for both conventions.  $\square$

*Remark 4.11.* Proposition 4.9 says that being a solution of the martingale problem for  $A$  is a statement about the *finite-dimensional distributions* of  $X$  only. Three consequences are used repeatedly:

- (a) any measurable modification of a solution is a solution;
- (b) a solution for  $A$  is a solution for the linear span  $A_S$ , by linearity of (19) in  $(f, g)$ , and for a closure of  $A_S$  that Lemma 4.12 makes precise;
- (c) if  $A^{(1)} \subset A^{(2)}$ , every solution for  $A^{(2)}$  is one for  $A^{(1)}$ .

The closure in (b) deserves to be stated carefully, because [7] take it in the bounded pointwise sense and that is more than the argument uses.

**Lemma 4.12** (Closure along a solution; (T0) + (E0) + clock). *Let  $X$  solve the martingale problem for  $A$  with respect to  $(\mathcal{G}_t)$ , and let  $(f, g) \in M(E) \times M(E)$  be such that  $Y^{f, g}$  is integrable. Suppose there are  $(f_n, g_n) \in A_S$  with, for all  $s \leq t$  in  $\mathbb{T}$ ,*

$$\begin{aligned} f_n(X(t)) &\longrightarrow f(X(t)) && \text{in } L^1(P), \\ \int_{\langle s, t \rangle_\iota} g_n(X(u)) q(du) &\longrightarrow \int_{\langle s, t \rangle_\iota} g(X(u)) q(du) && \text{in } L^1(P). \end{aligned} \tag{22}$$

*Then  $X$  solves the martingale problem for  $A \cup \{(f, g)\}$ .*

*Proof.* We argue directly, which avoids Proposition 4.9 and gives the statement for the filtration  $(\mathcal{G}_t)$  actually at hand rather than only for  ${}^*\mathcal{F}^X$ .

*Adaptedness.*  $f(X(t))$  is  $\mathcal{F}_t^X$ -measurable, and  $\int_{\langle 0, t \rangle_\iota} g(X(u))q(du)$  is  ${}^*\mathcal{F}_t^X$ -measurable by Lemma 4.8(a); both  $\sigma$ -fields are contained in  $\mathcal{G}_t$ , so  $Y^{f,g}$  is  $(\mathcal{G}_t)$ -adapted.

*The martingale property.* Since  $Y^{f',g'}$  is a  $(\mathcal{G}_t)$ -martingale for every  $(f', g') \in A$  and  $(f', g') \mapsto Y^{f',g'}$  is linear, the same holds for every  $(f_n, g_n) \in A_S$ ; note also that  $Y^{f_n, g_n}$  is then integrable, being a finite linear combination of integrable processes. Let  $s \leq t$  and let  $W$  be bounded and  $\mathcal{G}_s$ -measurable. Then  $E[(Y_t^{f_n, g_n} - Y_s^{f_n, g_n})W] = 0$ , and by (3)

$$Y_t^{f_n, g_n} - Y_s^{f_n, g_n} = (f_n(X(t)) - f_n(X(s))) - \int_{\langle s, t \rangle_\iota} g_n(X(u))q(du).$$

Each of the three terms converges in  $L^1(P)$  to its counterpart for  $(f, g)$ , by (22); multiplying by the bounded  $W$  preserves  $L^1$  convergence, so  $E[(Y_t^{f,g} - Y_s^{f,g})W] = 0$ . As  $W$  was arbitrary,  $E[Y_t^{f,g} | \mathcal{G}_s] = Y_s^{f,g}$ .  $\square$

*Remark 4.13.* The direct argument is worth the two extra lines. Routing through Proposition 4.9 would prove the conclusion only for  ${}^*\mathcal{F}^X$ , since that is the filtration the finite-dimensional characterization speaks of; the  $L^1$  limit, by contrast, is taken against an arbitrary bounded  $\mathcal{G}_s$ -measurable test variable and therefore delivers the martingale property for whichever filtration the hypothesis was stated with.

**Corollary 4.14** (Bounded pointwise convergence suffices; (T0) + (E0) + clock). *Let  $A \subset B(E) \times B(E)$  and let  $(f_n, g_n) \in A_S$  with  $\text{bp-lim}_n(f_n, g_n) = (f, g)$ . Then (22) holds — for every solution  $X$ , every  $P$  and every clock. Consequently a solution for  $A$  is a solution for the bp-closure of  $A_S$ , and if the bp-closures of  $A_S^{(1)}$  and  $A_S^{(2)}$  agree then the two martingale problems have the same solutions ([7], Proposition 4.3.1).*

*Proof.*  $|f_n(X(t))| \leq \sup_n \|f_n\| < \infty$  and  $f_n(X(t)) \rightarrow f(X(t))$  pointwise, so the first convergence is dominated convergence under the probability measure  $P$ . For the second,  $|g_n(X(u))| \leq \sup_n \|g_n\|$  and  $q(\langle s, t \rangle_\iota) \leq q(\mathbb{T}_{\leq t}) < \infty$  by Definition 2.3, so  $\int_{\langle s, t \rangle_\iota} |g_n - g|(X(u))q(du) \rightarrow 0$  by dominated convergence under  $q \otimes P$  — legitimate because  $(u, \omega) \mapsto g_n(X(u, \omega))$  is jointly measurable, the constant  $\sup_n \|g_n\|$  is  $q \otimes P$ -integrable on  $\langle s, t \rangle_\iota \times \Omega$ , and  $|\int (g_n - g)(X(u))q(du)| \leq \int |g_n - g|(X(u))q(du)$ . Both dominating constants depend on the sequence alone. Finally  $f$  and  $g$  are again bounded, so  $Y^{f,g}$  is integrable and Lemma 4.12 applies.  $\square$

*Remark 4.15* (What bounded pointwise convergence is, and is not, for). Corollary 4.14 isolates the rôle of the bp-topology, and it is worth stating, because the notion is easy to over-read.

*It is the strongest hypothesis that is independent of  $X$ .* The dominating constants in the proof depend on  $(f_n, g_n)$  alone — not on the law of  $X$ , not on the clock, not on the convention. That is exactly what makes Corollary 4.14 a statement about *operators*: two operators with the same bp-closure have the same solutions, with no solution mentioned. Lemma 4.12 is instead a statement about a *given* solution — weaker as an assertion about  $A$ , and stronger as a tool.

*It costs generality that the rest of Section 4 does not spend.* Definition 4.6 admits  $A \subset M(E) \times M(E)$ , and Proposition 4.9 assumes that  $Y^{f,g}$  is integrable, not that  $f$  and  $g$  are bounded. The bp-closure lives in  $B(E) \times B(E)$  and therefore silently narrows this; Lemma 4.12 does not.

*For the formalization* the difference is real. Lemma 4.12 is dominated convergence and nothing else. The bp apparatus, Definition 2.34 and Fact 2.35, is needed only for Corollary 4.14, and there only its first assertion.

**Definition 4.16** ([7], Section 4.4; (T0) + (E0) + clock). *Uniqueness* holds for the martingale problem for  $(A, \mu)$  if any two solutions have the same finite-dimensional distributions. The martingale problem for  $(A, \mu)$  is *well-posed* if a solution exists and uniqueness holds; the martingale problem for  $A$  is well-posed if it is well-posed for every  $\mu \in \mathcal{P}(E)$ . It is *well-posed in  $D_E[0, \infty)$*  (resp.  $C_E[0, \infty)$ ) if there is a unique  $P \in \mathcal{P}(D_E[0, \infty))$  (resp.  $\mathcal{P}(C_E[0, \infty))$ ) solving the  $D_E[0, \infty)$  (resp.  $C_E[0, \infty)$ ) martingale problem for  $(A, \mu)$ . Well-posedness in  $D_E[0, \infty)$  does not imply well-posedness; Theorem 5.6 shows that this gap is rare.

The following two results connect Definition 4.6 with the operator-theoretic notions of Section 2.4.

**Lemma 4.17** ([7], Lemma 4.3.2; (T3)). *Let  $X$  be a measurable process,  $\mathcal{G}_t \supset \mathcal{F}_t^X$ , and  $f, g \in B(E)$ . Then for fixed  $\lambda \in \mathbb{R}$  the process  $f(X(t)) - \int_0^t g(X(s)) ds$  is a  $(\mathcal{G}_t)$ -martingale if and only if*

$$e^{-\lambda t} f(X(t)) + \int_0^t e^{-\lambda u} (\lambda f(X(u)) - g(X(u))) du \quad (23)$$

*is a  $(\mathcal{G}_t)$ -martingale. Consequently, if  $(f, g) \in A$ ,  $\lambda > 0$  and  $X$  solves the martingale problem for  $A$  with respect to  $(\mathcal{G}_t)$ , then*

$$f(X(t)) = E \left[ \int_0^\infty e^{-\lambda s} (\lambda f(X(t+s)) - g(X(t+s))) ds \mid \mathcal{G}_t \right]. \quad (24)$$

*Proof.* Write  $Y_t = f(X(t)) - \int_0^t g(X(s)) ds$  and let  $Z_t$  denote (23).

*Step 1: an identity.* We claim

$$Z_t = e^{-\lambda t} Y_t + \lambda \int_0^t e^{-\lambda u} Y_u du. \quad (25)$$

Indeed, substituting  $Y_u = f(X(u)) - \int_0^u g(X(v)) dv$  into the right-hand side and using Fubini,

$$\lambda \int_0^t e^{-\lambda u} \int_0^u g(X(v)) dv du = \lambda \int_0^t g(X(v)) \int_v^t e^{-\lambda u} du dv = \int_0^t g(X(v)) (e^{-\lambda v} - e^{-\lambda t}) dv,$$

so the two terms involving  $g$  combine to  $-e^{-\lambda t} \int_0^t g dv - \int_0^t e^{-\lambda v} g dv + e^{-\lambda t} \int_0^t g dv = -\int_0^t e^{-\lambda v} g(X(v)) dv$ , and (25) is (23). Solving (25) for  $Y$  — put  $W_t = \int_0^t e^{-\lambda u} Y_u du$ , so that  $Z = W' + \lambda W$  and hence  $(e^{\lambda t} W)' = e^{\lambda t} Z_t$  — gives the inverse relation

$$Y_t = e^{\lambda t} Z_t - \lambda \int_0^t e^{\lambda u} Z_u du. \quad (26)$$

*Step 2: either relation transports the martingale property.* Suppose  $Y$  is a  $(\mathcal{G}_t)$ -martingale. For  $s \leq t$  we use (25) together with Fubini for conditional expectations, which is legitimate because  $\sup_{u \leq t} E|Y_u| < \infty$ :

$$\begin{aligned} E[Z_t \mid \mathcal{G}_s] &= e^{-\lambda t} Y_s + \lambda \int_0^s e^{-\lambda u} Y_u du + \lambda \int_s^t e^{-\lambda u} Y_s du \\ &= e^{-\lambda t} Y_s + \lambda \int_0^s e^{-\lambda u} Y_u du + Y_s (e^{-\lambda s} - e^{-\lambda t}) = Z_s. \end{aligned}$$

The converse is the same computation applied to (26), with  $e^{\lambda u}$  in place of  $e^{-\lambda u}$  and  $-\lambda$  in place of  $\lambda$ . This proves the equivalence.

*Step 3: the resolvent representation.* Let  $(f, g) \in A$  with  $f, g$  bounded and  $\lambda > 0$ , and let  $X$  solve the martingale problem for  $A$  with respect to  $(\mathcal{G}_t)$ , so that  $Z$  is a martingale by Step 2. For  $r > 0$ ,

$$Z_t = E[Z_{t+r} \mid \mathcal{G}_t] = E\left[e^{-\lambda(t+r)} f(X(t+r)) + \int_0^{t+r} e^{-\lambda u} (\lambda f - g)(X(u)) du \mid \mathcal{G}_t\right].$$

As  $r \rightarrow \infty$  the first term is bounded by  $e^{-\lambda(t+r)} \|f\| \rightarrow 0$  and the second increases to an integral dominated by  $\lambda^{-1} \|\lambda f - g\|$ , so dominated convergence for conditional expectations gives

$$Z_t = E\left[\int_0^\infty e^{-\lambda u} (\lambda f - g)(X(u)) du \mid \mathcal{G}_t\right].$$

Subtracting  $\int_0^t e^{-\lambda u} (\lambda f - g)(X(u)) du$ , which is  $\mathcal{G}_t$ -measurable, from both sides leaves  $e^{-\lambda t} f(X(t)) = E[\int_t^\infty e^{-\lambda u} (\lambda f - g)(X(u)) du \mid \mathcal{G}_t]$ , and the substitution  $u = t + s$  gives (24).  $\square$

*Remark 4.18.* Step 1 is the only place where  $\mathbb{T} = \mathbb{R}_+$  and  $q = \text{Lebesgue measure}$  are used, and they are used twice: for the exponential  $e^{-\lambda t}$ , which is the solution of  $\phi' = -\lambda\phi$ , and for the Fubini rearrangement. On  $\mathbb{T} = \mathbb{N}_0$  with  $q$  counting measure the analogue holds with  $e^{-\lambda}$  replaced by  $(1 + \lambda)^{-1}$  and the integrals by sums; we do not need it, and record only that Lemma 4.17 is (T3) for a reason of calculus, not of probability.

**Proposition 4.19** ([7], Proposition 4.3.5; (T3)). *Let  $A$  be a linear subset of  $B(E) \times B(E)$ . If for each  $x \in E$  there exists a solution  $X_x$  of the martingale problem for  $(A, \delta_x)$ , then  $A$  is dissipative in the sense of Definition 2.14.*

*Proof.* Given  $(f, g) \in A$  and  $\lambda > 0$ , (24) with  $t = 0$  gives  $f(x) = E[\int_0^\infty e^{-\lambda s} (\lambda f(X_x(s)) - g(X_x(s))) ds]$ , whence  $|f(x)| \leq \int_0^\infty e^{-\lambda s} \|\lambda f - g\| ds = \lambda^{-1} \|\lambda f - g\|$  for every  $x \in E$ .  $\square$

*Remark 4.20.* Only dissipative operators arise as generators of Markov processes. Indeed, if  $X$  is a Markov process with measurable transition semigroup  $\{T(t)\}$  on  $B(E)$  and full generator  $\hat{A}$  as in Definition 2.15, then  $X$  solves the martingale problem for  $\hat{A}$  ([7], Proposition 4.1.7), and  $\hat{A}$  is dissipative by Fact 2.16. Proposition 4.19 is the converse direction.

**Definition 4.21** ((T2b) + (E2)). Let  $D \subset \mathbb{T}$  be the countable dense set of (T2b). A process  $X$  satisfies the *compact containment condition* if for all  $\varepsilon > 0$  and  $T \in \mathbb{T}$  there is a compact  $K_{\varepsilon, T} \subset E$  with

$$P\{X(t) \in K_{\varepsilon, T} \text{ for all } t \in \mathbb{T}_{\leq T} \cap D\} > 1 - \varepsilon. \quad (27)$$

A family  $\{X_n\}$  satisfies the compact containment condition if (12) holds.

## 5 Sample path regularity: existence of a càdlàg modification

### 5.1 An abstract regularization theorem

[7] state Theorem 5.6 below for the Markovian martingale problem, that is, for the family  $\mathcal{X}_A(X)$  of (18). Their proof, however, never uses the operator  $A$ , nor the special filtration (4), nor the particular form  $\int_0^t g(X_s) ds$  of the compensator. What it uses is only that  $f(X)$  splits, for every  $f$  in a sufficiently rich class, into a martingale plus a process that is regular in time. Isolating that hypothesis gives a statement in the abstract setting of Definition 4.2, from which [7], Theorem 4.3.6 follows in one line.

[2] contain no counterpart of this result, and cannot: their Setting 4.1 *presupposes* a Polish path space  $F$  and right continuous test processes, which is exactly what a regularization theorem has to produce. Theorem 5.4 is therefore the statement that carries a solution *into* the [2] setting, with  $F = D_E[0, \infty)$ ; it is the missing first step of their programme rather than a consequence of it.

**Setting 5.1** ((T2b) + (E2) + clock,  $\iota = o$ ). As in Setting 4.1, but with no path space and no path regularity assumed. Let  $\mathbb{T}$  satisfy (T2b) with countable dense  $D \subset \mathbb{T}$ , let  $(\Omega, \mathcal{F}, \mathbb{F}, P)$  be a filtered probability space, let  $E$  satisfy (E2), let  $\mathcal{X}$  be a family of  $\mathbb{K}$ -valued  $\mathbb{F}$ -adapted processes with  $P \in \mathcal{M}(\mathcal{X})$ , and let  $X$  be an  $E$ -valued,  $\mathbb{F}$ -adapted, jointly measurable process. Neither the elements of  $\mathcal{X}$  nor  $X$  are assumed to be right continuous.

**Definition 5.2** ((T2b) + (E2)). A set  $\Phi \subset C_b(E)$  is a *regularizing class* for  $(X, \mathcal{X})$  if for every  $f \in \Phi$  there exist  $Y^f \in \mathcal{X}$  and an adapted  $\mathbb{K}$ -valued process  $C^f$  such that

- (R1)  $f(X(t)) = Y^f(t) + C^f(t)$   $P$ -a.s., for every  $t \in \mathbb{T}$ ;
- (R2)  $P$ -a.s.,  $t \mapsto C^f(t)$  has a limit along  $D$  from the right at every  $t \in \mathbb{T}$  and from the left at every  $t > 0$ ;
- (R3)  $C^f$  is right continuous in  $L^1$ , i.e.  $E[|C^f(s) - C^f(t)|] \rightarrow 0$  as  $s \downarrow t$ , for every  $t \in \mathbb{T}$ .

*Remark 5.3* (How restrictive is (R1)?). It looks like a statement about a generator, and it is not one; the question is worth settling before the theorem, because the answer changes what has to be checked.

(R1) *is not a hypothesis*. Given  $f$  and *any* choice of  $Y^f \in \mathcal{X}$ , put  $C^f := f(X) - Y^f$ . This is adapted, because  $X$  and  $Y^f$  are, and (R1) then holds by construction. So the definition contains exactly one existential with content — the choice of  $Y^f$  in the given family  $\mathcal{X}$ , which is a real constraint when  $\mathcal{X}$  is small — and all the force sits in (R2) and (R3), applied to the resulting remainder. Read that way:

$\Phi$  is regularizing if and only if, for each  $f \in \Phi$ , *some* member of  $\mathcal{X}$  approximates  $f(X)$  up to a process that is time-regular in the sense of (R2)–(R3).

*Why it looks Markovian.* In the Markovian case the choice is forced. With  $\mathcal{X} = \mathcal{X}_A(X)$  and  $(f, g) \in A$  the only candidate is  $Y^f = f(X(\cdot)) - \int_{(0, \cdot]} g(X(s))q(ds)$ , so that  $C_t^f = \int_{(0, t]} g(X(s))q(ds)$  is the compensator, and (R1) becomes the familiar decomposition. That is the shape one remembers. But nothing above mentions  $A$ , or  $g$ , or the fact that the compensator is generated pointwise; those enter only in Theorem 5.6.

*How (R2)–(R3) are verified, always.* By finite variation. If  $C^f$  has paths of finite variation on bounded intervals with  $E \int_{(0, T)} |dC^f| < \infty$  for every  $T$ , then (R2) holds because such a path is a difference of two monotone functions and therefore has one-sided limits everywhere, and (R3) holds because  $E|C^f(s) - C^f(t)| \leq E \int_{(t, s]} |dC^f| \rightarrow 0$ . Every example below is of this kind.

*Non-Markovian examples*, in increasing order of distance from the Markovian case.

- (a) *Time-dependent coefficient.*  $C_t^f = \int_{(0, t)} g(s, X_s)q(ds)$ , as in Example 6.16. Still generated pointwise, but the resulting Markov process is inhomogeneous.
- (b) *Path-dependent rate.* Setting 8.69:  $C_t^f = \int_0^t \mathcal{A}_s f ds$  with  $\Lambda$  and  $\mu$  *predictable functionals of the whole past*. Here  $|\mathcal{A}_s f| \leq 2\|f\|\Lambda_s$ , so (R3) amounts to  $E[\int_0^t \Lambda_s ds] = E[N_t] < \infty$  — which is exactly the hypothesis of Theorem 8.71(b). The two conditions coincide, and so do their failures: where the local theory is primary (Remark 8.72), (R3) is what fails. The Hawkes process of Example 8.74 is the instance.

- (c) *Semimartingale characteristics.* [10], Chapter II:  $C^f$  assembled from a predictable triplet  $(B, C, \nu)$  which are functionals of the path. Finite variation by construction.
- (d) *Control variables.* The [2] prototype  $C_t^f = \int_{[0,t]} g(s, L, X) q(ds)$  with a control variable  $L$ . This is the case that cannot be written with an operator  $A \subset C_b(E) \times B(E)$  at all; see Remark 5.5(i).

What is genuinely restrictive is therefore not (R1) but the two hypotheses that Theorem 5.4 places on  $\Phi$  — separating, and containing a countable subset that separates points — together with compact containment. Those are what a user has to check, and Remark 5.7 explains why the two hypotheses on  $\Phi$  are independent of each other.

A gap that is never used. (R2)–(R3) ask for less than finite variation: a deterministic continuous path of infinite variation satisfies both. No example here exploits that, exactly as with quasimartingales in Section 1.4; the definition is weaker than its instances because weakening it cost nothing, not because the slack was needed.

**Theorem 5.4** (Abstract càdlàg modification; (T2b) + (E2)). *In Setting 5.1, let  $\Phi$  be a regularizing class for  $(X, \mathcal{X})$  such that*

- (a)  $\Phi$  contains a countable subset separating the points of  $E$ , and
- (b)  $\Phi$  is separating (Definition 2.37),

and suppose that  $X$  satisfies the compact containment condition (27). Then  $X$  has a modification with sample paths in  $D_E[0, \infty)$ .

*Proof.* Let  $\{f_i\} \subset \Phi$  be countable and separating the points of  $E$ , and write  $Y^i = Y^{f_i}$ ,  $C^i = C^{f_i}$  for the associated data of Definition 5.2.

*Step 1: regularity along  $D$ .* Each  $Y^i$  is an  $(\mathbb{F}, P)$ -martingale, so Corollary 2.23, applied with the countable dense set  $D$ , provides for each  $i$  a set of full measure on which the limits of  $Y^i$  along  $D$  from above (and, for  $t > 0$ , from below) exist at every  $t$ . Intersect these over the countably many  $i$ ; intersect further with the sets of full measure furnished by (R2), one for each  $i$ ; and finally with the sets on which (R1) holds at the times  $t \in D$ , of which there are countably many. The result is  $\Omega' \subset \Omega$  with  $P(\Omega') = 1$ . Only countably many null sets have been discarded, and this is why (R1) may be assumed only  $t$  by  $t$ : the limits below are taken along  $D$ , so (28) is needed only for  $t \in D$ . For  $\omega \in \Omega'$ , every  $i$  and every  $t \in \mathbb{T}$  the one-sided limits of

$$f_i(X(t)) = Y^i(t) + C^i(t) \quad (28)$$

along  $D$  exist — as a sum of two summands each of which has them. This is the only place where (R2) is used, and it is weaker than continuity of  $C^i$ : the two summands are allowed to jump, as long as each does so in a càdlàg manner.

*Step 2: compactness.* First,  $D$  is cofinal in  $\mathbb{T}$ : if  $t$  is not maximal there is  $u > t$ , and (T2b) gives a point of  $D$  in  $(t, u)$ ; if  $t$  is maximal it is itself cofinal. So a countable cofinal set exists.

Fix such a  $T$  and let  $A_{m,T}$  be the event in (27) for  $\varepsilon = 1/m$ , so  $P(A_{m,T}) > 1 - 1/m$ . On  $A_{m,T}$  the set  $\{X(t, \omega) : t \in \mathbb{T}_{\leq T} \cap D\}$  is contained in the compact  $K_{1/m, T}$ , hence relatively compact. Now

$$P\left(\bigcup_m A_{m,T}\right) \geq \sup_m P(A_{m,T}) = 1,$$

so  $\bigcup_m A_{m,T}$  has full measure, and  $\Omega'' = \Omega' \cap \bigcap_T \bigcup_m A_{m,T}$  has full measure as a countable intersection. The union over  $m$  is essential: the events  $A_{m,T}$  are not nested and each has

probability only  $> 1 - 1/m$ , so intersecting them would prove nothing. For  $\omega \in \Omega''$  and every  $T$  in the cofinal set,  $\{X(t, \omega) : t \in \mathbb{T}_{\leq T} \cap D\}$  is relatively compact.

*Step 3: existence of one-sided limits.* Fix  $\omega \in \Omega''$  and a non-maximal  $t \in \mathbb{T}$ , and let  $s_n \in D$  with  $s_n \downarrow t$ , which exist by (T2b). (If  $t$  is maximal there is nothing to prove and we set  $Y(t, \omega) = X(t, \omega)$ .) By Step 2 the sequence  $(X(s_n, \omega))_n$  lies in a relatively compact set, so it has convergent subsequences. If  $x$  and  $x'$  are limits along two subsequences, then by Step 1 and the continuity of  $f_i$ ,

$$f_i(x) = \lim_{s \rightarrow t+, s \in D} f_i(X(s, \omega)) = f_i(x') \quad \text{for all } i,$$

and since  $\{f_i\}$  separates points,  $x = x'$ . Hence

$$Y(t, \omega) := \lim_{s \rightarrow t+, s \in D} X(s, \omega) \quad (29)$$

exists for all  $t \in \mathbb{T}$ , and similarly  $Y^-(t, \omega) = \lim_{s \rightarrow t-, s \in D} X(s, \omega)$  exists for  $t > 0$ . By Fact 2.21,  $Y(\cdot, \omega)$  is càdlàg, and lies in  $D_E[0, \infty)$  when  $\mathbb{T}$  satisfies (T3). Set  $Y(\cdot, \omega) \equiv x_0$  for  $\omega \notin \Omega''$  and some fixed  $x_0 \in E$ . Each  $Y(t, \cdot)$  is a random variable: on  $\Omega''$  it is the pointwise limit of the measurable maps  $X(s_n, \cdot)$  along a fixed sequence  $s_n \downarrow t$  in  $D$ , and a pointwise limit of measurable maps into a metrizable space is measurable where it exists.

*Step 4:  $Y$  is a modification of  $X$ .* Let  $f \in \Phi$  and  $t \in \mathbb{T}$ . For  $s > t$ , (R1) and the martingale property of  $Y^f$  give

$$E[f(X(s)) \mid \mathcal{F}_t] = E[Y^f(s) \mid \mathcal{F}_t] + E[C^f(s) \mid \mathcal{F}_t] = Y^f(t) + E[C^f(s) \mid \mathcal{F}_t],$$

and by (R3) and conditional Jensen,  $E[C^f(s) \mid \mathcal{F}_t] \rightarrow C^f(t)$  in  $L^1$  as  $s \downarrow t$ . Hence, using (R1) once more,  $E[f(X(s)) \mid \mathcal{F}_t] \rightarrow Y^f(t) + C^f(t) = f(X(t))$  in  $L^1$ . On the other hand  $X(s) \rightarrow Y(t)$  a.s. along  $D$  by (29), so  $f(X(s)) \rightarrow f(Y(t))$  a.s. and boundedly, whence  $E[f(X(s)) \mid \mathcal{F}_t] \rightarrow E[f(Y(t)) \mid \mathcal{F}_t]$  in  $L^1$ . Therefore

$$E[f(Y(t)) \mid \mathcal{F}_t] = f(X(t)) \quad \text{for all } f \in \Phi, t \in \mathbb{T}. \quad (30)$$

Since  $X(t)$  is  $\mathcal{F}_t$ -measurable and  $\Phi$  is separating, Fact 2.48 yields  $P\{Y(t) = X(t)\} = 1$  for every  $t \in \mathbb{T}$ .  $\square$

*Remark 5.5* (What the abstract form buys). Three hypotheses of [7], Theorem 4.3.6 disappear, and each of them is one that [2] would want gone.

- (i) *No operator.* The compensator  $C^f$  is an arbitrary adapted process subject to (R2) and (R3). In particular it may be *path dependent*: the [2] prototype  $C^f(t) = \int_{(0,t]} g(s, L, X) q(ds)$ , with a control variable  $L$  and a locally finite measure  $q$ , satisfies (R2) because it is of finite variation, and (R3) because  $E|C^f(s) - C^f(t)| \leq \int_{(t,s]} E|g(u, L, X)| q(du) \rightarrow 0$  under the integrability condition that [2] impose anyway. A statement of this generality cannot be made in the language of an operator  $A \subset C_b(E) \times B(E)$ .
- (ii) *Atoms are harmless.* Nothing above requires  $q$  to be atomless. (R2) asks for one-sided limits, not for continuity; the classical statement “ $t \mapsto \int_0^t g(X_s) ds$  is Lipschitz” is the special case  $q = \text{Lebesgue measure}$ . In (R3) an atom at  $t$  contributes to  $(0, t]$  but not to  $(t, s]$ , and so does not obstruct the limit — which is precisely why the half-open convention  $(0, t]$  is the right one here.
- (iii) *No special filtration.* (4) plays no rôle;  $\mathbb{F}$  is any filtration making  $X$  adapted and the  $Y^f$  martingales. The filtration  ${}^*\mathcal{F}^X$  is needed for Proposition 4.9, not here.

What is *not* gained is a local version; see Remark 5.8, whose argument is unaffected.

## 5.2 The Markovian case

**Theorem 5.6** ([7], Theorem 4.3.6; (T2b) + (E2) + clock,  $\iota = o$ ). *Assume (E2). Let  $A \subset C_b(E) \times B(E)$  and suppose that  $\mathcal{D}(A)$  is separating and contains a countable subset that separates points. Let  $X$  be a solution of the martingale problem for  $A$  satisfying the compact containment condition (27). Then  $X$  has a modification with sample paths in  $D_E[0, \infty)$ .*

*Proof.* Apply Theorem 5.4 with  $\mathcal{X} = \mathcal{X}_A(X)$ ,  $\mathbb{F} = (*\mathcal{F}_t^X)$  and  $\Phi = \mathcal{D}(A)$ : for  $f \in \mathcal{D}(A)$  pick  $g$  with  $(f, g) \in A$  and put  $Y^f = f(X(\cdot)) - \int_{(0, \cdot]} g(X(s)) q(ds) \in \mathcal{X}_A^o(X)$  and  $C^f(t) = \int_{(0, t]} g(X(s)) q(ds)$  — here, and only here, the convention must be  $\iota = o$ . Then (R1) holds by definition. Since  $g$  is bounded,  $C^f$  is of finite variation with  $|C^f(t) - C^f(s)| \leq \|g\| q((s, t])$ , so it has one-sided limits, which is (R2); and  $q((t, s]) \downarrow q(\emptyset) = 0$  as  $s \downarrow t$  by continuity from above, which is (R3). For  $q = \text{Lebesgue measure}$  this is the familiar statement that  $C^f$  is Lipschitz, but no atomlessness is needed: an atom of  $q$  at  $t$  lies in  $(0, t]$  and not in  $(t, s]$ , which is exactly why the convention of Remark 2.5 is the right one here. Hypotheses (a) and (b) are the assumptions on  $\mathcal{D}(A)$ .  $\square$

*Remark 5.7.* The two hypotheses on  $\mathcal{D}(A)$  — hypotheses (a) and (b) of Theorem 5.4 — play different rôles and should be kept apart in the formalization: “contains a countable subset separating points” is used in Step 3 of the proof of Theorem 5.4 to identify limit points, “separating” is used in Step 4 to identify  $Y(t)$  with  $X(t)$ . They are independent: neither implies the other.

*Remark 5.8.* Theorem 5.6 is stated for genuine martingales and this is not an artefact of the proof. For a local martingale problem the localizing sequence consists of stopping times, whose usefulness (via optional stopping) presupposes right continuity of the filtration, and typically of the paths — which is exactly the conclusion being sought. If one *postulates* a localizing sequence of *deterministic* times, or — equivalently for the present purpose — if  $A$  is replaced by  $A^K = \{(f, g) \in A : f, g \text{ bounded on } K\}$  for the compact sets  $K$  of (27), then the proof goes through verbatim. This is the honest local version of Theorem 5.6.

*Remark 5.9* (Verifying compact containment). Condition (27) is the price paid for allowing non-locally-compact Polish  $E$ . If  $E$  is locally compact and separable and  $\mathcal{D}(A)$  is dense in  $\hat{C}(E)$  (continuous functions vanishing at infinity) in the norm topology, then (27) may be dispensed with (see also Fact 2.38): pass to the one-point compactification  $E^\Delta$ , apply Theorem 5.6 there (where compact containment is trivial), and conclude that the modification has paths in  $D_{E^\Delta}[0, \infty)$ ; see [7], Corollary 4.3.7.

## 6 Uniqueness, the Markov property, and the strong Markov property

The theorem of this section is the workhorse of the whole theory: it reduces uniqueness for a martingale problem, which is a statement about all finite-dimensional distributions, to a statement about *one-dimensional* distributions only, and it produces the Markov property for free. No semigroup is assumed to exist, and  $\mathcal{D}(A)$  is not assumed to be separating.

As with Theorem 5.6, the operator plays no rôle in the proof. Section 6.1 extracts what does, which turns out to be a single lemma applied four times; Section 6.6 then recovers [7], Theorem 4.4.2. [2] have no counterpart of this result either — they prove no uniqueness theorem at all — but, unlike the situation in Section 5, the obstruction is not structural: the statement fits their setting once one adds the one ingredient a uniqueness theory needs and a convergence theory does not, namely a way of *restarting* the martingale problem.

## 6.1 Uniqueness without a Markov structure

Throughout this subsection we work on the path space, as in Setting 4.1;  $\mathbb{T}$  satisfies (T0) and  $E$  satisfies (E0) unless stated otherwise. Let  $F \subset E^{\mathbb{T}}$  be a path space with  $\sigma$ -field  $\mathcal{S} = \sigma(\pi_t : t \in \mathbb{T})$ , coordinate process  $\pi = (\pi_t)_{t \in \mathbb{T}}$  and canonical filtration  $\mathcal{F}_t^\circ = \sigma(\pi_u : u \leq t)$ , and let  $\mathcal{X}^\circ$  be a family of  $(\mathcal{F}_t^\circ)$ -adapted  $\mathbb{K}$ -valued functionals on  $F$ . Write

$$\mathcal{M}(\mathcal{X}^\circ) = \{P \in \mathcal{P}(F) : \text{every } Y^\circ \in \mathcal{X}^\circ \text{ is a } (\mathcal{F}_t^\circ, P)\text{-martingale}\},$$

and  $\mathcal{M}(\mathcal{X}^\circ, \mu) = \{P \in \mathcal{M}(\mathcal{X}^\circ) : P\pi_0^{-1} = \mu\}$ .

*Remark 6.1.* Nothing is lost by using the canonical filtration. If  $X$  is a process with paths in  $F$  solving the martingale problem with respect to *some* filtration  $(\mathcal{G}_t)$  to which it is adapted — for instance (4) — then, since each  $Y_t^\circ(X)$  is  $\sigma(X_u : u \leq t)$ -measurable and  $\sigma(X_u : u \leq t) \subset \mathcal{G}_t$ , the tower property makes  $Y^\circ(X)$  a martingale for the smaller filtration as well. Hence  $PX^{-1} \in \mathcal{M}(\mathcal{X}^\circ)$ , and  $\mathcal{M}(\mathcal{X}^\circ)$  is the set of laws of all solutions, for all filtrations at once.

The martingale property is an identity that is *linear* in the measure. This trivial observation is what carries the mixing steps of the classical proof, and it deserves to be recorded once.

**Lemma 6.2** (Mixture stability; (T0) + (E0)). *Let  $(\Theta, \mathcal{A}, \nu)$  be a probability space and  $\vartheta \mapsto P_\vartheta \in \mathcal{M}(\mathcal{X}^\circ)$  a measurable family, and put  $P = \int P_\vartheta \nu(d\vartheta)$ . If  $\int E^{P_\vartheta} |Y_t^\circ| \nu(d\vartheta) < \infty$  for every  $Y^\circ \in \mathcal{X}^\circ$  and  $t \in \mathbb{T}$ , then  $P \in \mathcal{M}(\mathcal{X}^\circ)$ . In particular  $\mathcal{M}(\mathcal{X}^\circ)$  is convex.*

*Proof.* For  $G \in \mathcal{F}_s^\circ$  the identity  $E^{P_\vartheta}[Y_t^\circ \mathbf{1}_G] = E^{P_\vartheta}[Y_s^\circ \mathbf{1}_G]$  holds for every  $\vartheta$ ; integrate it  $\nu(d\vartheta)$  and use Fubini, which is licensed by the integrability hypothesis.  $\square$

Its converse holds as well, and is what actually extends well-posedness from degenerate to general initial distributions. It is worth separating from Theorem 6.23, because — unlike the argument used in Corollary 7.20 — it does not go through the Markov property.

**Lemma 6.3** (Disintegration; (T0) + (E1)). *Suppose  $\mathcal{X}^\circ$  admits a determining set  $\mathcal{Z}^\circ$  and that membership in  $\mathcal{M}(\mathcal{X}^\circ)$  can be tested along countably many data, i.e. there are a countable  $\mathcal{X}_0^\circ \subset \mathcal{X}^\circ$ , a countable  $D \subset \mathbb{T}$  and countable sets  $\mathcal{Z}_s^\circ$  such that  $P \in \mathcal{M}(\mathcal{X}^\circ)$  if and only if*

$$E^P[(Y_t^\circ - Y_s^\circ)Z_s^\circ] = 0 \quad \text{for all } Y^\circ \in \mathcal{X}_0^\circ, s \leq t \text{ in } D, Z_s^\circ \in \mathcal{Z}_s^\circ. \quad (31)$$

*Let  $P \in \mathcal{M}(\mathcal{X}^\circ)$  with  $P\pi_0^{-1} = \mu$ . Then  $P(\cdot \mid \pi_0 = x) \in \mathcal{M}(\mathcal{X}^\circ, \delta_x)$  for  $\mu$ -a.e.  $x$ . In particular, if  $\mathcal{M}(\mathcal{X}^\circ, \delta_x) = \{P_x\}$  for  $\mu$ -a.e.  $x$ , then  $P = \int P_x \mu(dx)$ , and  $\mathcal{M}(\mathcal{X}^\circ, \mu)$  is a singleton.*

*Proof.* Fix  $Y^\circ \in \mathcal{X}_0^\circ$ ,  $s \leq t$  in  $D$  and  $Z_s^\circ \in \mathcal{Z}_s^\circ$ . For bounded measurable  $h : E \rightarrow \mathbb{R}$  the variable  $Z_s^\circ h(\pi_0)$  is bounded and  $\mathcal{F}_s^\circ$ -measurable, so the martingale property of  $Y^\circ$  under  $P$  gives  $E^P[(Y_t^\circ - Y_s^\circ)Z_s^\circ h(\pi_0)] = 0$ . As  $h$  is arbitrary,  $E^P[(Y_t^\circ - Y_s^\circ)Z_s^\circ \mid \pi_0] = 0$   $P$ -a.s. The data being countable, one null set serves all of them at once; off it,  $P(\cdot \mid \pi_0 = x)$  satisfies (31) and has initial law  $\delta_x$ .  $\square$

*Remark 6.4.* Lemma 6.3 together with Lemma 6.2 is [16], Theorem 32.10, in abstract form; [16] attributes it to Stroock and Varadhan and uses it exactly this way, to pass from  $P_x$  to  $P_\mu = \int P_x \mu(dx)$  for the local martingale problem of a diffusion. The countability hypothesis is his too: he replaces  $\hat{C}^\infty$  by a countable class of truncated coordinate functions and their products. Note that Corollary 7.20 currently justifies the same passage via Theorem 6.23(a); Lemma 6.3 is the better route, since it needs no Markov property.

Restarting a solution is available at once, and — this is worth stating separately, because it is easy to attribute to the Markov property what belongs to the tower property — it needs no shift and no Markov structure whatever.

**Lemma 6.5** (Restart with memory; (T0) + (E0)). *Let  $P \in \mathcal{M}(\mathcal{X}^\circ)$ , let  $r \in \mathbb{T}$  and let  $Z \geq 0$  be bounded and  $\mathcal{F}_r^\circ$ -measurable with  $E^P[Z] = 1$ . Then  $Z \cdot P$  is again a solution from  $r$  onwards: for every  $Y^\circ \in \mathcal{X}^\circ$  and all  $r \leq s \leq t$ ,*

$$E^{Z \cdot P}[Y_t^\circ \mid \mathcal{F}_s^\circ] = Y_s^\circ.$$

*Proof.* Let  $G \in \mathcal{F}_s^\circ$ . Since  $r \leq s$ , the variable  $Z\mathbf{1}_G$  is bounded and  $\mathcal{F}_s^\circ$ -measurable, so the martingale property of  $Y^\circ$  under  $P$  gives  $E^{Z \cdot P}[(Y_t^\circ - Y_s^\circ)\mathbf{1}_G] = E^P[(Y_t^\circ - Y_s^\circ)Z\mathbf{1}_G] = 0$ .  $\square$

*Remark 6.6* (Restarting is not where the Markov structure enters). Lemma 6.5 says that one may condition on any event of positive probability observed up to  $r$ , or reweight by any bounded  $\mathcal{F}_r^\circ$ -measurable density, and what remains after  $r$  is again a solution. Nothing is forgotten and nothing is assumed. If one wants the conditional laws themselves rather than a reweighting, Lemma 6.3 supplies them: for standard Borel  $F$  the family  $P(\cdot \mid \mathcal{F}_r^\circ)$  exists, and it consists of solutions from  $r$  onwards by the same two lines.

What a shift adds is therefore not the restart but a *re-indexing*. A hypothesis stated about *initial* distributions — as (38) is, and as [7], Theorem 4.4.2 is — can be applied to the restarted object only after that object has been turned into a process starting at time 0, and the map that does this is  $\theta_r$ , which discards the path before  $r$ . That is the whole of the Markov structure in Section 6.2: not the ability to begin again, but the demand that beginning again should look like beginning.

Definition 6.11 is the same idea without the re-indexing: the two measures are compared at the time  $s$  where they already agree, so nothing has to be transported back to 0 and no shift is needed. This is why Proposition 6.13 is Markov-free while Lemma 6.22 is not.

### An objection: make the state the path, and everything is Markov

The obvious way to make a path-dependent problem Markovian is to enlarge the state so that it *is* the past. The construction works, and it is worth carrying out, because the exact form of its failure is the best available justification for Section 6.1.

**Definition 6.7** (The path lift). Let  $X$  be a process with values in  $E$ . Its *path lift* is the process

$$\hat{X}_t := X|_{\mathbb{T}_{\leq t}} \in \hat{E}_t := E^{\mathbb{T}_{\leq t}}, \quad t \in \mathbb{T},$$

with  $\hat{E}_t$  carrying the product  $\sigma$ -field. This is a fibred state space in the sense of Definition 2.10, and it is never a constant one: the fibres grow with  $t$ . In the càdlàg setting one takes  $\hat{E}_t = D_E[0, t]$  instead.

**Lemma 6.8** (Every process lifts to a Markov process; (T0) + (E0)). *Let  $X$  be any process, with no hypothesis whatsoever.*

- (a)  $\sigma(\hat{X}_t) = \mathcal{F}_t^X$  for every  $t$ , and consequently  $\mathcal{F}_t^{\hat{X}} = \mathcal{F}_t^X = \sigma(\hat{X}_t)$ .
- (b)  $\hat{X}$  is a Markov process: for  $s \leq t$  and bounded measurable  $\varphi$  on  $\hat{E}_t$ ,

$$E[\varphi(\hat{X}_t) \mid \mathcal{F}_s^{\hat{X}}] = E[\varphi(\hat{X}_t) \mid \hat{X}_s].$$

- (c) On the canonical space, and for a strict stopping time  $T$ ,  $\sigma(\hat{X}_T) = \mathcal{F}_T^\circ$ , where  $\hat{X}_T$  is read as a value in the total space  $\Sigma \hat{E} = \coprod_t \hat{E}_t$ ; so  $\hat{X}$  is strong Markov as well.

*Proof.* (a)  $\hat{X}_t$  is the family  $(X_u)_{u \leq t}$ , so  $\sigma(\hat{X}_t) = \sigma(X_u : u \leq t) = \mathcal{F}_t^X$ ; and for  $u \leq t$  the variable  $\hat{X}_u$  is the restriction of  $\hat{X}_t$ , hence  $\sigma(\hat{X}_u) \subset \sigma(\hat{X}_t)$ , so  $\mathcal{F}_t^{\hat{X}} = \bigvee_{u \leq t} \sigma(\hat{X}_u) = \sigma(\hat{X}_t)$ .

(b) By (a) the two conditioning  $\sigma$ -fields are *equal*. There is nothing to prove.

(c) This is Definition 6.42: for a strict  $T$  one has  $\mathcal{F}_T^\circ = a_T^{-1}(\mathcal{S})$ , and  $a_T \omega = \omega(\cdot \wedge T)$  carries the same information as  $\hat{X}_T$  together with the value of  $T$ , which is recorded by the fibre index. This is exactly the point at which the total space of Definition 2.10 is needed.  $\square$

*Remark 6.9* (Why this settles nothing). Lemma 6.8(b) is a tautology — an identity of  $\sigma$ -fields, not a property of  $X$  — and the tautology propagates to everything the Markov layer proves. Three collapses, in increasing order of seriousness.

(i) *The Markov property.* Theorem 6.23(a) asserts of the lift only what Lemma 6.8 already gives for nothing.

(ii) *The uniqueness criterion.* The *one-dimensional* distributions of  $\hat{X}$  are the *finite-dimensional* distributions of  $X$ , since  $\hat{X}_t$  is the whole path on  $\mathbb{T}_{\leq t}$ . So hypothesis (38), read for the lift, is the *conclusion* of Theorem 6.23(b), and the induction over  $t_1 < \dots < t_n$  that carries that theorem collapses to its own conclusion.

(iii) *Even the Markov-free theory collapses.* In Definition 6.11 the test functions  $h$  range over all bounded measurable functions of  $\hat{\pi}_s$ , that is over all bounded  $\mathcal{F}_s^\circ$ -measurable variables — which absorbs the weight  $Z$ . So for the lift, propagation of agreement reads

$$P = Q \text{ on } \mathcal{F}_s^\circ \implies P = Q \text{ on } \mathcal{F}_t^\circ,$$

and at  $s = 0$  that is uniqueness itself. Proposition 6.13 becomes “if the laws agree then the laws agree”.

The principle behind all three is worth stating on its own, because it is what the whole of Section 6 trades on. A *Markov structure is a compression statement*: its content is the gap between  $\sigma(X_s)$  and  $\mathcal{F}_s^X$ , and the theorems are worth exactly as much as that gap is wide. The path lift attains the Markov property by closing the gap, and closing the gap is the same operation as deleting the theorems. “Every process is Markov” is therefore true, and true in a way that converts every hypothesis into its own conclusion.

Two further observations, both of which matter elsewhere.

*The triviality is filtration-relative.* Lemma 6.8 makes  $\hat{X}$  Markov with respect to *its own* filtration. Being a solution with respect to a larger  $\mathbb{G}$ , as in Lemma 6.17, is not automatic even for the lift: an ambient filtration may carry information about the future. This is one reason the restart lemma is stated on a general filtered space.

*The lift is not useless, only useless here.* Where the content of a theorem sits somewhere other than in the Markov property, lifting is a genuine device. In Section 8.2 the content is the kernel representability (D3), which is a statement about the *dual* process on an unchanged and typically small space; taking  $E_1$  to be a path or genealogy space then costs nothing on the side where the difficulty is not. That is exactly [3]’s outlook and the subject of Remark 8.8. Note also that the lift is never time-homogeneous, so it can only be treated in the fibred, inhomogeneous frame of Definition 2.10 — which is a second reason for setting that frame up.

*Remark 6.10* (Is the abstract martingale problem a special case of the Markovian one?). Lemma 6.8 invites the question, and the answer separates two things that are easy to run together.

*Formally yes, and emptily.* Every  $Y^\circ \in \mathcal{X}^\circ$  is  $\mathcal{F}_t^\circ$ -adapted, so by (a) and Doob–Dynkin there are measurable  $\varphi_t$  on  $\hat{E}_t$  with  $Y_t^\circ = \varphi_t(\hat{\pi}_t)$ . Read on the lift,  $Y^\circ$  therefore has the shape (5) of Definition 4.6 with  $g = 0$ . So the abstract martingale problem *is* a Markovian

one, on the lifted state space, with vanishing compensator. This is the same emptiness that Section 2.2 records when it observes that every process solves the martingale problem for *some* family of test processes, namely the family of its own martingales: the Markovian form is not a restriction on the process but on the *presentation*, and its content is that  $\mathcal{X}_A$  is a small family parametrized by  $\mathcal{D}(A)$  whose compensator is generated pointwise from  $g$ . Taking  $g = 0$  discards exactly that content.

*Informatively, with  $g \neq 0$ .* The useful lift is not the one that makes the compensator vanish but the one that keeps it and lets its integrand be a *path* functional. On the lift that is again literally Definition 4.6, and the manuscript already contains it: Setting 8.69 is the Markovian martingale problem on the lift, with

$$\hat{f}_t(\hat{x}) = f(\hat{x}(t)) \text{ acting on the tip,} \quad \hat{g}_u(\hat{x}) = \mathcal{A}_u f(\hat{x}) \text{ the path-dependent rate,}$$

and Examples 6.48, 6.49 and 8.74 are all of this kind. So the honest formulation is: the lift does not enlarge the class of *processes* — there was never a restriction there — but it does supply a useful *presentation*, provided one lifts the state and keeps the generator rather than lifting the state and deleting it.

This is not a device invented here. [15] lift stochastic Volterra processes to Markov processes on a space of forward curves, with a genuine generator and a generalized Feller semigroup, and [14] treat the same objects through their transform formula; Section 8.2 meets both in Remark 8.16.

*What still does not follow is any theorem.* Remark 6.9 applies verbatim to either version: on the lift the hypotheses of Section 6.2 become their own conclusions. And the classical Markov apparatus — a time-independent operator, a transition semigroup, Feller theory — is unavailable in either version, since the fibres  $\hat{E}_t$  grow with  $t$  and there is nothing for a semigroup to act on. What the lift is good for is recorded at the end of Remark 6.9 and carried out in Section 8.2.

So far nothing beyond a preorder has been used, and nothing Markovian. It is worth pushing that as far as it goes, because uniqueness of the finite-dimensional distributions is *not* a Markovian fact. What is Markovian is only the reduction of its hypothesis to the *one-dimensional* distributions, and that reduction is the subject of the next subsection. The condition the uniqueness argument actually consumes is the following.

**Definition 6.11** (Propagation of agreement; (T0) + (E0)). A set  $\mathcal{N} \subset \mathcal{P}(F)$  *propagates agreement* if for all  $P, Q \in \mathcal{N}$ , all  $s \leq t$  in  $\mathbb{T}$  and every bounded  $\mathcal{F}_s^\circ$ -measurable  $Z \geq 0$ ,

$$\begin{aligned} E^P[Z h(\pi_s)] &= E^Q[Z h(\pi_s)] \quad \text{for all } h \in B(E) \\ \implies E^P[Z f(\pi_t)] &= E^Q[Z f(\pi_t)] \quad \text{for all } f \in B(E). \end{aligned} \tag{32}$$

*Remark 6.12* (Why the weight  $Z$  cannot be dropped). It is tempting to state (32) in the simpler unconditional form

$$P = Q \text{ on } \mathcal{F}_s^\circ \implies P = Q \text{ on } \mathcal{F}_s^\circ \vee \sigma(\pi_t), \tag{33}$$

and an earlier version of this manuscript did. That is a mistake, and it is worth recording where the induction of Proposition 6.13 breaks under it. The induction produces agreement of the finite-dimensional distributions, i.e. agreement on  $\sigma(\pi_{t_1}, \dots, \pi_{t_n})$  for finite chains; (33) demands agreement on all of  $\mathcal{F}_{t_n}^\circ = \sigma(\pi_u : u \leq t_n)$ , which involves *all* times below  $t_n$  and is therefore strictly more than any stage of the induction delivers. Iterating (33) adds one future coordinate at a time but never closes the gap, since after adding  $\pi_{t_{n+1}}$  one is still short of  $\mathcal{F}_{t_{n+1}}^\circ$ .

Form (32) repairs this by carrying the past in the weight  $Z$  instead of in a  $\sigma$ -field: at stage  $n$  one takes  $Z = \prod_{k \leq n} f_k(\pi_{t_k})$ , and the hypothesis of (32) is then exactly the induction hypothesis. It is also exactly what the restart of Lemma 6.17 supplies, which is no accident: the two are the same statement read from opposite ends.

**Proposition 6.13** (Uniqueness; (T2a) + (E0)). *Let  $\mathcal{N} \subset \mathcal{P}(F)$  propagate agreement. Then  $\mathcal{N}$  contains at most one element with a given law of  $\pi_0$ . In particular, if  $\mathcal{M}(\mathcal{X}^\circ)$  propagates agreement then  $\mathcal{M}(\mathcal{X}^\circ, \mu)$  has at most one element for every  $\mu$ , and the same holds for  $\mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$ .*

*Proof.* Let  $P, Q \in \mathcal{N}$  with  $P\pi_0^{-1} = Q\pi_0^{-1}$ .

*Step 1: the finite-dimensional distributions.* We show by induction on  $m$  that

$$E^P \left[ \prod_{k=1}^m f_k(\pi_{t_k}) \right] = E^Q \left[ \prod_{k=1}^m f_k(\pi_{t_k}) \right] \quad (34)$$

for every chain  $t_1 \leq \dots \leq t_m$  in  $\mathbb{T}$  and all bounded measurable  $f_k \geq 0$ .

For  $m = 0$  both sides are 1. For  $m = 1$ , apply (32) with  $s = t = t_1$  and  $Z = 1$ : the hypothesis is  $P\pi_{t_1}^{-1} = Q\pi_{t_1}^{-1}$  — which for  $t_1 = 0$  is the assumption on the initial laws, and for general  $t_1$  follows once the case  $m = 1$  is known for  $s = 0$ , i.e. from (32) with  $s = 0$ ,  $t = t_1$ ,  $Z = 1$ .

Assume (34) for  $m = n$ , fix a chain  $t_1 \leq \dots \leq t_{n+1}$  and bounded  $f_1, \dots, f_{n+1} \geq 0$ , and put

$$Z = \prod_{k=1}^n f_k(\pi_{t_k}),$$

which is bounded, nonnegative and  $\mathcal{F}_{t_n}^\circ$ -measurable because  $t_k \leq t_n$  for  $k \leq n$ . For every bounded measurable  $h \geq 0$  the identity  $E^P[Z h(\pi_{t_n})] = E^Q[Z h(\pi_{t_n})]$  is (34) for  $m = n$  with  $f_n$  replaced by  $f_n h$ , hence holds; by linearity it holds for all  $h \in B(E)$ . Now (32), applied with  $s = t_n$  and  $t = t_{n+1}$ , gives  $E^P[Z f_{n+1}(\pi_{t_{n+1}})] = E^Q[Z f_{n+1}(\pi_{t_{n+1}})]$ , which is (34) for  $m = n + 1$ .

*Step 2: from the finite-dimensional distributions to the law.* By linearity, (34) holds for all  $f_k \in B(E)$ . Under (T2a) every finite subset of  $\mathbb{T}$  is a chain, so the collection

$$\mathcal{K} = \left\{ \prod_{k=1}^m f_k(\pi_{t_k}) : m \geq 0, t_k \in \mathbb{T}, f_k \in B(E) \right\}$$

is closed under multiplication and generates  $\sigma(\pi_t : t \in \mathbb{T}) = \mathcal{S}$  by (17). Two probability measures agreeing on a multiplicative generating class agree, by Dynkin's lemma. Hence  $P = Q$ .  $\square$

## 6.2 The Markov layer: shift systems

Everything so far is free of any Markov structure. What follows adds it, in the one form the arguments need, and it does two things: it makes (32) checkable from the one-dimensional distributions (Lemma 6.22), and it produces the Markov property itself (Theorem 6.23). It is what a convergence theory does not need and a uniqueness theory cannot do without.

**Definition 6.14** (Shift system; (T4) + (E0)). A family of measurable maps  $\theta_r : F \rightarrow F$ ,  $r \in \mathbb{T}$ , is a *shift* on  $F$  if  $\pi_t \circ \theta_r = \pi_{r+t}$  for all  $r, t \in \mathbb{T}$ .

A *shift system* for  $\mathcal{X}^\circ$  is a family  $(\mathcal{X}_r^\circ)_{r \in \mathbb{T}}$  of families of  $(\mathcal{F}_t^\circ)$ -adapted  $\mathbb{K}$ -valued processes on  $F$ , with  $\mathcal{X}_0^\circ = \mathcal{X}^\circ$ , such that for every  $r \in \mathbb{T}$  and every  $\hat{Y}^\circ \in \mathcal{X}_r^\circ$  there are  $Y^\circ \in \mathcal{X}^\circ$  and an  $\mathcal{F}_r^\circ$ -measurable  $\kappa$  with

$$\hat{Y}_t^\circ \circ \theta_r = Y_{r+t}^\circ - Y_r^\circ + \kappa, \quad t \in \mathbb{T}. \quad (35)$$

We call  $\mathcal{X}_r^\circ$  the *problem shifted to  $r$* , and say that  $\mathcal{X}^\circ$  is *shift stable* in the narrow sense if  $\mathcal{X}_r^\circ = \mathcal{X}^\circ$  for every  $r$  is a shift system.

The system is *measurable* if there is a set  $I$  and, for each  $i \in I$  and  $r \in \mathbb{T}$ , an element  $\hat{Y}^{\circ, r, i} \in \mathcal{X}_r^\circ$  exhausting  $\mathcal{X}_r^\circ$ , such that  $(\omega, r) \mapsto \hat{Y}_t^{\circ, r, i}(\omega)$  is  $\mathcal{B}(F) \otimes \mathcal{T}$ -measurable for every  $i$  and  $t$ . For an a.s. finite stopping time  $\tau$  one then writes  $\mathcal{X}_\tau^\circ = \{\hat{Y}^{\circ, \tau(\cdot), i} : i \in I\}$ .

*Remark 6.15.* Requiring  $\mathcal{X}_r^\circ = \mathcal{X}^\circ$  — which is what the first version of this manuscript did — is a real restriction and not a normalization: it forces the martingale problem to look the same from every instant, hence forces the clock to be shift invariant and the resulting Markov process to be *homogeneous*. Definition 6.14 follows [10], III.2.39 instead, where a whole family of shifted triplets  $(\rho_r B, \rho_r C, \rho_r \nu)$  is carried along and  $(\rho_r B)_t(\theta_r \omega) = B_{r+t}(\omega) - B_r(\omega)$ ; the measurability clause is their III.2.39(i). The cost is nil: the proofs below use only that the *shifted* process is a  $P$ -martingale, never that it belongs to the original family. The gain is the time-inhomogeneous case, and with it the removal of shift invariance of the clock from the hypotheses; see Remark 6.29(ii) and Example 6.16.

**Example 6.16** (The shifted problem for  $\mathcal{X}_A$ ; (T0) + (T4)). Let  $q$  be a clock on  $\mathbb{T}$ , let  $A \subset M(\mathbb{T} \times E) \times M(E)$  allow a time-dependent coefficient, and put

$$\mathcal{X}^\circ = \left\{ \left( f(\pi_t) - \int_{\langle 0, t \rangle_t} g(s, \pi_s) q(ds) \right)_{t \in \mathbb{T}} : (f, g) \in A \right\}.$$

Define the *pulled-back clock*  $q_r$  by  $q_r(A) = q(r + A)$ , i.e. the image of  $q$  restricted to  $\{v : v \geq r\}$  under  $v \mapsto v - r$ , and set

$$\mathcal{X}_r^\circ = \left\{ \left( f(\pi_t) - \int_{\langle 0, t \rangle_t} g(r + u, \pi_u) q_r(du) \right)_{t \in \mathbb{T}} : (f, g) \in A \right\}.$$

Then  $(\mathcal{X}_r^\circ)$  is a shift system, with  $\kappa = f(\pi_r)$ : substituting  $v = r + u$ ,

$$\begin{aligned} \hat{Y}_t^\circ(\theta_r \omega) &= f(\omega(r + t)) - \int_{\langle 0, t \rangle_t} g(r + u, \omega(r + u)) q_r(du) \\ &= f(\omega(r + t)) - \int_{\langle r, r+t \rangle_t} g(v, \omega(v)) q(dv), \end{aligned}$$

which is  $Y_{r+t}^\circ(\omega) - Y_r^\circ(\omega) + f(\omega(r))$ . Note  $q_0 = q$  and  $\mathcal{X}_0^\circ = \mathcal{X}^\circ$ .

The shifted problem always exists. It coincides with the original one, i.e.  $\mathcal{X}_r^\circ = \mathcal{X}^\circ$  for all  $r$ , precisely when  $q_r = q$  and  $g$  does not depend on time — that is, when the clock is shift invariant in the sense of Definition 2.3. The system is measurable as soon as  $(r, u, x) \mapsto g(r + u, x)$  is, which for  $q = \text{Lebesgue measure}$  and measurable  $g$  is automatic.

**Lemma 6.17** (Restart; (T0) + (T4) + (E0)). *Let  $(\mathcal{X}_r^\circ)$  be a shift system for  $\mathcal{X}^\circ$  and let  $Z^\circ$  be a determining set for every  $\mathcal{X}_r^\circ$  (Definition 4.3). Let  $X$  be a process with paths in  $F$  on a filtered probability space  $(\Omega, \mathcal{F}, \mathbb{G}, P)$ , solving the martingale problem for  $\mathcal{X}^\circ$  with respect to  $\mathbb{G}$ ; let  $r \in \mathbb{T}$  and let  $Z \geq 0$  be bounded and  $\mathcal{G}_r$ -measurable with  $E^P[Z] = 1$ . Then*

$$R := (Z \cdot P) \circ (X(r + \cdot))^{-1} \in \mathcal{M}(\mathcal{X}_r^\circ). \quad (36)$$

*In the canonical case  $\Omega = F$ ,  $\mathbb{G} = \mathbb{F}^\circ$ ,  $X = \pi$  this reads  $R = (Z \cdot P) \circ \theta_r^{-1}$  with  $Z$  being  $\mathcal{F}_r^\circ$ -measurable.*

*Remark 6.18.* The two levels matter and cannot be collapsed. In Theorem 6.23(a) the weight is  $Z = \mathbf{1}_{F_0}$  with  $F_0 \in \mathcal{G}_r$ , and  $\mathcal{G}_r$  may be strictly larger than  $\sigma(X(u) : u \leq r)$ ; such a  $Z$  is *not* a functional of the path and cannot be transported to  $F$ . What can be transported is the *law* of  $X(r + \cdot)$  under  $Z \cdot P$ , which is what (36) asserts.

*Proof.* Let  $\hat{Y}^\circ \in \mathcal{X}_r^\circ$ , let  $s \leq t$  and let  $Z_s^\circ \in \mathcal{Z}_s^\circ$ . Choose  $Y^\circ \in \mathcal{X}^\circ$  and  $\kappa$  as in (35), and evaluate everything along  $X$ ; to keep the notation readable write  $\tilde{V} = V(X)$  for a functional  $V$  on  $F$ , so that  $\tilde{Y}_u^\circ$  is a  $\mathbb{G}$ -martingale by hypothesis.

Since  $Z_s^\circ$  is  $\mathcal{F}_s^\circ$ -measurable and  $\pi_u \circ \theta_r = \pi_{r+u}$ , the variable  $(Z_s^\circ \circ \theta_r)(X) = Z_s^\circ(X(r+\cdot))$  is  $\sigma(X(u) : u \leq r+s)$ -measurable, hence  $\mathcal{G}_{r+s}$ -measurable; so is  $Z$ , because  $\mathcal{G}_r \subset \mathcal{G}_{r+s}$ , and so is  $\tilde{\kappa}$ . Hence  $W := Z \cdot (Z_s^\circ \circ \theta_r)(X)$  is bounded and  $\mathcal{G}_{r+s}$ -measurable, and the martingale property gives  $E^P[(\tilde{Y}_{r+t}^\circ - \tilde{Y}_{r+s}^\circ)W] = 0$ . Therefore

$$\begin{aligned} E^R[\hat{Y}_t^\circ Z_s^\circ] &= E^P[Z(\hat{Y}_t^\circ \circ \theta_r)(X)(Z_s^\circ \circ \theta_r)(X)] = E^P[(\tilde{Y}_{r+t}^\circ - \tilde{Y}_r^\circ + \tilde{\kappa})W] \\ &= E^P[(\tilde{Y}_{r+s}^\circ - \tilde{Y}_r^\circ + \tilde{\kappa})W] = E^R[\hat{Y}_s^\circ Z_s^\circ], \end{aligned}$$

the second equality by (35) and the third by the display above. As  $Z_s^\circ \in \mathcal{Z}_s^\circ$  was arbitrary, part (ii) of Definition 4.3 yields  $E^R[\hat{Y}_t^\circ \mid \mathcal{F}_s^\circ] = \hat{Y}_s^\circ$ ; integrability of  $\hat{Y}_t^\circ$  under  $R$  holds because  $Z$  is bounded and  $\tilde{Y}_{r+t}^\circ, \tilde{Y}_r^\circ, \tilde{\kappa} \in L^1(P)$ .  $\square$

Note where the two hypotheses act. Shift stability is used only to rewrite  $\hat{Y}_t^\circ \circ \theta_r$ ; the determining set is used only in the last step, and it is exactly the rôle that Proposition 4.9 plays in [7]. Everything else is the tower property.

*Remark 6.19* (Submartingale problems: a different direction, not a wider one). Replacing “ $Y^\circ$  is a  $P$ -martingale” by “ $Y^\circ$  is a  $P$ -submartingale” throughout defines the *submartingale problem*  $\mathcal{M}^{\text{sub}}(\mathcal{X}^\circ)$ , which is the standard formulation of a diffusion with reflecting boundary conditions ([17]). Three things have to be said about it, and the first two are easy to run together.

(i) *It is not a generalization in the direction of Remark 4.5, but transverse to it.* Formulating “submartingale” needs an order on the values, so it forces  $\mathbb{K} = \mathbb{R}$ ; and complexifying destroys the order. The two directions are mutually exclusive, and a table that lists both as “free” — as Section 1.4 does — must be read as offering a choice, not a sum.

(ii) *Over  $\mathbb{R}$  it is nevertheless a genuine generalization, and the precise sense is worth recording because it is what makes the label honest:*

$$\mathcal{M}(\mathcal{X}^\circ) = \mathcal{M}^{\text{sub}}(\mathcal{X}^\circ \cup (-\mathcal{X}^\circ)), \quad (37)$$

since  $Y^\circ$  and  $-Y^\circ$  are both submartingales exactly when  $Y^\circ$  is a martingale. The martingale problem is thus the submartingale problem for a *symmetric* test family, and every statement about  $\mathcal{M}^{\text{sub}}$  specializes to one about  $\mathcal{M}$  by symmetrizing.

(iii) *Which of the machinery survives, and at what cost.* Lemma 6.2 transfers with no change at all: its proof integrates  $E^{P_\nu}[(Y_t^\circ - Y_s^\circ)\mathbf{1}_G] = 0$  over  $\nu$ , and an inequality integrates just as well; note that it uses no determining set. Lemma 6.17 likewise, the weight  $W = Z \cdot (Z_s^\circ \circ \theta_r)(X)$  in its proof being nonnegative whenever  $Z$  and  $Z_s^\circ$  are.

What does *not* come for free is the determining set, and this is where the transfer stops being a matter of reading “ $\geq$ ” for “ $=$ ”. Definition 4.3(ii) has to be replaced by its one-sided form:  $E^P[(Y_t - Y_s)Z_s^\circ] \geq 0$  for all *nonnegative*  $Z_s^\circ \in \mathcal{Z}_s^\circ$  should imply  $E^P[Y_t \mid \mathcal{F}_s] \geq Y_s$ . That is a strictly stronger demand on  $\mathcal{Z}^\circ$  than the two-sided one, because a finite signed measure that is nonnegative on a  $\pi$ -system generating  $\mathcal{F}_s$  need not be nonnegative. It does hold for the family of Example 4.4, but for a reason that has to be supplied: Lemma 6.20 below.

**Lemma 6.20** (One-sided determination; (E2)). *Let  $\mathcal{C}$  be a semiring of subsets of  $\Omega$  with  $\Omega \in \mathcal{C}$ , generating the  $\sigma$ -field  $\mathcal{G}$ , and let  $\mu$  be a finite signed measure on  $\mathcal{G}$  with  $\mu(C) \geq 0$  for every  $C \in \mathcal{C}$ . Then  $\mu \geq 0$ .*

The hypothesis  $\Omega \in \mathcal{C}$  cannot be dropped: on  $\Omega = \{1, 2\}$  the family  $\{\emptyset, \{1\}\}$  is a semiring generating  $2^\Omega$ , and  $\mu = \delta_1 - 5\delta_2$  is nonnegative on it without being nonnegative.

Consequently, if  $E$  satisfies (E2), then the products  $\prod_{k \leq n} h_k(\pi_{t_k})$  with  $h_k \in C_b(E)$ ,  $h_k \geq 0$ , are one-sided determining in the sense of Remark 6.19(iii), so that Proposition 4.9 and Example 4.4 survive the passage to submartingale problems.

*Proof. The signed-measure statement.* Finite disjoint unions of elements of  $\mathcal{C}$  form an algebra  $\mathcal{A}$  — closure under complements is where  $\Omega \in \mathcal{C}$  enters, via the semiring property  $\Omega \setminus C = \bigsqcup_i C_i$  — and  $\mu \geq 0$  on  $\mathcal{A}$  by additivity. Let  $\mu = \mu^+ - \mu^-$  be the Hahn–Jordan decomposition with negative set  $N$ . Since  $|\mu|$  is finite and  $\mathcal{A}$  generates  $\mathcal{G}$ , for every  $\varepsilon > 0$  there is  $A \in \mathcal{A}$  with  $|\mu|(A \triangle N) < \varepsilon$ ; then  $\mu(A) \leq \mu(N) + \varepsilon = -\mu^-(\Omega) + \varepsilon$ , and  $\mu(A) \geq 0$  forces  $\mu^-(\Omega) < \varepsilon$ . As  $\varepsilon$  was arbitrary,  $\mu^- = 0$ .

*The application.* Put  $\mu(G) = E^P[(Y_t - Y_s)\mathbf{1}_G]$  for  $G \in \mathcal{F}_s$ , a finite signed measure. Given closed  $A_1, \dots, A_n \subset E$  choose  $h_k^m(x) = (1 - m d(x, A_k))^+ \in C_b(E)$ , which is legitimate under (E2) and decreases to  $\mathbf{1}_{A_k}$ ; then  $\prod_k h_k^m$  decreases to  $\mathbf{1}_{A_1 \times \dots \times A_n}$  and dominated convergence gives  $\mu \geq 0$  on closed measurable rectangles. By regularity of the finite measure  $|\mu| \circ (\pi_{t_1}, \dots, \pi_{t_n})^{-1}$  on a metric space this extends to all Borel rectangles, which form a semiring containing the full space (all  $B_k = E$ ) and generating  $\sigma(\pi_{t_1}, \dots, \pi_{t_n})$ ; the first part applies. Finally, the Borel rectangles over *all* finite subsets of  $\mathbb{T}_{\leq s}$  again form a semiring containing the full space — a product of two rectangles over different time sets is a rectangle over their union — and it generates  $\mathcal{F}_s$ , so one more application of the first part gives  $\mu \geq 0$  there.  $\square$

*Remark 6.21* (What is still lost). With Lemma 6.20 the structural half of Section 6.1 is available for submartingale problems. Two things are not.

*Disintegration.* Lemma 6.3 needs the countable test (31), so that one null set serves all data at once. In the one-sided form the test variables must range over a family rich enough for Lemma 6.20, and a countable such family exists only if  $E$  is second countable — which (E2) gives, so the lemma does transfer, but the hypothesis is no longer the innocuous one it was.

*Uniqueness.* Definition 6.11 compares two measures by *equality* of expectations, and an inequality propagates nothing: from  $E^P[Zf(\pi_s)] \geq E^Q[Zf(\pi_s)]$  one cannot conclude anything at  $t$ . A submartingale problem therefore needs a uniqueness criterion of its own, and supplying one is exactly the work done in the reflecting-boundary literature. This manuscript does not attempt it; Task 22 of the plan records it.

**Lemma 6.22** (Agreement propagates under a shift system; (T0) + (T4) + (E0)). *Let  $(\mathcal{X}_r^\circ)$  be a shift system for  $\mathcal{X}^\circ$  with  $\mathcal{Z}^\circ$  determining for each  $\mathcal{X}_r^\circ$ , and suppose that for every  $r \in \mathbb{T}$*

$$P, Q \in \mathcal{M}(\mathcal{X}_r^\circ), P\pi_0^{-1} = Q\pi_0^{-1} \implies P\pi_t^{-1} = Q\pi_t^{-1} \text{ for all } t \in \mathbb{T}. \quad (38)$$

*Then  $\mathcal{M}(\mathcal{X}^\circ)$  propagates agreement.*

*Proof.* Let  $P, Q \in \mathcal{M}(\mathcal{X}^\circ)$ , let  $s \leq t$ , and let  $Z \geq 0$  be bounded and  $\mathcal{F}_s^\circ$ -measurable with

$$E^P[Z h(\pi_s)] = E^Q[Z h(\pi_s)] \quad \text{for all } h \in B(E). \quad (39)$$

Taking  $h \equiv 1$  gives  $c := E^P[Z] = E^Q[Z]$ . If  $c = 0$  then  $Z = 0$  both  $P$ - and  $Q$ -a.s. and the conclusion is trivial, so assume  $c > 0$  and replace  $Z$  by  $Z/c$ , i.e. assume  $E^P[Z] = E^Q[Z] = 1$ .

By Lemma 6.17,  $R_P = (Z \cdot P) \circ \theta_s^{-1}$  and  $R_Q = (Z \cdot Q) \circ \theta_s^{-1}$  both lie in  $\mathcal{M}(\mathcal{X}_s^\circ)$  — the *same* shifted problem, which is what makes (38) applicable. Since  $\pi_0 \circ \theta_s = \pi_s$ ,

$$E^{R_P}[h(\pi_0)] = E^P[Z h(\pi_s)] = E^Q[Z h(\pi_s)] = E^{R_Q}[h(\pi_0)]$$

for every  $h \in B(E)$  by (39), so  $R_P$  and  $R_Q$  have the same initial law. Hence (38) gives  $E^{R_P}[f(\pi_u)] = E^{R_Q}[f(\pi_u)]$  for every  $u \in \mathbb{T}$  and  $f \in B(E)$ . Taking  $u = t - s$ , which exists by (T4), and using  $\pi_u \circ \theta_s = \pi_{s+u} = \pi_t$ , this reads  $E^P[Zf(\pi_t)] = E^Q[Zf(\pi_t)]$ . That is (32).  $\square$

**Theorem 6.23** (Abstract uniqueness and Markov property; (T0) + (T4), and (T2a) for (b)). *Let  $(\mathcal{X}_r^\circ)$  be a shift system for  $\mathcal{X}^\circ$ , let  $\mathcal{Z}^\circ$  be determining for every  $\mathcal{X}_r^\circ$ , and suppose (38) holds for every  $r \in \mathbb{T}$ . Then, subject to the integrability provisos of Lemmas 6.2 and 6.17:*

- (a) *every solution is Markov — in general time inhomogeneously so: if  $X$  has paths in  $F$  and solves the martingale problem with respect to  $(\mathcal{G}_t)$ , then for  $f \in B(E)$ ,  $r, t \in \mathbb{T}$ ,*

$$E[f(X(r+t)) \mid \mathcal{G}_r] = E[f(X(r+t)) \mid X(r)];$$

- (b) *if  $\mathbb{T}$  satisfies (T2a), then  $\mathcal{M}(\mathcal{X}^\circ, \mu)$  contains at most one element, for every  $\mu \in \mathcal{P}(E)$ .*

*Proof.* (a). Fix  $r \in \mathbb{T}$  and  $F_0 \in \mathcal{G}_r$  with  $P(F_0) > 0$ , and put

$$Z_1 = \frac{\mathbf{1}_{F_0}}{P(F_0)}, \quad Z_2 = \frac{E[\mathbf{1}_{F_0} \mid X(r)]}{P(F_0)}.$$

Both are nonnegative, bounded by  $1/P(F_0)$ , of mean 1, and  $\mathcal{G}_r$ -measurable —  $Z_2$  is even  $\sigma(X(r))$ -measurable, while  $Z_1$  need not be a functional of the path, which is why Lemma 6.17 is stated on  $(\Omega, \mathcal{F}, \mathbb{G}, P)$  and not on  $F$ ; see Remark 6.18. Applying Lemma 6.17 at  $r$ , the two laws  $R_i = (Z_i \cdot P) \circ (X(r+\cdot))^{-1}$  lie in  $\mathcal{M}(\mathcal{X}_r^\circ)$  — the *same* shifted problem for  $i = 1$  and  $i = 2$ , which is all that (38) needs. A direct computation gives, for  $B \in \mathcal{B}(E)$ ,

$$R_1 \pi_0^{-1}(B) = \frac{E[\mathbf{1}_{F_0} \mathbf{1}_B(X(r))]}{P(F_0)} = R_2 \pi_0^{-1}(B),$$

the middle expression for  $i = 2$  because  $E[\mathbf{1}_{F_0} \mathbf{1}_B(X(r))] = E[E[\mathbf{1}_{F_0} \mid X(r)] \mathbf{1}_B(X(r))]$ . So (38) applies and yields  $E^{R_1}[f(\pi_t)] = E^{R_2}[f(\pi_t)]$ , i.e.

$$E[\mathbf{1}_{F_0} E[f(X(r+t)) \mid \mathcal{G}_r]] = E[\mathbf{1}_{F_0} E[f(X(r+t)) \mid X(r)]],$$

where on the right we used the  $\sigma(X(r))$ -measurability of  $Z_2$  once more. Since  $F_0 \in \mathcal{G}_r$  was arbitrary, the Markov property follows.

- (b). Lemma 6.22 and Proposition 6.13.  $\square$

*Remark 6.24* (Uniqueness is not a Markovian fact). The decomposition above is worth reading off, because the classical statement hides it.

|                                             | needs        | Markovian? |
|---------------------------------------------|--------------|------------|
| Uniqueness (Proposition 6.13)               | (32)         | no         |
| (32) from one-dimensional laws (Lemma 6.22) | shift system | yes        |
| Markov property (Theorem 6.23(a))           | shift system | yes        |

Two consequences. First, Proposition 6.13 uses no shift, no  $\theta_r$ , no re-indexing and *no* bundle (T4): only (T2a), for the chains of Remark 6.25. [7], Theorem 4.4.2 looks Markovian because its hypothesis is stated *unconditionally*, on the one-dimensional distributions; and an unconditional hypothesis can be applied at a later time only after the restarted process

has been re-indexed to start at 0, which is what  $\theta_r$  does and where the shift enters. Restarting *itself* is free, by Lemma 6.5. State the hypothesis conditionally, as (32) does, and the Markov structure disappears from the uniqueness half.

Second, (32) is available in genuinely non-Markovian situations, where no shift system exists:

- *Path-dependent equations.* Pathwise uniqueness gives more than uniqueness in law: it gives uniqueness in law of the equation *restarted from a prescribed past*, and that is what (32) asks for. Plain uniqueness in law is *not* enough — (32) is a conditional statement, and Remark 6.12 explains why the unconditional one does not suffice. The coefficients may depend on the whole past.
- *Change of measure.* If  $\mathcal{X}^\circ$  arises from a family with (32) by an absolutely continuous change of measure with a path-dependent density, the property is inherited.
- *Semimartingale problems.* [10], Chapter III prove uniqueness for triplets  $(B, C, \nu)$  that are predictable functionals of the path, with no Markov structure in sight.

None of these is developed here; the point is only that Proposition 6.13 is the statement they would use, and Theorem 6.23(b) is not.

*Remark 6.25* (Why (b) needs a linear order). Part (a) uses nothing about  $\mathbb{T}$  beyond (T0) and (T4): the proof needs  $\theta_r$ , the inclusion  $\mathcal{F}_r^\circ \subset \mathcal{F}_{r+s}^\circ$  and the martingale property at  $r+s \leq r+t$ , all of which hold in a preordered monoid. There is no topology and no linear order in it.

Part (b) is different, and the reason is the induction of Proposition 6.13, not the shift. That induction advances along  $\mathcal{F}_{t_1}^\circ \subset \mathcal{F}_{t_2}^\circ \subset \dots$ , so the times must form a *chain*. On a linearly ordered  $\mathbb{T}$  every finite set of times is a chain and nothing is lost. On a merely partially ordered  $\mathbb{T}$  one obtains equality of the finite-dimensional distributions along chains only, and these do not determine the law: on  $\mathbb{T} = \mathbb{R}_+^2$  the times  $(1, 0)$  and  $(0, 1)$  are incomparable, and no chain sees their joint distribution. Replacing  $t_n$  by  $t_1 \vee \dots \vee t_n$ , which (T1) would allow, does not repair this, since the next time  $t_{n+1}$  need not dominate that join.

So the slogan “uniqueness of the one-dimensional distributions implies uniqueness” is *false* for a multiparameter index, and the obstruction is the induction, not the martingale problem. This is the sharpest instance of the general pattern of Section 2.2: the Markov property is algebraic and survives on a poset, uniqueness is order-theoretic and does not.

For the strong Markov property one restarts at a stopping time. The only extra input is optional sampling — which is where path regularity finally enters — and it has to be handled with care, because the random times  $\tau + s$  are unbounded and Fact 2.25 does not apply to them as it stands. The following truncation lemma is the correct tool, and its integrability hypothesis is the honest one. An earlier version of this manuscript asked instead for uniformly integrable increments *on bounded intervals*. That is a statement about deterministic windows  $[s, t] \subset \mathbb{T}_{\leq T}$ , and it says nothing about the random window  $[\tau + s, \tau + t]$ , which is unbounded when  $\tau$  is; in particular it does not deliver the integrability that the truncation argument needs in order to pass to the limit. What Lemma 6.26 shows is that this integrability is not merely necessary for the bookkeeping but *sufficient* for the conclusion — no uniform integrability of the martingale itself is required.

**Lemma 6.26** (Optional sampling after a finite stopping time; (T2b) + (T4)). *Let  $\mathbb{T}$  satisfy (T2b) and (T4), let  $Y$  be a right continuous  $(\mathcal{G}_t)$ -martingale, let  $\tau$  be an a.s. finite*

$(\mathcal{G}_t)$ -stopping time, and let  $s \leq t$  in  $\mathbb{T}$  be such that

$$Y_{\tau+t} - Y_{\tau+s} \in L^1(P). \quad (40)$$

Then  $E[(Y_{\tau+t} - Y_{\tau+s})W] = 0$  for every bounded  $\mathcal{G}_{\tau+s}$ -measurable  $W$ .

*Proof.* By (T2b) there is a countable cofinal increasing sequence  $(T_m)$  in  $\mathbb{T}$  (Step 2 of the proof of Theorem 5.4), so  $\mathbf{1}_{\{\tau \leq T_m\}} \uparrow 1$  a.s. Fix  $m$  and put  $\sigma_1 = (\tau \wedge T_m) + s \leq \sigma_2 = (\tau \wedge T_m) + t$ . Both are  $(\mathcal{G}_u)$ -stopping times:  $\tau \wedge T_m$  is one by Fact 2.19, and adding a constant  $s$  preserves the property because by (T4) the map  $u \mapsto u - s$  is defined and order preserving on  $\{u \geq s\}$ , so  $\{\sigma_1 \leq u\} = \{\tau \wedge T_m \leq u - s\} \in \mathcal{G}_{u-s} \subset \mathcal{G}_u$  for  $u \geq s$ , while for  $u < s$  the event is empty since  $\sigma_1 \geq s$ . Both are bounded by  $T_m + t$ . Optional sampling at bounded stopping times — the case  $\tau_2 \leq T$  of Fact 2.25, which needs only right continuity and integrability of  $Y$  — gives  $E[Y_{\sigma_2} | \mathcal{G}_{\sigma_1}] = Y_{\sigma_1}$ . The variable  $W\mathbf{1}_{\{\tau \leq T_m\}}$  is  $\mathcal{G}_{\sigma_1}$ -measurable. Indeed, take  $W = \mathbf{1}_G$  with  $G \in \mathcal{G}_{\tau+s}$ ; since on  $\{\tau \leq T_m\}$  the times  $\sigma_1$  and  $\tau + s$  agree,

$$G \cap \{\tau \leq T_m\} \cap \{\sigma_1 \leq u\} = (G \cap \{\tau + s \leq u\}) \cap \{\tau \leq T_m\},$$

and this lies in  $\mathcal{G}_u$  in both of the two cases the linear order allows: for  $u \geq T_m$  because  $G \cap \{\tau + s \leq u\} \in \mathcal{G}_u$  by definition of  $\mathcal{G}_{\tau+s}$  and  $\{\tau \leq T_m\} \in \mathcal{G}_{T_m} \subset \mathcal{G}_u$ ; for  $u < T_m$  because then  $\{\tau + s \leq u\} \subset \{\tau \leq u\} \subset \{\tau \leq T_m\}$  — using  $\tau \leq \tau + s$ , i.e.  $0 \leq s$  and order preservation — so the second intersecting event is redundant and the set equals  $G \cap \{\tau + s \leq u\} \in \mathcal{G}_u$ . Hence

$$E[(Y_{\sigma_2} - Y_{\sigma_1})W\mathbf{1}_{\{\tau \leq T_m\}}] = 0,$$

and on  $\{\tau \leq T_m\}$  the integrand equals  $(Y_{\tau+t} - Y_{\tau+s})W$ , while off it the indicator vanishes. So  $E[(Y_{\tau+t} - Y_{\tau+s})W\mathbf{1}_{\{\tau \leq T_m\}}] = 0$  for every  $m$ , and dominated convergence — the dominator is  $|Y_{\tau+t} - Y_{\tau+s}| \|W\|$ , integrable by (40) — gives the assertion.  $\square$

**Theorem 6.27** (Abstract strong Markov property; (T2b) + (T4)). *In the situation of Theorem 6.23, assume in addition that  $F \subset D_E[0, \infty)$ , that  $(\theta_r)$  extends to a measurable map  $(\omega, r) \mapsto \theta_r \omega$ , that the shift system is measurable in the sense of Definition 6.14, that every  $Y^\circ \in \mathcal{X}^\circ$  has right continuous paths, and that (40) holds for  $Y = Y^\circ(X)$  at the stopping time  $\tau$  below, for every  $Y^\circ \in \mathcal{X}^\circ$  and  $s \leq t$ . Let  $X$  have paths in  $F$ , solve the martingale problem with respect to  $(\mathcal{G}_t)$ , and let  $\tau$  be an a.s. finite  $(\mathcal{G}_t)$ -stopping time. Then*

$$E[f(X(\tau+t)) | \mathcal{G}_\tau] = E[f(X(\tau+t)) | X(\tau)], \quad f \in B(E), t \in \mathbb{T}.$$

*If moreover  $\tau$  takes countably many values and there is a measurable family  $(P_{x,r})_{x \in E, r \in \mathbb{T}}$  with  $P_{x,r} \in \mathcal{M}(\mathcal{X}_r^\circ, \delta_x)$ , then*

$$E[f(X(\tau+t)) | \mathcal{G}_\tau] = (T_{\tau, \tau+t}f)(X(\tau)), \quad (T_{r, r+t}f)(x) = \int f(\pi_t) dP_{x,r},$$

*a two-parameter family of kernels satisfying the Chapman–Kolmogorov relation  $T_{r,s}T_{s,t} = T_{r,t}$  for  $r \leq s \leq t$ .*

*Proof. Step 1: the restart identity at a stopping time.* The proof of Lemma 6.17 rests on the single identity  $E^P[(\tilde{Y}_{r+t}^\circ - \tilde{Y}_{r+s}^\circ)W] = 0$  for bounded  $\mathcal{G}_{r+s}$ -measurable  $W$ . We need it with  $r$  replaced by  $\tau$ . Now  $\tau + s$  and  $\tau + t$  are  $(\mathcal{G}_u)$ -stopping times: by (T4) the map  $u \mapsto u - s$  is defined on  $\{u \geq s\}$  and order preserving, so  $\{\tau + s \leq u\} = \{\tau \leq u - s\} \in \mathcal{G}_{u-s} \subset \mathcal{G}_u$

for  $u \geq s$ , and the event is empty otherwise, since by linearity of the order  $u < s$  forces  $\tau + s \geq s > u$ . Since  $\tilde{Y}^\circ$  is a right continuous martingale and (40) holds, Lemma 6.26 gives the identity for every bounded  $\mathcal{G}_{\tau+s}$ -measurable  $W$ .

*Step 2: the first assertion.* With  $\mathcal{X}_\tau^\circ$  in place of  $\mathcal{X}_r^\circ$  — meaningful because the shift system is measurable, so that  $\hat{Y}^{\circ, \tau(\omega)}$  is jointly measurable — the remainder of the proof of Lemma 6.17 and of Theorem 6.23(a) is unchanged.

*Step 3: the second assertion.* Let  $\tau$  take the countably many values  $r_1, r_2, \dots$ , and fix  $j$ . It suffices to prove the identity on  $\{\tau = r_j\}$ , since these events partition  $\Omega$  and lie in  $\mathcal{G}_\tau$ . Let  $F_0 \in \mathcal{G}_\tau$  with  $F_0 \subset \{\tau = r_j\}$  and  $P(F_0) > 0$ , write  $\mu_{F_0} = P\{X(\tau) \in \cdot \mid F_0\}$ , and put

$$R_1 = (Z_1 \cdot P) \circ (X(\tau + \cdot))^{-1} \quad \text{with } Z_1 = \mathbf{1}_{F_0}/P(F_0), \quad R_2 = \int P_{x, r_j} \mu_{F_0}(\mathrm{d}x).$$

Both lie in  $\mathcal{M}(\mathcal{X}_{r_j}^\circ)$ : the first by Steps 1–2, the second by Lemma 6.2, and it is here that  $F_0$  must be carried by a single value of  $\tau$  — a mixture of solutions of *different* shifted problems need solve none of them. Since  $R_1 \pi_0^{-1} = \mu_{F_0} = R_2 \pi_0^{-1}$ , (38) applies and gives the assertion on  $F_0$ . The Chapman–Kolmogorov relation follows by applying the first assertion twice.  $\square$

*Remark 6.28* (On the two restrictions). Two hypotheses in Theorem 6.27 deserve comment.

*Countably many values.* In the time-homogeneous case  $\mathcal{X}_r^\circ = \mathcal{X}^\circ$  the restriction is vacuous, because then all the measures  $P_{x, r} = P_x$  solve one and the same problem and Step 3 needs no partition. It is only in the inhomogeneous case that the mixture must be confined to a single  $r$ . A general  $\tau$  is then reached by approximation from above by countably-valued stopping times  $\tau_n \downarrow \tau$ ; this requires right continuity of  $\mathbb{G}$  and of  $r \mapsto P_{x, r}$  in a suitable sense, and is not developed here.

*The integrability (40).* This is what Step 1 needs, and it concerns the increment over the *random window*  $\langle \tau + s, \tau + t \rangle_\iota$ , not over a bounded interval. For the family  $\mathcal{X}_A$  with bounded  $f$  and  $g$  the increment is bounded by  $2\|f\| + \|g\| q(\langle \tau + s, \tau + t \rangle_\iota)$ ; if the clock is shift invariant this equals  $2\|f\| + \|g\| q(\langle s, t \rangle_\iota)$ , a *constant*, so (40) is automatic — this is the case of Theorem 6.51. For a general clock it requires  $E[q(\langle \tau + s, \tau + t \rangle_\iota)] < \infty$  and is a genuine restriction.

*Remark 6.29* (What the abstract form buys). (i) *One lemma, four applications.* In [7] the conditioning construction in the proof of the Markov property, its variant in the induction step of the uniqueness proof, and the two constructions in parts (b) and (c) look like four separate arguments. They are one: Lemma 6.17 with, in turn,

$$\begin{aligned} Z &= \mathbf{1}_{F_0}, & Z &= E[\mathbf{1}_{F_0} \mid X(r)], \\ Z &= \mathbf{1}_G \text{ for } G \in \mathcal{F}_s^\circ, & Z &= \mathbf{1}_{F_0} \text{ at a stopping time,} \end{aligned}$$

all normalized to mean 1. The only property of  $Z$  that is ever used is that it is nonnegative and  $\mathcal{F}_r^\circ$ -measurable. The third of these is Lemma 6.22; [7] use  $Z = \prod_k f_k(\pi_{t_k})$  there, which is the same lemma applied to a tilted measure instead of a conditioned one, and is unnecessary once the induction is arranged as in Proposition 6.13.

(ii) *The shift system is the real hypothesis, and it costs nothing.* Definition 6.14 is where the Markov property comes from, and it is checkable. For  $\mathcal{X}_A$  it never fails: whatever the clock  $q$  and however  $g$  depends on time, the shifted problem exists and is given explicitly by Example 6.16, with the pulled-back clock  $q_r = q(r + \cdot)$  and the shifted coefficient  $g(r + \cdot, \cdot)$ . What *does* depend on  $q$  and  $g$  is whether the shifted problem is the *same* problem:  $\mathcal{X}_r^\circ = \mathcal{X}^\circ$  holds if and only if  $q$  is shift invariant and  $g$  is time

independent, and only then is the Markov process homogeneous. The first version of this manuscript required  $\mathcal{X}_r^\circ = \mathcal{X}^\circ$  and therefore had to assume shift invariance of the clock; carrying the shifted family instead — as [10], III.2.39 do — removes that hypothesis and yields the inhomogeneous case for free. [7] never have to say any of this because their clock is Lebesgue measure and their  $A$  does not depend on time.

- (iii) *No operator, no boundedness, no path regularity.* Parts (a) and (b) use neither  $C_b(E)$  nor  $D_E[0, \infty)$ ;  $F$  may be any Polish path space, for instance  $L_{\text{loc}}^p$ . Path regularity enters only in Theorem 6.27, and there only through optional sampling.
- (iv) *Mixtures.* Lemma 6.2 says  $\mathcal{M}(\mathcal{X}^\circ)$  is convex and stable under measurable mixtures. This is what makes the passage from  $\delta_x$  to a general  $\mu$  legitimate — in [7] it is invoked silently in part (c) and in Corollary 7.20.

### 6.3 The local martingale problem

Everything in Section 6.1 is stated for  $\mathcal{M}(\mathcal{X}^\circ)$ . The local analogue is not a corollary, and the reason is worth stating plainly: “ $Y$  is a local martingale” is not a condition linear in  $P$ , because the localizing sequence may depend on  $P$ . Lemma 6.2 and Lemma 6.3 rest on that linearity, and Lemma 6.17 rests on a martingale identity that a local martingale does not supply. All three therefore fail for  $\mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$  without a further hypothesis.

The hypothesis that repairs them is due to Stroock and Varadhan and is used in this form by [16], Theorems 32.10 and 32.11: insist that the localization be performed by stopping times that are *functionals of the path*, hence the same for every  $P$ , and that the family of such times be stable under the shift.

**Definition 6.30** (Localizing system; (T2b) + (T4)). A *localizing system* for  $\mathcal{X}^\circ$  is a family  $\Sigma$  of *strict*  $(\mathcal{F}_t^\circ)$ -stopping times on  $F$  — functionals of the path, not of any measure — such that the following hold. If the solution is presented on an ambient space  $(\Omega, \mathcal{F}, \mathbb{G}, P)$  rather than on  $F$ , all three conditions are to be read with  $\mathcal{F}^\circ$  replaced by  $\mathbb{G}$  and  $Y^\circ$  by  $\tilde{Y} = Y^\circ(X)$ ; strictness of the times in  $\Sigma$  is what makes  $\sigma(X)$  a  $\mathbb{G}$ -stopping time for every such presentation.

- (L1) *Uniform localization.*  $\Sigma$  contains a sequence  $\tau_1 \leq \tau_2 \leq \dots$  with  $\tau_n \uparrow \infty$  pointwise, and

$$P \in \mathcal{M}_{\text{loc}}(\mathcal{X}^\circ) \iff Y^{\circ, \tau_n} := Y_{\cdot \wedge \tau_n}^\circ \text{ is a } P\text{-martingale for all } Y^\circ \in \mathcal{X}^\circ, n \in \mathbb{N}.$$

- (L2) *Shift covariance.*  $\sigma \in \Sigma$  and  $r \in \mathbb{T}$  imply  $r + \sigma \circ \theta_r \in \Sigma$ .

- (L3) *Integrable increments after a restart.* For every  $P \in \mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$ ,  $Y^\circ \in \mathcal{X}^\circ$ ,  $r \in \mathbb{T}$  and  $\sigma \in \Sigma$  with  $\sigma \geq r$ , the process  $t \mapsto Y_{(r+t) \wedge \sigma}^\circ - Y_r^\circ$  is a martingale for the filtration  $(\mathcal{F}_{r+t}^\circ)_{t \in \mathbb{T}}$  under  $P$ .

$\Sigma$  is *strongly* shift covariant if (L2) and (L3) hold with  $r$  replaced by any a.s. finite  $(\mathcal{F}_t^\circ)$ -stopping time  $\tau$ , the time  $\tau + \sigma \circ \theta_\tau$  being required to be a stopping time.

*Remark 6.31.* Three comments on Definition 6.30.

(L1) *is the whole point.* It turns membership in  $\mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$  into a family of identities  $E^P[(Y_t^{\circ, \tau_n} - Y_s^{\circ, \tau_n})\mathbf{1}_G] = 0$ , each of which is linear in  $P$ . The implication “ $\Leftarrow$ ” is automatic because  $\tau_n \uparrow \infty$ ; “ $\Rightarrow$ ” is the assumption, and it says that *one* sequence localizes every test process under every solution.

(L3) is not implied by (L1). Stopping a  $P$ -local martingale at  $\sigma \in \Sigma$  leaves a local martingale, not a martingale:  $Y^{\circ, \sigma}$  is unstopped before  $r$ , where nothing controls it. What (L3) asks is only that the *increments after  $r$*  be integrable, which is the same kind of hypothesis as in Theorem 6.27 and is usually visible at once.

*The diffusion case.* For  $\mathcal{X}^\circ = \{f(\pi) - f(\pi_0) - \int Af(\pi)\}$  on  $F = C_{\mathbb{R}_+, \mathbb{R}^d}$  take  $\Sigma$  generated by the exit times  $\tau_n = \inf\{t : |\pi_t| \geq n\}$  and their shifts. Then (L1) is [16], Theorem 32.10, (L2) holds because  $r + \tau_n \circ \theta_r = \inf\{u \geq r : |\pi_u| \geq n\}$  is again an exit time, and (L3) holds because between  $r$  and  $\sigma$  the path stays in  $\bar{B}_n$ , so that the increments are bounded by  $2\|f\|_{\bar{B}_n} + q(\mathbb{T}_{\leq T})\|Af\|_{\bar{B}_n}$ . [16] uses  $\tau_n^f = \inf\{t : |M_t^f| \geq n\}$  instead, which serves equally well and has the advantage of not referring to the state space.

With Definition 6.30 in hand the three lemmas of Section 6.1 all go through, and their proofs are the earlier ones with  $Y^\circ$  replaced by  $Y^{\circ, \tau_n}$ . We give them in full, since the point of the exercise is that nothing is being waved through.

**Lemma 6.32** (Local mixtures and disintegration; (T0) + (E1)). *Parts (a) and (b) need only (E0); (E1) is used in (c).*

- (a)  $\mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$  is convex. No hypothesis is needed for this.
- (b) If  $\Sigma$  satisfies (L1), then  $\mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$  is stable under measurable mixtures  $P = \int P_\vartheta \nu(d\vartheta)$  subject to  $\int E^{P_\vartheta} |Y_t^{\circ, \tau_n}| \nu(d\vartheta) < \infty$ .
- (c) If in addition membership can be tested along countably many data as in (31), then for  $P \in \mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$  one has  $P(\cdot \mid \pi_0 = x) \in \mathcal{M}_{\text{loc}}(\mathcal{X}^\circ, \delta_x)$  for  $P\pi_0^{-1}$ -a.e.  $x$ .

*Proof.* (a) Let  $P, P' \in \mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$ ,  $Q = \alpha P + (1 - \alpha)P'$ , and let  $Y^\circ \in \mathcal{X}^\circ$  be localized by  $(\tau_n)$  under  $P$  and by  $(\tau'_n)$  under  $P'$ . Put  $\sigma_n = \tau_n \wedge \tau'_n$ . Then  $Y^{\circ, \sigma_n}$  is a martingale under  $P$  and under  $P'$  — stopping a martingale at a further stopping time leaves a martingale — hence under  $Q$ , and  $\sigma_n \rightarrow \infty$   $Q$ -a.s. because  $\tau_n \rightarrow \infty$   $P$ -a.s. and  $\tau'_n \rightarrow \infty$   $P'$ -a.s. Here the two localizing sequences are allowed to differ; it is only for (b) that this ceases to be possible.

(b), (c) By (L1),  $\mathcal{M}_{\text{loc}}(\mathcal{X}^\circ) = \mathcal{M}(\mathcal{X}_\bullet^\circ)$  with  $\mathcal{X}_\bullet^\circ = \{Y^{\circ, \tau_n} : Y^\circ \in \mathcal{X}^\circ, n \in \mathbb{N}\}$ , a family of  $(\mathcal{F}_t^\circ)$ -adapted processes — stopping a process at a stopping time preserves adaptedness once the process is progressively measurable, which under (T2b) a càdlàg adapted process is. Now apply Lemma 6.2 respectively Lemma 6.3 to  $\mathcal{X}_\bullet^\circ$ ; neither proof uses anything about the family beyond adaptedness, and the countability hypothesis for (c) is inherited because  $n$  ranges over  $\mathbb{N}$ .  $\square$

*Remark 6.33.* The contrast between (a) and (b) is the whole point of (L1), and it is easy to get wrong. A *finite* convex combination costs nothing, because two localizing sequences can be combined by  $\tau_n \wedge \tau'_n$ . A mixture over a continuum cannot: one would need  $\inf_\vartheta \tau_n^\vartheta$ , which need not tend to infinity and need not even be measurable. That is exactly where a localizing sequence common to all  $P$  becomes indispensable, and it is why [16] states his Theorem 32.10 for a *kernel*  $x \mapsto P_x$  and takes the trouble to exhibit one.

The hypothesis (L1) is less exotic than it looks: in the two places in the literature where it is used it is not assumed but *constructed*, and always by the same device — localizing along the test process itself rather than along the state space.

**Lemma 6.34** (Uniform localization for bounded jumps; (T2b)). *Suppose every  $Y^\circ \in \mathcal{X}^\circ$  is càdlàg with  $Y_0^\circ = 0$  and has jumps bounded by a constant  $c_{Y^\circ}$ . Write  $S_t^{Y^\circ} = \sup_{s \leq t} |Y_s^\circ|$  for the running supremum and put*

$$\tau_n^{Y^\circ} = \inf\{t \in \mathbb{T} : S_t^{Y^\circ} \geq n\}, \quad n \in \mathbb{N}.$$

Then each  $\tau_n^{Y^\circ}$  is a strict  $(\mathcal{F}_t^\circ)$ -stopping time, and  $\Sigma_0 = \{\tau_n^{Y^\circ} : Y^\circ \in \mathcal{X}^\circ, n \in \mathbb{N}\}$  satisfies (L1).

*Proof.* Fix  $Y = Y^\circ$  and drop the superscript.

*Step 1:*  $S$  is càdlàg, nondecreasing and adapted. Monotonicity is clear. Right continuity: given  $t$  and  $\varepsilon > 0$ , right continuity of  $Y$  gives  $u > t$  with  $|Y_s| \leq |Y_t| + \varepsilon \leq S_t + \varepsilon$  for  $s \in [t, u)$ , whence  $S_s \leq S_t + \varepsilon$  there. Left limits exist because  $S$  is monotone. For adaptedness note that, again by right continuity of  $Y$  and the density hypothesis in (T2b),

$$S_t = \sup\{|Y_s| : s \in (D \cap \mathbb{T}_{\leq t}) \cup \{t\}\}, \quad (41)$$

a countable supremum of  $\mathcal{F}_t^\circ$ -measurable functions. Indeed “ $\geq$ ” is trivial, and for “ $\leq$ ” let  $s \leq t$ ; if  $s = t$  there is nothing to prove, and if  $s < t$  then  $s$  is non-maximal in  $\mathbb{T}_{\leq t}$ , so by (T2b) there are  $s_k \in D$  with  $s < s_k$ ,  $s_k \rightarrow s$  from the right and  $s_k \leq t$  (replace  $s_k$  by  $s_k \wedge t$ , which lies in  $D$  or equals  $t$ ), and  $|Y_{s_k}| \rightarrow |Y_s|$ .

*Step 2:*  $\tau_n$  is a strict stopping time. We claim

$$\{\tau_n \leq t\} = \{S_t \geq n\}, \quad (42)$$

which lies in  $\mathcal{F}_t^\circ$  by (41). If  $S_t \geq n$  then  $t$  belongs to the set whose infimum is  $\tau_n$ , so  $\tau_n \leq t$ . Conversely let  $\tau_n \leq t$ . If  $t$  is maximal, then  $S_u \geq n$  for some  $u \leq t$  and monotonicity gives  $S_t \geq n$ . Otherwise, for every  $u > t$  there is  $v < u$  with  $S_v \geq n$ , hence  $S_u \geq n$ ; letting  $u \downarrow t$  along  $D$  — possible by (T2b) — right continuity of  $S$  gives  $S_t \geq n$ . This is where the *closed* threshold  $S_t \geq n$  matters:  $\inf\{t : |Y_t| > n\}$ , the choice of [10], satisfies (42) only with  $\{S_t > n\}$  replaced by an event that is not  $\mathcal{F}_t^\circ$ -measurable, and is a stopping time for  $\mathcal{F}_{t+}^\circ$  only. See Remark 6.35.

*Step 3:* the stopped process is bounded. On  $\{s < \tau_n\}$  we have  $S_s < n$ , hence  $|Y_s| < n$ ; consequently  $|Y_{\tau_n-}| \leq n$ , and  $|Y_{\tau_n}| \leq |Y_{\tau_n-}| + |\Delta Y_{\tau_n}| \leq n + c_Y$ . Thus  $|Y^{\tau_n}| \leq n + c_Y$  everywhere.

*Step 4:* (L1). A bounded local martingale is a uniformly integrable martingale, so for every  $P \in \mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$  the process  $Y^{\tau_n}$  is a  $P$ -martingale; this is the implication “ $\Rightarrow$ ”. For “ $\Leftarrow$ ” note that  $\tau_n \uparrow \infty$  because a càdlàg path is bounded on bounded intervals, so  $S_t < \infty$  for each  $t$  and hence  $\tau_n > t$  for  $n > S_t$ .  $\square$

*Remark 6.35* (Strict versus right continuous débuts). The distinction in Step 2 is not pedantry, and it is the one place in the local theory where the raw filtration bites. For a càdlàg adapted  $Y$  and an *open* set  $G$  the début  $\inf\{t : Y_t \in G\}$  satisfies  $\{\tau < t\} = \bigcup_{s \in D, s < t} \{Y_s \in G\}$  and is therefore a stopping time for  $\mathcal{F}_{t+}^\circ$ , but in general not for  $\mathcal{F}_t^\circ$ : a path with  $|Y_s| \leq n$  for  $s \leq t$  and  $|Y_s| > n$  for  $s > t$  has  $\tau = t$ , yet it agrees on  $\mathbb{T}_{\leq t}$  with a path that never exceeds  $n$ . Replacing the threshold by the running supremum repairs this, because  $\{S_t \geq n\}$  is decided by the path up to and including  $t$ . For *continuous* paths and a closed set the début is strict without any such device, which is why the exit times of Remark 6.31 cause no trouble. In the formalization this is the difference between `IsStoppingTime f` and `IsStoppingTime (f.rightLimit)` and has to be tracked.

*Remark 6.36.* Lemma 6.34 is the argument of [10], III.2.8, where it proves that the set of solutions of a semimartingale martingale problem is convex: they take  $T_n = \inf\{t : |Y_t| > n\}$  and observe that  $Y_0 = 0$  together with bounded  $|\Delta Y|$  makes  $Y^{T_n}$  bounded. [10] work throughout with a filtration satisfying the usual conditions, so for them this  $T_n$  is a stopping time; here the raw filtration forces the running-supremum variant of Lemma 6.34, and the boundedness argument is unaffected. [16] uses the same device,  $\tau_n^f = \inf\{t : |M_t^f| \geq n\}$ , in his Theorem 32.10. Both localize along the *test process*, not along the state space; the

exit times from balls in Remark 6.31 are the less canonical choice, since they refer to  $E$  and presuppose that  $f$  and  $g$  are bounded on compacts.

**Lemma 6.37** (Local restart; (T2b) + (T4)). *Let  $(\mathcal{X}_r^\circ)$  be a shift system for  $\mathcal{X}^\circ$  with  $\mathcal{Z}^\circ$  determining for each  $\mathcal{X}_r^\circ$ , and let  $\Sigma$  be a localizing system. Let  $X$  be a process on  $(\Omega, \mathcal{F}, \mathbb{G}, P)$  solving the local martingale problem for  $\mathcal{X}^\circ$  with respect to  $\mathbb{G}$  and  $\Sigma$ , let  $r \in \mathbb{T}$ , and let  $Z \geq 0$  be bounded and  $\mathcal{G}_r$ -measurable with  $E^P[Z] = 1$ . Then*

$$R := (Z \cdot P) \circ (X(r + \cdot))^{-1} \in \mathcal{M}_{\text{loc}}(\mathcal{X}_r^\circ).$$

*On the canonical space this is  $R = (Z \cdot P) \circ \theta_r^{-1}$ . The same holds with  $r$  replaced by an a.s. finite  $\mathbb{G}$ -stopping time  $\tau$  and  $\mathcal{G}_r$  by  $\mathcal{G}_\tau$ , provided  $\Sigma$  is strongly shift covariant.*

*Proof.* By (L1) it suffices to show that  $\hat{Y}^{\circ, \tau_n}$  is an  $R$ -martingale for every  $\hat{Y}^\circ \in \mathcal{X}_r^\circ$  and  $n \in \mathbb{N}$ , and by the determining property it suffices to show, for  $s \leq t$  and  $Z_s^\circ \in \mathcal{Z}_s^\circ$ ,

$$E^R[(\hat{Y}_t^{\circ, \tau_n} - \hat{Y}_s^{\circ, \tau_n})Z_s^\circ] = 0. \quad (43)$$

*Step 1: the shifted stopped process.* Let  $Y^\circ \in \mathcal{X}^\circ$  and the  $\mathcal{F}_r^\circ$ -measurable  $\kappa$  be as in (35), and write  $\tilde{Y} = Y^\circ(X)$ ,  $\tilde{\kappa} = \kappa(X)$ ,  $\tilde{\tau}_n = \tau_n(X(r + \cdot))$ ; these are defined because the elements of  $\Sigma$  are path functionals, and  $\tilde{\tau}_n$  is a  $(\mathcal{G}_{r+t})_t$ -stopping time because they are strict. Put  $\tilde{\sigma} := r + \tilde{\tau}_n$ , the ambient realization of  $\sigma = r + \tau_n \circ \theta_r \in \Sigma$  (by (L2)), so that  $\tilde{\sigma} \geq r$ . For  $\omega \in \Omega$  write  $u = t \wedge \tilde{\tau}_n(\omega)$ . Then, by (35),

$$\hat{Y}_t^{\circ, \tau_n}(X(r + \cdot)(\omega)) = \hat{Y}_u^\circ(X(r + \cdot)(\omega)) = \tilde{Y}_{r+u}(\omega) - \tilde{Y}_r(\omega) + \tilde{\kappa}(\omega),$$

and,  $r + \cdot$  being an order embedding that commutes with  $\wedge$  (Lemma 2.2, using (T2a), which (T2b) supplies),

$$r + u = r + (t \wedge \tilde{\tau}_n(\omega)) = (r + t) \wedge (r + \tilde{\tau}_n(\omega)) = (r + t) \wedge \tilde{\sigma}(\omega).$$

Hence

$$\hat{Y}_t^{\circ, \tau_n} \circ X(r + \cdot) = \tilde{Y}_{(r+t) \wedge \tilde{\sigma}} - \tilde{Y}_r + \tilde{\kappa}, \quad t \in \mathbb{T}. \quad (44)$$

This is [16]'s display (17) in the proof of his Theorem 32.11, and it is the only computation in the local theory that is not already in Section 6.1.

*Step 2: conclusion.* Put  $W := Z \cdot (Z_s^\circ \circ X(r + \cdot))$ . As in the proof of Lemma 6.17,  $W$  is bounded and  $\mathcal{G}_{r+s}$ -measurable:  $Z_s^\circ$  is  $\mathcal{F}_s^\circ$ -measurable and  $\pi_u \circ X(r + \cdot) = X(r + u)$  is  $\mathcal{G}_{r+u}$ -measurable, while  $Z$  is bounded,  $\mathcal{G}_r$ -measurable and  $\mathcal{G}_r \subset \mathcal{G}_{r+s}$ . *Boundedness of  $Z$  is used here and cannot be dropped:* (L3) supplies a martingale identity against bounded test variables only, and the stopped increments need not be in any  $L^p$  with  $p > 1$ . Using (44) twice, the terms  $-\tilde{Y}_r + \tilde{\kappa}$  cancel, and

$$E^R[(\hat{Y}_t^{\circ, \tau_n} - \hat{Y}_s^{\circ, \tau_n})Z_s^\circ] = E^P[(\tilde{Y}_{(r+t) \wedge \tilde{\sigma}} - \tilde{Y}_{(r+s) \wedge \tilde{\sigma}})W] = 0$$

by (L3), which says exactly that  $t \mapsto \tilde{Y}_{(r+t) \wedge \tilde{\sigma}} - \tilde{Y}_r$  is a  $(\mathcal{G}_{r+t})$ -martingale under  $P$ . This is (43).

For a stopping time  $\tau$  the argument is unchanged, with  $\mathcal{G}_{\tau+s}$  in place of  $\mathcal{G}_{r+s}$  and with  $\tilde{\sigma} = \tau + \tau_n(X(\tau + \cdot))$ , a stopping time by hypothesis.  $\square$

**Theorem 6.38** (Local uniqueness, Markov and strong Markov property). *Let  $(\mathcal{X}_r^\circ)$  be a shift system for  $\mathcal{X}^\circ$  with  $\mathcal{Z}^\circ$  determining for each  $\mathcal{X}_r^\circ$ , let  $\Sigma$  be a localizing system, and suppose that for every  $r \in \mathbb{T}$*

$$P, Q \in \mathcal{M}_{\text{loc}}(\mathcal{X}_r^\circ), \quad P\pi_0^{-1} = Q\pi_0^{-1} \implies P\pi_t^{-1} = Q\pi_t^{-1} \quad \text{for all } t \in \mathbb{T}. \quad (45)$$

Then Theorem 6.23 holds verbatim with  $\mathcal{M}$  replaced by  $\mathcal{M}_{\text{loc}}$  throughout — for the uniqueness half nothing has to be redone, since Proposition 6.13 is a statement about an arbitrary family of measures and applies to  $\mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$  as it stands; and if  $\Sigma$  is strongly shift covariant and the  $Y^{\circ, \tau_n}$  are right continuous and satisfy the integrability (40) at  $\tau$ , so does Theorem 6.27.

*Proof.* The proof of Theorem 6.23 uses exactly three things: Lemma 6.17, applied with the densities of Remark 6.29(i); the hypothesis (38); and (17). Replace the first by Lemma 6.37 and the second by (45); the third is a statement about  $F$  and is untouched. Every density used is an indicator or a conditional expectation of one, hence nonnegative, bounded by  $1/P(F_0)$  and  $\mathcal{G}_r$ -measurable, so Lemma 6.37 applies to each; and Proposition 6.13 is about  $\mathcal{M}$  only, so it carries over with  $\mathcal{M}_{\text{loc}}$  unchanged.

For the strong Markov property, the proof of Theorem 6.27 consists of the stopping time version of Lemma 6.17, which is the second assertion of Lemma 6.37, together with Lemma 6.2 for the mixture  $\int P_x \mu_{F_0}(dx)$ ; the latter is Lemma 6.32(b). Optional sampling (Fact 2.25) is applied to the  $Y^{\circ, \tau_n}$ , which are genuine martingales by (L1).  $\square$

*Remark 6.39* (What [10] do differently). Three points where [10], Chapter III go beyond the account above.

(i) *The shifted problem — adopted.* [10], III.2.39 carry a whole family of shifted triplets  $(\rho_t B, \rho_t C, \rho_t \nu)$  with  $(\rho_t B)_s(\theta_t \omega) = B_{t+s}(\omega) - B_t(\omega)$  and likewise for  $C$  and  $\nu$ , rather than demanding that the shifted problem be the original one. Definition 6.14 is that idea in the present notation, and Example 6.16 is the corresponding computation for  $\mathcal{X}_A$ . The earlier version of this manuscript required  $\mathcal{X}_r^\circ = \mathcal{X}^\circ$  and consequently had to assume that the clock be shift invariant and  $g$  time independent; both hypotheses are now gone, and the conclusion is the Markov property in its time-inhomogeneous form. The change costs nothing, because Lemmas 6.17 and 6.37 use only that the *shifted* process is a  $P$ -martingale.

(ii) *Local uniqueness.* [10], Definition III.2.37 isolate a notion strictly stronger than the uniqueness of (45): *local uniqueness* holds if for every strict stopping time  $T$  any two solutions of the *stopped* martingale problem agree on  $\mathcal{F}_T$ . It implies uniqueness ( $T \equiv \infty$ ) and is what absolute-continuity and limit-theorem arguments actually require. Their Theorem III.2.40 shows that uniqueness *does* imply local uniqueness once the problem is of Markovian type in the sense of (i) — precisely, once there is a kernel  $P_{x,t}$  solving the shifted problem. This is carried out in Section 6.4 — and, as it turns out, without the Markovian type: Theorem 6.45 needs only a restart kernel, which their construction supplies but which does not presuppose a shift.

(iii) *Strict stopping times.* The canonical filtration  $\mathcal{F}_t^\circ = \sigma(\pi_u : u \leq t)$  is not right continuous, and [10], III.2.35 therefore distinguish *strict* stopping times — those with  $\{T \leq t\} \in \mathcal{F}_t^\circ$  for the raw filtration — from stopping times for the completed right continuous filtration, with  $\mathcal{F}_T^\circ \subset \mathcal{F}_T$  and  $\mathcal{F}_{T-} \subset \mathcal{F}_T^\circ$  on  $\{T > 0\}$ . Definition 6.30 *demands* that the times in  $\Sigma$  be strict, and this is a genuine restriction, not a consequence of their being path functionals: the debut  $\inf\{t : |Y_t| > n\}$  is a path functional and is not strict (Remark 6.35). Strictness is what Lemma 6.34 has to work for, and it is also what allows a localizing system defined on  $F$  to be transported to an ambient filtration  $\mathbb{G}$ , as Lemma 6.37 does. The distinction should be made explicit in the formalization, where Mathlib’s `IsStoppingTime` is relative to whatever filtration it is given.

*Remark 6.40.* Two things are worth extracting.

First, the entire local theory rests on the single identity (44), which says that shifting a stopped test process produces a test process stopped at the *shifted* time. That is (L2),

and it is the local form of shift stability. Everything else is either the global argument verbatim or the linearity that (L1) restores.

Second, this route localizes the *test processes* and leaves the operator alone. [7], Section 4.6 — and hence Remark 6.53 — instead replaces  $A$  by  $A^{(m)} = \{(f\mathbf{1}_{K_m}, g\mathbf{1}_{K_m})\}$ . In the abstract setting the present route is the right one, because  $\mathcal{X}^\circ$  is primitive here and  $A$  is not; it is also the one that [16] takes.

What is *not* obtained is a local version of Theorem 5.4: Definition 6.30 presupposes stopping times, which presuppose the path regularity that Theorem 5.4 exists to produce. See Remark 5.8.

## 6.4 Local uniqueness

Theorem 6.38 concludes that two solutions with the same initial law have the same one-dimensional — hence all — distributions. [10], Definition III.2.37 isolate a strictly stronger property, and it is the one that absolute-continuity and limit arguments actually use.

**Definition 6.41** (Local uniqueness; (T2b) + (T4)). Write  $\mathcal{X}^{\circ, T} = \{Y^{\circ, T} : Y^\circ \in \mathcal{X}^\circ\}$  for the family stopped at a stopping time  $T$ . *Local uniqueness* holds for  $\mathcal{X}^\circ$  if for every strict stopping time  $T$  (Remark 6.39(iii)) and any two  $P, P' \in \mathcal{M}(\mathcal{X}^{\circ, T})$  with  $P\pi_0^{-1} = P'\pi_0^{-1}$  one has  $P = P'$  on  $\mathcal{F}_T$ .

Taking  $T \equiv \infty$  recovers uniqueness, so local uniqueness is at least as strong; [10] note that it is strictly stronger in general. [10], Theorem III.2.40 derives it from uniqueness for problems of Markovian type, and it is natural to expect a shift system to be needed. It is not. What is needed are two properties of the path space — one of the stopping time, one supplying solutions after it — and neither mentions a shift.

**Definition 6.42** (Strict stopping times; (T2b)). A stopping time  $T$  on  $F$  is *strict* if  $F$  is stable under the stopping operator  $a_T : F \rightarrow F$ ,  $\pi_t \circ a_T = \pi_{t \wedge T}$ , and  $\mathcal{F}_T^\circ = a_T^{-1}(\mathcal{S})$ . Equivalently, a set lies in  $\mathcal{F}_T^\circ$  exactly when membership depends on the path only through  $a_T$ . This is [10], Lemma III.2.43 for the canonical space, and it is what “strict” buys. Note the consequence  $T \circ a_T = T$ : a strict stopping time is determined by the stopped path.

The second ingredient is a supply of solutions that continue a given past. In the Markovian case one takes the restarted process to depend on the current *state*; the whole point of what follows is that it may depend on the whole *stopped path*, and that nothing is lost by letting it.

**Definition 6.43** (Restart kernel; (T2b)). Let  $T$  be a strict stopping time. A *restart kernel at  $T$*  is a kernel  $\alpha \mapsto Q_\alpha$  from  $(F, \mathcal{F}_T^\circ)$  to  $(F, \mathcal{S})$  such that

- (R1)  $Q_\alpha\{\beta \in F : a_T\beta = a_T\alpha\} = 1$  — the past is kept;
- (R2) for every  $Y^\circ \in \mathcal{X}^\circ$  and all  $T(\alpha) \leq s \leq t$ ,  $E^{Q_\alpha}[Y_t^\circ \mid \mathcal{F}_s^\circ] = Y_s^\circ$  — it is a solution from  $T$  onwards, in the sense of Lemma 6.5.

By (R1) and Definition 6.42,  $T$  is  $Q_\alpha$ -a.s. equal to the constant  $T(\alpha)$ , so the times in (R2) are deterministic: indeed  $T \circ a_T = T$  gives  $T(\beta) = T(a_T\beta) = T(a_T\alpha) = T(\alpha)$  on the set of full measure in (R1). Two further points. First, the event in (R1) is measurable:  $\{(\alpha, \beta) : a_T\alpha = a_T\beta\}$  is the preimage of the diagonal of  $F \times F$  under  $a_T \times a_T$ , and the diagonal is measurable because  $\mathcal{S}$  is countably generated and separating — which is (E1). Second, (R2) is a martingale property only *from  $T(\alpha)$  onwards*; nothing is asked of  $Y^\circ$  before that time, where the path is frozen at  $a_T\alpha$  anyway.

**Lemma 6.44** (Pasting with memory; (T2b)). *Let  $T$  be a strict stopping time, let  $P \in \mathcal{M}(\mathcal{X}^{\circ, T})$  and let  $(Q_\alpha)$  be a restart kernel at  $T$ . Assume the integrability condition*

$$\int_F E^{Q_\alpha} [|Y_t^\circ|] P(d\alpha) < \infty \quad \text{for every } Y^\circ \in \mathcal{X}^\circ \text{ and } t \in \mathbb{T}.$$

Then

$$Q := \int_F Q_\alpha P(d\alpha) \in \mathcal{M}(\mathcal{X}^\circ), \quad \text{and} \quad Q = P \text{ on } \mathcal{F}_T^\circ.$$

*Proof.* Agreement on  $\mathcal{F}_T^\circ$ . Let  $A \in \mathcal{F}_T^\circ$ . By Definition 6.42,  $\mathbf{1}_A = \mathbf{1}_A \circ a_T$ , and by (R1),  $Q_\alpha(A) = \mathbf{1}_A(\alpha)$ . Hence  $Q(A) = P(A)$ .

*The martingale property.* Fix  $Y^\circ \in \mathcal{X}^\circ$ ,  $s \leq t$  and a bounded  $\mathcal{F}_s^\circ$ -measurable  $W$ , and split as before,

$$Y_t^\circ - Y_s^\circ = \underbrace{(Y_t^{\circ, T} - Y_s^{\circ, T})}_{(I)} + \underbrace{(Y_t^\circ - Y_{t \wedge T}^\circ) - (Y_s^\circ - Y_{s \wedge T}^\circ)}_{(II)}.$$

*The term (I).* We claim that for every  $\alpha$

$$E^{Q_\alpha}[(I)W] = (I)(\alpha) W(\alpha) \mathbf{1}_{\{T(\alpha) > s\}}. \quad (46)$$

Fix  $\alpha$  and recall from Definition 6.43 that  $Q_\alpha$ -a.s.  $a_T\beta = a_T\alpha$  and  $T(\beta) = T(\alpha)$ .

If  $T(\alpha) \leq s$ , then  $Q_\alpha$ -a.s.  $(I)(\beta) = Y_{T(\beta)}^\circ(\beta) - Y_{T(\beta)}^\circ(\beta) = 0$  and both sides of (46) vanish.

If  $T(\alpha) > s$ , then  $Q_\alpha$ -a.s.  $W(\beta) = W(\alpha)$ , because  $W$  is  $\mathcal{F}_s^\circ$ -measurable, hence a function of  $\beta$  restricted to  $\mathbb{T}_{\leq s} \subset \mathbb{T}_{\leq T(\beta)}$ , on which  $\beta$  and  $\alpha$  agree. Likewise  $(I)(\beta) = (I)(\alpha)$ , because  $Y^\circ$  is adapted, so  $Y_{u \wedge T}^\circ$  is a function of the path on  $\mathbb{T}_{\leq T}$ , i.e. of  $a_T$ . This proves (46).

Integrating (46) over  $P(d\alpha)$  and using that  $\mathbf{1}_{\{T > s\}}W$  is bounded and  $\mathcal{F}_s^\circ$ -measurable —  $\{T > s\}$  lies in  $\mathcal{F}_s^\circ$  because  $T$  is a stopping time — we get  $E^Q[(I)W] = E^P[(Y_t^{\circ, T} - Y_s^{\circ, T})\mathbf{1}_{\{T > s\}}W] = 0$  because  $Y^{\circ, T}$  is a  $P$ -martingale.

*The term (II).* Fix  $\alpha$  and write  $r = T(\alpha)$ , which by Definition 6.43 is the  $Q_\alpha$ -a.s. value of  $T$ .

- If  $r \leq s$ , then  $(II) = Y_t^\circ - Y_s^\circ$  and  $W$  is bounded and  $\mathcal{F}_s^\circ$ -measurable with  $s \geq r$ , so (R2) gives  $E^{Q_\alpha}[(II)W] = 0$ .
- If  $r > s$ , then  $(II) = Y_t^\circ - Y_{t \wedge r}^\circ$ , which vanishes for  $t \leq r$ ; for  $t > r$  it is  $Y_t^\circ - Y_r^\circ$ , and  $W$  is  $\mathcal{F}_r^\circ$ -measurable on this event because  $s < r$  and  $\mathcal{F}_s^\circ \subset \mathcal{F}_r^\circ$ . Again (R2), now at the time  $r$ , gives  $E^{Q_\alpha}[(II)W] = 0$ .

Integrating over  $P(d\alpha)$  gives  $E^Q[(II)W] = 0$ . □

**Theorem 6.45** (Uniqueness implies local uniqueness; [10], Theorem III.2.40). *Suppose that a restart kernel exists at every strict stopping time, and that  $\mathcal{M}(\mathcal{X}^\circ, \mu)$  contains at most one element for every  $\mu \in \mathcal{P}(E)$ . Then local uniqueness holds in the sense of Definition 6.41.*

*Proof.* Let  $T$  be a strict stopping time and  $P \in \mathcal{M}(\mathcal{X}^{\circ, T})$  with  $P\pi_0^{-1} = \mu$ . Lemma 6.44 produces  $Q \in \mathcal{M}(\mathcal{X}^\circ)$  with  $Q = P$  on  $\mathcal{F}_T^\circ$ ; in particular  $Q\pi_0^{-1} = \mu$ , since  $\pi_0$  is  $\mathcal{F}_T^\circ$ -measurable. By hypothesis  $Q$  is the unique element of  $\mathcal{M}(\mathcal{X}^\circ, \mu)$ , hence depends on  $\mu$  alone; therefore so does  $P|_{\mathcal{F}_T^\circ} = Q|_{\mathcal{F}_T^\circ}$ . □

**Corollary 6.46** (The Markovian case). *Let  $(\mathcal{X}_r^\circ)$  be a shift system, measurable in the sense of Definition 6.14 and full, i.e. satisfying in addition that for every  $Y^\circ \in \mathcal{X}^\circ$  and  $r$  there are  $\hat{Y}^\circ \in \mathcal{X}_r^\circ$  and an  $\mathcal{F}_r^\circ$ -measurable  $\kappa$  with  $\hat{Y}_u^\circ \circ \theta_r = Y_{r+u}^\circ - Y_r^\circ + \kappa$ . Let  $(P_{x,r})$  be a measurable family with  $P_{x,r} \in \mathcal{M}(\mathcal{X}_r^\circ, \delta_x)$ , and suppose  $F$  admits concatenation at  $T$ : a measurable  $\gamma(\alpha, \beta)$  agreeing with  $\alpha$  before  $T(\alpha)$ , with  $\beta$  shifted after it, and with  $\theta_T \gamma(\alpha, \beta) = \beta$ . Then*

$$Q_\alpha = \text{law of } \gamma(\alpha, \beta), \quad \beta \sim P_{\alpha_{T(\alpha)}, T(\alpha)},$$

*is a restart kernel at  $T$ , and Theorem 6.45 applies.*

*Proof.* (R1) holds because  $a_T \gamma(\alpha, \beta) = a_T \alpha$ . For (R2) write  $r = T(\alpha)$ . Fullness gives  $\hat{Y}^\circ \in \mathcal{X}_r^\circ$  and an  $\mathcal{F}_r^\circ$ -measurable  $\kappa$  with  $Y_{r+u}^\circ - Y_r^\circ = \hat{Y}_u^\circ \circ \theta_r - \kappa$ . Under  $Q_\alpha$  the shifted path  $\theta_r$  has law  $P_{\alpha_r, r}$ , under which  $\hat{Y}^\circ$  is a martingale for  $(\mathcal{F}_u^\circ)_u$ , and  $\kappa$  is  $Q_\alpha$ -a.s. the constant  $\kappa(\alpha)$ , being  $\mathcal{F}_r^\circ$ -measurable and hence a function of  $a_T$ .

It remains to transport the filtrations. Under  $Q_\alpha$  the path on  $\mathbb{T}_{\leq r}$  is  $Q_\alpha$ -a.s. the deterministic  $a_T \alpha$ , so for  $u \in \mathbb{T}$  the  $\sigma$ -field  $\mathcal{F}_{r+u}^\circ$  coincides  $Q_\alpha$ -a.s. with  $\theta_r^{-1} \mathcal{F}_u^\circ$ : the generators  $\pi_v$  with  $v \leq r+u$  are either deterministic ( $v \leq r$ ) or of the form  $\pi_w \circ \theta_r$  with  $w = v - r \leq u$ , the latter by Lemma 2.2. Hence for  $r \leq s \leq t$ , writing  $s = r + u$  and  $t = r + v$ ,

$$E^{Q_\alpha} [Y_t^\circ - Y_s^\circ \mid \mathcal{F}_s^\circ] = E^{P_{\alpha_r, r}} [\hat{Y}_v^\circ - \hat{Y}_u^\circ \mid \mathcal{F}_u^\circ] \circ \theta_r = 0,$$

which is (R2). □

*Remark 6.47* (What memory buys, and what it does not). Three comments.

*The shift has left the statement.* Definition 6.43, Lemma 6.44 and Theorem 6.45 mention no  $\theta_r$ , no shifted family and no concatenation; the restarted measures are indexed by the *stopped path*, not by the state, and condition (R2) is exactly the conclusion of Lemma 6.5. The Markovian version, with a state-indexed kernel and concatenation, is now the corollary it should be. The proof is also shorter: because a strict stopping time satisfies  $T \circ a_T = T$ , it is  $Q_\alpha$ -a.s. *deterministic*, so no optional sampling and no case analysis with  $\theta_T$  are needed.

*What is genuinely assumed.* The existence of a restart kernel is a hypothesis and cannot be otherwise. Unlike the restart of Lemma 6.5, which merely reweights a solution one already has, pasting must *supply* solutions after  $T$  for every possible past; and the conditional distributions of a solution of the full problem are of no use here, since producing such a solution is the point. This is the one place in Section 6 where something must be given from outside.

*Why strict stopping times.* Definition 6.42 is used twice: to know that  $\mathcal{F}_T^\circ$ -measurable functions see only  $a_T$ , and to know that  $T$  is determined by the stopped path. For the raw filtration  $\mathcal{F}_t^\circ = \sigma(\pi_u : u \leq t)$  this is [10], Lemma III.2.43; for a completed or right continuous filtration it is false, which is the practical reason for [10]’s insistence on the distinction.

## 6.5 Non-Markovian test objects

The point of the arrangement above is that it can be tested. We run it against the two examples of [2] that are not Markovian, and record for each which results apply and which fail.

**Example 6.48** (Volterra equations; [2], Example 3.13). Let  $F = L_{\text{loc}}^p(\mathbb{R}_+, \mathbb{R}^d) \times D_{\mathbb{R}^k}[0, \infty)$  with coordinate process  $(\pi^X, \pi^Z)$ , let  $K : \mathbb{R}_+ \rightarrow \mathbb{R}^{d \times k}$  be a convolution kernel and  $g_0 : \mathbb{R}_+ \rightarrow \mathbb{R}^d$ , and let

$$Lf(x, z) = \langle b(x), \nabla f(z) \rangle + \frac{1}{2} \text{tr}(a(x) \nabla^2 f(z)) + \int (f(z+y) - f(z) - \langle h(y), \nabla f(z) \rangle) \nu(x, dy).$$

Take for  $\mathcal{X}^\circ$  the processes  $f(\pi_t^Z) - \int_0^t Lf(\pi_s^X, \pi_s^Z) ds$ ,  $f \in C_b^2(\mathbb{R}^k)$ , together with the requirement that  $\int_0^t \pi_s^X ds = \int_0^t g_0(s) ds + \int_0^t K_{t-s} \pi_s^Z ds$  for all  $t$ . Then  $\mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$  is the set of laws of solutions of the Volterra equation.

This is not a Markov problem, and not by accident: the kernel  $K_{t-s}$  makes  $\pi_t^X$  depend on the whole past of  $\pi^Z$ . In the language of Section 6.2, *there is no shift system*: shifting by  $r$  turns  $K_{t-s}$  into a kernel that also sees  $\pi^Z$  before  $r$ , which no member of a family  $\mathcal{X}_r^\circ$  of the form required by Definition 6.14 can reproduce. What survives is exactly what Section 6.1 proves:

- Definitions 4.2 and 4.3 apply as they stand, on a path space that is not  $D_E[0, \infty)$  and on which  $\pi^X$  is not even right continuous;
- Lemma 6.2 (mixtures) and Lemma 6.3 (disintegration) apply, the latter because  $F$  is Polish, hence (E1);
- **Proposition 6.13 applies**, provided the Volterra equation is uniquely solvable in law *from any prescribed past* — which under Lipschitz conditions on  $b, a, \nu$  it is, by the Picard iteration, since the kernel term  $\int_{[0,t)} K_{t-s} dZ_s$  splits into a known contribution from the past and a contraction on the future. That is (32), and Proposition 6.13 turns it into uniqueness of the whole law with no Markov structure. Plain uniqueness in law would not suffice; see Remark 6.12;
- Theorem 8.40 applies, and [2], Section 4.2 is precisely a stability theorem for Volterra equations obtained from it;
- Theorem 6.23 does *not* apply, and the conclusion is false: a Volterra process is not Markov. Nor do Theorem 6.27 or Section 7, both of which use the shift. Section 6.4 applies as soon as a restart kernel is available.

**Example 6.49** (Semimartingales with path-dependent characteristics). Let  $F = D_E[0, \infty)$  and let  $(B, C, \nu)$  be a triplet of *predictable functionals of the path*, as in [10], Chapter III. The associated  $\mathcal{X}^\circ$  consists of the processes of [10], III.2.7, and  $\mathcal{M}_{\text{loc}}(\mathcal{X}^\circ)$  is the set of laws of semimartingales with these characteristics. Again there is no shift system unless the triplet happens to be shift covariant in the sense of [10], III.2.39, which for a genuinely path-dependent triplet it is not.

Here Section 6.3 is available in full — the localizing system of Lemma 6.34 is constructed from the test processes themselves and needs no shift, which is exactly the argument [10] give in their III.2.8 — and so is Section 6.4, provided a restart kernel (Definition 6.43) can be exhibited. For a path-dependent triplet that is a genuine requirement rather than a formality: one must produce, for every stopped path, a semimartingale continuing it with the prescribed characteristics. But it is a requirement on the *data*, not a Markov hypothesis, and [10] verify it in their setting.

*Remark 6.50* (Where the Markov structure is used, in one place). Collecting the two examples with the results of Sections 5–8:

| Needs no Markov structure                 | Needs a shift system                  |
|-------------------------------------------|---------------------------------------|
| Definitions 4.2, 4.3, 5.2                 | Definition 6.14                       |
| Theorem 5.4 (càdlàg modification)         | Lemma 6.22                            |
| Lemma 6.2, Lemma 6.3                      | Example 6.16                          |
| Lemma 6.5 ( <b>restart with memory</b> )  | Theorem 6.23 (Markov property)        |
| Definition 6.11, Proposition 6.13         | Lemma 6.17 (restart with re-indexing) |
| ( <b>uniqueness</b> )                     |                                       |
| Definition 6.30, Lemma 6.34, Lemma 6.32   | Lemma 6.37, Theorem 6.38              |
| Lemma 7.1 (chain identity)                | Corollary 6.46                        |
| Definition 6.43, Lemma 6.44, Theorem 6.45 | —                                     |
| ( <b>local uniqueness</b> )               |                                       |
| Theorem 8.40, Theorem 8.48                | Proposition 7.3 and Section 7.2       |

The left column is the whole of Sections 4 and 8, all of Section 5, the uniqueness half of Section 6 and the analytic core of Section 7. The right column is: the shift system and what is built on it. Every entry there produces a statement whose *conclusion* is Markovian — the Markov property, the reduction to one-dimensional distributions, pasting, duality — so nothing in it could have been avoided by a better argument.

## 6.6 The Markovian case

**Theorem 6.51** ([7], Theorem 4.4.2; (T2b) + (T4) + shift invariant clock, for homogeneity). *Assume (E2) and let  $A \subset B(E) \times B(E)$ . Suppose that for each  $\mu \in \mathcal{P}(E)$  any two solutions  $X, Y$  of the martingale problem for  $(A, \mu)$  have the same one-dimensional distributions, that is, for each  $t \in \mathbb{T}$ ,*

$$P\{X(t) \in \Gamma\} = P\{Y(t) \in \Gamma\}, \quad \Gamma \in \mathcal{B}(E). \quad (47)$$

Then the following hold.

- (a) Any solution of the martingale problem for  $A$  with respect to a filtration  $(\mathcal{G}_t)$  is a Markov process with respect to  $(\mathcal{G}_t)$ , and any two solutions of the martingale problem for  $(A, \mu)$  have the same finite-dimensional distributions. In particular, (47) implies uniqueness.
- (b) If moreover  $A \subset C_b(E) \times B(E)$ ,  $X$  is a solution of the martingale problem for  $A$  with respect to  $(\mathcal{G}_t)$ , and  $X$  has sample paths in  $D_E[0, \infty)$ , then for every a.s. finite  $(\mathcal{G}_t)$ -stopping time  $\tau$ ,

$$E[f(X(\tau + t)) \mid \mathcal{G}_\tau] = E[f(X(\tau + t)) \mid X(\tau)], \quad f \in B(E), t \in \mathbb{T}. \quad (48)$$

- (c) If, in addition to the conditions of (b), for each  $x \in E$  there exists a solution  $P_x \in \mathcal{P}(D_E[0, \infty))$  of the  $D_E[0, \infty)$  martingale problem for  $(A, \delta_x)$  such that  $x \mapsto P_x(B)$  is Borel measurable for each  $B \in \mathcal{B}(D_E[0, \infty))$ , then, setting  $T(t)f(x) = \int f(\omega(t))P_x(d\omega)$ ,

$$E[f(X(\tau + t)) \mid \mathcal{G}_\tau] = T(t)f(X(\tau)) \quad (49)$$

for all  $f \in B(E)$ ,  $t \in \mathbb{T}$  and a.s. finite  $(\mathcal{G}_t)$ -stopping times  $\tau$ ; that is,  $X$  is strong Markov.

*Proof.* We check the hypotheses of Theorems 6.23 and 6.27 for  $\mathcal{X}^\circ = \mathcal{X}_A$ , i.e. for

$$\mathcal{X}^\circ = \left\{ (f(\pi_t) - \int_{\langle 0, t \rangle_\iota} g(\pi_s) q(ds))_{t \in \mathbb{T}} : (f, g) \in A \right\}$$

on the path space  $F = E^{\mathbb{T}}$  (respectively  $F = D_E[0, \infty)$  in (b)–(c)).

*Determining set.* Proposition 4.9 says precisely that the products  $\prod_k h_k(\pi_{t_k})$ , over arbitrary finite subsets  $\{t_k\} \subset \mathbb{T}_{\leq s}$ , form a determining set for  $\mathcal{X}_A$  in the sense of Definition 4.3: (19) for all such products is equivalent to the martingale property. Note that this is stronger than what Definition 4.3 asks — it delivers the martingale property for  ${}^*\mathcal{F}^X$ , not merely for the canonical filtration.

*Shift system.* Example 6.16 exhibits one, with  $\theta_r \omega = \omega(r + \cdot)$ , the pulled-back clock  $q_r$  and  $\kappa = f(\pi_r)$ . Since  $A \subset C_b(E) \times B(E)$  does not depend on time and  $q$  is assumed shift invariant,  $q_r = q$  and  $\mathcal{X}_r^\circ = \mathcal{X}^\circ$  for every  $r$ ; the shift system is the constant one, and the Markov process obtained is homogeneous. Dropping either assumption leaves Theorem 6.23 intact and yields a time-inhomogeneous Markov process instead — this is the only point at which [7], Theorem 4.4.2 is genuinely less general than Theorem 6.23.

*Integrability.*  $f$  and  $g$  are bounded, so every  $Y^\circ \in \mathcal{X}^\circ$  satisfies  $|Y_t^\circ| \leq \|f\| + \|g\| q(\mathbb{T}_{\leq t})$ , which is finite by Definition 2.3; all the integrability provisos are automatic, and the increments of  $Y^\circ$  on  $\mathbb{T}_{\leq T}$  are bounded, hence uniformly integrable.

*Conclusion.* Hypothesis (47) is (38) by Remark 6.1, so Theorem 6.23(a) gives the Markov property and (b) the equality of the finite-dimensional distributions, which is (a). For (b) and (c),  $A \subset C_b(E) \times B(E)$  and paths in  $D_E[0, \infty)$  make each  $Y^\circ(X)$  a right continuous martingale whose increment over  $\langle \tau + s, \tau + t \rangle_\iota$  is bounded by  $2\|f\| + \|g\| q(\langle s, t \rangle_\iota)$ , the clock being shift invariant here; so (40) is automatic and Theorem 6.27 applies; its second assertion is (49) with  $T(t)f(x) = \int f(\omega(t))P_x(d\omega)$ . Its restriction to stopping times with countably many values is vacuous here: the shift system is the constant one, so all the measures  $P_{x,r} = P_x$  solve one and the same problem and the partition of Step 3 there is unnecessary. See Remark 6.28.  $\square$

**Corollary 6.52** ([7], Corollary 4.4.3; (T3)). *Assume (E3) and let  $A \subset B(E) \times B(E)$ . Suppose that for each  $\mu \in \mathcal{P}(E)$  any two solutions  $X, Y$  of the martingale problem for  $(A, \mu)$  with sample paths in  $D_E[0, \infty)$  (respectively  $C_E[0, \infty)$ ) satisfy (47). Then for each  $\mu \in \mathcal{P}(E)$  any two such solutions have the same distribution on  $D_E[0, \infty)$  (respectively  $C_E[0, \infty)$ ).*

*Proof.* The processes  $\tilde{X}, \tilde{Y}$  constructed in the proof of Theorem 6.51 again have sample paths in  $D_E[0, \infty)$  (resp.  $C_E[0, \infty)$ ) whenever  $X$  and  $Y$  do, so the induction goes through and  $X, Y$  have the same finite-dimensional distributions. By Theorem 3.13 these determine the law on  $D_E[0, \infty)$  (resp.  $C_E[0, \infty)$ ).  $\square$

*Remark 6.53.* For the *local* martingale problem, Theorem 6.51 holds in the following localized form. Suppose  $A \subset M(E) \times M(E)$  and let  $(K_m)$  be an increasing sequence of compact sets with  $\tau_m = \inf\{t : X(t) \notin K_m \text{ or } X(t-) \notin K_m\}$  and  $\tau_m \rightarrow \infty$  a.s. for every solution  $X$ . If for each  $\mu$  and each  $m$  any two solutions of the local martingale problem for  $(A, \mu)$  agree in the one-dimensional distributions of the *stopped* processes  $X(\cdot \wedge \tau_m)$ , then the conclusions (a)–(c) hold. The proof is the above one applied to the stopped martingale problem for  $A^{(m)} = \{(f\mathbf{1}_{K_m}, g\mathbf{1}_{K_m}) : (f, g) \in A\}$ , followed by  $m \rightarrow \infty$ . This is the content of [7], Section 4.6. It is worth stating separately in the formalization, since the reduction “local  $\rightsquigarrow$  global via stopping” is reused in Section 8.

This route modifies the operator. Section 6.3 gives the alternative, which leaves  $A$  alone and stops the test process instead; it is the one that fits the abstract setting, and the one [16] uses.

## 7 Uniqueness via duality

Duality reduces the computation of  $E[f(X(t), y)]$  for a process  $X$  to the computation of  $E[f(x, Y(t))]$  for a second, typically simpler, process  $Y$ . Combined with Theorem 6.51 it is one of the main practical routes to uniqueness.

Here, unlike in Sections 5 and 6, there is nothing to strip away on the probabilistic side: Theorem 7.12 below already assumes only that  $X$  and  $Y$  are *measurable* processes, uses no path regularity, no Polish structure and no boundedness, and would be stated verbatim in the [2] setting. What can be generalized is the *clock*. Section 7.1 therefore isolates the analytic core as a statement about a function of two time variables, valid for an arbitrary measure  $q$ , and reads off exactly which  $q$  make the duality relation true. The answer turns out to be sharp, and it is not the answer one gets in Sections 5 and 6.

### 7.1 The anti-diagonal lemma

Duality is the assertion that a certain function of two time variables is constant on anti-diagonals. The following identity is that assertion stripped of all analysis: it is an exact telescoping, with no hypothesis beyond the two increment representations, and every version of the duality lemma is obtained from it by choosing a chain.

**Lemma 7.1** (Chain identity; (T0) + (E0) + clock). *Let  $\mathbb{T}$  be a preordered set with a least element 0, let  $q$  be a measure on  $\mathbb{T}$  for which the sets  $[s, s') := \mathbb{T}_{<s'} \setminus \mathbb{T}_{<s}$  are measurable, and let  $\Phi, \gamma_1, \gamma_2 : \mathbb{T} \times \mathbb{T} \rightarrow \mathbb{R}$  satisfy, for all  $s \leq s'$  and  $t \leq t'$ ,*

$$\Phi(s', t) - \Phi(s, t) = \int_{[s, s')} \gamma_1(r, t) q(dr), \quad \Phi(s, t') - \Phi(s, t) = \int_{[t, t')} \gamma_2(s, r) q(dr), \quad (50)$$

*all integrals being assumed to exist. Let  $t \in \mathbb{T}$  and let*

$$0 = s_0 \leq s_1 \leq \dots \leq s_m = t, \quad t = t_0 \geq t_1 \geq \dots \geq t_m = 0$$

*be a staircase from  $(0, t)$  to  $(t, 0)$ . Then*

$$\Phi(t, 0) - \Phi(0, t) = \sum_{k=0}^{m-1} \left[ \int_{[s_k, s_{k+1})} \gamma_1(r, t_{k+1}) q(dr) - \int_{[t_{k+1}, t_k)} \gamma_2(s_k, r) q(dr) \right]. \quad (51)$$

*Proof.* Telescope along the staircase:  $\Phi(t, 0) - \Phi(0, t) = \sum_k [\Phi(s_{k+1}, t_{k+1}) - \Phi(s_k, t_k)]$ , the sum collapsing because  $(s_0, t_0) = (0, t)$  and  $(s_m, t_m) = (t, 0)$ . Split each summand through the corner  $(s_k, t_{k+1})$ :

$$[\Phi(s_{k+1}, t_{k+1}) - \Phi(s_k, t_{k+1})] - [\Phi(s_k, t_k) - \Phi(s_k, t_{k+1})].$$

Now apply the first representation in (50) to the first bracket, which is legitimate because  $s_k \leq s_{k+1}$ , and the second to the second, legitimate because  $t_{k+1} \leq t_k$ .  $\square$

*Remark 7.2* (What the staircase form buys). The version of (51) one meets in the literature takes  $t_k = t - s_k$ , the *anti-diagonal* staircase, and therefore needs (T4) so that  $t - s_k$  is defined. Nothing in the proof requires this: the two coordinates may be stepped through independently, and the identity is then a statement about an arbitrary preorder with a clock. Which staircase one chooses is the whole art of the section, and the point of Corollary 7.4 below is that the anti-diagonal one is rarely the right choice: the correct staircase is the one that is anti-diagonal *in clock time* — available whenever the clock is atomless, and, as Remark 7.5 shows, genuinely obstructed by atoms.

The duality relation is the case in which the two brackets in (51) cancel. Under the balance condition  $\gamma_1 = \gamma_2 = \gamma$  the  $k$ -th summand is

$$\int_{[s_k, s_{k+1})} \gamma(r, t_{k+1}) q(dr) - \int_{[t_{k+1}, t_k)} \gamma(s_k, r) q(dr), \quad (52)$$

in which the two integrals run over the block  $[s_k, s_{k+1})$  and over the partner block  $[t_{k+1}, t_k)$ . So the duality relation holds as soon as the staircase can be chosen so that a block and its partner receive the same  $q$ -mass, at every scale. On the anti-diagonal staircase  $t_k = t - s_k$  that is translation invariance of  $q$ ; but the anti-diagonal staircase is not forced, and it is not the right one.

**Proposition 7.3** (Haar clocks; (T2a) + (T4),  $\iota = p$ ). *Assume (50) with  $\gamma_1 = \gamma_2 = \gamma$ .*

- (a) *If  $\mathbb{T} = \mathbb{N}_0$  and  $q$  is counting measure, then  $\Phi(t, 0) = \Phi(0, t)$  for every  $t$ , for every  $\gamma$  whatsoever, and no integrability hypothesis is needed.*
- (b) *If  $\mathbb{T} = \mathbb{R}_+$  and  $q$  is Lebesgue measure, then  $\Phi(t, 0) = \Phi(0, t)$  for almost every  $t$ , under the integrability hypothesis (56); this is Lemma 7.11 below.*

*Proof.* (a) Take the anti-diagonal staircase  $s_k = k$ ,  $t_k = t - k$ . Then  $[s_k, s_{k+1}) = \{k\}$  and  $[t_{k+1}, t_k) = \{t - k - 1\}$ , so (52) reads  $\gamma(k, t - k - 1) - \gamma(k, t - k - 1) = 0$ . Every summand vanishes identically; no regularity of  $\gamma$  is used.

(b) Refine the anti-diagonal staircase; see Lemma 7.11. □

The two cases of Proposition 7.3 are the ones in the literature, and they are the ones for which the anti-diagonal staircase works. It is tempting to conclude — as an earlier version of this manuscript did — that duality *requires* a translation invariant clock, on the grounds that (52) does not cancel otherwise. That inference is invalid: the failure of one staircase to cancel says nothing about  $\Phi(t, 0) - \Phi(0, t)$ . The two results below show that translation invariance is not necessary — for an atomless clock by a reduction to Lemma 7.11, for a purely atomic one by the linear algebra of Proposition 7.8. Whether *every* clock admits duality remains open, and Remark 7.5 explains where the two arguments stop.

**Corollary 7.4** (Atomless clocks, by time change; (T3) + (E0) + clock,  $\iota = p$ ). *Assume (T3) and let  $q$  be an atomless clock. Let  $\Phi, \gamma$  satisfy (50) with  $\gamma_1 = \gamma_2 = \gamma$ , with  $\gamma$  measurable for the order  $\sigma$ -field in each variable — automatic if  $\mathcal{T}$  is taken to be that  $\sigma$ -field, which Definition 2.3 permits. Put  $Q(s) = q(\mathbb{T}_{\leq s})$  and  $Q^\leftarrow(z) = \sup\{t : Q(t) \leq z\}$ . If  $\Psi = \Phi(Q^\leftarrow(\cdot), Q^\leftarrow(\cdot))$  and  $\psi = \gamma(Q^\leftarrow(\cdot), Q^\leftarrow(\cdot))$  satisfy the integrability hypothesis (56), then*

$$\Phi(t, 0) = \Phi(0, t) \quad \text{for } Q\text{-almost every } t \in \mathbb{T}.$$

*Proof.* Since  $q$  is atomless,  $Q$  is continuous; by (T3) its domain is an interval, so  $Q$  maps  $\mathbb{T}_{\leq t^*}$  onto  $[0, L]$ ,  $L = Q(t^*)$ , and  $Q^\leftarrow$  is a right inverse:  $Q(Q^\leftarrow(z)) = z$ . Monotonicity of  $Q$  gives  $Q^\leftarrow(z) \geq a \iff z \geq Q(a)$ , whence the change-of-variables formula

$$\int_{[s, s')} g \, dq = \int_{[Q(s), Q(s'))} g(Q^\leftarrow(z)) \, dz \quad (53)$$

for order-measurable  $g$ : for  $g = \mathbf{1}_{[a, a')}$  both sides equal  $Q(s' \wedge a') - Q(s \vee a)$ , and Fact 2.46 does the rest.

Now  $q([s, s']) = 0$  forces  $\Phi(s, \cdot) = \Phi(s', \cdot)$  as well as  $\Phi(\cdot, s) = \Phi(\cdot, s')$ , by (50); hence  $\Psi(Q(s), Q(t)) = \Phi(s, t)$ . And for  $x \leq x'$ , writing  $s = Q^{\leftarrow}(x)$  and  $s' = Q^{\leftarrow}(x')$ ,

$$\Psi(x', y) - \Psi(x, y) = \int_{[s, s']} \gamma(r, Q^{\leftarrow}(y)) q(dr) = \int_x^{x'} \psi(z, y) dz$$

by (53), and symmetrically in  $y$ . So  $\Psi$  satisfies the hypotheses of Lemma 7.11 on  $[0, L]^2$  with  $\nabla \Psi = (\psi, \psi)$ , which gives  $\Psi(\ell, 0) = \Psi(0, \ell)$  for almost every  $\ell$ ; and  $\Psi(L, 0) = \Phi(t^*, 0)$ ,  $\Psi(0, L) = \Phi(0, t^*)$ .  $\square$

*Remark 7.5* (How much this is, and how much it is not). Corollary 7.4 is a change of variables and should be read as one. Under (T3) an atomless clock is the image of Lebesgue measure under the continuous increasing map  $\tau$ , so a martingale problem with such a clock is a *deterministic time change* of one with Lebesgue clock, and the corollary says no more than that duality is invariant under that time change. It is worth stating because it settles the question Proposition 7.3 raises — which clocks admit duality — for everything except atoms, and because the earlier version of this manuscript got that question wrong in the opposite direction. It is not worth calling a theorem, and an intermediate version of this manuscript did, under the name “every clock admits duality”.

*Atoms are not a change of variables, and this is not a gap in the argument.* With an atom at  $a$  the map  $Q$  jumps,  $\tau$  is no longer a right inverse, and the interval  $[Q(a), Q(a) + q(\{a\})]$  has no preimage. Filling it by interpolation — which the intermediate version attempted — is impossible. Take  $\mathbb{T} = [0, 2]$ ,  $q = \delta_1$ ,  $\iota = p$ , and  $\Phi(s, t) = M(i(s), i(t))$  with  $i(s) = \mathbf{1}_{\{s > 1\}}$  and  $M(0, 0) = 0$ ,  $M(1, 0) = M(0, 1) = c$ ,  $M(1, 1) = d$ . Both representations (50) hold with the single function  $\gamma$  pinned by  $\gamma(1, t) = M(1, i(t)) - M(0, i(t))$  and  $\gamma(s, 1) = M(i(s), 1) - M(i(s), 0)$ , consistently at  $(1, 1)$ , and  $d$  is *free*; any extension to  $[0, 1]^2$  satisfying (54) below would force  $d = 2c$ .

The reason is structural. A pair  $\Psi, \psi$  with

$$\Psi(x', y) - \Psi(x, y) = \int_x^{x'} \psi(z, y) dz, \quad \Psi(x, y') - \Psi(x, y) = \int_y^{y'} \psi(x, z) dz \quad (54)$$

satisfies  $\partial_x \Psi = \partial_y \Psi$ , hence is a function of  $x + y$ , and therefore has  $\Psi(L, 0) = \Psi(0, L)$  built in. Producing one across a gap *presupposes* the conclusion. So the atomic case is not a harder instance of the same argument; it is a different question, and Proposition 7.8 answers it by different means.

*Status of Section 7.1.*

| Clock                             | Status                                |
|-----------------------------------|---------------------------------------|
| Haar (Lebesgue, counting)         | proved, Proposition 7.3               |
| atomless, (T3)                    | proved, Corollary 7.4 (a time change) |
| purely atomic, atoms a chain      | proved, Proposition 7.8               |
| purely atomic, atoms incomparable | verified symbolically; not proved     |
| order-dense atoms, or mixed       | open                                  |

Nothing in this manuscript depends on the last two rows: Theorem 7.12 and Corollary 7.20 use Lebesgue measure, Corollary 7.16 counting measure, and Section 8.2 uses only the additivity (3).

*Remark 7.6* (The role of Haar measure, correctly stated). Corollary 7.4 confirms that translation invariance of  $q$  is not necessary for duality; it is a property of a particular

*proof*, namely it is what makes the anti-diagonal staircase  $t_k = t - s_k$  coincide with the staircase that is anti-diagonal in clock time,

$$Q(s_k) + Q(t_k) = Q(t) \quad \text{for all } k,$$

which is the one (52) actually wants. Equivalently: a martingale problem with clock  $q$  is a time change of one with Lebesgue clock, and duality is invariant under that time change. This is the same phenomenon as in Section 8.8, seen from the analytic side.

Two consequences worth recording. First, the conclusion is genuinely “ $Q$ -almost every  $t$ ”: on an interval where  $q$  vanishes,  $\Phi$  is constant in both arguments and there is nothing to prove. Second, for a purely atomic clock — the case Corollary 7.4 excludes — the conclusion holds at *every*  $t$  and without any integrability hypothesis, exactly as in Proposition 7.3(a); see Proposition 7.8. The two are complementary and neither covers a mixed clock.

For a purely atomic clock there is a second, purely algebraic route to the conclusion, and it is instructive because it does not go through Lemma 7.11 at all. Its combinatorial core is the following statement about a finite grid, in which the clock has become a list of masses and nothing else is left of the index set.

**Lemma 7.7** (Atom grid). *Let  $M \geq 1$ , let  $m_1, \dots, m_{M-1}$  be non-zero reals, and let  $\Phi : \{1, \dots, M\}^2 \rightarrow \mathbb{R}$  satisfy*

$$m_j(\Phi(i+1, j) - \Phi(i, j)) = m_i(\Phi(i, j+1) - \Phi(i, j)), \quad 1 \leq i, j \leq M-1. \quad (55)$$

*Then  $\Phi(i, j) = \Phi(j, i)$  for all  $i, j$ .*

*Proof.* Equation (55) is linear in  $\Phi$  and invariant under transposition: written at  $(j, i)$  it reads  $m_i(\Phi(j+1, i) - \Phi(j, i)) = m_j(\Phi(j, i+1) - \Phi(j, i))$ , which is (55) at  $(i, j)$  for the function  $(i, j) \mapsto \Phi(j, i)$ . So

$$w(i, j) := \Phi(i, j) - \Phi(j, i)$$

satisfies (55) as well, and  $w(j, i) = -w(i, j)$ . We show  $w \equiv 0$  by induction on the distance  $d = |i - j|$  from the diagonal, the induction hypothesis being used at the two levels  $d$  and  $d - 1$  at once.

For  $d = 0$ ,  $w(i, i) = 0$  by antisymmetry. For  $d = 1$  and  $1 \leq j \leq M - 1$ , equation (55) at  $(j, j)$  reads  $m_j(w(j+1, j) - w(j, j)) = m_j(w(j, j+1) - w(j, j))$ ; since  $m_j \neq 0$  and  $w(j, j) = 0$  this gives  $w(j+1, j) = w(j, j+1) = -w(j+1, j)$ , hence  $w(j+1, j) = w(j, j+1) = 0$ .

Let  $d \geq 1$  and suppose  $w$  vanishes at distances  $d - 1$  and  $d$ . Let  $j \geq 1$  with  $j + d + 1 \leq M$  and put  $i = j + d$ ; then  $i \leq M - 1$  and  $j \leq M - d - 1 \leq M - 1$ , so (55) is available at  $(i, j)$  and reads

$$m_j(w(j+d+1, j) - w(j+d, j)) = m_{j+d}(w(j+d, j+1) - w(j+d, j)).$$

Here  $w(j+d, j) = 0$  at distance  $d$ , and  $(j+d) - (j+1) = d - 1$ , so  $w(j+d, j+1) = 0$  at distance  $d - 1$ . Hence  $m_j w(j+d+1, j) = 0$ , and  $m_j \neq 0$  gives  $w(j+d+1, j) = 0$ ; antisymmetry gives  $w(j, j+d+1) = 0$ . Every pair at distance  $d + 1$  in  $\{1, \dots, M\}^2$  is of this form, which completes the induction.  $\square$

The induction sweeps the *diagonals*  $i - j = d$  outwards, not the anti-diagonals; that it needs two consecutive levels at once is why no single staircase in (51) exhibits the cancellation — a staircase sees one anti-diagonal at a time. Note also that positivity of the masses is nowhere used, only  $m_i \neq 0$ , and that the list stops at  $m_{M-1}$ : the mass of the last atom never enters.

**Proposition 7.8** (Atomic clocks; (T0) + (E0) + clock, either  $\iota$ ). *Let  $q = \sum_k m_k \delta_{a_k}$  be a purely atomic clock and let  $\Phi, \gamma : \mathbb{T} \times \mathbb{T} \rightarrow \mathbb{R}$  satisfy (50) with  $\gamma_1 = \gamma_2 = \gamma$ . Let  $t^* \in \mathbb{T}$  and assume that the atoms of  $q$  in  $\mathbb{T}_{\leq t^*}$  are finitely many and pairwise comparable, in the sense that of any two distinct such atoms  $a, b$  either  $a \in \mathbb{T}_{< b}$  or  $b \in \mathbb{T}_{< a}$ . Then*

$$\Phi(t^*, 0) = \Phi(0, t^*).$$

*This holds at every such  $t^*$ , for either convention  $\iota$ , and under no integrability hypothesis beyond the existence of the integrals in (50) and no regularity of  $\gamma$  whatsoever.*

*Proof.* Take  $\iota = p$  first. Let  $a_1, \dots, a_N$  be the atoms lying in  $\mathbb{T}_{\leq t^*}$ , indexed so that  $a_i \in \mathbb{T}_{< a_j}$  for  $i < j$ , which the comparability hypothesis permits. Put  $u_0 = 0$ ,  $u_i = a_i$  for  $1 \leq i \leq N$  and  $u_M = t^*$  with  $M = N + 1$ ; this is an increasing chain. Since  $\mathbb{T}_{< 0} = \emptyset$ , the set  $[u_0, u_1)$  carries no atom, and for  $1 \leq i \leq N$  the set  $[u_i, u_{i+1}) = \mathbb{T}_{< u_{i+1}} \setminus \mathbb{T}_{< u_i}$  carries the single atom  $a_i$ : the atoms in  $\mathbb{T}_{< u_{i+1}}$  are exactly  $a_1, \dots, a_i$  and those in  $\mathbb{T}_{< u_i}$  exactly  $a_1, \dots, a_{i-1}$ , and likewise with  $a_{N+1}$  read as  $t^*$ .

Write  $\hat{\Phi}(i, j) = \Phi(u_i, u_j)$  and  $\hat{\gamma}(i, j) = \gamma(u_i, u_j)$  on  $\{0, \dots, M\}^2$ , and set  $\hat{m}_0 = 0$ ,  $\hat{m}_i = m_i$  for  $1 \leq i \leq N$ . Because  $u_i = a_i$  carries the mass of the step from  $u_i$  to  $u_{i+1}$ , (50) becomes

$$\hat{\Phi}(i+1, j) - \hat{\Phi}(i, j) = \hat{m}_i \hat{\gamma}(i, j), \quad \hat{\Phi}(i, j+1) - \hat{\Phi}(i, j) = \hat{m}_j \hat{\gamma}(i, j)$$

for  $0 \leq i, j \leq M - 1$ . Multiplying the first by  $\hat{m}_j$  and the second by  $\hat{m}_i$  makes both right-hand sides  $\hat{m}_i \hat{m}_j \hat{\gamma}(i, j)$ , so  $\hat{\Phi}$  satisfies (55) on  $1 \leq i, j \leq M - 1$  with the non-zero masses  $\hat{m}_1, \dots, \hat{m}_{M-1}$ ; note that  $\hat{\gamma}$  has disappeared. Lemma 7.7 gives  $\hat{\Phi}(i, j) = \hat{\Phi}(j, i)$  on  $\{1, \dots, M\}^2$ . Finally  $\hat{m}_0 = 0$  gives  $\hat{\Phi}(1, j) = \hat{\Phi}(0, j)$  and  $\hat{\Phi}(i, 1) = \hat{\Phi}(i, 0)$ , whence

$$\Phi(t^*, 0) = \hat{\Phi}(M, 0) = \hat{\Phi}(M, 1) = \hat{\Phi}(1, M) = \hat{\Phi}(0, M) = \Phi(0, t^*).$$

For  $\iota = o$  use the shorter chain  $u_0 = 0$ ,  $u_i = a_i$  ( $1 \leq i \leq N$ ), and put  $M = N$ . Now  $(u_{i-1}, u_i] = \mathbb{T}_{\leq u_i} \setminus \mathbb{T}_{\leq u_{i-1}}$  carries the single atom  $a_i$ , so (50) reads  $\hat{\Phi}(i, j) - \hat{\Phi}(i-1, j) = m_i \hat{\gamma}(i, j)$  and symmetrically, for  $1 \leq i, j \leq M$ , and eliminating  $\hat{\gamma}$  as before gives

$$m_j (\hat{\Phi}(i, j) - \hat{\Phi}(i-1, j)) = m_i (\hat{\Phi}(i, j) - \hat{\Phi}(i, j-1)), \quad 1 \leq i, j \leq M.$$

Reflect the grid:  $\Phi^\sharp(i, j) = \hat{\Phi}(M-i, M-j)$  and  $m_i^\sharp = m_{M-i}$ . Substituting  $(M-i, M-j)$  for  $(i, j)$  turns the display into  $m_j^\sharp (\Phi^\sharp(i+1, j) - \Phi^\sharp(i, j)) = m_i^\sharp (\Phi^\sharp(i, j+1) - \Phi^\sharp(i, j))$  for  $0 \leq i, j \leq M-1$ , which is (55) on the shifted index  $\{1, \dots, M+1\}$  with the non-zero masses  $m_0^\sharp, \dots, m_{M-1}^\sharp = m_M, \dots, m_1$ . Lemma 7.7 makes  $\Phi^\sharp$ , hence  $\hat{\Phi}$ , symmetric on  $\{0, \dots, M\}^2$ ; in particular  $\hat{\Phi}(M, 0) = \hat{\Phi}(0, M)$ . Since  $(a_N, t^*]$  carries no atom,  $\Phi(t^*, \cdot) = \Phi(a_N, \cdot)$  and  $\Phi(\cdot, t^*) = \Phi(\cdot, a_N)$ , so this is  $\Phi(t^*, 0) = \Phi(0, t^*)$ . If  $N = 0$  then  $(0, t^*]$  carries no atom and both sides equal  $\Phi(0, 0)$  outright.  $\square$

*Remark 7.9* (Atomic clocks, and why the conventions do not clash here). Four comments on Proposition 7.8.

*It proves more than duality.* What the grid argument delivers is  $\Phi(s, t) = \Phi(t, s)$  at every pair of points of the chain, not merely at  $(t^*, 0)$ ; equivalently,  $\gamma(s, t) = \gamma(t, s)$  there, which is what (52) wants and what the anti-diagonal staircase fails to see. For equal masses (55) says precisely that  $\Phi$  is constant on anti-diagonals of the atom grid, so the whole content sits in unequal masses. The symmetry is the exact discrete counterpart of the atomless case, where (54) makes  $\Psi$  a function of  $x + y$ ; but here it holds at every point and without an integrability hypothesis, as Remark 7.6 predicts.

*The conventions agree, and now for a reason.* With  $\iota = \mathbf{o}$  the  $k$ -th summand of (52) on the anti-diagonal staircase becomes  $\gamma(k+1, t-k-1) - \gamma(k, t-k)$  and the sum telescopes to  $\gamma(t, 0) - \gamma(0, t)$  rather than to 0. The proof above shows that the two conventions differ by nothing more than the reflection  $(i, j) \mapsto (M-i, M-j)$  of the grid together with a reversal of the list of masses, under which (55) is invariant; so the same lemma settles both, and  $\gamma(t, 0) = \gamma(0, t)$  falls out. Section 7 is therefore convention-agnostic, and the bundle table records it as such.

*What the comparability hypothesis is for.* The proposition needs a chain of atoms below  $t^*$ , not a linearly ordered  $\mathbb{T}$ : it lives on (T0), and under (T2a) the hypothesis is automatic. This is why duality is not confined to a linearly ordered index even though Corollary 7.4 lives on (T3). Atoms that are *incomparable* are a different matter and are not covered; for the smallest instance —  $\mathbb{T} = \{0, a, b, t^*\}$  with  $a, b$  incomparable and an atom at each — the three relations along  $[0, t^*)$ ,  $[a, t^*)$  and  $[b, t^*)$  force  $m_a \gamma(a, t) = m_b \gamma(b, t) = 0$  and the conclusion is immediate, and the two-dimensional index  $\{0, 1, 2\}^2$  has been verified symbolically, but no argument covering a general preorder is offered here. Nor can one run through the symmetry of the first comment: on  $\{0, 1, 2\}^2$  the conclusion  $\Phi(t, 0) = \Phi(0, t)$  holds at every  $t$  while  $\Phi(s, t) = \Phi(t, s)$  fails, at the incomparable pair  $((1, 2), (2, 1))$  among others. Symmetry is a phenomenon of the chain, not of the atomic clock.

*What is not covered.* Finitely many atoms below each  $t$  excludes an order-dense set of atoms, and with it the possibility of transferring the argument: where the atoms are dense, no interval  $[s, s']$  carries exactly one of them, (55) has no discrete analogue, and there is no grid to induct along. That is the same obstruction as in the mixed case, seen from the atomic side.

*Remark 7.10* (Scope of Section 7). It is worth saying plainly what Section 7 needs, since the matter is easy to get wrong.

It needs the *balance condition*  $\gamma_1 = \gamma_2$  — in the probabilistic form of Theorem 7.12, that the two generators act on the same duality function  $f$  so as to produce the same  $g = h$  — and it needs integrability. It does *not* need a translation invariant clock: both Corollary 7.4 (atomless clocks) and Proposition 7.8 (atomic ones) cover clocks that are not Haar. Nor does the conclusion need a linearly ordered index: the atomic argument asks only that the atoms below a point form a chain. The opposite claim, made in an earlier version of this manuscript on the strength of the non-cancellation of (52) for  $\mathbb{T} = \mathbb{R}_+^d$ , was a fallacy — the non-cancellation of a particular staircase is not the failure of the conclusion.

What genuinely remains open is the *mixed* case, a clock with both an atomic and a diffuse part, together with an order-dense set of atoms and a set of atoms that is not a chain; Remark 7.5 explains why the two available arguments do not combine. Nothing in this manuscript depends on it: the applications in Section 8 use Lebesgue clocks, and Section 8.2 uses (3) only.

## 7.2 The continuous-time theory

**Lemma 7.11** ([7], Lemma 4.4.10; (T3)). *Let  $\Phi : [0, \infty) \times [0, \infty) \rightarrow \mathbb{R}$  be absolutely continuous in  $s$  for each fixed  $t$  and absolutely continuous in  $t$  for each fixed  $s$ , and set  $(\gamma_1, \gamma_2) = \nabla \Phi$ . Suppose that*

$$\int_0^T \int_0^T |\gamma_i(s, t)| \, ds \, dt < \infty, \quad i = 1, 2, \quad T > 0. \quad (56)$$

*Then for almost every  $t \geq 0$ ,*

$$\Phi(t, 0) - \Phi(0, t) = \int_0^t (\gamma_1(s, t-s) - \gamma_2(s, t-s)) \, ds. \quad (57)$$

*Proof.* By (56) and Fubini,

$$\begin{aligned}
& \int_0^T \int_0^t (\gamma_1(t-s, s) - \gamma_2(s, t-s)) \, ds \, dt \\
&= \int_0^T \int_0^t \gamma_1(t-s, s) \, ds \, dt - \int_0^T \int_0^t \gamma_2(s, t-s) \, ds \, dt \\
&= \int_0^T \int_s^T \gamma_1(t-s, s) \, dt \, ds - \int_0^T \int_s^T \gamma_2(s, t-s) \, dt \, ds \\
&= \int_0^T (\Phi(T-s, s) - \Phi(0, s)) \, ds - \int_0^T (\Phi(s, T-s) - \Phi(s, 0)) \, ds \\
&= \int_0^T (\Phi(s, 0) - \Phi(0, s)) \, ds,
\end{aligned}$$

where the third equality uses absolute continuity in the first resp. second variable, and the fourth is the substitution  $s \mapsto T-s$  in the first two integrals. Differentiating with respect to  $T$  gives (57) for a.e.  $T$ .  $\square$

**Theorem 7.12** ([7], Theorem 4.4.11; (T3)). *Let  $X$  and  $Y$  be independent measurable processes with values in  $E_1$  and  $E_2$  respectively. Let  $f, g, h \in M(E_1 \times E_2)$ ,  $\alpha \in M(E_1)$  and  $\beta \in M(E_2)$ . Suppose that for each  $T > 0$  there exist an integrable random variable  $\Gamma_T$  and a constant  $C_T$  such that*

$$\begin{aligned}
& \sup_{r,s,t \leq T} (|\alpha(X(r))| + 1) |f(X(s), Y(t))| \leq \Gamma_T, \\
& \sup_{r,s,t \leq T} (|\beta(Y(r))| + 1) |f(X(s), Y(t))| \leq \Gamma_T, \\
& \sup_{r,s,t \leq T} (|\alpha(X(r))| + 1) |g(X(s), Y(t))| \leq \Gamma_T, \\
& \sup_{r,s,t \leq T} (|\beta(Y(r))| + 1) |h(X(s), Y(t))| \leq \Gamma_T,
\end{aligned} \tag{58}$$

and

$$\int_0^T |\alpha(X(u))| \, du + \int_0^T |\beta(Y(u))| \, du \leq C_T. \tag{59}$$

Suppose that

$$f(X(t), y) - \int_0^t g(X(s), y) \, ds \tag{60}$$

is a  $(*\mathcal{F}_t^X)$ -martingale for each  $y \in E_2$ , and

$$f(x, Y(t)) - \int_0^t h(x, Y(s)) \, ds \tag{61}$$

is a  $(*\mathcal{F}_t^Y)$ -martingale for each  $x \in E_1$  (the integrals in (60) and (61) being assumed to exist). Then for almost every  $t \geq 0$ ,

$$\begin{aligned}
& E \left[ f(X(t), Y(0)) \exp \left\{ \int_0^t \alpha(X(u)) \, du \right\} \right] - E \left[ f(X(0), Y(t)) \exp \left\{ \int_0^t \beta(Y(u)) \, du \right\} \right] \\
&= E \left[ \int_0^t \{ g(X(s), Y(t-s)) - h(X(s), Y(t-s)) + (\alpha(X(s)) - \beta(Y(t-s))) f(X(s), Y(t-s)) \} \right. \\
&\quad \left. \times \exp \left\{ \int_0^s \alpha(X(u)) \, du + \int_0^{t-s} \beta(Y(u)) \, du \right\} \, ds \right]. \tag{62}
\end{aligned}$$

*Proof.* Set

$$\Phi(s, t) = E \left[ f(X(s), Y(t)) \exp \left\{ \int_0^s \alpha(X(u)) du \right\} \exp \left\{ \int_0^t \beta(Y(u)) du \right\} \right]. \quad (63)$$

*Step 1: freezing  $Y$ .* Write  $e_\alpha(s) = \exp\{\int_0^s \alpha(X(u)) du\}$  and  $e_\beta(t) = \exp\{\int_0^t \beta(Y(u)) du\}$ , and fix  $t$ . Both  $e_\beta(t)$  and  $Y(t)$  are  $\sigma(Y)$ -measurable, and  $\sigma(Y)$  is independent of  $\sigma(X)$ ; so every expectation below may be computed by conditioning on  $\sigma(Y)$ , freezing  $Y(t) = y$  and  $e_\beta(t)$ , and using that  $M_r^y = f(X(r), y) - \int_0^r g(X(v), y) dv$  is a  $(*\mathcal{F}_r^X)$ -martingale under the conditioned measure — which it is, because conditioning on an independent  $\sigma$ -field leaves the law of  $X$  and the filtration  $*\mathcal{F}^X$  unchanged. We suppress the conditioning from the notation.

*Step 2: the increment identity.* For  $h > 0$  split

$$\begin{aligned} & f(X(s+h), y)e_\alpha(s+h) - f(X(s), y)e_\alpha(s) \\ &= \underbrace{[f(X(s+h), y) - f(X(s), y)]e_\alpha(s)}_{(A)} + \underbrace{f(X(s+h), y)[e_\alpha(s+h) - e_\alpha(s)]}_{(B)}. \end{aligned} \quad (64)$$

For (A), write  $f(X(s+h), y) - f(X(s), y) = M_{s+h}^y - M_s^y + \int_s^{s+h} g(X(v), y) dv$ . Since  $e_\alpha(s)$  is  $*\mathcal{F}_s^X$ -measurable and bounded by  $e^{C_T}$ , the martingale term contributes 0, so

$$\begin{aligned} E[(A)] &= \int_s^{s+h} E[g(X(r), y)e_\alpha(s)] dr \\ &= \int_s^{s+h} E[g(X(r), y)e_\alpha(r)] dr \\ &\quad + \int_s^{s+h} E[g(X(r), y)(e_\alpha(s) - e_\alpha(r))] dr. \end{aligned}$$

For (B), the chain rule gives  $e_\alpha(s+h) - e_\alpha(s) = \int_s^{s+h} \alpha(X(r))e_\alpha(r) dr$ . For each  $r \in [s, s+h]$  write

$$f(X(s+h), y) = f(X(r), y) + (M_{s+h}^y - M_r^y) + \int_r^{s+h} g(X(v), y) dv.$$

The martingale term again contributes 0, since  $\alpha(X(r))e_\alpha(r)$  is  $*\mathcal{F}_r^X$ -measurable, and one obtains

$$\begin{aligned} E[(B)] &= \int_s^{s+h} E[f(X(r), y)\alpha(X(r))e_\alpha(r)] dr \\ &\quad + \int_s^{s+h} \int_r^{s+h} E[g(X(v), y)\alpha(X(r))e_\alpha(r)] dv dr. \end{aligned}$$

Multiplying by  $e_\beta(t)$ , integrating over the law of  $Y$  and using Fubini — legitimate because

all integrands are dominated by  $\Gamma_T e^{C_T}$  by (58)–(59) — we obtain

$$\begin{aligned}
\Phi(s+h, t) - \Phi(s, t) &= \underbrace{\int_s^{s+h} E[f\alpha(X(r), Y(t))e_\alpha(r)e_\beta(t)] \, dr}_{T_1} \\
&\quad + \underbrace{\int_s^{s+h} \int_r^{s+h} E[g(X(v), Y(t))\alpha(X(r))e_\alpha(r)e_\beta(t)] \, dv \, dr}_{T_2} \\
&\quad + \underbrace{\int_s^{s+h} E[g(X(r), Y(t))e_\alpha(r)e_\beta(t)] \, dr}_{T_3} \\
&\quad + \underbrace{\int_s^{s+h} E[g(X(r), Y(t))(e_\alpha(s) - e_\alpha(r))e_\beta(t)] \, dr}_{T_4}, \quad (65)
\end{aligned}$$

where  $f\alpha$  abbreviates  $(x, y) \mapsto f(x, y)\alpha(x)$ .

*Step 3:*  $T_2$  and  $T_4$  are  $O(h^2)$ . Let  $t, s+h \leq T$ . The third line of (58) gives  $|\alpha(X(r))| |g(X(v), Y(t))| \leq \Gamma_T$ , and (59) gives  $e_\alpha(r)e_\beta(t) \leq e^{C_T}$ . Hence

$$|T_2| \leq e^{C_T} E[\Gamma_T] \int_s^{s+h} (s+h-r) \, dr = \frac{1}{2} h^2 e^{C_T} E[\Gamma_T].$$

For  $T_4$  use  $e_\alpha(s) - e_\alpha(r) = -\int_s^r \alpha(X(u))e_\alpha(u) \, du$  and the same two bounds:

$$|T_4| \leq \int_s^{s+h} \int_s^r E[|g(X(r), Y(t))| |\alpha(X(u))| e_\alpha(u)e_\beta(t)] \, du \, dr \leq \frac{1}{2} h^2 e^{C_T} E[\Gamma_T].$$

*Step 4: passing to the limit.* Since  $T_1 + T_3 = \int_s^{s+h} E[(f\alpha + g)(X(r), Y(t))e_\alpha(r)e_\beta(t)] \, dr$ , summing (65) over a partition  $0 = s_0 < \dots < s_m = s$  gives

$$\left| \Phi(s, t) - \Phi(0, t) - \int_0^s E[(f\alpha + g)(X(r), Y(t))e_\alpha(r)e_\beta(t)] \, dr \right| \leq \frac{1}{2} e^{C_T} E[\Gamma_T] \sum_{i=1}^m (s_i - s_{i-1})^2$$

— the main terms telescoping *exactly*, with no limit needed — and the right-hand side is at most  $\frac{1}{2} e^{C_T} E[\Gamma_T] s \max_i (s_i - s_{i-1}) \rightarrow 0$ . This is

$$\Phi(s, t) - \Phi(0, t) = \int_0^s E[(f(X(r), Y(t))\alpha(X(r)) + g(X(r), Y(t)))e_\alpha(r)e_\beta(t)] \, dr. \quad (66)$$

Symmetrically,

$$\Phi(s, t) - \Phi(s, 0) = \int_0^t E[(f(X(s), Y(r))\beta(Y(r)) + h(X(s), Y(r)))e_\alpha(s)e_\beta(r)] \, dr. \quad (67)$$

Thus  $\Phi$  satisfies the hypotheses of Lemma 7.11, with  $\gamma_1$  and  $\gamma_2$  the integrands in (66) and (67); the integrability (56) follows from (58) and (59). Now (57) is exactly (62).  $\square$

*Remark 7.13* ([7], Remark 4.4.12). Equation (62) can frequently be extended to more general  $\alpha$  and  $\beta$  by approximating them by bounded functions.

**Corollary 7.14** (Duality relation; [7], Corollary 4.4.13; (T3)). *If, in addition to the conditions of Theorem 7.12,*

$$g(x, y) + \alpha(x)f(x, y) = h(x, y) + \beta(y)f(x, y), \quad (x, y) \in E_1 \times E_2, \quad (68)$$

*then for all  $t \geq 0$*

$$E\left[f(X(t), Y(0)) \exp\left\{\int_0^t \alpha(X(u)) du\right\}\right] = E\left[f(X(0), Y(t)) \exp\left\{\int_0^t \beta(Y(u)) du\right\}\right]. \quad (69)$$

*Proof.* Under (68) the right-hand side of (62) vanishes, so (69) holds for almost every  $t$ . It extends to all  $t$  because  $t \mapsto F(t, 0)$  and  $t \mapsto F(0, t)$  are continuous, which follows from (66) and (67).  $\square$

*Remark 7.15.* Corollary 7.14 is Proposition 7.3(b) applied to the  $\Phi$  of (63), with  $\gamma_1$  and  $\gamma_2$  the integrands of (66) and (67); the balance condition (68) is exactly  $\gamma_1 = \gamma_2$ . The whole of Theorem 7.12 is thus Lemma 7.1 plus the computation (65) that produces the increment representations (50) from the martingale hypotheses. Only the latter uses probability; and it uses no path regularity, no topology on  $E_1$ ,  $E_2$  and no boundedness, so it is already stated at the level of generality of [2]. What it does use, and what Remark 7.10 isolates, is the balance condition and integrability — not, as one might guess from the shape of (52), translation invariance of the clock; see Corollary 7.4 and Proposition 7.8.

The discrete counterpart, which [7] do not state, now costs nothing.

**Corollary 7.16** (Duality for Markov chains;  $\mathbb{T} = \mathbb{N}_0$ , (E0),  $\iota = p$ ). *Let  $P$  and  $Q$  be transition kernels on measurable spaces  $E_1$  and  $E_2$ , and let  $f \in B(E_1 \times E_2)$  satisfy the duality relation*

$$\int f(x', y) P(x, dx') = \int f(x, y') Q(y, dy'), \quad (x, y) \in E_1 \times E_2. \quad (70)$$

*Let  $X$  and  $Y$  be independent Markov chains with kernels  $P$  and  $Q$  and  $X(0) = x$ ,  $Y(0) = y$ . Then*

$$E[f(X(n), y)] = E[f(x, Y(n))], \quad n \in \mathbb{N}_0. \quad (71)$$

*Proof.* Put  $\Phi(n, m) = E[f(X(n), Y(m))]$ . Writing  $P$  for the action of the kernel on the first variable and  $Q$  for its action on the second, the Markov property and independence give  $\Phi(n+1, m) - \Phi(n, m) = E[((P - I)f)(X(n), Y(m))]$  and  $\Phi(n, m+1) - \Phi(n, m) = E[((Q - I)f)(X(n), Y(m))]$ , which is (50) for  $q =$  counting measure with  $\gamma_1(n, m) = E[((P - I)f)(X(n), Y(m))]$  and  $\gamma_2$  its analogue. By (70),  $(P - I)f = (Q - I)f$ , so  $\gamma_1 = \gamma_2$ , and Proposition 7.3(a) applies.  $\square$

*Remark 7.17.* In the language of Section 4, (70) says that  $(f(\cdot, y), g(\cdot, y)) \in A$  and  $(f(x, \cdot), g(x, \cdot)) \in B$  hold with the same  $g = (P - I)f = (Q - I)f$ , where  $A$  and  $B$  denote the one-step generators; the associated test process is then the Doob decomposition

$$f(X(n), y) - \sum_{k < n} g(X(k), y),$$

i.e. the compensator in the predictable convention  $[0, t)$ . So Corollary 7.16 is the martingale-problem duality theorem for  $\mathbb{T} = \mathbb{N}_0$ ; the predictable convention is the one in which its proof is shortest, not the only one in which it is true (Remark 7.9).

**Corollary 7.18** (Stopped version; [7], Corollary 4.4.14; (T3)). *Let  $(\mathcal{F}_t)$  and  $(\mathcal{G}_t)$  be independent filtrations, let  $X$  be  $(\mathcal{F}_t)$ -progressive with  $(\mathcal{F}_t)$ -stopping time  $\tau$ , and let  $Y$  be  $(\mathcal{G}_t)$ -progressive with  $(\mathcal{G}_t)$ -stopping time  $\sigma$ . Suppose (58) and (59) hold with  $X, Y$  replaced by  $X(\cdot \wedge \tau)$ ,  $Y(\cdot \wedge \sigma)$ , and that  $f(X(t \wedge \tau), y) - \int_0^{t \wedge \tau} g(X(s), y) ds$  is an  $(\mathcal{F}_t)$ -martingale for each  $y$ , and  $f(x, Y(t \wedge \sigma)) - \int_0^{t \wedge \sigma} h(x, Y(s)) ds$  is a  $(\mathcal{G}_t)$ -martingale for each  $x$ . Then the analogue of (62) holds with  $X, Y$  replaced by the stopped processes and the integrand multiplied by the indicators  $\mathbf{1}_{\{s \leq \tau\}}$  resp.  $\mathbf{1}_{\{t-s \leq \sigma\}}$ .*

*Proof.* Rewrite the martingale in the hypothesis as  $f(X(t \wedge \tau), y) - \int_0^t \mathbf{1}_{\{s \leq \tau\}} g(X(s), y) ds$  and repeat the proof of Theorem 7.12.  $\square$

*Remark 7.19.* Corollary 7.18 is the practically relevant form: the estimates (58) are often unavailable globally but easy on the exit times  $\tau$  from compact sets. One then lets  $\tau, \sigma \rightarrow \infty$  along an exhausting sequence; the integrand on the right of the stopped version of (62) tends to zero, and the only difficulty is to justify the interchange of limits and expectations ([7], Remark 4.4.16). This is also the route by which duality is applied to *local* martingale problems.

We can now close the circle with Section 6.

**Corollary 7.20** (Uniqueness via duality; (T3)). *Assume (E2), let  $A \subset B(E) \times B(E)$ , let  $(E_2, \mathcal{E}_2)$  be a measurable space, and let  $B \subset M(E_2) \times M(E_2)$ . Suppose there is  $f \in M(E \times E_2)$  such that:*

- (i)  $\{f(\cdot, y) : y \in E_2\} \subset B(E)$  is separating for  $\mathcal{P}(E)$ ;
- (ii)  $(f(\cdot, y), g(\cdot, y)) \in A$  for every  $y \in E_2$ , and  $(f(x, \cdot), h(x, \cdot)) \in B$  for every  $x \in E$ , with  $g = h$ ;
- (iii) for every  $y \in E_2$  there is a solution  $Y^y$  of the martingale problem for  $(B, \delta_y)$ , and for every solution  $X$  of the martingale problem for  $A$  with  $X$  independent of  $Y^y$  the hypotheses (58)–(59) hold with  $\alpha = \beta = 0$ .

*Then for every  $\mu \in \mathcal{P}(E)$  the martingale problem for  $(A, \mu)$  has at most one solution, every solution is Markov, and under the additional hypotheses of Theorem 6.51(b)–(c) it is strong Markov.*

*Proof.* Let  $X$  solve the martingale problem for  $(A, \delta_x)$  and let  $Y^y$  be as in (iii), independent of  $X$ . By (ii), (68) holds with  $\alpha = \beta = 0$ , so Corollary 7.14 gives

$$E[f(X(t), y)] = E[f(x, Y^y(t))], \quad t \geq 0. \quad (72)$$

The right-hand side does not depend on  $X$ . Hence  $\int f(\cdot, y) d\nu_t$  is the same for the one-dimensional distributions  $\nu_t = PX(t)^{-1}$  of any two solutions, and by (i) these coincide. For general  $\mu$ , integrate (72) over  $\mu(dx)$ , which is legitimate because the one-dimensional distributions of a solution for  $(A, \mu)$  are mixtures of those for  $(A, \delta_x)$  by Theorem 6.51(a). Now apply Theorem 6.51.  $\square$

## 8 Existence

Sections 6 and 7 are a uniqueness theory, and by themselves they are vacuous: nothing so far exhibits a single solution of a single martingale problem. This section collects the five routes to existence, in increasing order of cost.

- (a) *From a transition semigroup* (Section 8.1). Immediate, and correspondingly weak: it presupposes the object one usually wants to construct.
- (b) *From a dual process* (Section 8.2). Produces the semigroup that (a) presupposes, out of a *known* process on a second state space and a duality function. Cheap, because Section 7 has already done the analytic work; conditional on one hypothesis, kernel representability, which is where all the difficulty is concentrated.
- (c) *Explicit construction* (Sections 8.3 and 8.11). Jump processes are built by hand out of a Markov chain and exponential holding times, and the martingale property is verified by a two-line computation with the exponential distribution; the same computation survives a *path-dependent* rate, which gives Hawkes processes. These are the only places in the manuscript where a solution is written down.
- (d) *From stochastic differential equations* (Section 8.4). The most important route in practice and the most expensive here, since it presupposes stochastic integration; we state the chain of results and prove none of it.
- (e) *By convergence* (Sections 8.5–8.7). Weak limits of solutions of approximating problems. Conditional, like (a), but on much less.

Routes (c) and (d) are the two that produce a solution out of nothing, and it is worth noticing before we start that they are the same argument at two levels of difficulty: both are Picard–Lindelöf. In (c) it is the Picard iteration for the linear equation  $\dot{v} = Av$  in  $B(E)$ , with  $A$  bounded (Remark 8.25); in (d) it is the Picard iteration for a Lipschitz stochastic differential equation. What differs is the space on which the contraction acts. Route (b) is of a different nature: it produces nothing out of nothing either, but what it presupposes lives on a *different* state space, which in the applications is a simpler one — a finite particle system dual to a measure-valued diffusion, say.

## 8.1 From a transition semigroup

**Proposition 8.1** ([7], Proposition 4.1.7; (T3) + (E0)). *Let  $X$  be a Markov process with measurable transition semigroup  $\{T(t)\}$  on  $B(E)$  and full generator  $\hat{A}$  in the sense of Definition 2.15. Then  $X$  solves the martingale problem for  $\hat{A}$ .*

This is Remark 4.20, restated as an existence statement, and it is recorded only for completeness: it turns a Markov process into a solution, and producing the Markov process is the whole difficulty. The next section produces one; the three after that do not presuppose one.

## 8.2 From a dual process

There is a route that produces exactly the object Proposition 8.1 presupposes, and it produces it out of the material that Section 7 has already assembled. It is due to [3], and its point is that no approximating sequence of processes of a *different kind* — jump processes to approximate diffusions, and the like — is needed: the dual process, which one has anyway for uniqueness, delivers existence as well.

The idea in one line: if  $Y$  is a known process on  $E_2$  and  $H$  is a duality function, then the right-hand side  $E^y[H(x, Y_t)]$  of the duality relation is a candidate for the left-hand side  $E^x[H(X_t, y)]$  of a process  $X$  that does not yet exist; if that candidate is *represented by a kernel* in  $x$ , the kernels form a transition semigroup and  $X$  exists. Everything else is

bookkeeping, and all of it survives an arbitrary *shift invariant* clock — including one with atoms, the obstruction of Remark 7.5 being confined to Section 7.1.

**Setting 8.2** (Duality data; (T2a) + (T4) + shift invariant clock + (E1) on  $E_1$ , (E0) on  $E_2$ ). Let  $\mathbb{T}$  satisfy (T2a) and (T4) — the linear order is used exactly once, in Step 1 of Theorem 8.5, and Remark 8.8(ii) explains why it cannot be dropped there — let  $q$  be a clock which is *shift invariant*,  $q(r + \cdot) = q(\cdot)$ , and fix  $\iota$ . Let  $(E_1, \mathcal{E}_1)$  satisfy (E1) and  $(E_2, \mathcal{E}_2)$  satisfy (E0). Let

$$H, g : E_1 \times E_2 \rightarrow \mathbb{R}, \quad \beta : E_2 \rightarrow \mathbb{R}$$

be bounded and measurable, and let  $(P^y)_{y \in E_2}$  be a time-homogeneous Markov family on a path space  $F_2$  over  $E_2$  with  $P^y\{\pi_0 = y\} = 1$ . Write

$$V_t = \exp\left\{-\int_{\langle 0, t \rangle_\iota} \beta(\pi_u) q(\mathrm{d}u)\right\}, \quad (73)$$

the *Feynman–Kac factor*, and note  $0 < V_t \leq e^{\|\beta\|q(\mathbb{T}_{\leq t})} < \infty$ , the finiteness being Definition 2.3. Define, for  $\varphi \in \{H, g\}$ ,

$$(P_t \varphi)(x, y) := E^y[\varphi(x, \pi_t) V_t], \quad x \in E_1, y \in E_2, t \in \mathbb{T}. \quad (74)$$

We assume throughout:

(D1) *Balance*. For every  $x \in E_1$  the process

$$t \mapsto H(x, \pi_t) V_t - \int_{\langle 0, t \rangle_\iota} g(x, \pi_u) V_u q(\mathrm{d}u)$$

is a  $P^y$ -martingale, for every  $y \in E_2$ .

(D2) *Separation*.  $\text{span}\{H(\cdot, y) : y \in E_2\}$  is separating on  $\mathcal{P}(E_1)$ .

(D3) *Kernel representability*. There is a family  $(\mu_t)_{t \in \mathbb{T}}$  of probability kernels on  $E_1$  with  $(t, x) \mapsto \mu_t(x, \Gamma)$  jointly measurable for every  $\Gamma \in \mathcal{E}_1$ , such that

$$(P_t \varphi)(x, y) = \int_{E_1} \mu_t(x, \mathrm{d}x') \varphi(x', y), \quad \varphi \in \{H, g\},$$

for all  $x, y, t$ .

Three comments before the statements. (D1) is the balance condition of Section 7 in integrated form: it says that the pair  $(H(x, \cdot), g(x, \cdot))$ , twisted by the Feynman–Kac factor, is a test process for  $Y$  — in the differential notation of [3] it reads  $G_1 H(\cdot, y)(x) = G_2 H(x, \cdot)(y) + \beta(y) H(x, y)$  with  $g = G_1 H(\cdot, y)$ . (D2) is a mild non-degeneracy of the duality function. (D3) is the only substantial hypothesis, and it is the one that cannot be checked by bookkeeping; Proposition 8.7 gives the standard sufficient condition.

**Lemma 8.3** (The semigroup identity; (T4) + shift invariant clock). *In Setting 8.2, for  $\varphi \in \{H, g\}$ ,  $s, t \in \mathbb{T}$ ,*

$$(P_{s+t} \varphi)(x, y) = E^y[(P_t \varphi)(x, \pi_s) V_s]. \quad (75)$$

*Proof. Step 1: the factor splits.* By clock additivity (3),

$$V_{s+t} = V_s \cdot \exp \left\{ - \int_{\langle s, s+t \rangle_\iota} \beta(\pi_v) q(\mathrm{d}v) \right\},$$

and substituting  $v = s + u$ , which is a bijection of  $\langle 0, t \rangle_\iota$  onto  $\langle s, s+t \rangle_\iota$  by Lemma 2.2, the second factor equals  $\exp \{ - \int_{\langle 0, t \rangle_\iota} \beta(\pi_{s+u}) q(\mathrm{d}u) \} = V_t \circ \theta_s$ . *This is the one place where shift invariance of  $q$  is used:* without it the substitution changes the measure and the shifted factor is not  $V_t$ .

*Step 2: the Markov property.*  $\varphi(x, \pi_{s+t}) = (\varphi(x, \pi_t)) \circ \theta_s$  and  $V_s$  is  $\mathcal{F}_s^\circ$ -measurable, so

$$(P_{s+t}\varphi)(x, y) = E^y[\varphi(x, \pi_t) \circ \theta_s \cdot (V_t \circ \theta_s) \cdot V_s] = E^y[E^{\pi_s}[\varphi(x, \pi_t)V_t]V_s],$$

by the Markov property of  $(P^y)$  and its time homogeneity. The inner expectation is  $(P_t\varphi)(x, \pi_s)$ , which is (75). All integrands are bounded by Setting 8.2, so every expectation is finite.  $\square$

**Proposition 8.4** (The kernels form a transition semigroup; + (E1)). *In Setting 8.2,  $\mu_0(x, \cdot) = \delta_x$  and*

$$\mu_{s+t}(x, \cdot) = \int_{E_1} \mu_s(x, \mathrm{d}x') \mu_t(x', \cdot), \quad s, t \in \mathbb{T}, \quad x \in E_1. \quad (76)$$

*Proof. Initial condition.*  $\langle 0, 0 \rangle_\iota = \emptyset$  gives  $V_0 = 1$ , and  $P^y\{\pi_0 = y\} = 1$  gives  $(P_0H)(x, y) = H(x, y)$ . So  $\int \mu_0(x, \mathrm{d}x') H(x', y) = H(x, y)$  for every  $y$ , and (D2) forces  $\mu_0(x, \cdot) = \delta_x$ .

*Chapman–Kolmogorov.* Fix  $x, y, s, t$  and compute, using (D3) at  $x'$ , then Fubini — legitimate because  $H$  is bounded and  $\mu_s(x, \cdot)$  is a probability measure — then (D3) at  $x$ , then Lemma 8.3:

$$\begin{aligned} \int \mu_s(x, \mathrm{d}x') \int \mu_t(x', \mathrm{d}x'') H(x'', y) &= \int \mu_s(x, \mathrm{d}x') (P_t H)(x', y) \\ &= \int \mu_s(x, \mathrm{d}x') E^y[H(x', \pi_t) V_t] \\ &= E^y \left[ \left( \int \mu_s(x, \mathrm{d}x') H(x', \pi_t) \right) V_t \right] \\ &= E^y[(P_s H)(x, \pi_t) V_t] \\ &= (P_{t+s} H)(x, y) = \int \mu_{s+t}(x, \mathrm{d}x'') H(x'', y), \end{aligned}$$

the commutativity of  $\mathbb{T}$  being used once, in  $t + s = s + t$ . Since this holds for every  $y$ , (D2) gives (76).  $\square$

**Theorem 8.5** (Existence by duality; [3], Theorem 2.1; (T2a) + (T4) + shift invariant clock + (E1)). *In Setting 8.2, let  $\nu \in \mathcal{P}(E_1)$ . Then there is a Markov process  $X$  on  $E_1$  with initial distribution  $\nu$  and transition kernels  $(\mu_t)_{t \in \mathbb{T}}$ , and it solves the martingale problem for*

$$\mathcal{X}^{(1)} = \left\{ H(\pi_t, y) - \int_{\langle 0, t \rangle_\iota} g(\pi_u, y) q(\mathrm{d}u) : y \in E_2 \right\}. \quad (77)$$

*Moreover  $X$  and  $Y$  are in duality with potential  $\beta$ :*

$$E[H(X_t, y) \mid X_0 = x] = E^y[H(x, \pi_t) V_t], \quad x \in E_1, \quad y \in E_2, \quad t \in \mathbb{T}. \quad (78)$$

*Proof. Step 1: the process.* By Proposition 8.4 the family  $(\mu_t)$  is a measurable transition semigroup on  $E_1$ , measurability being (D3). Under (T2a) every finite subset  $t_1 \leq \dots \leq t_n$  of  $\mathbb{T}$  is a chain, so the finite-dimensional distributions  $\nu \otimes \mu_{t_1} \otimes \mu_{t_2-t_1} \otimes \dots$  are defined for *all* finite subsets and form a projective family, consistency being Chapman–Kolmogorov plus marginalization; *this is the one step that needs the linear order*, see Remark 8.8(ii). Since  $E_1$  satisfies (E1) it is standard Borel, so the Kolmogorov extension theorem (Fact 2.47) applies; this is [7], Theorem 4.1.1. Let  $X$  be the resulting Markov process. Then (78) is (D3) for  $\varphi = H$ .

*Step 2: the one-dimensional identity.* Fix  $x$  and  $y$  and take  $P^y$ -expectations in (D1) between 0 and  $t$ ; Fubini is legitimate because  $g$  and  $V$  are bounded on  $\mathbb{T}_{\leq t}$  and  $q(\mathbb{T}_{\leq t}) < \infty$ . This gives

$$(P_t H)(x, y) = H(x, y) + \int_{\langle 0, t \rangle_\iota} (P_u g)(x, y) q(du).$$

Now apply (D3) to both sides — *here it is used for  $g$  as well as for  $H$ , which is why Setting 8.2 asks for both*:

$$\int \mu_t(x, dx') H(x', y) = H(x, y) + \int_{\langle 0, t \rangle_\iota} \left( \int \mu_u(x, dx') g(x', y) \right) q(du). \quad (79)$$

*Step 3: the martingale property.* Let  $s \leq t$  in  $\mathbb{T}$ . By the Markov property of  $X$  and (79) applied at  $X_s$  with  $t - s$  in place of  $t$ ,

$$\begin{aligned} E[H(X_t, y) \mid \mathcal{F}_s^X] &= \int \mu_{t-s}(X_s, dx') H(x', y) \\ &= H(X_s, y) + \int_{\langle 0, t-s \rangle_\iota} E[g(X_{s+u}, y) \mid \mathcal{F}_s^X] q(du), \end{aligned}$$

using the Markov property once more inside the integral. Substituting  $v = s + u$  and using shift invariance of  $q$  together with Lemma 2.2, the last integral is  $E[\int_{\langle s, t \rangle_\iota} g(X_v, y) q(dv) \mid \mathcal{F}_s^X]$ , Fubini being available as in Step 2. Hence

$$E\left[H(X_t, y) - \int_{\langle 0, t \rangle_\iota} g(X_v, y) q(dv) \mid \mathcal{F}_s^X\right] = H(X_s, y) - \int_{\langle 0, s \rangle_\iota} g(X_v, y) q(dv).$$

Testing against products  $\prod_k h_k(X_{t_k})$  with  $t_k \in \mathbb{T}_{\leq s}$  and invoking Proposition 4.9 upgrades this from  $\mathcal{F}^X$  to the filtration  $(\mathcal{F}_t^X)$  of Definition 4.6. So  $X \in \mathcal{M}(\mathcal{X}^{(1)}, \nu)$ .  $\square$

**Corollary 8.6** (Well-posedness). *Assume in addition that  $\mathcal{X}^{(1)}$  admits a shift system with  $\mathcal{Z}^\circ$  determining, and that Section 7 applies to the pair  $(\mathcal{X}^{(1)}, Y)$  — for instance under the integrability hypothesis of Corollary 7.4 for an atomless clock, or Proposition 7.8 for an atomic one. Then the martingale problem for  $\mathcal{X}^{(1)}$  is well posed for every  $\nu \in \mathcal{P}(E_1)$ , and its solution is the Markov process of Theorem 8.5.*

*Proof.* Existence is Theorem 8.5. For uniqueness let  $\tilde{X} \in \mathcal{M}(\mathcal{X}^{(1)}, \nu)$ . The pair  $(\tilde{X}, Y)$  satisfies the hypotheses of Section 7 with the same  $H$  and the balance condition (D1), so the duality relation (78) holds for  $\tilde{X}$  as well; by (D2) this determines the one-dimensional distributions of  $\tilde{X}$ , and Theorem 6.23 concludes. This is the closing of the circle: *the same duality relation is used twice, once forwards to construct the process and once backwards to show it is the only one.*  $\square$

**Proposition 8.7** (Riesz–Markov: a sufficient condition for (D3); (E3)). *Suppose  $E_1$  is compact metrizable,  $H \in C_b(E_1 \times E_2)$ , and  $\mathcal{H}_1 := \text{span}\{H(\cdot, y) : y \in E_2\}$  is a convergence*

determining algebra containing the constants. If for each  $t$  the map  $H(\cdot, y) \mapsto (P_t H)(\cdot, y)$  is positive —  $\varphi \geq 0$  implies  $P_t \varphi \geq 0$  — and  $P_t \mathbf{1} = \mathbf{1}$ , then the kernels of (D3) exist for  $\varphi = H$ .

*Proof.* Fix  $t$  and  $x$ . By hypothesis  $\mathcal{H}_1$  is a subalgebra of  $C(E_1)$  containing the constants and separating points — being convergence determining — so it is dense in  $C(E_1)$  by Stone–Weierstrass (Fact 2.39). The map  $\varphi \mapsto (P_t \varphi)(x, \cdot)$  is linear, positive and unital on  $\mathcal{H}_1$ , hence bounded, hence extends to a positive unital linear functional on  $C(E_1)$ ; the Riesz–Markov theorem provides  $\mu_t(x, \cdot) \in \mathcal{P}(E_1)$  representing it. Measurability in  $(t, x)$  follows because  $(t, x) \mapsto \mu_t(x, \cdot)$  is then continuous into  $\mathcal{P}(E_1)$  with the weak topology,  $\mathcal{H}_1$  being convergence determining; this is [3], Proposition 2.5.  $\square$

*Remark 8.8* (What is new here, and what is [3]’s). The architecture is [3], Theorem 2.1. Three things are more general here, and in each case the source’s hypothesis turns out to have been used for convenience rather than of necessity.

- (i) *The clock.* [3] have  $\mathbb{T} = \mathbb{R}_+$  with  $q = \lambda$ . The proofs above use the clock only through the additivity (3) and the local finiteness  $q(\mathbb{T}_{\leq t}) < \infty$ , so an arbitrary shift invariant clock is admissible — in particular  $\mathbb{T} = \mathbb{N}_0$  with counting measure, where Theorem 8.5 becomes a statement about a Markov chain constructed from its dual, and  $\iota = p$  makes it the Doob decomposition of Remark 8.27.
- (ii) *The index — and here the generalization has a sharp edge.* The kernel layer uses no linear order: Lemma 8.3 needs (T4), and Chapman–Kolmogorov needs commutativity of  $+$ , so the two-parameter semigroup  $(\mu_t)$  exists on  $\mathbb{T} = \mathbb{R}_+^2$ . But Step 1 of Theorem 8.5 does *not* survive: to apply Kolmogorov one must exhibit finite-dimensional distributions over all finite subsets of  $\mathbb{T}$ , and at an antichain —  $(1, 0)$  and  $(0, 1)$  in  $\mathbb{R}_+^2$  — the joint law is simply not determined by one-parameter kernels; nor is some consistent choice guaranteed to exist, the compatibility conditions at  $(1, 0), (0, 1), (1, 1)$  being a genuine constraint that  $(\mu_t)$  does not resolve. So Theorem 8.5 is stated under (T2a), and what survives a partial order is the semigroup layer (Lemma 8.3, Proposition 8.4), not the process. An earlier version of this manuscript claimed more; the corrected claim is the exact analogue of Remark 6.25, with the chain obstruction now sitting in the construction instead of the induction.
- (iii) *The state spaces.* Theorem 8.5 needs (E1) on  $E_1$  — for the Kolmogorov extension in Step 1 — and only (E0) on  $E_2$ . The compactness of [3] enters solely in Proposition 8.7, i.e. in *checking* (D3), not in the theorem.

What does *not* generalize is the reach of the method. The relation (78) is a statement about one-dimensional distributions from a fixed initial state, so it presupposes exactly the Markov structure that Sections 6.1 and 6.4 took pains to avoid, and it has nothing to say about the path-dependent problems of Examples 6.48 and 6.49 or about Theorem 8.71. The natural repair is [3]’s own outlook, their Section 5: dualize the *historical* process, taking  $E_1$  to be a path or genealogy space. That changes  $E_1$  from compact to not even locally compact and puts the weight back on (D3); it is a project, not a remark. Remark 6.9 explains why this is the one place where lifting to the path is worth doing: the content of Theorem 8.5 sits in (D3), a statement about the dual on an unchanged space, and not in a Markov property that lifting would trivialize.

*The time-inhomogeneous version.* Dropping shift invariance of  $q$ , or homogeneity of  $Y$ , replaces  $(P_t)$  by a two-parameter family  $(P_{r,t})$  and  $Y$  by a shift system in the sense of Definition 6.14; the dual then runs on the reflected clock, which by Remark 7.6 is the

$q$ -reflection  $s \mapsto Q^{-1}(Q(t) - Q(s))$  and not  $t - s$ . Lemma 8.3 becomes  $P_{r,s}P_{s,t} = P_{r,t}$  and Proposition 8.4 the corresponding two-parameter Chapman–Kolmogorov relation, exactly as in Theorem 6.27. Nothing in the proofs resists; it is not carried out here only to keep the notation readable.

### Towards a historical duality

Remark 8.8 left one thing open: the duality relation (78) speaks about one-dimensional distributions from a fixed initial state, and therefore does not reach the path-dependent problems of Section 6.1. [3]’s own proposal is to dualize the *historical* process. This subsection sets out what that would require, proves the one step of it that is not conditional, and names a test object. It is a programme, not a theorem, and is marked as such.

**Setting 8.9** (The historical variant; sketch). Let  $E$  be a state space and let  $E_1$  be the *fibred* state space of Definition 2.10 given by the path lift of Definition 6.7,

$$E_{1,t} = \hat{E}_t = E^{\mathbb{T}^{\leq t}} \quad (\text{or } D_E[0, t]), \quad \rho_{r,t} : \hat{E}_t \rightarrow \hat{E}_r \text{ the restriction.}$$

The duality function becomes a section on the total space,  $H : \Sigma \hat{E} \times E_2 \rightarrow \mathbb{R}$ , and Setting 8.2 becomes its two-parameter form: kernels  $\mu_{r,t}$  from  $\hat{E}_r$  to  $\hat{E}_t$ , operators  $(P_{r,t}\varphi)(\hat{x}, y)$ , and the Chapman–Kolmogorov relation  $\mu_{r,s}\mu_{s,t} = \mu_{r,t}$  of Proposition 8.4 in its two-parameter shape.

Three things change, and they change in the direction the manuscript has already prepared for. The state space is fibred, which is D45’s reason for Definition 2.10. The problem is irreducibly time-inhomogeneous — the fibres grow — so the two-parameter version sketched in Remark 8.8 is not optional here but forced. And the dual typically runs *backwards* from the horizon, so the reflection that appears is the  $q$ -reflection of Remark 7.6.

The step that is not conditional is the following, and it is worth isolating because it answers a question left open in Section 6.4.

**Lemma 8.10** (Kernel representability delivers a restart kernel). *In Setting 8.9, let  $(\mu_{r,t})$  satisfy the two-parameter form of (D3), and suppose that for  $r \leq t$  there is a subfamily  $E_2^{(r)} \subset E_2$  with*

- (i)  $H_t(\hat{x}', y) = H_r(\rho_{r,t}\hat{x}', y)$  for all  $\hat{x}' \in \hat{E}_t$  and  $y \in E_2^{(r)}$  — these duality functions read only the past up to  $r$ ;
- (ii)  $(P_{r,t}H)(\hat{x}, y) = H_r(\hat{x}, y)$  for  $y \in E_2^{(r)}$  — they are at rest under the dual dynamics;
- (iii)  $\text{span}\{H_r(\cdot, y) : y \in E_2^{(r)}\}$  is separating on  $\mathcal{P}(\hat{E}_r)$ .

Then

$$\mu_{r,t}(\hat{x}, \cdot) \circ \rho_{r,t}^{-1} = \delta_{\hat{x}}, \quad \hat{x} \in \hat{E}_r,$$

that is, the kernel keeps the past. Consequently  $\mu_{r,t}$  satisfies (R1) of Definition 6.43 at the deterministic time  $r$ ; and if Theorem 8.5 applies in its two-parameter form, it satisfies (R2) as well, so it is a restart kernel.

*Proof.* Fix  $\hat{x} \in \hat{E}_r$  and  $y \in E_2^{(r)}$ . By (i), (D3) and (ii), in that order,

$$\int_{\hat{E}_t} H_r(\rho_{r,t}\hat{x}', y) \mu_{r,t}(\hat{x}, d\hat{x}') = \int_{\hat{E}_t} H_t(\hat{x}', y) \mu_{r,t}(\hat{x}, d\hat{x}') = (P_{r,t}H)(\hat{x}, y) = H_r(\hat{x}, y).$$

The left-hand side is  $\int H_r(\cdot, y)$  against the image measure  $\mu_{r,t}(\hat{x}, \cdot) \circ \rho_{r,t}^{-1} \in \mathcal{P}(\hat{E}_r)$ , and the right-hand side is  $\int H_r(\cdot, y) d\delta_{\hat{x}}$ . As this holds for every  $y \in E_2^{(r)}$ , hypothesis (iii) identifies the two measures.  $\square$

*Remark 8.11* (What Lemma 8.10 would buy). Remark 6.47 records that the existence of a restart kernel is “the one place in Section 6 where something must be given from outside”, and Section 6.4 has no example of one that is not Markovian: Corollary 6.46 builds it from a shift system, which is exactly what a path-dependent problem does not have. Lemma 8.10 says that a historical duality would supply one — from the *dual*, and with no shift anywhere.

The consequence is not small. Theorem 6.45 would then give local uniqueness in the sense of [10], Definition III.2.37 for a path-dependent martingale problem constructed by duality, and Proposition 6.13 would give uniqueness in law; both without a Markov structure, on a problem whose solution was produced by a Markovian device on an enlarged space. That is the circle Sections 6.1, 6.4 and 8.2 were built to close, and it is the reason the programme is worth carrying out.

*Remark 8.12* (The obstruction, named). What blocks the programme is (D3), and specifically the route to it. Proposition 8.7 obtains the kernels from positivity by Riesz–Markov, and needs  $E_1$  compact. A path space  $\hat{E}_t$  is Polish but neither compact nor, for genealogy-valued states, locally compact, so that route is unavailable — this is precisely the difficulty [3] flag in their Section 5, and it is not a technicality that a better argument removes.

The substitute is a *transform* characterization: instead of representing a positive linear functional on  $C(E_1)$  by a measure, one recognizes a given function of  $y$  as a Laplace transform, or a Laplace functional, and reads off the law. [3]’s Proposition 2.8 is the abstract version, and their Example 6 is the concrete one, where the check is that  $y \mapsto Y_t^y$  is negative definite. For a state space of point configurations the corresponding statement is the classical characterization of a point process by its Laplace functional, and it is available. For a state space of genealogies it is not, and that is where the programme currently stops.

### Carrying it out: the Hawkes process

The manuscript already contains a path-dependent process whose historical dual is classical, *deterministic*, and of exactly the shape Setting 8.9 asks for. This subsection verifies the hypotheses.

**Setting 8.13** (The Hawkes duality data; (T3)). Let  $N$  be the Hawkes process of Example 8.74, with immigration rate  $\mu_0 > 0$  and offspring kernel  $\phi \in L_{\text{loc}}^1(\mathbb{R}_+)$ ,  $\phi \geq 0$ . Put

$$\begin{aligned}\hat{E}_t &= \{\text{counting measures on } [0, t] \text{ with finitely many atoms}\}, \\ E_2 &= \{f : \mathbb{R}_+ \rightarrow [0, \infty) \text{ bounded measurable}\}, \\ H_t(\hat{x}, f) &= \exp\left\{-\int_{[0,t]} f d\hat{x}\right\}, \quad \Lambda_t(\hat{x}) = \mu_0 + \int_{[0,t]} \phi(t-s) \hat{x}(ds).\end{aligned}$$

**Lemma 8.14** (The dual flow; (T3)). *For every  $t > 0$  and  $f \in E_2$  the backward equation*

$$1 - w_f^{(t)}(s) = e^{-f(s)} \exp\left\{-\int_s^t \phi(u-s) w_f^{(t)}(u) du\right\}, \quad s \in [0, t], \quad (80)$$

*has a unique solution  $w_f^{(t)} : [0, t] \rightarrow [0, 1]$ . Put, for  $r \leq t$ ,*

$$\Theta_{r,t} f(v) = f(v) + \int_r^t \phi(u-v) w_f^{(t)}(u) du \quad (v \in [0, r]), \quad V_{r,t}(f) = \exp\left\{-\mu_0 \int_r^t w_f^{(t)}\right\}.$$

Then for  $r \leq s \leq t$

$$w_f^{(t)} = w_{\Theta_{s,t}f}^{(s)} \quad \text{on } [0, s], \quad (81)$$

and consequently

$$\Theta_{r,t} = \Theta_{r,s} \circ \Theta_{s,t}, \quad V_{r,t}(f) = V_{s,t}(f) \cdot V_{r,s}(\Theta_{s,t}f). \quad (82)$$

*Proof. Existence and uniqueness.* The right-hand side of (80) lies in  $(0, 1]$ , so any solution takes values in  $[0, 1]$ . The map  $w \mapsto 1 - e^{-f} \exp\{-\phi * w\}$  is monotone and maps  $[0, 1]$ -valued functions to  $[0, 1]$ -valued functions, so backward iteration from  $w \equiv 0$  increases to a solution; and on an interval of length  $\delta$  with  $\int_0^\delta \phi < 1$  it is a contraction in the supremum norm, which gives uniqueness there and then, stepping backwards, on all of  $[0, t]$ .

(81). For  $s' \in [0, s]$  split the integral in (80) at  $s$ :

$$1 - w_f^{(t)}(s') = e^{-f(s')} \exp\left\{-\int_s^t \phi(u - s') w_f^{(t)}(u) du\right\} \cdot \exp\left\{-\int_{s'}^s \phi(u - s') w_f^{(t)}(u) du\right\}.$$

The first two factors are  $e^{-\Theta_{s,t}f(s')}$  by the definition of  $\Theta$ . So  $w_f^{(t)}$  restricted to  $[0, s]$  satisfies (80) with horizon  $s$  and datum  $\Theta_{s,t}f$ , and uniqueness gives (81).

(82). Both assertions are (81) inserted into the definitions, the first after splitting  $\int_r^t = \int_r^s + \int_s^t$ .  $\square$

**Proposition 8.15** (The conditional duality relation; (T3)). *Let  $N$  be a Hawkes process as in Setting 8.13 and let  $\hat{N}$  be its path lift. Then for  $r \leq t$  and  $f \in E_2$ ,*

$$E[H_t(\hat{N}_t, f) \mid \hat{N}_r = \hat{x}] = H_r(\hat{x}, \Theta_{r,t}f) \cdot V_{r,t}(f). \quad (83)$$

*Proof.* By the Hawkes–Oakes cluster representation, conditionally on the past  $\hat{x}$  the points in  $(r, t]$  form a Poisson cluster process whose immigrant intensity at  $u \in (r, t]$  is  $\mu_0 + \int_{[0,r]} \phi(u - s) \hat{x}(ds)$  — the constant immigration plus the offspring of the past — each immigrant at  $u$  generating an independent cluster with Laplace functional  $1 - w_f^{(t)}(u)$  by (80). The Poisson superposition formula  $E[\prod_{u \in \Pi} h(u)] = \exp\{-\int (1 - h)\}$  therefore gives

$$E[e^{-\int_{(r,t]} f dN} \mid \hat{N}_r = \hat{x}] = \exp\left\{-\int_r^t \left(\mu_0 + \int_{[0,r]} \phi(u - v) \hat{x}(dv)\right) w_f^{(t)}(u) du\right\},$$

and multiplying by the deterministic factor  $e^{-\int_{[0,r]} f d\hat{x}}$  and using Fubini to rewrite the double integral as  $\int_{[0,r]} (\Theta_{r,t}f - f) d\hat{x}$  gives (83).  $\square$

Equation (83) is (78) in its two-parameter form, with a *deterministic* dual — so  $E^y[\cdot]$  is evaluation — whose dynamics is the flow  $\Theta$  and whose Feynman–Kac factor is  $V$ . It is the memory-carrying analogue of [3]’s Example 6, where the dual solves  $\dot{Y} = -\Psi(Y)$  for the branching mechanism  $\Psi$ ; here the branching mechanism has become a Volterra kernel and the dual runs backwards.

*Remark 8.16* (This duality is classical, and under three names). Nothing in Lemma 8.14 or Proposition 8.15 is new, and it is worth saying under which names it is already available, because they are not obviously the same name.

*As a cluster representation.* [8] represent a self-exciting process as a Poisson cluster process; the Laplace functional of a Poisson cluster process is  $\exp\{-\int \lambda(1 - G_c)\}$  with  $G_c$  that of a cluster, and for a self-exciting cluster  $G_c$  satisfies a fixed-point equation. That

is (80) and the second half of (83), and it is textbook point-process theory; see [4], the chapter on Poisson cluster processes.

*As an exponential-affine transform formula.* Substituting  $\chi(s) = \int_s^t \phi(u-s) w_f^{(t)}(u) du$ , so that  $w_f^{(t)} = 1 - e^{-f-\chi}$ , turns (80) into

$$\chi(s) = \int_s^t \phi(u-s) (1 - e^{-f(u)-\chi(u)}) du, \quad (84)$$

a backward *Volterra–Riccati* equation whose nonlinearity  $x \mapsto 1 - e^{-x}$  is the branching mechanism of Poisson offspring. For the exponential kernel  $\phi = \alpha\beta e^{-\beta\cdot}$  the pair  $(N, \Lambda)$  is a finite-dimensional affine Markov process and (84) collapses to a Riccati ODE; this is [5]. For a general kernel it is the affine Volterra theory of [14], and the rough Heston analogue is [6].

*As a Markovian lift.* [15] lift stochastic Volterra processes to Markov processes on a space of forward curves and characterize them by a generalized Feller semigroup. That is Definition 6.7 carried out properly — the lift that *keeps* the generator, in the sense of Remark 6.10 — and (84) is the transform formula of the lifted process.

*What is not classical* is only the packaging. The verification below reads the transform formula as (D1)–(D3) of Setting 8.2, and Corollary 8.18 reads it as a restart kernel in the sense of [10], Definition III.2.37. Both are rearrangements of known facts, and neither is claimed as new mathematics; what they buy is recorded in Remark 8.19.

**Proposition 8.17** (The remaining hypotheses; (T3)). *In Setting 8.13:*

(a) Balance, in integrated form. For every  $\hat{x} \in \hat{E}_r$ ,  $f \in E_2$  and  $r \leq t$ ,

$$H_r(\hat{x}, \Theta_{r,t}f) V_{r,t}(f) - H_r(\hat{x}, f) = \int_r^t E[\Lambda_u(\hat{N}_u)(e^{-f(u)} - 1) H_u(\hat{N}_u, f) \mid \hat{N}_r = \hat{x}] du, \quad (85)$$

which is the two-parameter form of (D1) with  $g_u(\hat{x}', f) = \Lambda_u(\hat{x}')(e^{-f(u)} - 1) H_u(\hat{x}', f)$  — exactly  $\mathcal{A}_u$  of (107) with  $\mu(u, \hat{x}', \cdot) = \delta_{\hat{x}'(u-)+1}$  applied to  $H_u(\cdot, f)$ . So the martingale problem (77) the duality produces is the Hawkes martingale problem of Example 8.74. At a point  $r$  at which  $f$  is continuous and which is a Lebesgue point of  $u \mapsto \Lambda_u(\hat{x})$  — automatic for a.e.  $r$ , and for every  $r$  when  $\phi$  is continuous and bounded — dividing (85) by  $t - r$  and letting  $t \downarrow r$  recovers the generator relation

$$\frac{d}{dt} \Big|_{t=r+} H_r(\hat{x}, \Theta_{r,t}f) V_{r,t}(f) = \Lambda_r(\hat{x})(e^{-f(r)} - 1) H_r(\hat{x}, f).$$

(b) (D2).  $\text{span}\{H_r(\cdot, f) : f \in E_2\}$  is separating on  $\mathcal{P}(\hat{E}_r)$ , Laplace functionals determining the law of a point process.

(c) (D3).  $1 - w_f^{(t)}$  is the Laplace functional of a point process on  $[0, t]$ , and consequently the right-hand side of (83) is the Laplace functional of a point process on  $[0, t]$  extending  $\hat{x}$ . The kernels  $\mu_{r,t}(\hat{x}, \cdot)$  of (D3) are its laws.

*Proof.* (a) The integrated identity comes from the process, not from the flow. By Proposition 8.15 the left-hand side of (85) is  $E[H_t(\hat{N}_t, f) - H_r(\hat{N}_r, f) \mid \hat{N}_r = \hat{x}]$ . Now  $M_t := e^{-\int_{[0,t]} f dN}$  jumps by  $M_{u-}(e^{-f(u)} - 1)$  at each point  $u$  of  $N$  and is constant in between, so

$$M_t - M_r = \int_{(r,t]} M_{u-}(e^{-f(u)} - 1) dN_u,$$

an exact pathwise identity; and compensating the integral — the same bracket-by-bracket computation as in Steps 1–2 of the proof of Theorem 8.71, with the bounded integrand  $M_{u-}(e^{-f(u)} - 1)$  in place of  $f(Y_{n+1}) - f(Y_n)$ , and with  $E[N_t] < \infty$  from Example 8.74 licensing the expectations — gives

$$E[M_t - M_r \mid \mathcal{F}_r^N] = E\left[\int_r^t \Lambda_u(\hat{N}_u) M_u(e^{-f(u)} - 1) du \mid \mathcal{F}_r^N\right],$$

where  $M_{u-}$  may be replaced by  $M_u$  under the  $du$ -integral, the paths being piecewise constant with finitely many jumps. This is (85).

*The generator relation.* Write  $A(t) = \int_r^t \Lambda_u^{\hat{x}} w_f^{(t)}(u) du$  with  $\Lambda_u^{\hat{x}} = \mu_0 + \int_{[0,r]} \phi(u - v) \hat{x}(dv)$ , so that the left-hand side of the derivative claim is  $H_r(\hat{x}, f) \cdot \frac{d}{dt} e^{-A(t)}$  at  $t = r$ . As  $t \downarrow r$ ,  $w_f^{(t)}(u) \rightarrow 1 - e^{-f(u)}$  uniformly on  $[r, t]$  — from (80), since  $\int_u^t \phi(v - u) w(v) dv \leq \int_0^{t-r} \phi \rightarrow 0$  — so

$$\frac{A(t)}{t - r} = \frac{1}{t - r} \int_r^t \Lambda_u^{\hat{x}} (1 - e^{-f(u)}) du + o(1) \rightarrow \Lambda_r^{\hat{x}} (1 - e^{-f(r)}),$$

the convergence of the average being exactly the Lebesgue-point hypothesis on  $\Lambda^{\hat{x}}$  together with continuity of  $f$  at  $r$ ; note that  $u \mapsto \Lambda_u^{\hat{x}}$  is locally integrable but, for  $\phi$  merely locally integrable, not continuous, which is why the hypothesis cannot be dropped. Also  $\Lambda_r^{\hat{x}} = \Lambda_r(\hat{x})$ , and the claim follows from  $\frac{d}{dt} e^{-A(t)}|_{t=r} = -A'(r^+)$ , since  $A(r) = 0$ . That the integrand of (85) is  $\mathcal{A}_u H_u(\cdot, f)$  is (107): adding one point at  $u$  multiplies  $H_u(\hat{x}', f)$  by  $e^{-f(u)}$ .

(b) Standard.

(c) Let  $G_0(s) = e^{-f(s)}$  and  $G_{n+1}(s) = e^{-f(s)} \exp\{-\int_s^t \phi(u - s)(1 - G_n(u)) du\}$ . By induction  $G_n$  is the Laplace functional of the cluster started at  $s$  and truncated after  $n$  generations:  $G_0$  is that of the single point  $\delta_s$ , and the recursion superposes onto it a Poisson process of intensity  $\phi(\cdot - s)$  of independent  $n$ -generation clusters, whose Laplace functional is the exponential factor. The truncated clusters increase with  $n$ , so  $G_n$  decreases, to  $1 - w_f^{(t)}$  by uniqueness in Lemma 8.14. The limiting cluster has finitely many points a.s., its expected size being  $\sum_{n \geq 0} \|\phi^{*n}\|_{L^1[0,t]}$ , which is finite because the Volterra resolvent of  $\phi$  exists in  $L_{\text{loc}}^1$  for every  $\phi \in L_{\text{loc}}^1$ . So it is a point process and  $1 - w_f^{(t)}$  is its Laplace functional; the second assertion follows by one more Poisson superposition, as in the proof of Proposition 8.15. *Note that this argument does not presuppose the existence of  $N$ :* it constructs the cluster from the branching recursion alone.  $\square$

**Corollary 8.18** (The restart kernel). *Take  $E_2^{(r)} = \{f \in E_2 : f = 0 \text{ on } (r, \infty)\}$ . Then  $w_f^{(t)} \equiv 0$  on  $[r, t]$ , hence  $\Theta_{r,t} f = f$  and  $V_{r,t}(f) = 1$ ; hypotheses (i)–(iii) of Lemma 8.10 hold, and therefore*

$$\mu_{r,t}(\hat{x}, \cdot) \circ \rho_{r,t}^{-1} = \delta_{\hat{x}} :$$

*the kernels of Proposition 8.17(c) form a restart kernel in the sense of Definition 6.43.*

*Proof.* For  $f$  vanishing on  $(r, t]$ , equation (80) on  $[r, t]$  reads  $1 - w(s) = \exp\{-\int_s^t \phi(u - s)w(u) du\}$ , which  $w \equiv 0$  solves; uniqueness gives  $w_f^{(t)} \equiv 0$  there, and the two displays follow. Hypothesis (i) holds because  $H_t(\hat{x}', f)$  then depends on  $\hat{x}'$  only through its restriction to  $[0, r]$ , (ii) is  $\Theta_{r,t} f = f$  together with  $V_{r,t}(f) = 1$ , and (iii) is Proposition 8.17(b) applied on  $\hat{E}_r$ .  $\square$

*Remark 8.19* (What this settles, and what it does not). *Settled.* Setting 8.9 is not empty: there is a path-dependent martingale problem — the Hawkes problem of Example 8.74, whose rate is a functional of the whole past — for which the historical duality data exist, the dual is deterministic, and (D1)–(D3) all hold. The dual dynamics is a backward nonlinear Volterra flow, and the two-parameter structure of Remark 8.8 is not a convenience here but forced, the fibres  $\hat{E}_t$  growing with  $t$ .

*The payoff is Corollary 8.18, not existence.* The verification of (D3) constructs the cluster, and assembling the Hawkes process from clusters is the Hawkes–Oakes construction; so on the existence side the duality route reproduces a known argument rather than replacing one. What is genuinely new is the restart kernel. Remark 6.47 records that its existence is “the one place in Section 6 where something must be given from outside”, and Section 6.4 had no non-Markovian instance: Corollary 6.46 builds one from a shift system, which a path-dependent problem does not have. Corollary 8.18 is such an instance, and with it Theorem 6.45 gives local uniqueness in the sense of [10], Definition III.2.37 for the Hawkes martingale problem, with no shift anywhere.

*Not settled.* The general programme. What made the verification possible is that the state space is one of point configurations, where Laplace functionals both separate ((D2)) and characterize ((D3)); Remark 8.12 explains that for a genealogy space neither is available, and nothing here helps there.

*Novelty.* None is claimed on the analytic side: see Remark 8.16. The duality is the exponential-affine transform formula for Hawkes processes, and what is done here is to check that it satisfies (D1)–(D3) and to extract Corollary 8.18 from it.

*Numerics.* Equation (83) and the cocycle (82) were checked against a cluster simulation and against a discretization of (80); the relative errors were  $4 \cdot 10^{-4}$  and  $10^{-4}$ , against a Monte Carlo standard error of  $2 \cdot 10^{-3}$ .

### 8.3 Jump processes

Throughout this subsection  $\mathbb{T} = \mathbb{R}_+$  with  $q$  Lebesgue measure, so that the two conventions of Definition 2.3 agree and  $\iota$  plays no rôle. The state space needs only (E0) for the construction and the martingale property; (E2) enters only when one wants to say that the paths lie in  $D_E[0, \infty)$ .

**Setting 8.20** (Jump data; (T3) + (E0)). Let  $\lambda : E \rightarrow [0, \infty)$  be measurable (the *rate*) and let  $\mu$  be a transition kernel on  $E$  (the *jump distribution*). Define

$$Af(x) = \lambda(x) \int_E (f(y) - f(x)) \mu(x, dy), \quad f \in B(E), \quad (86)$$

and  $A = \{(f, Af) : f \in B(E)\}$ . If  $\bar{\lambda} := \sup_x \lambda(x) < \infty$  then  $A$  maps  $B(E)$  into itself with  $\|Af\| \leq 2\bar{\lambda}\|f\|$ ; in general  $Af$  need not be bounded.

**Definition 8.21** (The construction; (T3) + (E0)). Let  $\nu \in \mathcal{P}(E)$ . On some probability space let  $(Y_n)_{n \geq 0}$  be a Markov chain with initial distribution  $\nu$  and transition kernel  $\mu$ , and let  $\varepsilon_1, \varepsilon_2, \dots$  be i.i.d. exponential random variables of parameter 1, independent of  $(Y_n)$ . Put

$$\tau_0 = 0, \quad \tau_{n+1} = \tau_n + \frac{\varepsilon_{n+1}}{\lambda(Y_n)}, \quad \zeta = \lim_n \tau_n, \quad (87)$$

with the convention  $1/0 = \infty$ , and

$$X_t = Y_n \quad \text{for } \tau_n \leq t < \tau_{n+1}, \quad t < \zeta. \quad (88)$$

We call  $\zeta$  the *explosion time* and  $(\tau_n)$  the *jump times*.

Three observations before the theorem. The paths of  $X$  are piecewise constant and right continuous by construction, hence lie in  $D_E[0, \infty)$  as soon as  $E$  carries a topology, i.e. under (E2). If  $\lambda(Y_n) = 0$  then  $\tau_{n+1} = \infty$  and  $X$  is absorbed at  $Y_n$ . And each  $\tau_n$  is a functional of the path — the time of its  $n$ -th jump — which is what will make it a legitimate localizing sequence in the sense of (L1).

**Lemma 8.22** (No explosion for bounded rates; (T3) + (E0)). *If  $\bar{\lambda} < \infty$  then  $\zeta = \infty$  almost surely.*

*Proof.*  $\tau_{n+1} - \tau_n \geq \varepsilon_{n+1}/\bar{\lambda}$ , and  $\sum_n \varepsilon_{n+1}/\bar{\lambda} = \infty$  a.s. by the strong law of large numbers, the  $\varepsilon_n$  being i.i.d. with mean 1.  $\square$

**Theorem 8.23** (Jump processes solve their martingale problem; (T3) + (E0)). *Let  $\bar{\lambda} < \infty$  and let  $X$  be as in Definition 8.21. Then  $X$  solves the martingale problem for  $(A, \nu)$  of (86).*

*Proof.* Fix  $f \in B(E)$  and write  $\mu f(x) = \int f(y)\mu(x, dy)$ , so that  $Af = \lambda \cdot (\mu f - f)$ . Let  $\mathcal{G}_t = \sigma(X_u : u \leq t)$  and

$$M_t = f(X_t) - f(X_0) - \int_0^t Af(X_r) dr.$$

*Step 1: the paths make both terms explicit.* Since  $X$  is constant on  $[\tau_n, \tau_{n+1})$ ,

$$\begin{aligned} \int_0^t Af(X_r) dr &= \sum_{n \geq 0} Af(Y_n)((\tau_{n+1} \wedge t) - (\tau_n \wedge t)), \\ f(X_t) - f(X_0) &= \sum_{n \geq 1} (f(Y_n) - f(Y_{n-1}))\mathbf{1}_{\{\tau_n \leq t\}}, \end{aligned}$$

both sums being finite a.s. by Lemma 8.22. Hence

$$M_t = \sum_{n \geq 0} D_n, \quad D_n = (f(Y_{n+1}) - f(Y_n))\mathbf{1}_{\{\tau_{n+1} \leq t\}} - Af(Y_n)((\tau_{n+1} \wedge t) - (\tau_n \wedge t)). \quad (89)$$

*Step 2: each bracket is a martingale difference.* Let  $\mathcal{H}_n = \sigma(Y_0, \dots, Y_n, \tau_1, \dots, \tau_n)$ . Conditionally on  $\mathcal{H}_n$ , and on the event  $\{\tau_n \leq t\}$ , put  $x = Y_n$ ,  $\lambda = \lambda(x)$  and  $a = t - \tau_n \geq 0$ . By construction  $\tau_{n+1} - \tau_n$  is exponential with parameter  $\lambda$  and independent of  $Y_{n+1} \sim \mu(x, \cdot)$ , so

$$\begin{aligned} E[(f(Y_{n+1}) - f(Y_n))\mathbf{1}_{\{\tau_{n+1} \leq t\}} \mid \mathcal{H}_n] &= (\mu f(x) - f(x))(1 - e^{-\lambda a}), \\ E[(\tau_{n+1} \wedge t) - \tau_n \mid \mathcal{H}_n] &= E[\text{Exp}(\lambda) \wedge a] = \frac{1 - e^{-\lambda a}}{\lambda}. \end{aligned}$$

Multiplying the second identity by  $Af(x) = \lambda(\mu f(x) - f(x))$  gives exactly the first. If  $\lambda = 0$  both displays are read as 0 — then  $\tau_{n+1} = \infty$ , so the first term of  $D_n$  vanishes, and  $Af(x) = 0$ , so the second does too. Hence  $E[D_n \mid \mathcal{H}_n] = 0$  on  $\{\tau_n \leq t\}$ , and  $D_n = 0$  on  $\{\tau_n > t\}$ , an event in  $\mathcal{H}_n$ . *This cancellation — the factor  $\lambda$  in (86) against the mean  $1/\lambda$  of the holding time — is the whole content of the theorem.*

*Step 3: summation.* The family  $(D_n)$  is dominated:

$$\sum_{n \geq 0} |D_n| \leq 2\|f\|N_t + \|Af\|t, \quad N_t := \#\{n \geq 1 : \tau_n \leq t\},$$

because the second terms telescope. Since  $\tau_{n+1} - \tau_n \geq \varepsilon_{n+1}/\bar{\lambda}$ , the variable  $N_t$  is stochastically dominated by a Poisson variable of parameter  $\bar{\lambda}t$  and is in particular integrable. So dominated convergence applies to (89), and Step 2 gives

$$E[M_t] = \sum_{n \geq 0} E[E[D_n | \mathcal{H}_n]] = 0 \quad \text{for every } t \geq 0 \quad (90)$$

and every initial law  $\nu$ .

*Step 4: the Markov property of the construction.* Fix  $s \geq 0$  and let  $N_s$  be as above. On  $\{N_s = n\}$  the  $\sigma$ -field  $\mathcal{G}_s = \sigma(X_u : u \leq s)$  is generated by  $Y_0, \dots, Y_n, \tau_1, \dots, \tau_n$  and the event  $\{\tau_{n+1} > s\}$ . Conditionally on these, the residual holding time  $\tau_{n+1} - s$  is again exponential with parameter  $\lambda(Y_n)$  — memorylessness — and is independent of  $(Y_{n+1}, Y_{n+2}, \dots)$  and of  $\varepsilon_{n+2}, \varepsilon_{n+3}, \dots$ . Hence, conditionally on  $\mathcal{G}_s$ , the process  $(X_{s+u})_{u \geq 0}$  is again a process of Definition 8.21, with the same  $\lambda$  and  $\mu$  and with initial state  $X_s$ .

*Step 5: conclusion.* For  $s \leq t$ ,

$$M_t - M_s = (f(X_t) - f(X_s)) - \int_s^t Af(X_r) dr$$

is the variable (89) at time  $t - s$  for the restarted process of Step 4. By Step 4 and (90) applied with  $\nu = \delta_{X_s}$ ,

$$E[M_t - M_s | \mathcal{G}_s] = 0.$$

As  $|M_t| \leq 2\|f\| + 2\bar{\lambda}\|f\|t$  is bounded on bounded intervals,  $M$  is a  $(\mathcal{G}_t)$ -martingale, and since  $*\mathcal{F}_t^X = \mathcal{F}_t^X = \mathcal{G}_t$  for the right continuous  $X$  (Section 2.1), the same holds for the filtration of Definition 4.6.  $\square$

The construction gives existence. Uniqueness comes from Section 6, and the input it needs is a Gronwall estimate.

**Proposition 8.24** (Well-posedness for bounded rates; (T3) + (E2)). *Let  $\bar{\lambda} < \infty$ . Then the martingale problem for  $(A, \nu)$  is well posed for every  $\nu \in \mathcal{P}(E)$ , every solution is a Markov process, and if  $E$  satisfies (E2) it is strong Markov.*

*Proof.* Existence is Theorem 8.23. For uniqueness it suffices, by Theorem 6.23, to show that the one-dimensional distributions are determined by  $\nu$ , and since  $B(E)$  is separating it suffices to determine  $E[f(X_t)]$  for every  $f \in B(E)$ . We show that they are given by the exponential of a bounded operator.

*Step 1:  $A$  is bounded on  $B(E)$ .* For  $f \in B(E)$ ,  $\|Af\| \leq 2\bar{\lambda}\|f\|$  by Setting 8.20, so  $A^k f \in B(E)$  is defined for every  $k \geq 0$  with

$$\|A^k f\| \leq (2\bar{\lambda})^k \|f\|. \quad (91)$$

This is the only place where  $\bar{\lambda} < \infty$  is used, and it is what makes the iteration below converge.

*Step 2: one step of the iteration.* Let  $X$  be any solution of the martingale problem for  $(A, \nu)$  and let  $h \in B(E)$ . Applying the martingale property to the pair  $(h, Ah) \in A$  between 0 and  $t$  and taking expectations gives

$$E[h(X_t)] = E[h(X_0)] + \int_0^t E[Ah(X_r)] dr, \quad (92)$$

Fubini being legitimate because  $Ah$  is bounded. Note that (92) is available for *every*  $h \in B(E)$ , which is what allows it to be iterated; a martingale problem posed on a smaller domain would not.

*Step 3: the iteration.* Insert (92) into itself, starting from  $h = f$  and using it at each stage for  $h = A^k f$ :

$$E[f(X_t)] = \sum_{k=0}^n \frac{t^k}{k!} E[A^k f(X_0)] + R_n, \quad R_n = \int_{\Delta_{n+1}(t)} E[A^{n+1} f(X_{r_{n+1}})] dr,$$

where  $\Delta_{n+1}(t) = \{0 \leq r_{n+1} \leq \dots \leq r_1 \leq t\}$  is the simplex of volume  $t^{n+1}/(n+1)!$ . By (91),

$$|R_n| \leq \frac{t^{n+1}}{(n+1)!} (2\bar{\lambda})^{n+1} \|f\| \xrightarrow{n \rightarrow \infty} 0.$$

Hence

$$E[f(X_t)] = \sum_{k \geq 0} \frac{t^k}{k!} \int_E A^k f d\nu = \int_E (e^{tA} f) d\nu, \quad (93)$$

the series converging absolutely in  $B(E)$  by (91). The right hand side depends only on  $A$  and  $\nu$ , so the one-dimensional distributions are determined and Theorem 6.23 applies.

The strong Markov property is Theorem 6.27, whose hypotheses hold because the paths are piecewise constant — so the shift system is measurable — and  $M$  has increments bounded on bounded intervals.  $\square$

*Remark 8.25* (Which contraction, and where Gronwall really sits). The proof above is Picard iteration for the linear equation  $\dot{v} = Av$  in the Banach space  $B(E)$ , and (91) is the estimate that makes the Picard operator a contraction on every bounded time interval; Gronwall's lemma would serve equally well in place of the explicit remainder  $R_n$ . It is worth contrasting it with the *first-jump equation*

$$v(t, x) = e^{-\lambda(x)t} f(x) + \int_0^t \lambda(x) e^{-\lambda(x)s} (\mu v(t-s, \cdot))(x) ds, \quad (94)$$

which one is tempted to use instead, and which yields the same conclusion by Gronwall. The temptation should be resisted at this point: (94) presupposes that the solution *is* a jump process with the prescribed first jump law, and that is a statement about the path structure which the martingale problem does not hand over — Proposition 4.9 characterizes finite-dimensional distributions, not jump times. Equation (94) is a valid identity for the constructed process of Definition 8.21, where it follows by conditioning on  $\tau_1$ ; but as an argument for *uniqueness* it would be circular. The iteration of Step 3 avoids the issue because it uses nothing but the martingale property.

*Remark 8.26* (Explosion, and why localization is not academic). Drop the assumption  $\bar{\lambda} < \infty$ . Then  $\zeta$  may be finite with positive probability — take  $E = \mathbb{N}$ ,  $\mu(n, \cdot) = \delta_{n+1}$  and  $\lambda(n) = 2^n$ , so that  $E[\zeta] = \sum_n 2^{-n} < \infty$  — and (88) defines  $X$  only on  $[0, \zeta)$ . There is then no solution of the *global* martingale problem for  $A$  on  $\mathbb{R}_+$ , and the honest statement is the local one:

- Step 2 of the proof of Theorem 8.23 is a finite computation and is unaffected, so  $M^{\tau_n}$  is a martingale for every  $n$ ;
- $\tau_n$  is a functional of the path, the same for every solution, and  $\tau_n \uparrow \zeta$ ;
- hence  $(\tau_n)$  is a localizing system in the sense of Definition 6.30, and (L1) holds by construction rather than by assumption — exactly as Lemma 6.34 predicts, since the stopped test processes are bounded.

So  $X$  solves the local martingale problem for  $A$  on  $[0, \zeta)$ , and Section 6.3 applies verbatim. This is the example that Definition 6.30 was written for; up to this point it had none.

*Remark 8.27* (Discrete time, and the two conventions). Deleting the holding times from Definition 8.21 leaves the chain  $(Y_n)$  itself, which is the case  $\mathbb{T} = \mathbb{N}_0$  with  $q$  counting measure. There  $A = \mu - I$  under the predictable convention  $\iota = p$ , so that  $M_n = f(Y_n) - f(Y_0) - \sum_{k < n} (\mu f - f)(Y_k)$  is the Doob decomposition; under  $\iota = o$  one would obtain the condition  $\mu f - f = \mu(\mu f - f)$  instead, which is not what one wants. This is the concrete form of Remark 2.5, and it is also the case in which Corollary 7.16 lives.

*Remark 8.28* (What this example is for). Besides being the only explicit solution in the manuscript, the construction exercises three of its abstractions and is the reason they were made. The state space needs only (E0):  $\lambda$  measurable and  $\mu$  a kernel, no topology, no metric, no Polish structure — so jump processes on a standard Borel  $E$ , for instance on a space of measures or of distributions, are covered by Theorem 8.23 as it stands. The localizing system of Remark 8.26 is a genuine instance of (L1)–(L3). And the discrete case of Remark 8.27 is a genuine instance of a clock with atoms.

## 8.4 From stochastic differential equations

For  $E = \mathbb{R}^d$  and continuous paths there is a route that is more important than all the others put together, and we state it without proofs. Let  $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$  and  $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$  be measurable, put  $a = \sigma \sigma^\top$ , and let

$$Af(x) = \frac{1}{2} a^{ij}(x) \partial_{ij} f(x) + b^i(x) \partial_i f(x), \quad f \in C_c^\infty(\mathbb{R}^d). \quad (95)$$

A *weak solution* of the equation  $(\sigma, b)$  is a pair  $(X, B)$  on some filtered probability space with  $B$  an  $m$ -dimensional Brownian motion and  $X_t = X_0 + \int_0^t \sigma(X_s) dB_s + \int_0^t b(X_s) ds$ .

*Fact 8.29* (Picard–Lindelöf for SDEs). If  $\sigma$  and  $b$  are Lipschitz, then for every  $x$  the equation  $(\sigma, b)$  has a strong solution with  $X_0 = x$ , and pathwise uniqueness holds. The proof is Picard iteration in  $L^2$ , with the Itô isometry in place of the triangle inequality.

*Fact 8.30* (Yamada–Watanabe). Pathwise uniqueness together with weak existence implies uniqueness in law.

*Fact 8.31* (Stroock–Varadhan; [16], Theorem 32.7). For progressive  $\sigma, b$  and  $P \in \mathcal{P}(C_{\mathbb{R}_+, \mathbb{R}^d})$  the following are equivalent:  $P$  is the law of a weak solution of  $(\sigma, b)$ ; and  $P$  solves the local martingale problem for  $A$  of (95).

**Corollary 8.32** (Well-posedness for Lipschitz coefficients; (T3),  $E = \mathbb{R}^d$ ). *If  $\sigma$  and  $b$  are Lipschitz, the local martingale problem for  $A$  of (95) is well posed for every initial distribution, and its solution is a strong Markov process.*

*Proof.* Facts 8.29 and 8.30 give weak existence and uniqueness in law; Fact 8.31 transfers both to the local martingale problem. The Markov properties are Theorems 6.38 and 6.27, applied with the localizing system of Remark 6.31 — which is where (L1)–(L3) were verified for exactly this operator — or directly [16], Theorem 32.11.  $\square$

*Remark 8.33* (What a proof would cost). Section 2.5 contains nothing about stochastic integration, and this is the only place in the manuscript that reaches outside [7], Chapters 1–3. A self-contained treatment of Corollary 8.32 would need the Itô integral against a continuous local martingale, quadratic variation and the Kunita–Watanabe inequality, Itô’s formula, and Lévy’s characterization of Brownian motion — the last being what turns

a solution of the martingale problem back into a solution of the equation in Fact 8.31. That is a chapter, not a section, and it is deliberately not attempted here.

For the formalization the situation is less bad than it looks: the **brownian-motion** development of Remark 9.1 is building stochastic integration in Lean, so the dependency may become available rather than having to be created. Until then, Section 8.3 is the route that can actually be carried out.

## 8.5 By convergence: the Ethier–Kurtz version

The remaining route obtains solutions as weak limits of solutions of approximating martingale problems. We state the classical result first, and then the abstract theorem of which it is a corollary; its proof below is a forward reference, and deliberately so — the point of the arrangement is that Lemma 8.34 has no proof of its own.

**Lemma 8.34** ([7], Lemma 4.5.1; (T3)). *Assume (E3), let  $A \subset C_b(E) \times C_b(E)$  and  $A_n \subset B(E) \times B(E)$  for  $n \in \mathbb{N}$ . Suppose that for each  $(f, g) \in A$  there exist  $(f_n, g_n) \in A_n$  with*

$$\lim_{n \rightarrow \infty} \|f_n - f\| = 0, \quad \lim_{n \rightarrow \infty} \|g_n - g\| = 0. \quad (96)$$

*If for each  $n$  the process  $X_n$  is a solution of the martingale problem for  $A_n$  with sample paths in  $D_E[0, \infty)$ , and if  $X_n \Rightarrow X$  in  $D_E[0, \infty)$ , then  $X$  is a solution of the martingale problem for  $A$ .*

*Proof.* We verify (C1)–(C3) of Theorem 8.40, with  $F = D_E[0, \infty)$  — Polish by (E3) and Theorem 3.12 — and  $\mathcal{X} = \mathcal{X}_A(X)$ , whose canonical version is  $Y_t^\circ(\omega) = f(\omega(t)) - \int_0^t g(\omega(u)) du$  for  $(f, g) \in A$ . Fix  $(f, g) \in A$  and an approximating sequence  $(f_n, g_n)$  as in (96), and put

$$D = \{u \geq 0 : P\{X(u) = X(u-)\} = 1\},$$

which has countable complement by Fact 2.41 and is therefore dense;  $\mathbb{T} = \mathbb{R}_+$  has no greatest element, so the proviso in Theorem 8.40 on  $D$  is vacuous.

(C1) and (C2). The first is the assumed weak convergence  $X_n \Rightarrow X$  on  $D_E[0, \infty)$ . For the second take  $\mathcal{Z}^\circ$  to be the determining set of Example 4.4 built from  $D$ , i.e. the products  $Z_s^\circ = \prod_{i \leq k} h_i(\pi_{t_i})$  with  $h_i \in C_b(E)$  and  $t_i \in D \cap \mathbb{T}_{\leq s}$ .

(a). For  $t \in D$  the evaluation  $\omega \mapsto \omega(t)$  is  $J_1$ -continuous at every  $\omega$  with  $\omega(t) = \omega(t-)$ , hence  $PX^{-1}$ -a.s. The integral term is continuous at every  $\omega$ : if  $\omega_m \rightarrow \omega$  in  $D_E[0, \infty)$ , then  $\omega_m(u) \rightarrow \omega(u)$  for every  $u$  outside the at most countable set of discontinuities of  $\omega$ , hence Lebesgue almost everywhere, so  $g(\omega_m(u)) \rightarrow g(\omega(u))$  by continuity of  $g$  and bounded convergence gives  $\int_0^t g(\omega_m(u)) du \rightarrow \int_0^t g(\omega(u)) du$ . So  $Y_t^\circ$  is  $P$ -continuous at  $X$ , and so is  $Y_t^\circ Z_s^\circ$ , the factor  $Z_s^\circ$  being a product of evaluations at times in  $D$ .

(b).  $f$  and  $g$  are bounded, so  $|Y_r^\circ| \leq \|f\| + t\|g\|$  for  $r \leq t$ , and a uniformly bounded family is uniformly integrable.

(c). Write  $Y_t^{\circ, n}(\omega) = f_n(\omega(t)) - \int_0^t g_n(\omega(u)) du$ . Since  $X_n$  solves the martingale problem for  $A_n$  and  $(f_n, g_n) \in A_n$ , Proposition 4.9 gives

$$E^{P^n}[(Y_t^{\circ, n}(X_n) - Y_s^{\circ, n}(X_n))Z_s^\circ(X_n)] = 0$$

for all  $s \leq t$  and  $Z_s^\circ \in \mathcal{Z}_s^\circ$ . Moreover

$$|Y_r^{\circ, n}(\omega) - Y_r^\circ(\omega)| \leq \|f_n - f\| + r\|g_n - g\| \quad \text{for every } \omega \in D_E[0, \infty),$$

a bound uniform in  $\omega$ , so with  $C = \sup_i \|h_i\|^k$ ,

$$\left| E^{P^n}[(Y_t^\circ(X_n) - Y_s^\circ(X_n))Z_s^\circ(X_n)] \right| \leq 2C(\|f_n - f\| + t\|g_n - g\|) \xrightarrow{n \rightarrow \infty} 0,$$

which is (c). This is the only place where (96) is used, and it is used only as a uniform bound — which is why Theorem 8.44 can replace it by the far weaker (100)–(101).

*Conclusion.* The paths of  $X$  lie in  $D_E[0, \infty)$ , so every  $Y \in \mathcal{X}$  is right continuous, and  $\mathbb{T} = \mathbb{R}_+$  satisfies (T2b) with  $D$  dense. Theorem 8.40 therefore gives  $P \in \mathcal{M}(\mathcal{X})$ , i.e.  $X$  solves the martingale problem for  $A$ .  $\square$

*Remark 8.35* (Why this is a corollary and not a theorem). [7] prove Lemma 8.34 directly, and an earlier version of this manuscript reproduced that proof. It is worth being explicit that nothing is lost by not doing so, because the two arguments are the same argument: [7]’s “let  $D(X)$  be the continuity times, take limits there, extend by right continuity” is precisely Steps 1 and 3 of Theorem 8.40, and their appeal to bounded convergence is its Steps 0 and 2 in the special case where (b) holds because  $f$  and  $g$  are bounded. Reproducing it would duplicate the abstract proof in a notation that hides which hypothesis does what.

What the derivation makes visible is the division of labour that the direct proof conceals. Of the four hypotheses of Lemma 8.34, weak convergence is (C1);  $A \subset C_b \times C_b$  is what buys (a); boundedness of  $f$  and  $g$  is what buys (b); and the uniform approximation (96) is what buys (c) — one hypothesis per condition, with no overlap. Each can then be weakened separately, and Theorem 8.44 weakens the last two while Section 8.9 weakens (a).

**Corollary 8.36** (Approximation along the paths). *Lemma 8.34 remains true if (96) is replaced by the requirement that for each  $(f, g) \in A$  there be  $(f_n, g_n) \in A_n$  with*

$$\lim_{n \rightarrow \infty} E^{P^n} [|(f_n - f)(X_n(t))|] = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} E^{P^n} \left[ \int_0^t |(g_n - g)(X_n(u))| du \right] = 0 \quad (97)$$

for every  $t \geq 0$ . In particular the conclusion holds if

- (a)  $\sup_{n \in \mathbb{N}} (\|f_n\| + \|g_n\|) < \infty$ ;
- (b)  $f_n \rightarrow f$  and  $g_n \rightarrow g$  uniformly on compact subsets of  $E$ ;
- (c)  $\{X_n\}$  satisfies the compact containment condition (12).

*Proof.* Only (c) used (96) in the proof of Lemma 8.34; (a) and (b) concern  $f$  and  $g$  alone and are untouched. Keeping the notation there, and writing  $\Delta_r^n = Y_r^\circ - Y_r^{\circ, n}$ , one has for  $s \leq t$

$$\Delta_t^n(\omega) - \Delta_s^n(\omega) = (f - f_n)(\omega(t)) - (f - f_n)(\omega(s)) - \int_s^t (g - g_n)(\omega(u)) du,$$

so with  $C = \sup_i \|h_i\|^k \geq \|Z_s^\circ\|$  and  $E^{P^n}[(Y_t^{\circ, n}(X_n) - Y_s^{\circ, n}(X_n))Z_s^\circ(X_n)] = 0$  as before,

$$\begin{aligned} \left| E^{P^n} [(Y_t^\circ(X_n) - Y_s^\circ(X_n))Z_s^\circ(X_n)] \right| &\leq C \sum_{r \in \{s, t\}} E^{P^n} [|(f_n - f)(X_n(r))|] \\ &\quad + C E^{P^n} \left[ \int_0^t |(g_n - g)(X_n(u))| du \right], \end{aligned}$$

which tends to 0 by (97). Note that no uniform bound on  $f_n$  or  $g_n$  is needed: (b) is bought by boundedness of  $f$  and  $g$ , not of  $f_n$  and  $g_n$ .

For the sufficient condition, let  $t > 0$  and  $\eta > 0$ , put  $M = \sup_n (\|f_n\| + \|g_n\|) + \|f\| + \|g\|$ , which is finite by (a) and  $A \subset C_b \times C_b$ , and use (c) to choose a compact  $\Gamma \subset E$  with

$\inf_n P\{X_n(u) \in \Gamma \text{ for all } u \leq t\} \geq 1 - \eta$ . On that event both integrands are at most  $\sup_\Gamma |f_n - f|$  resp.  $\sup_\Gamma |g_n - g|$ , and off it at most  $M$ ; hence

$$\limsup_{n \rightarrow \infty} E^{P^n} [|(f_n - f)(X_n(t))|] \leq M\eta, \quad \limsup_{n \rightarrow \infty} E^{P^n} \left[ \int_0^t |(g_n - g)(X_n(u))| du \right] \leq tM\eta,$$

using (b) for the terms on  $\Gamma$ . Let  $\eta \downarrow 0$ .  $\square$

*Remark 8.37* (Three rungs of approximation). Corollary 8.36 sits strictly between [7] and [2]:

$$(96) \implies (97) \implies (100)\text{--}(101) \text{ for } \xi^n = f_n(X^n), \varphi^n = g_n(X^n),$$

the second implication because the test variables  $\prod_i h_i(X_{t_i}^n)$  are bounded and (97) controls absolute values. The three rungs differ in what they are conditions *on*: (96) is a condition on  $E$ , (97) a condition on the occupation measures  $\int_0^\cdot P\{X_n(u) \in dx\} du$  and the one-dimensional marginals, and (100)–(101) a condition on the joint laws which moreover permits  $\xi^n, \varphi^n$  that are not of the form  $h(X^n)$  at all. The middle rung is the one that costs nothing: its proof is the proof of Lemma 8.34 with the estimate taken under  $E$  instead of over  $\omega \in D_E[0, \infty)$ , and it is what one wants in practice, since approximating operators are typically obtained by localization and then converge only locally uniformly. It is also free in situ: (c) is the hypothesis one is assuming anyway in Remark 8.38 in order to have convergence at all.

*Remark 8.38* ([7], Remark 4.5.2). Lemma 8.34 presupposes convergence. Suppose  $(E, r)$  is Polish and  $\mathcal{D}(A)$  contains an algebra that separates points and vanishes nowhere; by Fact 2.39 such an algebra is separating, and it is dense in  $C_b(E)$  in the topology of uniform convergence on compact sets. Then  $\{X_n\}$  is relatively compact provided (96) holds for each  $(f, g) \in A$  and the compact containment condition (12) holds. This is Facts 2.43, 2.44, 2.42 and 2.33. Combining with Lemma 8.34: every limit point of  $\{X_n\}$  solves the martingale problem for  $A$ ; in particular the martingale problem for  $A$  has a solution with sample paths in  $D_E[0, \infty)$ . If in addition uniqueness holds (e.g. by Theorem 6.51 or Corollary 7.20), then  $X_n \Rightarrow X$  for the unique solution  $X$ .

## 8.6 By convergence: the abstract theorem

Theorem 8.44 below is an instance of one statement, which uses neither an operator, nor càdlàg paths, nor the Skorokhod topology, nor any structure on the time index beyond a dense subset — and none at all if that subset is all of  $\mathbb{T}$ . It is [2], Theorem 3.14 and Corollary 3.17, and it is the cheapest genuine theorem in this manuscript; we give it with its proof, since it is what the formalization should target (step (F4) of Section 9). Lemma 8.34 is a special case of it directly, as its proof shows, and a fortiori of Theorem 8.44; so all of Section 8 rests on Theorem 8.40.

**Definition 8.39** ( $P$ -continuity at  $X$ ; (T0) + (E3)). A Borel map  $\psi : F \rightarrow \mathbb{R}$  is  $P$ -continuous at  $X$  if there is  $C \in \mathcal{B}(F)$  with  $P\{X \in C\} = 1$  such that  $\psi(\alpha_m) \rightarrow \psi(\alpha)$  whenever  $\alpha_m \rightarrow \alpha$  in  $F$  with  $\alpha \in C$ .

**Theorem 8.40** (Abstract convergence; [2], Theorem 3.14 and Corollary 3.17; (T0) + (E3)). *In Setting 4.1, let  $\mathcal{X}$  be canonical for  $X$ , let  $D \subset \mathbb{T}$ , and for each  $n$  let  $B^n = (\Omega^n, \mathcal{F}^n, \mathbb{P}^n, P^n)$  be a filtered probability space carrying a process  $X^n$  with paths in  $F$ . Assume:*

(C1)  $X^n \Rightarrow X$  weakly on  $F$ ;

(C2)  $\mathcal{X}$  admits a determining set  $\mathcal{Z}^\circ$  (Definition 4.3);

(C3) for every  $Y \in \mathcal{X}$  there is a canonical version  $(Y_t^\circ)$  such that for all  $t \in D$ ,  $s \in D \cap \mathbb{T}_{\leq t}$  and  $Z_s^\circ \in \mathcal{Z}_s^\circ$ :

- (a)  $Y_t^\circ$  and  $Y_t^\circ Z_s^\circ$  are  $P$ -continuous at  $X$ ;
- (b)  $\{Y_r^\circ(X^n) : r \in D \cap \mathbb{T}_{\leq t}, n \in \mathbb{N}\}$  is uniformly integrable;
- (c)  $\lim_{n \rightarrow \infty} E^{P^n}[(Y_t^\circ(X^n) - Y_s^\circ(X^n))Z_s^\circ(X^n)] = 0$ .

Then  $E^P[Y_t | \mathcal{F}_s] = Y_s$  for all  $s \leq t$  in  $D$  and all  $Y \in \mathcal{X}$ . If moreover  $D = \mathbb{T}$ , then  $P \in \mathcal{M}(\mathcal{X})$ . The same conclusion holds for a proper  $D \subsetneq \mathbb{T}$  provided  $\mathbb{T}$  satisfies (T2b) with  $D$  its countable dense subset,  $D$  contains the greatest element of  $\mathbb{T}$  if there is one, and every  $Y \in \mathcal{X}$  is right continuous.

*Proof.* Fix  $Y \in \mathcal{X}$  with canonical version  $Y^\circ$ , and  $t \in D$ .

*Step 0: integrability.* For  $r \in D \cap \mathbb{T}_{\leq t}$  the map  $|Y_r^\circ|$  is  $P$ -continuous at  $X$  by (a), so  $|Y_r^\circ(X^n)| \Rightarrow |Y_r^\circ(X)|$  by (C1) and Fact 2.31; the family is uniformly integrable by (b), so Fact 2.45 gives  $E^{P^n}|Y_r^\circ(X^n)| \rightarrow E^P|Y_r^\circ(X)| < \infty$ . In particular  $Y_r = Y_r^\circ(X) \in L^1(P)$ .

*Step 1: the martingale identity on  $D$ .* Let  $s \in D \cap \mathbb{T}_{\leq t}$  and  $Z_s^\circ \in \mathcal{Z}_s^\circ$ , and put  $\psi = (Y_t^\circ - Y_s^\circ)Z_s^\circ : F \rightarrow \mathbb{R}$ . Both  $Y_t^\circ Z_s^\circ$  and  $Y_s^\circ Z_s^\circ$  are  $P$ -continuous at  $X$  by (a) — the second by applying (a) with  $t$  replaced by  $s$ , legitimate since  $s \in D$  and  $s \in D \cap \mathbb{T}_{\leq s}$  — and Definition 8.39 is stable under differences, the exceptional sets being intersected; so  $\psi$  is  $P$ -continuous at  $X$ . Hence  $\psi(X^n) \Rightarrow \psi(X)$  by (C1) and Fact 2.31. The family  $\{\psi(X^n)\}_n$  is uniformly integrable, being dominated by  $\|Z_s^\circ\|(|Y_t^\circ(X^n)| + |Y_s^\circ(X^n)|)$  and  $Z_s^\circ$  bounded by Definition 4.3(i). So Fact 2.45 applies and, with (c),

$$E^P[(Y_t - Y_s)Z_s^\circ(X)] = E^P[\psi(X)] = \lim_{n \rightarrow \infty} E^{P^n}[\psi(X^n)] = 0,$$

the first equality because  $\mathcal{X}$  is canonical. As  $Z_s^\circ \in \mathcal{Z}_s^\circ$  was arbitrary and  $Y_s, Y_t \in L^1(P)$  by Step 0, part (ii) of Definition 4.3 yields  $E^P[Y_t | \mathcal{F}_s] = Y_s$ . If  $D = \mathbb{T}$  this is already the assertion.

*Step 2: uniform integrability under  $P$ .* Let  $\varphi_N(x) = |x| - |x| \wedge N$ . For  $r \in D \cap \mathbb{T}_{\leq t}$  the map  $\varphi_N \circ Y_r^\circ$  is  $P$ -continuous at  $X$ , and  $\{\varphi_N(Y_r^\circ(X^n))\}_n$  is uniformly integrable because it is dominated by  $|Y_r^\circ(X^n)|$ . So Fact 2.45 applies twice and gives

$$E^P[\varphi_N(Y_r)] = \lim_{n \rightarrow \infty} E^{P^n}[\varphi_N(Y_r^\circ(X^n))] \leq \sup_{n \in \mathbb{N}} E^{P^n}[\varphi_N(Y_r^\circ(X^n))].$$

Taking the supremum over  $r \in D \cap \mathbb{T}_{\leq t}$  and letting  $N \rightarrow \infty$ , the right-hand side tends to 0 by (b) and the second characterization of uniform integrability in Fact 2.45. Hence  $\{Y_r : r \in D \cap \mathbb{T}_{\leq t}\}$  is uniformly integrable under  $P$ .

*Step 3: from  $D$  to  $\mathbb{T}$ .* Assume (T2b) and  $Y$  right continuous, and let  $s < t$  in  $\mathbb{T}$  be arbitrary; the case  $s = t$  is trivial. Choose  $t_m$  as follows: if  $t$  is the greatest element of  $\mathbb{T}$ , put  $t_m = t$ , which lies in  $D$  by hypothesis; otherwise (T2b) provides  $t_m \downarrow t$  in  $D$  with  $t_m > t$ . Since  $s < t$ , the same argument gives  $s_m \downarrow s$  in  $D$  with  $s \leq s_m < t_m$  for all large  $m$ . Let  $G \in \mathcal{F}_s$ . Then  $G \in \mathcal{F}_{s_m}$  because  $s \leq s_m$ , so Step 1 gives  $E^P[Y_{t_m} \mathbf{1}_G] = E^P[Y_{s_m} \mathbf{1}_G]$ . By right continuity  $Y_{t_m} \rightarrow Y_t$  and  $Y_{s_m} \rightarrow Y_s$   $P$ -a.s. All the indices involved lie in  $D \cap \mathbb{T}_{\leq t_1}$ , so both families are uniformly integrable by Step 2 applied at  $t_1$ ; Fact 2.45 applied to  $Y_{t_m} \mathbf{1}_G$  and  $Y_{s_m} \mathbf{1}_G$  yields  $E^P[Y_t \mathbf{1}_G] = E^P[Y_s \mathbf{1}_G]$ . As  $G \in \mathcal{F}_s$  was arbitrary,  $E^P[Y_t | \mathcal{F}_s] = Y_s$ , and  $P \in \mathcal{M}(\mathcal{X})$ .  $\square$

*Remark 8.41* ((C1) and (a) are one hypothesis, and it is not topological). The proof invoked (C1) and (a) only through Fact 2.31, and only in order to know that certain *real* random variables converge in distribution. Collecting them: for  $t \in D$ ,  $s, r \in D \cap \mathbb{T}_{\leq t}$  and  $Z_s^\circ \in \mathcal{Z}_s^\circ$ , what Steps 0–2 used is

$$Y_r^\circ(X^n) \Rightarrow Y_r^\circ(X) \quad \text{and} \quad (Y_t^\circ(X^n) - Y_s^\circ(X^n))Z_s^\circ(X^n) \Rightarrow (Y_t^\circ(X) - Y_s^\circ(X))Z_s^\circ(X), \quad (98)$$

and nothing else — Step 2 needs only  $\varphi_N \circ Y_r^\circ$ , a bounded continuous function of the first. Replacing (C1) and (a) by (98) therefore leaves the theorem and its proof standing verbatim.

In that form the path space  $F$  carries no topology at all: it is the measurable path space of Setting 4.1 and nothing more is asked of it. This is worth isolating for two reasons. First, it locates the only topological hypothesis in Section 8: (C1) presupposes a topology on  $F$  and Fact 2.31 presupposes it separable and metric, whereas (98) presupposes neither, so a formalization can carry Theorem 8.40 on a bare measurable space and pay for a topology only where one verifies (98). Second, (98) is what one is reduced to verifying whenever the topology on  $F$  is too weak for the coordinates  $\pi_t$  to be continuous — which is the situation of Section 8.9, where they fail at the atoms of the clock, and the situation of Section 8.10, where they fail everywhere.

*Remark 8.42* (Where the index enters). Steps 0–2 use no structure on  $\mathbb{T}$  whatsoever beyond the preorder:  $D$  is any subset, and  $\mathbb{T}_{\leq t}$  is a down-set. The only place where more is needed is Step 3, and it is needed only to pass from  $D$  to its complement. With  $D = \mathbb{T}$  — in particular for  $\mathbb{T} = \mathbb{N}_0$ , where every point is isolated from the right and no approximation is available or necessary — Step 3 is empty and the theorem holds under (T0). This is why step (F4) of Section 9 can be done immediately after (F3), before any of the Skorokhod theory.

*Remark 8.43* (What has to be checked, and what does not). Of the three hypotheses, (C1) is the input, (C2) is a property of the path space and is supplied once and for all by Example 4.4 or by Proposition 4.9, and (C3) is the work. Within (C3), (a) is where the topology of  $F$  enters — and it is the only place, which is what Lemma 8.54 makes precise and Theorem 8.55 exploits; for  $F = D_E[0, \infty)$  it is the reason for the “countable complement” hypothesis of Theorem 8.44, since  $\alpha \mapsto \alpha(t)$  is  $J_1$ -continuous exactly at paths not jumping at  $t$ . Condition (c) is where the approximating martingales are used, and (b) is what makes the two limits interchangeable.

## 8.7 By convergence: the general version

The following theorem is [2], Theorem 4.1; it is the specialization of the abstract convergence theorem [2], Corollary 3.17, to the Markovian martingale problem, and it generalizes Lemma 8.34 — which Theorem 8.40 has already delivered — in three independent directions:

- (1) the approximating objects are *not* required to solve a martingale problem for an operator; they are arbitrary pairs  $(\xi^n, \varphi^n)$  of progressively measurable processes such that  $\xi^n - \int_0^\cdot \varphi_s^n ds$  is a martingale;
- (2) the approximation (96) in the uniform norm is replaced by the much weaker *asymptotic* conditions (100) and (101), which are conditions on the laws only;
- (3) uniform boundedness is replaced by uniform integrability (99).

**Theorem 8.44** ([2], Theorem 4.1; (T3)). Assume (E3). Let  $B = (\Omega, \mathcal{F}, \mathbb{F}, P)$  and  $B^n = (\Omega^n, \mathcal{F}^n, \mathbb{F}^n, P^n)$  be filtered probability spaces carrying  $E$ -valued càdlàg adapted processes  $X$  and  $X^n$ , and assume that  $\mathbb{F}$  is generated by  $X$ . Let  $A \subset C_b(E) \times C_b(E)$  and let  $\mathcal{X} = \mathcal{X}_A(X)$  be as in (18). For each  $n$ , let  $\mathcal{X}^n$  be a set of pairs  $(\xi, \varphi)$  of  $\mathbb{K}$ -valued progressively measurable processes on  $B^n$  such that  $\sup_{s \leq T} E^{P^n}[|\xi_s| + |\varphi_s|] < \infty$  for all  $T > 0$  and such that  $\xi - \int_0^\cdot \varphi_s ds$  is a martingale on  $B^n$ .

Suppose that  $X^n \Rightarrow X$  weakly on  $D_E[0, \infty)$  with the Skorokhod  $J_1$  topology, and that there is a set  $\Gamma \subset \mathbb{R}_+$  with countable complement such that for each  $(f, g) \in A$  and  $T > 0$  there is a sequence  $(\xi^n, \varphi^n) \in \mathcal{X}^n$  with

$$\sup_{n \in \mathbb{N}} \sup_{s \leq T} E^{P^n}[|\xi_s^n| + |\varphi_s^n|] < \infty, \quad (99)$$

$$\lim_{n \rightarrow \infty} E^{P^n} \left[ (\xi_t^n - f(X_t^n)) \prod_{i=1}^k h_i(X_{t_i}^n) \right] = 0, \quad (100)$$

$$\lim_{n \rightarrow \infty} E^{P^n} \left[ \int_s^t (\varphi_u^n - g(X_u^n)) du \prod_{i=1}^k h_i(X_{t_i}^n) \right] = 0, \quad (101)$$

for all  $k \in \mathbb{N}$ ,  $t_1, \dots, t_k \in \Gamma \cap [0, t]$ ,  $t \in \Gamma \cap [0, T]$ ,  $h_1, \dots, h_k \in C_b(E)$ . Then  $P \in \mathcal{M}(\mathcal{X})$ , i.e.  $X$  solves the martingale problem for  $A$ .

*Proof.* We verify (C1)–(C3) of Theorem 8.40.

The set  $D$ , and (C2). Set  $D = \{t \in \Gamma : P\{X_t \neq X_{t-}\} = 0\}$ . By Fact 2.41 and the hypothesis on  $\Gamma$ ,  $D^c$  is countable, so  $D$  is dense. Let  $\mathcal{Z}^\circ$  be the determining set of Example 4.4 built from  $D$ ; this is (C2). Hypothesis (C1) is the assumed weak convergence  $X^n \Rightarrow X$  on  $D_E[0, \infty)$ .

(b). Let  $Y = f(X) - \int_0^\cdot g(X_s) ds \in \mathcal{X}$  with canonical version  $Y_t^\circ(\omega) = f(\omega(t)) - \int_0^t g(\omega(s)) ds$ . Since  $f, g$  are bounded,  $Y_r^\circ$  is bounded on  $r \in [0, t]$ , so  $\{Y_r^\circ(X^n) : r \in D \cap [0, t], n \in \mathbb{N}\}$  is uniformly integrable.

(a). The evaluation  $\omega \mapsto \omega(t)$  is  $J_1$ -continuous at every  $\omega$  with  $\omega(t) = \omega(t-)$ , so the definition of  $D$  makes  $Y_t^\circ$  and  $Y_t^\circ Z_s^\circ$   $P$ -continuous at  $X$  in the sense of Definition 8.39, for  $s, t \in D$ . This is the only point at which the Skorokhod topology is used, and it is what the hypothesis on  $\Gamma$  is for.

(c). It remains to verify, for  $s, t \in D$  with  $s \leq t$  and  $Z_s^\circ = \prod_i h_i(\cdot(t_i))$ ,

$$\lim_{n \rightarrow \infty} E^{P^n} [(Y_t^\circ(X^n) - Y_s^\circ(X^n)) Z_s^\circ(X^n)] = 0. \quad (102)$$

Put  $Y^n = \xi^n - \int_0^\cdot \varphi_u^n du$ , a martingale on  $B^n$ . Because  $Z_s^\circ(X^n)$  is  $\mathcal{F}_s^n$ -measurable,  $E^{P^n}[(Y_t^n - Y_s^n) Z_s^\circ(X^n)] = 0$ , so

$$E^{P^n} [(Y_t^\circ(X^n) - Y_s^\circ(X^n)) Z_s^\circ(X^n)] = E^{P^n} [(Y_t^\circ(X^n) - Y_t^n + Y_s^n - Y_s^\circ(X^n)) Z_s^\circ(X^n)],$$

and the right-hand side tends to 0 by (100) and (101), since  $Y_t^\circ(X^n) - Y_t^n = (f(X_t^n) - \xi_t^n) + \int_0^t (\varphi_u^n - g(X_u^n)) du$ . Thus (102) holds, which is (c).

Theorem 8.40 now gives  $E^P[Y_t | \mathcal{F}_s] = Y_s$  for  $s \leq t$  in  $D$ , and its Step 3 — which uses the uniform integrability established in its Step 2 from (b), together with the right continuity of  $Y$  — extends this to arbitrary  $s < t$ . Hypothesis (99) is what makes all the expectations in (100) and (101) finite, uniformly in  $n$ ; it is not what supplies (b), which here comes for free from the boundedness of  $f$  and  $g$ .  $\square$

*Remark 8.45.* Theorem 8.44 concludes that  $P$  solves the *global* martingale problem. To obtain a solution of a *local* martingale problem in the limit, apply the theorem to a localized family of test processes, i.e. to  $\mathcal{X}^{(m)} = \{Y^{\tau_m} : Y \in \mathcal{X}\}$  for a suitable exhausting sequence of stopping times  $\tau_m$ , and let  $m \rightarrow \infty$ . This is how [2] obtain their semimartingale results (their Section 4.3), and it is the same reduction as in Remark 6.53.

*Remark 8.46* (Path spaces other than  $D_E[0, \infty)$ ). Theorem 8.40 is stated for an arbitrary Polish path space  $F \subset E^{\mathbb{T}}$ , and Theorem 8.44 is the instance  $F = D_E[0, \infty)$ . Nothing in the abstract theorem refers to càdlàg paths, to a Markov structure, or to the Skorokhod topology;  $F$  may for instance be  $L_{\text{loc}}^p(\mathbb{R}_+, \mathbb{R}^d)$ , which is the state space [2] use for solutions of Volterra equations (their Example 3.13), or a product of such spaces. What changes from one  $F$  to another is only which maps are  $P$ -continuous at  $X$ , i.e. hypothesis (a) — see Remark 8.43. [2] also give a version in which the uniform integrability of (b) is dropped (their Corollary 3.21) and one with control variables and weak-strong convergence (their Theorem 3.14). The part of the latter that concerns different clocks is Section 8.8, the part that concerns weak-strong convergence is Section 8.9. Theorem 8.55 is a third instance of the same freedom in  $F$  — there the enlargement is by finitely or countably many coordinates rather than by a change of topology — and it is the one that does the work here.

## 8.8 Approximation on different clocks

Nothing so far requires the approximating processes to run on the same clock as the limit, and this subsection makes that precise. The observation to begin with is negative and structural: *Theorem 8.40 does not mention the approximating martingale problems at all.* Its hypotheses (C1)–(C3) speak only of the canonical version  $Y^\circ$  of the *limit* test process, evaluated along  $X^n$ . Where the approximating martingales come from, on which clock they are compensated, and with which convention, is not part of the theorem — it is part of the verification of (c). So different clocks are not a difficulty for the theorem; they are a difficulty for one of its hypotheses, and the following isolates it.

**Setting 8.47** (Approximants with their own clocks; (T3) + (E3)). In the situation of Theorem 8.40, let the limit test process be

$$Y_t^\circ = f(\pi_t) - \int_{\langle 0, t \rangle_\iota} g(u, \pi_u) q(du),$$

and for each  $n$  let  $q^n$  be a clock on  $\mathbb{T}$ ,  $\iota_n$  a convention, and  $\xi^n, \psi^n$  progressively measurable processes on  $B^n$  such that

$$Y_t^n = \xi_t^n - \int_{\langle 0, t \rangle_{\iota_n}} \psi_u^n q^n(du) \quad (103)$$

is a  $P^n$ -martingale. Write, for  $s \leq t$ ,

$$\Delta_{s,t}^n := \int_{\langle s, t \rangle_{\iota_n}} \psi_u^n q^n(du) - \int_{\langle s, t \rangle_\iota} \psi_u^n q(du) \quad (104)$$

for the *clock discrepancy* along the integrands.

**Theorem 8.48** (Clock change; (T3) + (E3)). Assume (C1), (C2), (a) and (b) of Theorem 8.40, and suppose that for all  $s \leq t$  in  $D$  and all  $Z_s^\circ \in \mathcal{Z}_s^\circ$ :

$$(K1) \sup_n \sup_{r \leq T} E^{P^n} [|\xi_r^n| + |\psi_r^n|] < \infty \text{ for every } T;$$

$$(K2) \lim_n E^{P^n} [(\xi_t^n - f(X_t^n)) Z_s^\circ(X^n)] = 0;$$

$$(K3) \lim_n E^{P^n} \left[ \int_{\langle s, t \rangle_{\iota}} (\psi_u^n - g(u, X_u^n)) q(du) Z_s^\circ(X^n) \right] = 0;$$

$$(K4) \lim_n E^{P^n} [\Delta_{s,t}^n Z_s^\circ(X^n)] = 0.$$

Then (c) holds, and consequently  $P \in \mathcal{M}(\mathcal{X})$ .

*Proof.* Fix  $s \leq t$  in  $D$  and  $Z_s^\circ \in \mathcal{Z}_s^\circ$ . Since  $Y^n$  is a  $P^n$ -martingale and  $Z_s^\circ(X^n)$  is  $\mathcal{F}_s^n$ -measurable,  $E^{P^n}[(Y_t^n - Y_s^n)Z_s^\circ(X^n)] = 0$ , so

$$E^{P^n}[(Y_t^\circ(X^n) - Y_s^\circ(X^n))Z_s^\circ(X^n)] = E^{P^n}[(R_t^n - R_s^n)Z_s^\circ(X^n)], \quad R_r^n := Y_r^\circ(X^n) - Y_r^n. \quad (105)$$

By (103) and the definition of  $Y^\circ$ ,

$$R_r^n = (f(X_r^n) - \xi_r^n) + \int_{\langle 0, r \rangle_{\iota_n}} \psi_u^n q^n(du) - \int_{\langle 0, r \rangle_{\iota}} g(u, X_u^n) q(du),$$

Inserting  $\pm \int_{\langle 0, r \rangle_{\iota}} \psi_u^n q(du)$  splits the last two terms into  $\Delta_{0,r}^n$  and

$$\int_{\langle 0, r \rangle_{\iota}} (\psi_u^n - g(u, X_u^n)) q(du).$$

Taking the difference at  $r = t$  and  $r = s$ , and using the additivity (3) of both interval families,

$$R_t^n - R_s^n = (f(X_t^n) - \xi_t^n) - (f(X_s^n) - \xi_s^n) + \Delta_{s,t}^n + \int_{\langle s, t \rangle_{\iota}} (\psi_u^n - g(u, X_u^n)) q(du).$$

Each of the four terms has vanishing limit against  $Z_s^\circ(X^n)$ , by (K2) twice, (K4) and (K3); (K1) makes all the expectations finite. So the left-hand side of (105) tends to 0, which is (c), and Theorem 8.40 applies.  $\square$

*Remark 8.49.* Condition (K4) is *not* weak convergence  $q^n \Rightarrow q$ . It is convergence of the clocks *paired with the integrands*, and it does not follow from  $q^n \Rightarrow q$  unless the family  $\{\psi^n\}$  is equicontinuous in a suitable sense — which for the piecewise constant  $\psi^n$  of Example 8.50 it is not. This is why [2], Assumption 5.3 states it as a hypothesis, their condition (5.10), rather than deriving it; and Theorem 8.48 is their Theorem 5.4, stripped of the control variables. Note also that (K4) absorbs the difference between  $\iota_n$  and  $\iota$ : the two conventions for one clock differ by a boundary term, and (104) does not care which of the two discrepancies it is measuring.

**Example 8.50** (Rescaled Markov chains; the invariance principle). Let  $E = \mathbb{R}^d$ ,  $F = D_E[0, \infty)$ ,  $q = \lambda$  for the limit, and for the  $n$ -th approximant

$$q^n = \frac{1}{n} \sum_{k \geq 1} \delta_{k/n}, \quad \iota_n = 0.$$

Let  $(\Xi_k^n)_{k \geq 0}$  be a Markov chain with one-step kernel  $P_n$  and put  $X_t^n = \Xi_{\lfloor nt \rfloor}^n$ , a piecewise constant element of  $D_E[0, \infty)$ . With  $\xi^n = f(X^n)$  and the *predictable* integrand  $\psi_u^n = n(P_n f - f)(X_{u-}^n)$  we get, since  $X_{k/n-}^n = \Xi_{k-1}^n$ ,

$$\int_{(0,t]} \psi_u^n q^n(du) = \frac{1}{n} \sum_{1 \leq k \leq nt} \psi_{k/n}^n = \sum_{j < \lfloor nt \rfloor} (P_n f - f)(\Xi_j^n),$$

so that (103) reads  $Y_t^n = f(\Xi_{[nt]}^n) - \sum_{j < [nt]} (P_n f - f)(\Xi_j^n)$ . This is the Doob decomposition of  $f(\Xi^n)$  read along the embedded grid, it is constant on each  $[k/n, (k+1)/n)$ , and it is a genuine  $(\mathcal{F}_t^{X^n})_{t \geq 0}$ -martingale.

Then (K2) holds trivially,  $\xi^n - f(X^n)$  being identically zero; (K3) is the classical requirement that the rescaled generators  $n(P_n f - f)$  converge to  $Af$  — the discrepancy between  $X_{u-}^n$  and  $X_u^n$  is immaterial there, the two differing only on the Lebesgue null set  $\frac{1}{n}\mathbb{N}$ ; and (K4) holds because  $\psi^n$  is constant on each interval  $(\frac{k-1}{n}, \frac{k}{n}]$ , on which the atomic and the Lebesgue integral give the same value  $(P_n f - f)(\Xi_{k-1}^n)$ , so that the two integrals agree except on the two boundary intervals and  $|\Delta_{s,t}^n| \leq \frac{2}{n} \sup_u |\psi_u^n| = 2\|P_n f - f\|$ , which tends to 0 as soon as  $n(P_n f - f)$  is bounded.

So “a rescaled Markov chain converges to a diffusion” is, in this formalism, a statement about clocks:  $q^n \rightarrow q$  along the integrands, with the chain’s clock atomic and the limit’s atomless.

*Remark 8.51* (Why the convention flips on embedding). Remark 8.27 says that on  $\mathbb{T} = \mathbb{N}_0$  with counting measure the Doob decomposition is the *predictable* convention  $\iota = p$ , and Example 8.50 uses  $\iota_n = o$ . There is no contradiction, and the point is worth isolating because it is easy to get wrong — an earlier version of this manuscript did.

On  $\mathbb{T} = \mathbb{N}_0$  there is nothing between two time points, and  $f(\Xi_m) - \sum_{k < m} (P_n f - f)(\Xi_k)$  is a martingale for the only filtration there is. Embed the grid into  $\mathbb{R}_+$ , however, and the same formula with  $\iota_n = p$  and  $\psi_u^n = n(P_n f - f)(X_u^n)$  gives

$$\int_{[0,t)} \psi_u^n q^n(du) = \sum_{k < nt} (P_n f - f)(\Xi_k^n),$$

whose value for  $t \in (\frac{m}{n}, \frac{m+1}{n})$  is  $\sum_{k \leq m} (P_n f - f)(\Xi_k^n)$ , one summand *more* than at  $t = m/n$ . So  $Y^n$  drops by  $(P_n f - f)(\Xi_m^n)$  immediately after  $m/n$  while  $\mathcal{F}^{X^n}$  does not change, and  $Y^n$  is a martingale along the grid but *not* in continuous time. The mismatch is exactly the one Section 5 warns about: with  $\iota = p$  an atom of the clock at  $u$  is compensated *after*  $u$ , whereas  $f(X^n)$  jumps *at*  $u$ .

The repair is the one the semimartingale literature makes by reflex: take  $\iota_n = o$  and evaluate the integrand at  $u-$ . Then atom and jump occur at the same instant and cancel. This is the sharpest argument in the manuscript for carrying both conventions: the same object needs  $p$  intrinsically and  $o$  after embedding, and a manuscript committed to one of them could not state Example 8.50 at all.

*Remark 8.52* (What this example costs, and what paid for it). Three earlier decisions are needed to write Example 8.50 down at all, and none of them was made with it in view.

- (i) *Atoms* (D7 of the progress log, Theorem 5.4). Without a clock that may have atoms, the chain’s martingale problem is not a martingale problem; one would have to interpolate the chain into continuous time, which is precisely the manoeuvre the clock makes unnecessary.
- (ii) *Clocks as measures on  $\mathbb{T}$*  (Definition 2.3). Different clocks on one  $\mathbb{T}$  subsume different time indices, as long as the latter embed:  $\frac{1}{n}\mathbb{N}_0 \subset \mathbb{R}_+$  carries  $q^n$ , and no separate index set is needed.
- (iii) *Both conventions* (Remark 2.5). The chain needs  $\iota = p$  intrinsically and  $\iota = o$  once its grid is embedded in  $\mathbb{R}_+$  (Remark 8.51); the limit does not care. Had one convention been fixed, Example 8.50 could not be stated as it is. That the two differ by  $\psi_t^n q^n(\{t\}) = (P_n f - f)(X_{t-}^n) = O(1/n)$ , and so *wash out in the limit*, is what

makes the flip harmless; it is a retrospective argument for carrying both rather than choosing.

What is *not* supplied here is tightness: Theorem 8.48 assumes (C1), and establishing  $X^n \Rightarrow X$  on  $D_E[0, \infty)$  is the Skorokhod half of the theory, cited in Remark 8.38 and not proved.

## 8.9 Relaxing the continuity hypothesis

Theorem 8.40 has two hypotheses that can fail in practice, and this manuscript now has a section for each. Section 8.8 relaxes (c), the approximation condition. This one relaxes (a), the continuity condition — and it has to, because (a) fails outright in precisely the situation that the clock of Definition 2.3 was widened to admit.

There are two repairs, and they are not of the same weight. The first, given here in full, enlarges the path space by the values at the atoms of the clock and needs nothing beyond the continuous mapping theorem; it settles every case that arises in this manuscript. The second is the weak-strong convergence of [2], which is what the literature states and what is needed when the times at which continuity fails are *random*. Remark 8.58 sets the two side by side.

**Example 8.53** (Why (a) is not a technicality; [2], Section 4.3). Take  $\mathbb{T} = \mathbb{R}_+$ ,  $E = \mathbb{R}$ ,  $F = D_{\mathbb{R}}[0, \infty)$ ,  $g(x) = x$  and the atomic clock  $q = \delta_1$ . The compensator of Definition 4.6 is then the functional

$$F_t(\omega) = \int_{(0,t]_\iota} \omega(s) q(ds) = \omega(1) \mathbf{1}_{\{t>1\}} \quad (\iota = p), \quad \omega(1) \mathbf{1}_{\{t \geq 1\}} \quad (\iota = o),$$

since  $[0, t)$  contains the atom 1 exactly when  $t > 1$ , and  $(0, t]$  exactly when  $t \geq 1$ . Either way  $F_t$  is, for every  $t$  beyond the atom, the evaluation  $\omega \mapsto \omega(1)$ , and *that* map is  $J_1$ -discontinuous at every path jumping at 1: with

$$\omega_n = \mathbf{1}_{[1+1/n, \infty)} \longrightarrow \omega = \mathbf{1}_{[1, \infty)} \quad \text{in } J_1,$$

one has  $F_t(\omega_n) = \omega_n(1) = 0 \not\rightarrow 1 = \omega(1) = F_t(\omega)$ . For  $t$  before the atom  $F_t \equiv 0$  is continuous, so the times at which  $F_t$  is continuous at  $\omega$  form the bounded set  $\mathbb{T}_{\leq 1}$ .

Consequently, if the limit law  $P$  charges paths jumping at the atom —  $P\{X_1 \neq X_{1-}\} > 0$ , which is the generic situation when the clock has an atom at 1 — then hypothesis (a) of Theorem 8.40 fails for *every*  $t > 1$ , a set of times of full measure. This is a different failure from the one Theorem 8.44 handles: there the bad times form a countable set, namely the fixed discontinuities of  $X$  itself, and a dense  $\Gamma$  avoids them; here the bad *index*  $t$  ranges over a half-line because the discontinuity of the functional sits at the atom of the clock and not at  $t$ . No choice of  $\Gamma$  helps. The cause is exactly the atom of  $q$ , and hence exactly the generality that Remark 2.5 and Section 8.3 insist on. Processes with fixed times of discontinuity are not a curiosity here; they are the case this manuscript admits and [7] do not.

### An elementary repair: enlarge the path space by the atoms

Before importing machinery it is worth asking how much of it the failure in Example 8.53 actually calls for. The answer is: none. The discontinuity there sits at a *deterministic* place — an atom of the clock — and a functional that is discontinuous only because it evaluates the path at finitely or countably many prescribed times becomes continuous as soon as those values are carried along as separate coordinates. That is the whole idea, and it costs one lemma.

**Lemma 8.54** (Where continuity is used in Theorem 8.40). *In the proof of Theorem 8.40, hypotheses (C1) and (a) are used only in the following way: to conclude, from a Borel  $\psi : F \rightarrow \mathbb{R}$  which is  $P$ -continuous at  $X$ , that  $\psi(X^n) \Rightarrow \psi(X)$ . They are used for  $\psi \in \{|Y_r^\circ|, \varphi_N \circ Y_r^\circ, (Y_t^\circ - Y_s^\circ)Z_s^\circ\}$  and for nothing else. In particular neither hypothesis enters the filtration, the determining set, or the conclusion, all of which are statements about  $F$  and  $P$  alone.*

*Proof.* Inspection. Step 0 applies Fact 2.31 to  $|Y_r^\circ|$  and then Fact 2.45; Step 1 does the same for  $(Y_t^\circ - Y_s^\circ)Z_s^\circ$ ; Step 2 for  $\varphi_N \circ Y_r^\circ$ . Step 3 uses right continuity and Step 2, and no continuity on  $F$  at all. The conclusion  $E^P[Y_t | \mathcal{F}_s] = Y_s$  and the hypothesis (C2) refer only to  $F$ .  $\square$

Lemma 8.54 says that the continuity may be demanded on *any* space through which the relevant functionals factor, provided the convergence (C1) is available there. So we enlarge.

**Theorem 8.55** (Abstract convergence, augmented form; (T0) + (E3)). *In the situation of Theorem 8.40, let  $G$  be Polish, let  $\gamma : F \rightarrow G$  be Borel, and put*

$$\hat{F} = F \times G, \quad \hat{\gamma} : F \rightarrow \hat{F}, \quad \omega \mapsto (\omega, \gamma(\omega)), \quad \hat{X}^n = \hat{\gamma}(X^n), \quad \hat{X} = \hat{\gamma}(X).$$

Replace (C1) and (a) by

(C1')  $\hat{X}^n \Rightarrow \hat{X}$  weakly on  $\hat{F}$ ;

(C3a') there are Borel  $\hat{Y}_t^\circ : \hat{F} \rightarrow \mathbb{R}$  with  $\hat{Y}_t^\circ \circ \hat{\gamma} = Y_t^\circ$  such that  $\hat{Y}_t^\circ$  and  $\hat{Y}_t^\circ(Z_s^\circ \circ \text{pr}_F)$  are  $\hat{P}$ -continuous at  $\hat{X}$ , where  $\hat{P} = P \circ \hat{X}^{-1}$ .

Keep (C2), (b) and (c) unchanged. Then the conclusion of Theorem 8.40 holds verbatim.

*Proof.* By Lemma 8.54 it suffices to produce  $\psi(X^n) \Rightarrow \psi(X)$  for the three functionals listed there. Each factors through  $\hat{\gamma}$ :  $\psi = \hat{\psi} \circ \hat{\gamma}$  with  $\hat{\psi} \in \{|\hat{Y}_r^\circ|, \varphi_N \circ \hat{Y}_r^\circ, (\hat{Y}_t^\circ - \hat{Y}_s^\circ)(Z_s^\circ \circ \text{pr}_F)\}$ , by the defining property  $\hat{Y}_t^\circ \circ \hat{\gamma} = Y_t^\circ$  and because  $\text{pr}_F \circ \hat{\gamma} = \text{id}_F$ . Each  $\hat{\psi}$  is  $\hat{P}$ -continuous at  $\hat{X}$  — for the first and second because  $x \mapsto |x|$  and  $\varphi_N$  are continuous, for the third by (C3a') and stability of Definition 8.39 under differences. So  $\hat{\psi}(\hat{X}^n) \Rightarrow \hat{\psi}(\hat{X})$  by (C1') and Fact 2.31, and  $\hat{\psi}(\hat{X}^n) = \psi(X^n)$ ,  $\hat{\psi}(\hat{X}) = \psi(X)$ . Every other step of the proof of Theorem 8.40 is untouched.  $\square$

Note that (C1') implies (C1), by projection; the augmented statement is a genuine strengthening of the input and a genuine weakening of the work. The point is what  $G$  has to be for an atomic clock.

**Proposition 8.56** (Atomic clocks need only the values at the atoms; (T3) + (E2)). *Let  $F = D_E[0, \infty)$ , let  $q = q_a + q_c$  with  $q_a = \sum_{a \in A} m_a \delta_a$  for a countable  $A \subset \mathbb{T}$  and  $q_c$  atomless, and let*

$$Y_t^\circ(\omega) = f(\omega(t)) - \int_{\langle 0, t \rangle_t} g(u, \omega(u)) q(du)$$

with  $f \in C_b(E)$ ,  $g$  bounded,  $g(u, \cdot) \in C_b(E)$  for each  $u$  and  $u \mapsto g(u, x)$  measurable. Put  $G = E^A$  with the product topology,  $\gamma(\omega) = (\omega(a))_{a \in A}$ , and

$$\hat{Y}_t^\circ(\omega, \xi) = f(\omega(t)) - \int_{\langle 0, t \rangle_t} g(u, \omega(u)) q_c(du) - \sum_{a \in A \cap \langle 0, t \rangle_t} m_a g(a, \xi_a). \quad (106)$$

Then  $\hat{Y}_t^\circ \circ \hat{\gamma} = Y_t^\circ$ , and (C3a') holds for every  $t$  outside the countable set  $\{t : P\{X_t \neq X_{t-}\} > 0\}$  — the same exceptional set as in the atomless case, and in particular independent of  $A$ .

*Proof.* The identity  $\hat{Y}_t^\circ \circ \hat{\gamma} = Y_t^\circ$  is the splitting  $q = q_a + q_c$ , the atomic integral being  $\sum_{a \in A \cap \langle 0, t \rangle_\iota} m_a g(a, \omega(a))$ ; note that the index set  $A \cap \langle 0, t \rangle_\iota$  is *deterministic*, which is the whole point. We check continuity of the three summands of (106) on  $\hat{F} = D_E[0, \infty) \times E^A$ .

*The atomic sum.* Each  $(\omega, \xi) \mapsto g(a, \xi_a)$  is continuous, the coordinate projections of a product being continuous and  $g(a, \cdot)$  being continuous. The series converges uniformly on  $\hat{F}$ , since  $\sum_{a \in A \cap \langle 0, t \rangle_\iota} m_a \leq q(\mathbb{T}_{\leq t}) < \infty$  by Definition 2.3 and  $|g| \leq \|g\|$ . A uniform limit of continuous functions is continuous, so this summand is continuous *everywhere* on  $\hat{F}$  — no exceptional set at all. *This is where the augmentation does its work:* on  $D_E[0, \infty)$  alone the same sum is discontinuous at every path jumping at an atom, which is Example 8.53.

*The diffuse integral.* Let  $\omega_m \rightarrow \omega$  in  $J_1$ . Then  $\omega_m(u) \rightarrow \omega(u)$  for every  $u$  outside the at most countable set of discontinuities of  $\omega$ , hence  $q_c$ -almost everywhere,  $q_c$  being atomless; by continuity of  $g(u, \cdot)$  and bounded convergence the integral converges. So this summand too is continuous everywhere, for *every*  $\omega$ , and it does not use the second coordinate.

*The evaluation  $f(\omega(t))$ .* This is the classical term: it is  $J_1$ -continuous at every  $\omega$  with  $\omega(t) = \omega(t-)$ , so it is  $\hat{P}$ -continuous at  $\hat{X}$  as soon as  $P\{X_t = X_{t-}\} = 1$ , and by Fact 2.41 the set of  $t$  failing this is countable.

Finally  $Z_s^\circ \circ \text{pr}_F$  is bounded and, for  $s$  in the same exceptional-free set, continuous at  $\hat{X}$   $\hat{P}$ -a.s. by Example 4.4; products of  $\hat{P}$ -continuous functions are  $\hat{P}$ -continuous.  $\square$

*Remark 8.57* (What (C1') asks, concretely). For  $G = E^A$  the hypothesis (C1') reads:  $X^n \Rightarrow X$  in  $J_1$ , *jointly with convergence of the values at the atoms of the clock*. This is ordinary weak convergence on a product of Polish spaces, so Prohorov (Fact 2.33) applies and tightness is the tightness of  $\{X^n\}$  in  $D_E[0, \infty)$  together with that of the finitely or countably many  $E$ -valued families  $\{X_a^n\}$ , which for  $E$  Polish is automatic once  $\{X^n\}$  is tight in  $D_E[0, \infty)$ . What is *not* automatic is the identification of the limit: one must know that  $X_a^n \Rightarrow X_a$  and not, say,  $X_a^n \Rightarrow X_{a-}$ . That is exactly the content of Example 8.53, and it is a condition on the approximating scheme — it holds whenever the  $X^n$  jump *at* the atoms rather than near them, which is the case for schemes whose grid contains  $A$ , and in particular in Example 8.50.

It is worth being explicit that (C1') is strictly stronger than (C1): the sequence  $\omega_n = \mathbf{1}_{[1+1/n, \infty)}$  of Example 8.53 converges in  $J_1$  but not in  $D_E[0, \infty) \times E^{\{1\}}$ . Something has to be strengthened, and the question is only what.

*Remark 8.58* (Augmentation versus weak-strong convergence). Proposition 8.56 settles Example 8.53 with the continuous mapping theorem and Vitali's theorem, both of which the manuscript already has (Facts 2.31 and 2.45). Set against the alternative below, the ledger reads:

| Augmentation (Theorem 8.55)                                   | Weak-strong (Theorem 8.62)                                            |
|---------------------------------------------------------------|-----------------------------------------------------------------------|
| one Borel map $\gamma : F \rightarrow G$                      | a control space $U$ , Definitions 8.59 and 8.60                       |
| ordinary weak convergence on $F \times G$<br>Facts 2.31, 2.45 | a new mode of convergence<br>Fact 8.61 (Jacod–Mémmin), cited unproved |
| discontinuity at <i>deterministic</i> times                   | discontinuity at <i>random</i> times                                  |

The last line is the real division, and it is what decides which to use. The atoms of a clock are fixed in advance, so the bad set of paths is  $\{\omega : \omega(a) \neq \omega(a-) \text{ for some } a \in A\}$  with  $A$  deterministic, and carrying the finitely or countably many numbers  $(\omega(a))_{a \in A}$  removes it. [2] need more because in their Section 5.3 the times at which continuity fails are the jump

times of the approximating processes, which are random and different for each  $n$ ; no fixed countable family of coordinates captures them, and the exceptional set must be allowed to depend on a control variable.

For everything in this manuscript the deterministic case is the case that occurs. Theorem 8.55 is therefore the tool to reach for, and Theorem 8.62 is recorded for completeness and because it is what the literature states.

### The general repair of [2]

The repair of [2] is to weaken not the continuity but the *space on which it is demanded*: a continuity that holds along random sections is enough, provided the sections are indexed by a variable that converges in a stronger sense than weakly. This is the setting we have so far silently avoided by taking the control space to be a singleton.

**Definition 8.59** (Weak-strong convergence; [2], Definition 2.1). Let  $(U, \mathcal{U})$  be a measurable space, let  $F$  be Polish, and put  $S = U \times F$  with  $\mathcal{S} = \mathcal{U} \otimes \mathcal{B}(F)$ . Let  $C_S$  be the set of bounded  $\mathcal{S}$ -measurable  $f : S \rightarrow \mathbb{R}$  such that  $\omega \mapsto f(\alpha, \omega)$  is continuous for every  $\alpha \in U$ . Probability measures  $P^n$  on  $S$  converge to  $P$  in the *weak-strong* sense,  $P^n \rightarrow_{\text{ws}} P$ , if  $E^{P^n}[f] \rightarrow E^P[f]$  for every  $f \in C_S$ .

Continuity is thus required in the second variable only; in the first, measurability suffices. If  $U$  is a singleton this is ordinary weak convergence ([2], Remark 2.2(i)), which is why nothing of it has appeared until now.

**Definition 8.60** ( $(P^n, P)$ -continuity; [2], Definition 2.7). An  $\mathcal{S}$ -measurable  $g : S \rightarrow \mathbb{R}$  is  $(P^n, P)$ -continuous if there is  $A \in \mathcal{S}$  with  $P^n(A) \rightarrow 1$  and  $P(A) = 1$  such that

$$\{(\alpha, \omega) \in A : A_\alpha \ni \zeta \mapsto g(\alpha, \zeta) \text{ is discontinuous at } \omega\}$$

is  $P$ -null, where  $A_\alpha = \{\zeta : (\alpha, \zeta) \in A\}$ .

Two relaxations at once: continuity is asked for only along the sections  $A_\alpha$ , and only on a set of asymptotically full measure. For a singleton  $U$  and  $A = U \times C$  this is Definition 8.39.

*Fact 8.61* (Continuous mapping, Jacod–Mémín; [2], Theorem 2.9). If  $P^n \rightarrow_{\text{ws}} P$  and  $g : S \rightarrow \mathbb{R}$  is  $(P^n, P)$ -continuous with  $\sup_n E^{P^n}[|g| \mathbf{1}_{\{|g|>a\}}] \rightarrow 0$  as  $a \rightarrow \infty$ , then  $E^{P^n}[g] \rightarrow E^P[g]$ .

**Theorem 8.62** (Abstract convergence with control variables; [2], Theorem 3.14). *Theorem 8.40 holds with  $X^n, X$  replaced by pairs  $(L^n, X^n), (L, X)$  taking values in  $S = U \times F$ , with (C1) replaced by  $(L^n, X^n) \rightarrow_{\text{ws}} (L, X)$  and the  $P$ -continuity of (a) replaced by  $(Q^n, Q)$ -continuity, where  $Q^n$  and  $Q$  are the laws of  $(L^n, X^n)$  and  $(L, X)$ .*

*Proof, or rather: what has to be changed.* Nothing, beyond two citations. Steps 0, 1 and 2 of the proof of Theorem 8.40 each consist of applying the continuous mapping theorem (Fact 2.31) to obtain weak convergence and then Fact 2.45 to upgrade it to convergence of expectations. Fact 8.61 performs both steps at once under the weaker hypotheses, and is the only place where the difference between weak and weak-strong convergence is felt. Step 3 uses no continuity at all. All the work is therefore in Fact 8.61, which we do not prove.  $\square$

*Remark 8.63* (What the control variable does). The point of  $U$  is that the exceptional set may depend on  $\alpha$ . In [2], Section 5.3 the control variables  $L^n$  are càdlàg processes chosen so that, with probability tending to one, the jump times of  $X^n$  are dominated by those

of  $L^n$  (their Assumption 5.1). On the resulting sections the Skorokhod  $J_1$  and the local uniform topology coincide (their Proposition 2.12), and the functional of Example 8.53 — discontinuous in  $J_1$  at every  $t \geq 1$  — becomes continuous. One may then “treat the control variables as deterministic”, which is the practical content of Definition 8.59.

So the answer to why weak-strong convergence has been absent is that Theorem 8.40 is [2], Corollary 3.17, the case of a singleton control space, and that in that case (a) is available whenever the limit has no fixed times of discontinuity. Atoms in the clock destroy that, but — and this is the conclusion of Remark 8.58 — they destroy it at deterministic times, which Theorem 8.55 handles with the continuous mapping theorem. So the price atoms exact is *not* a new mode of convergence but a joint convergence hypothesis, (C1'). Weak-strong convergence is needed for a different reason: random times of discontinuity, which no fixed family of coordinates captures. In Section 7 atoms do cost something, though not the conclusion: they obstruct the time change of Corollary 7.4, so that the atomic case has to be treated by a separate argument (Remark 7.5 and Proposition 7.8).

## 8.10 Convergence in measure

Every convergence statement so far has been made for the Skorokhod  $J_1$  topology, and  $J_1$  asks for a great deal: the approximating processes must place their jumps at nearly the right times and of nearly the right sizes. A sequence can fail it for reasons that have nothing to do with the martingale problem — two jumps of  $X_n$  merging into one jump of  $X$  is the standard example, and  $J_1$  also fails when the jump times are perturbed by a vanishing but non-uniform amount. What survives such perturbations is convergence *in measure* of the paths. [12] showed that this weaker mode has, for quasimartingales, a compactness theory that  $J_1$  does not have, and Remark 8.41 makes it cheap to record that the abstract theorem transfers to it. The point of this subsection is that the transfer costs one page and no new machinery: it is the last of the three relaxations of Theorem 8.40 — (c) in Section 8.8, (a) in Section 8.9, and now (C1) itself.

*Fact 8.64* (Pseudo-paths; [12], Section 1 and Lemma 1). Let  $\lambda(du) = e^{-u} du$  on  $\mathbb{R}_+$  and let  $E$  be Polish. The *pseudo-path* of a Borel map  $w : \mathbb{R}_+ \rightarrow E$  is the image of  $\lambda$  under  $u \mapsto (u, w(u))$ , a probability measure on  $[0, \infty] \times \hat{E}$  for a compact metrizable  $\hat{E} \supset E$ . This assignment identifies two paths exactly when they agree  $\lambda$ -a.e., so it is injective on  $D_E[0, \infty)$ , and the topology it induces on  $D_E[0, \infty)$  — the *pseudo-path topology* — is

- (i) exactly the topology of convergence in  $\lambda$ -measure, hence in Lebesgue measure on compact time intervals;
- (ii) metrizable and separable, being induced from the weak topology on the laws on a compact metric space; but *not* Polish (B. V. Rao; [12], Appendix),  $D_E[0, \infty)$  being merely Borel in that compact space;
- (iii) of the same Borel  $\sigma$ -field on  $D_E[0, \infty)$  as  $J_1$ , namely  $\sigma(\pi_u : u \in \mathbb{R}_+)$  ([12], footnote 2).

The coordinates  $\pi_u$  are, by contrast, discontinuous at every point of  $D_E[0, \infty)$ .

*Fact 8.65* (Tightness; [12], Theorem 4, and [11]). For a càdlàg adapted  $X$  with  $X_u \in L^1$  and the convention  $X_\infty = 0$ , let

$$V(X) = \sup_{\tau} E \left[ \sum_i |E[X_{t_{i+1}} - X_{t_i} | \mathcal{F}_{t_i}]| \right] \quad (\tau = \{0 = t_0 < t_1 < \dots < t_n < \infty\})$$

be its *conditional variation*, so that  $V(X) \geq \sup_u E|X_u|$ , and call  $X$  a quasimartingale if  $V(X) < \infty$ . If  $P^n$  are laws on  $D_E[0, \infty)$  under which the coordinate process is a quasimartingale with  $\sup_n V < \infty$ , then a subsequence of  $(P^n)$  converges weakly on  $D_E[0, \infty)$

in the pseudo-path topology to a law  $P$  under which the coordinate process is again a quasimartingale — in particular is càdlàg. [11] extends the criterion to a separable metric state space and identifies the resulting convergence as  $J_1$ -convergence of time-changed processes.

The point of Fact 8.65 is what it does *not* require: no modulus of continuity, no Aldous condition, no compact containment (12). A single first-moment bound on the drift, uniform in  $n$ , replaces the whole apparatus behind Remark 8.38. This is the reason to want the following.

**Theorem 8.66** (Lemma 8.34 for convergence in measure; (T3)). *Assume (E3), let  $A \subset C_b(E) \times C_b(E)$  and  $A_n \subset B(E) \times B(E)$ , and suppose that for each  $(f, g) \in A$  there are  $(f_n, g_n) \in A_n$  satisfying (96), or more generally (97). Let  $X_n$  solve the martingale problem for  $A_n$  with sample paths in  $D_E[0, \infty)$ , let  $X$  have sample paths in  $D_E[0, \infty)$ , and suppose*

$$X_n \Rightarrow X \quad \text{on } D_E[0, \infty) \text{ in the pseudo-path topology.}$$

*Then  $X$  is a solution of the martingale problem for  $A$ .*

*Proof.* The conclusion is a property of the law of  $X$  alone, and each hypothesis on  $X_n$  is a property of the law of  $X_n$  alone — for the martingale property this is Proposition 4.9. We may therefore replace the sequence by any subsequence, and may realize all the processes on whatever space is convenient. Fix  $(f, g) \in A$  with approximants  $(f_n, g_n)$ , and let  $Y_t^\circ(\omega) = f(\omega(t)) - \int_0^t g(\omega(u)) du$  and  $Y^{\circ, n}$  be as in the proof of Lemma 8.34.

*Step 1: an almost sure realization.* By Fact 8.64(ii) the pseudo-path topology on  $D_E[0, \infty)$  is separable metric, so Fact 2.32 provides  $D_E[0, \infty)$ -valued random variables  $\tilde{X}_n, \tilde{X}$  on one probability space, with the laws of  $X_n$  and  $X$  — the same laws on the same  $\sigma$ -field, by Fact 8.64(iii) — and  $\tilde{X}_n \rightarrow \tilde{X}$  a.s. in the pseudo-path topology. Being càdlàg and measurable in  $\omega$  for each  $u$ , each  $\tilde{X}_n$  is jointly measurable in  $(\omega, u)$ .

*Step 2: from convergence in measure to convergence a.e.* Write  $\rho$  for a metric on  $E$  bounded by 1. By Fact 8.64(i), for a.e.  $\omega$  the paths  $\tilde{X}_n(\omega, \cdot)$  converge to  $\tilde{X}(\omega, \cdot)$  in  $\lambda$ -measure, so  $\int \rho(\tilde{X}_n, \tilde{X}) d\lambda \rightarrow 0$  a.s.; the integrals being bounded by 1, dominated convergence gives  $E \int \rho(\tilde{X}_n, \tilde{X}) d\lambda \rightarrow 0$ , i.e.  $\tilde{X}_n \rightarrow \tilde{X}$  in  $P \otimes \lambda$ -measure. Pass to a subsequence along which the convergence is  $P \otimes \lambda$ -a.e. Fubini now gives both halves of what is needed:

- (a) there is  $I \subset \mathbb{R}_+$  of full Lebesgue measure —  $\lambda$  and Lebesgue measure being equivalent — such that  $\tilde{X}_n(r) \rightarrow \tilde{X}(r)$  a.s. for every  $r \in I$ ;
- (b) for a.e.  $\omega$  the paths converge Lebesgue-a.e., that is  $\tilde{X}_n(\omega, u) \rightarrow \tilde{X}(\omega, u)$  for a.e.  $u$ ; since  $g$  is bounded and continuous, dominated convergence then gives  $\int_0^t |g(\tilde{X}_n(\omega, u)) - g(\tilde{X}(\omega, u))| du \rightarrow 0$  for every  $t$ .

Let  $D \subset I$  be countable and dense; such a set exists because  $I$  has full measure, hence is dense, and  $\mathbb{R}_+$  is second countable.

*Step 3: the hypotheses of Theorem 8.40.* We verify (98) of Remark 8.41 together with (C2), (b) and (c), on the set  $D$ .

For (C2) take the determining set of Example 4.4, built from  $D$ . This is legitimate by part (iii) of Fact 8.64, which says that the  $\sigma$ -field in play is the one that example is about. Its elements are  $Z_s^\circ = \prod_{i \leq k} h_i(\pi_{t_i})$  with  $t_i \in D \cap [0, s]$  and  $h_i \in C_b(E)$ .

For (98), let  $r \in D$ . By (a) and continuity of  $f$ ,  $f(\tilde{X}_n(r)) \rightarrow f(\tilde{X}(r))$  a.s., and by (b) the integral term converges a.s. as well, so  $Y_r^\circ(\tilde{X}_n) \rightarrow Y_r^\circ(\tilde{X})$  a.s. Likewise  $Z_s^\circ(\tilde{X}_n) \rightarrow Z_s^\circ(\tilde{X})$  a.s., since every  $t_i$  lies in  $D \subset I$ . Products of a.s. convergent sequences converge a.s., hence

in distribution, which is (98). Note that this is where the argument differs from every earlier one in this section: the coordinates are nowhere continuous, so (a) is unavailable and *only* the weakened form (98) can be verified — and it is verified not by a continuous mapping theorem but by exhibiting the convergence on a common space.

(b) holds because  $|Y_r^\circ| \leq \|f\| + t\|g\|$  for  $r \leq t$ , and (c) is verified exactly as in Corollary 8.36, an estimate that involves no topology.

*Conclusion.* Theorem 8.40, as amended by Remark 8.41, gives the martingale identity on  $D$ . Now  $\mathbb{T} = \mathbb{R}_+$  satisfies (T2b) with the countable dense set  $D$  and has no greatest element, and  $Y$  is right continuous because  $X$  has paths in  $D_E[0, \infty)$ ; so the last clause of Theorem 8.40 upgrades the identity to all of  $\mathbb{T}$ .  $\square$

*Remark 8.67* (What convergence in measure costs). Four things, of which only the last is serious.

*The dense set is not explicit.* Under  $J_1$  one takes  $D$  to be the continuity set of the limit and knows by Fact 2.41 that its complement is countable. Here  $D$  is an unspecified countable subset of an unspecified Lebesgue-full set, and there is no analogue of Fact 2.41: [12], Theorem 5 gives full measure and no more. This costs nothing because Theorem 8.40 was written for an arbitrary dense  $D$  from the start, and the passage from  $D$  to  $\mathbb{T}$  is its Step 3 — the same regularization that Section 5 performs in general. Note the order of business: one does *not* first prove a limit theorem at all times and then regularize; one proves it a.e. and lets right continuity do the rest.

*The path space is not Polish.* It is separable metric, which is all that Facts 2.31 and 2.32 were stated for, and this manuscript never used more. It does mean that a formalization must have the continuous mapping theorem and the Skorokhod representation at that generality and not only for Polish spaces (Section 9.1).

*Càdlàg paths of the limit must be known separately.* In the pseudo-path topology  $D_E[0, \infty)$  is not closed, and the limit of pseudo-paths of càdlàg functions need not be one; this is why it is a hypothesis of Theorem 8.66 and a conclusion of Fact 8.65, which supplies it from the quasimartingale bound.

*Nothing about jumps passes to the limit.* Convergence in measure is blind to any functional that is not an  $du$ -average or an a.e.- $u$  evaluation:  $\sup_{u \leq t} \rho(X_n(u), x_0)$ , the jump measure, the number of jumps in  $[0, t]$ , the value at a prescribed time. So the mode is exactly right for identifying a limit through a martingale problem — which is an a.e.- $u$  statement, as this proof makes visible — and wrong for anything that reads the paths pointwise. Section 8.9 is the sharp illustration: the functional of Example 8.53 is the evaluation at the atom of the clock, so a clock with atoms is precisely a martingale problem that convergence in measure cannot see, and no augmentation repairs it, since Theorem 8.55 augments by coordinates and coordinates are what fail.

*Remark 8.68* (Three topologies, and what a formalization should do). Between  $J_1$  and convergence in measure sits Jakubowski's  $S$ -topology [9], which retains finite-dimensional convergence outside a *countable* set — so Fact 2.41 survives — and for which both Prohorov theorems hold, at the price of not being metrizable, only sequential. For the present purpose the ordering is clear. Theorem 8.40 in the form of Remark 8.41 asks for no topology at all, so the choice is not a choice about the theorem but about which convergence one can verify in a given application; and a non-metrizable topology is a poor thing to formalize, since Facts 2.31 and 2.32 would both have to be replaced. The recommendation is therefore to formalize Theorem 8.40 with (98) in place of (C1) and (a) — costing nothing, since its proof is unchanged — and to treat  $J_1$ , weak-strong convergence and the pseudo-path topology as three ways of discharging one hypothesis, none of which the theorem needs to

know about. [11] makes the relation between the outer two precise: convergence in the pseudo-path topology is  $J_1$ -convergence after a random time change, which is the same move as the clock change of Section 8.8 with the clock allowed to depend on the path.

### 8.11 Hawkes processes and their Volterra limit

This last subsection is a single example, and it is here because it is the first place where every strand of the manuscript is used at once: the explicit construction of Section 8.3, the localization of Section 6.3, the *non-Markovian* uniqueness of Section 6.1, and the convergence theorem of Section 8.6. Throughout,  $\mathbb{T} = \mathbb{R}_+$  and  $q$  is Lebesgue measure.

#### Jump processes with a path-dependent rate

Section 8.3 constructed jump processes from a rate  $\lambda(x)$  and a kernel  $\mu(x, \cdot)$  depending on the current state. Nothing in that construction needs the dependence to be Markovian, and one gain is worth the change of notation: the holding time ceases to be exponential, and the computation that carried Theorem 8.23 survives intact.

**Setting 8.69** (Path-dependent jump data; (T3) + (E0)). Let  $F$  be the set of piecewise constant right continuous paths  $\mathbb{T} \rightarrow E$  with finitely many jumps on bounded intervals, and let

$$\Lambda : \mathbb{T} \times F \rightarrow [0, \infty), \quad \mu : \mathbb{T} \times F \rightarrow \mathcal{P}(E)$$

be *predictable*:  $\Lambda(t, \omega)$  and  $\mu(t, \omega, \cdot)$  depend on  $\omega$  only through  $\omega|_{[0, t)}$ , and  $t \mapsto \Lambda(t, \omega)$  is locally integrable for every  $\omega$ . Put

$$\mathcal{A}_t f(\omega) = \Lambda(t, \omega) \int_E (f(y) - f(\omega(t-))) \mu(t, \omega, dy), \quad f \in B(E), \quad (107)$$

and let  $\mathcal{X}^\circ$  consist of the processes  $f(\pi_t) - \int_0^t \mathcal{A}_s f ds$ ,  $f \in B(E)$ .

**Definition 8.70** (The construction; (T3) + (E0)). Let  $\varepsilon_1, \varepsilon_2, \dots$  be i.i.d. exponential of parameter 1. Set  $\tau_0 = 0$ , let  $Y_0 \sim \nu$ , and given  $(\tau_n, Y_n)$  let  $\omega^{(n)}$  be the path built so far, continued constantly at  $Y_n$ ; put

$$\tau_{n+1} = \inf \left\{ t > \tau_n : \int_{\tau_n}^t \Lambda(u, \omega^{(n)}) du > \varepsilon_{n+1} \right\}, \quad (108)$$

and draw  $Y_{n+1} \sim \mu(\tau_{n+1}, \omega^{(n)}, \cdot)$ . Define  $X$  by (88) and  $\zeta = \lim_n \tau_n$  as before.

Writing  $A_n(u) = \int_{\tau_n}^u \Lambda(v, \omega^{(n)}) dv$ , the construction gives, conditionally on  $\mathcal{H}_n = \sigma(Y_0, \dots, Y_n, \tau_1, \dots, \tau_n)$ , the survival function  $P(\tau_{n+1} > u \mid \mathcal{H}_n) = e^{-A_n(u)}$  and the jump-time density  $\Lambda(u, \omega^{(n)}) e^{-A_n(u)}$ . For  $\Lambda$  constant this is the exponential law of Definition 8.21.

**Theorem 8.71** (Path-dependent jump processes; (T3) + (E0)). *Let  $X$  be as in Definition 8.70 and let  $N_t = \#\{n \geq 1 : \tau_n \leq t\}$  be its jump counter.*

- (a)  *$X$  solves the local martingale problem for  $\mathcal{X}^\circ$  of Setting 8.69 on  $[0, \zeta)$ , with initial law  $\nu$  and with  $(\tau_n)$  a localizing system in the sense of Definition 6.30. No hypothesis beyond Setting 8.69 is needed.*
- (b) *If  $E[N_t] < \infty$  for every  $t$  — equivalently, by (a),  $E[\int_0^t \Lambda(u, X) du] < \infty$  — then  $\zeta = \infty$  a.s. and  $X$  solves the global martingale problem for  $\mathcal{X}^\circ$ .*

*Remark 8.72* (Why the local statement is the primary one). In Setting 8.20 the bound  $\bar{\lambda} < \infty$  gave  $E[N_t] \leq \bar{\lambda}t$  for free, and the global statement was the natural one. With a path-dependent rate this is no longer so, and the failure is not about explosion: even when  $\zeta = \infty$  a.s., the compensator  $\int_0^t \Lambda(u, X) du$  need not be integrable, so  $f(X_t) - \int_0^t \mathcal{A}_s f ds$  need not lie in  $L^1$  and the phrase “is a martingale” has no content. Stopping at  $\tau_n$  repairs this *unconditionally*, because

$$\int_{\tau_k}^{\tau_{k+1}} \Lambda(u, \omega^{(k)}) du = \varepsilon_{k+1} \quad \text{on } \{\tau_{k+1} < \infty\} \quad (109)$$

by the very definition (108) of  $\tau_{k+1}$ , so that  $\int_0^{t \wedge \tau_n} \Lambda du \leq \varepsilon_1 + \dots + \varepsilon_n$  is integrable. This is a clean instance of the phenomenon Definition 6.30 exists for, and unlike Remark 8.26 it is not about paths leaving every compact set.

*Proof.* The decomposition of Step 1 in the proof of Theorem 8.23 is unchanged, so the first task is to show  $E[D_n \mid \mathcal{H}_n] = 0$  on  $\{\tau_n \leq t\}$  for

$$D_n = (f(Y_{n+1}) - f(Y_n)) \mathbf{1}_{\{\tau_{n+1} \leq t\}} - \int_{\tau_n \wedge t}^{\tau_{n+1} \wedge t} \mathcal{A}_s f ds.$$

Write  $\Lambda_u = \Lambda(u, \omega^{(n)})$  and  $\mu_u f = \int f d\mu(u, \omega^{(n)}, \cdot)$ , and note that on  $\{\tau_{n+1} > s\}$  the path agrees with  $\omega^{(n)}$ , so that  $\mathcal{A}_s f = \Lambda_s(\mu_s f - f(Y_n))$  there. Then

$$\begin{aligned} E[(f(Y_{n+1}) - f(Y_n)) \mathbf{1}_{\{\tau_{n+1} \leq t\}} \mid \mathcal{H}_n] &= \int_{\tau_n}^t \Lambda_u e^{-A_n(u)} (\mu_u f - f(Y_n)) du, \\ E\left[\int_{\tau_n}^{\tau_{n+1} \wedge t} \mathcal{A}_s f ds \mid \mathcal{H}_n\right] &= \int_{\tau_n}^t \Lambda_s (\mu_s f - f(Y_n)) P(\tau_{n+1} > s \mid \mathcal{H}_n) ds, \end{aligned}$$

the first because the jump time has density  $\Lambda_u e^{-A_n(u)}$  and the jump distribution at time  $u$  is  $\mu(u, \omega^{(n)}, \cdot)$ , the second by Fubini. Since  $P(\tau_{n+1} > s \mid \mathcal{H}_n) = e^{-A_n(s)}$ , the two expressions *coincide*, and  $D_n = 0$  on  $\{\tau_n > t\}$ . Both displays are finite because  $|\mu_u f - f(Y_n)| \leq 2\|f\|$  and  $\int_{\tau_n}^t \Lambda_u e^{-A_n(u)} du \leq 1$ .

(a). Stop at  $\tau_m$ . Then (89) has only the  $m$  summands  $D_0, \dots, D_{m-1}$ , each integrable by the previous paragraph, so no domination argument is needed and  $E[M_{t \wedge \tau_m}] = 0$  for every  $t$ . For the increment from  $s$  to  $t$ , observe that conditionally on  $\mathcal{F}_s^\circ$  and on  $\{N_s = n\}$  the residual jump time satisfies

$$P(\tau_{n+1} > u \mid \mathcal{F}_s^\circ) = \exp\left\{-\int_s^u \Lambda(v, \omega^{(n)}) dv\right\}, \quad u \geq s,$$

by (108) and the lack of memory of  $\varepsilon_{n+1}$  — which here concerns the *compensator*  $A_n$ , not the time; and that the subsequent construction is (108) again, with  $\tau_n$  replaced by  $s$  and with the past  $\omega|_{[0, s]}$  carried along. The computation above then applies with  $\tau_n$  replaced by  $s$  and  $\mathcal{H}_n$  by  $\mathcal{F}_s^\circ$ , and gives  $E[M_{t \wedge \tau_m} - M_{s \wedge \tau_m} \mid \mathcal{F}_s^\circ] = 0$ . Note that the restart carries the whole past, not merely  $X_s$ : this is Lemma 6.5, and it is the reason Theorem 8.71 needs no shift system. Since  $\tau_m \uparrow \zeta$  and each  $\tau_m$  is a functional of the path — indeed a *strict* stopping time, being the  $m$ -th jump time of a right continuous piecewise constant path —  $(\tau_m)$  is a localizing system, with (L1) holding by construction, (L2) because the  $m$ -th jump time after  $r$  is a jump time, and (L3) by (109).

(b). Assume  $E[N_t] < \infty$ . Then  $\tau_m \uparrow \infty$  a.s., so  $\zeta = \infty$ . Moreover

$$\sum_{n \geq 0} |D_n| \leq 2\|f\|N_t + 2\|f\| \int_0^t \Lambda(u, X) du,$$

and by (109) the last integral equals  $\sum_{k < N_t} \varepsilon_{k+1} + \int_{\tau_{N_t}}^t \Lambda \, du \leq \sum_{k \leq N_t} \varepsilon_{k+1}$ , whose expectation is  $E[N_t] + 1 < \infty$  by Wald's identity, the  $\varepsilon_k$  being i.i.d. of mean 1 and  $N_t + 1$  a stopping time for their natural filtration. So dominated convergence applies to (89), and letting  $m \rightarrow \infty$  in (a) turns the local statement into the global one.  $\square$

*Remark 8.73.* The cancellation is the same one as in Theorem 8.23: the rate in the generator against the survival function of the holding time. What has gone is only the Markov structure —  $\Lambda$  and  $\mu$  may depend on the whole past — and with it the shift system. So Theorem 8.71 produces solutions to which Proposition 6.13 applies and Theorem 6.23 does not, which is the situation Remark 6.50 describes and which has so far been illustrated only by the cited Examples 6.48 and 6.49.

**Example 8.74** (Hawkes processes). Take  $E = \mathbb{N}_0$ ,  $\mu(t, \omega, \cdot) = \delta_{\omega(t-)+1}$  and

$$\Lambda(t, \omega) = \mu_0 + \int_{[0,t)} \phi(t-s) \, d\omega_s, \quad \mu_0 > 0, \quad \phi : \mathbb{R}_+ \rightarrow [0, \infty), \quad (110)$$

with  $\phi$  locally integrable. Then  $X = N$  is a *Hawkes process*, and Theorem 8.71 says that  $f(N_t) - \int_0^t \Lambda_s(f(N_s + 1) - f(N_s)) \, ds$  is a martingale for every bounded  $f$ ; taking  $f = \text{id}$  on bounded sets recovers the defining property that  $N - \int_0^\cdot \Lambda_s \, ds$  is a martingale.

A *linear Hawkes process never explodes*. The mean intensity  $m(t) = E[\Lambda_t]$  satisfies the renewal equation  $m = \mu_0 + \phi * m$ , and a Volterra kernel  $\phi \in L_{\text{loc}}^1$  has a resolvent in  $L_{\text{loc}}^1$  whatever its mass: on an interval short enough that  $\int_0^\delta \phi < 1$  the Neumann series converges, and one steps forward. So  $E[N_t] = \int_0^t m < \infty$  for *every*  $t$  and every  $\phi \geq 0$ , and Theorem 8.71(b) always applies. An earlier version of this manuscript asserted the opposite for  $\|\phi\|_1 \geq 1$ ; that was a confusion of exponential growth with explosion.

What  $\|\phi\|_1 = 1$  separates is not existence but *stability*: for  $\|\phi\|_1 < 1$  the bound is uniform in  $t$ ,  $\sup_t E[\Lambda_t] \leq \mu_0/(1 - \|\phi\|_1)$ , and for  $\|\phi\|_1 > 1$  it is not,  $m$  growing exponentially. The scaling limit below lives at  $\|\phi\|_1 \uparrow 1$ , where the two regimes meet.

## The Volterra limit

The observation that makes the limit theorem an application rather than an analogy is that (110) is a Volterra equation.

*Remark 8.75* (A Hawkes process is a Volterra SDE). Put  $Z = N$  and  $X = \Lambda$ . Then (110) reads

$$X_t = g_0(t) + \int_{[0,t)} K_{t-s} \, dZ_s, \quad g_0 \equiv \mu_0, \quad K = \phi,$$

which is verbatim the constraint of Example 6.48, and  $Z$  is a semimartingale whose characteristics are driven by  $X$ :  $a = 0$  and  $\nu(x, dy) = x \delta_1(dy)$ . A Hawkes process is therefore a solution of a Volterra stochastic differential equation with pure-jump noise, and the pair  $(\Lambda, N)$  lies in the path space  $L_{\text{loc}}^p(\mathbb{R}_+, \mathbb{R}) \times D_{\mathbb{R}}[0, \infty)$  of Example 6.48. This identification is standard: Hawkes processes are the pure-jump members of the affine Volterra class of [14], and Section 8.2 uses their transform formula (Remark 8.16). Approximants and limit below are thus elements of one and the same solution set, for two triplets of the same shape — one purely discontinuous, one continuous.

**Theorem 8.76** (Hawkes to Volterra, conditionally on tightness; (T3) + (E3)). *For each  $n$  let  $(\Lambda^n, N^n)$  be a Hawkes process with kernel  $\phi^n$  and baseline  $\mu_0^n$ , rescaled in time, and let  $(X^n, Z^n)$  be the corresponding pair of Remark 8.75. Let  $\mathcal{X}^\circ$  be the Volterra family of Example 6.48 for a kernel  $K$ , a triplet  $(b, a, \nu)$  and  $g_0$ . Suppose*

- (a)  $(X^n, Z^n) \Rightarrow (X, Z)$  weakly on  $L^p_{\text{loc}}(\mathbb{R}_+, \mathbb{R}) \times D_{\mathbb{R}}[0, \infty)$ ;
- (b) the rescaled generators converge in the sense of (c), i.e. for  $f \in C_b^2$  and  $s \leq t$  in a dense  $D$ ,  $\lim_n E[(\int_s^t (\mathcal{A}_u^n f - Lf(X_u^n, Z_u^n)) du) Z_s^\circ(X^n, Z^n)] = 0$ ;
- (c)  $\{f(Z_r^n) - \int_0^r Lf(X_u^n, Z_u^n) du : r \in D \cap [0, t], n \in \mathbb{N}\}$  is uniformly integrable.

Then  $(X, Z)$  solves the Volterra martingale problem  $\mathcal{X}^\circ$ . If moreover that problem propagates agreement in the sense of Definition 6.11 — uniqueness in law from any prescribed past — then  $(X, Z)$  is determined, and the whole sequence converges to it.

*Proof.* Hypotheses (a)–(c) are (C1), (c) and (b) of Theorem 8.40; (C2) is Example 4.4 on the second coordinate together with the integrated form of the Volterra constraint, and (a) holds because the first coordinate enters only through Lebesgue integrals — which is why Example 6.48 states the constraint in integrated form — and the second through evaluations at times where the limit does not jump. The approximating martingales required by (c) are those of Theorem 8.71. Theorem 8.40 then gives  $(X, Z) \in \mathcal{M}(\mathcal{X}^\circ)$ . The final assertion is Proposition 6.13, whose hypothesis (32) is uniqueness in law of the Volterra equation restarted from a prescribed past; it upgrades to uniqueness of the whole law with no Markov structure — which is essential, since a Volterra process has none. See Example 6.48 and Remark 6.12.  $\square$

*Remark 8.77* (What is proved here and what is cited). Hypothesis (a) is tightness, and it is not supplied — here as everywhere in this section (Remark 8.52). For the Hawkes scaling it is the substance of the theorem: the interesting regime is  $\|\phi^n\|_1 \uparrow 1$  with a heavy-tailed  $\phi$ , the kernel rescales to  $K(t) = t^{\alpha-1}/\Gamma(\alpha)$ , and the limit is a rough stochastic Volterra equation. That is the content of [13], and it is cited, not proved; the present theorem is the identification step, and it is the part the abstract machinery does supply.

Three remarks on scope. First, *neither weak-strong convergence nor the augmentation of Theorem 8.55 is needed*: the clock is Lebesgue, so there are no atoms, the jumps of  $N^n$  have height one and the limit is continuous; the limit therefore has no fixed times of discontinuity and (a) holds on a dense set with countable complement. Section 8.9 stays out of it entirely. Second, *the clock does not change* if one rescales the continuous-time Hawkes process, so Theorem 8.40 suffices and Section 8.8 is not used; it would be, if one approximated instead by the discrete branching representation, whose clock is atomic. Third, and this is the point of the example: *no result of Section 6.2 is used anywhere*. Neither the approximants nor the limit has a shift system, and none is wanted.

## 9 Notes for the formalization

### 9.1 What is available and what has to be built

Section 2 lists the prerequisites; the following is a stocktaking of which of them Mathlib already supplies.

*Available.* Filtrations, adapted and progressively measurable processes, stopping times and  $\mathcal{F}_\tau$  (Definition 2.18, Fact 2.19); martingales and the optional stopping theorem (Definition 2.20, Fact 2.25) — and note that `MeasureTheory.Martingale` is stated for values in an arbitrary real Banach space, so it covers  $\mathbb{K} = \mathbb{C}$  with nothing to do; Doob’s inequalities (Fact 2.26); Polish spaces; weak convergence of measures with the portmanteau theorem, the Lévy–Prokhorov metric and Prohorov’s theorem (Facts 2.30, 2.32, 2.33); uniform integrability (Fact 2.45); monotone class theorems (Fact 2.46); the Riesz–Markov–Kakutani

representation of a positive linear functional on compactly supported continuous functions, in the form `integral_rieszMeasure`, which is what Proposition 8.7 needs; and part of what Fact 2.47 needs — see the list there.

*To be built.*

- the Skorokhod space with the  $J_1$  metric, and the facts that it is Polish, that its Borel field is generated by the coordinates, and the characterization of its compact sets. These are no longer cited but built: Section 3, and Remark 3.19 there is the stocktaking for them;
- regularization of submartingales along a dense set (Facts 2.21, 2.22) — but see Remark 9.1: this exists already, outside Mathlib;
- local martingales in the [7] sense (Definition 2.27, Fact 2.28);
- separating and convergence determining classes and their Stone–Weierstrass theory (Definition 2.37, Facts 2.38, 2.39, 2.40, 2.48) — the separating half of 2.40 only, its path space half being Theorem 3.13;
- bounded pointwise convergence and bp-closures (Definition 2.34, Fact 2.35) — but only for Corollary 4.14, which is itself optional: the working statement is Lemma 4.12, and that is dominated convergence;
- the continuous mapping theorem in the form of Fact 2.31 and the Skorokhod representation (Fact 2.32);
- convergence of finite-dimensional distributions in  $D_E[0, \infty)$  and the relative compactness criteria (Facts 2.42, 2.43, 2.44);
- dissipative operators and the full generator of a measurable contraction semigroup (Definitions 2.14, 2.15, Fact 2.16); these are needed only for Proposition 4.19 and Remark 4.20, which are themselves optional context;
- the Kolmogorov extension theorem for an uncountable index (Fact 2.47). Mathlib has the scaffolding and the sequential case but not the theorem; this is the one external prerequisite of Section 8.2, and for  $\mathbb{T} = \mathbb{N}_0$  it is already there.

*Remark 9.1* (Prior art: the `brownian-motion` project). The càdlàg half of (F7) is already formalized, not in Mathlib but in the Lean development accompanying [1], in the repository `RemyDegenne/brownian-motion` on GitHub. Its file `Cadlag.lean` defines `IsRightContinuous` for a preorder and `IsCadlag` as a structure, and develops jump sets, the discreteness of large jumps, local boundedness and the usual closure properties, with no remaining gaps. Its `CadlagModification.lean` then constructs the càdlàg modification; the blueprint chapter proves that a submartingale has a càdlàg modification if and only if  $t \mapsto E[X_t]$  is right continuous, and that a martingale always has one. These are precisely Facts 2.21 and 2.22, and they are proved for *quasimartingales*, which is more general than what Theorem 5.4 needs.

Three consequences for the present plan. First, (F7) loses its most expensive prerequisite. Second, the index there is a preorder with an order topology, which agrees with the bundle (T2b) of Definition 2.1 — the two developments make the same choice independently. Third, what is *not* there is the Skorokhod space: no  $J_1$  metric, no  $\mathcal{B}(D_E[0, \infty)) = \sigma(\pi_t)$ , no compactness criterion. So (F8) is untouched there, and Theorems 3.12 and 3.15 are proved here instead.

This dictates a natural order of work. Each item names the bundle of Definition 2.1 it targets; the point of doing so is that (F1), (F4) and half of (F3) live under [Preorder  $\iota$ ], which is how Mathlib already indexes `Filtration` and `Martingale`, and can therefore be stated without any bespoke index theory.

- (F1) *Targets (T0) + (E0) + clock.* Definitions only: Definition 4.2, Definition 4.6, the filtration (4), Definition 4.21 (which needs (T2b)), and Proposition 4.9. The clock is a `MeasureTheory.Measure` on  $\mathbb{T}$  together with the hypothesis that every down-set is measurable and of finite mass; the interval algebra (2) is then definitional. These need nothing beyond what Mathlib has and should be done first; note that Proposition 4.9 is what turns every subsequent statement into a statement about finite-dimensional distributions.
  
- (F2) *Targets (T3) + (E0).* The jump processes of Section 8.3. Theorem 8.23 needs a Markov chain, i.i.d. exponential variables and the computation of Step 2, and nothing else; Proposition 8.24 adds the Picard iteration (93) for the bounded operator  $A$ , which in Lean is the exponential series of a bounded linear map and needs no analysis beyond `NormedSpace`. It is by a wide margin the cheapest theorem with real content in this manuscript, it is the only one that exhibits a solution, and it supplies the missing example for Definition 6.30. It is placed here, before any of the theory, as a way of testing the definitions of (F1) against something concrete.
  
- (F3) *Targets (T0) + (T4) + (E0); (T2a) for the uniqueness half, (E1) for Lemma 6.3.* Lemma 6.2, Lemma 6.17 and Theorem 6.23, then Theorem 6.51 and Corollary 6.52 as corollaries. Modulo (17) this is pure measure theory, and Theorem 6.23 needs no path regularity at all, so it is the best entry point after (F1). Almost all of the work sits in Lemma 6.17, which is four lines; the two theorems that follow are bookkeeping. Theorem 6.27 additionally needs Fact 2.25 and càdlàg paths and should be deferred until after (F7), and so should the local theory of Section 6.3 — Definition 6.30 is about stopping times. Note that Lemma 6.32(a) and Lemma 6.34 are the only genuinely new proofs there; Lemma 6.37 is Lemma 6.17 plus the one identity (44), and Theorem 6.38 is bookkeeping.
  
- (F4) *Targets (T0) + (E3), with  $D = \mathbb{T}$ .* Theorem 8.40, the abstract convergence theorem. It needs an abstract Polish path space, a determining set,  $P$ -continuity at  $X$ , uniform integrability and Vitali's theorem — and *no* Skorokhod topology, no càdlàg paths and no Markov structure. It is placed here, immediately after (F3), and not with the Skorokhod theory of (F8) where the ordering of the literature would put it: it is one of the cheapest genuine theorems in this list, not part of the most expensive one. Take Lemma 8.54 and Theorem 8.55 with it; they are two lines each once Theorem 8.40 is in place.
  
- (F5) *Targets (T0) + (E0) for Lemma 7.1, (T2a) +  $\iota = p$  for Corollary 7.4 (which needs (T3) and an atomless clock), (T3) + (E2) for the rest.* Lemma 7.1 — an exact telescoping identity, and the cheapest item in this list — then Proposition 7.3, Corollary 7.4, Corollary 7.16, and finally Lemma 7.11, Theorem 7.12 and their corollaries. Self-contained: Fubini, absolute continuity, dominated convergence. No Skorokhod space is needed. Note that Corollary 7.16 needs none of the analysis and can be formalized directly after Lemma 7.1. Together with (F3) this already yields Corollary 7.20, a complete uniqueness theory.

- (F6) *Targets (T0) + (T4) + (E1) + a shift invariant clock.* Section 8.2: Lemma 8.3, Proposition 8.4 and Theorem 8.5. Everything in the proofs is the Markov property, Fubini and (3). The one external input is Fact 2.47, and it is a real one: Mathlib has the scaffolding and the sequential Ionescu–Tulcea case but not the Kolmogorov extension theorem for an uncountable index. So this item splits: for  $\mathbb{T} = \mathbb{N}_0$  it can be done at once, for  $\mathbb{T} = \mathbb{R}_+$  it waits on that theorem. It uses (F3) for Theorem 6.23 and (F5) for Corollary 8.6, but Theorem 8.5 itself needs neither. Proposition 8.7 is separate; Mathlib’s `integral_rieszMeasure` should carry it, but the theorem is stated with (D3) as a hypothesis so that it can be skipped.
- (F7) *Targets (T2b) + (E2),  $\iota = 0$ .* Fact 2.21 and Fact 2.22, then Theorem 5.4 and, as a one-line corollary, Theorem 5.6. This requires the notion of a càdlàg path but *not* the Skorokhod topology. Formalize the abstract theorem, not the Markovian one: Definition 5.2 is three hypotheses on a pair of processes, whereas  $\mathcal{D}(A)$  drags in the operator, its domain and the filtration (4), none of which the proof uses.
- (F8) *Targets (T3) + (E3).* The Skorokhod space and its topology, then Lemma 8.34, Remark 8.38 and Theorem 8.44 as instances of (F4). This is by far the largest item. Theorem 8.55 belongs with (F4), not here: it is Lemma 8.54 plus a change of the ambient space, costs nothing once (F4) is done, and needs no Skorokhod theory. Only Proposition 8.56, which instantiates it for  $F = D_E[0, \infty)$ , does. Section 8.9 beyond that — Definitions 8.59 and 8.60 and Fact 8.61 — should be skipped: nothing in this manuscript needs it, by Remark 8.58.

## 9.2 Design decisions

- (a) *Operators as relations.*  $A$  is a subset of  $M(E) \times M(E)$ , not a function. In Lean this is  $A : \text{Set } ((E \rightarrow R) \times (E \rightarrow R))$  together with measurability side conditions, or a structure bundling the pair with its measurability. Keeping  $A$  multivalued costs nothing and is needed for Remark 4.11(b).
- (b) *Two notions of solution.* Definition 4.6 defines “solution with respect to  $(\mathcal{G}_t)$ ” as the primitive and “solution” as the special case  $\mathcal{G}_t = \mathcal{F}_t^X$ . Doing it the other way round duplicates every statement.
- (c) *Compact containment as a predicate.* Definition 4.21 occurs in Theorem 5.6, in Remark 8.38 and implicitly in Remarks 6.53 and 8.45. It should be one definition with the two variants (27) (over  $D$ , single process) and (12) (over  $[0, T]$ , family) related by a lemma for right continuous processes.
- (d) *Local martingale problems: localize the test processes, not the operator.* Define the local martingale problem at the level of Definition 4.2 and localize by stopping the elements of  $\mathcal{X}^\circ$ , as Definition 6.30 does. The alternative — replacing  $A$  by  $A^{(m)} = \{(f\mathbf{1}_{K_m}, g\mathbf{1}_{K_m})\}$  and passing to the limit, which is [7], Section 4.6 and Remark 6.53 — should *not* be the primitive: in the abstract setting  $\mathcal{X}^\circ$  is primitive and  $A$  is not, so the operator route has nothing to modify. See Remark 6.40. Concretely, the reusable object in Lean is the structure (L1)–(L3) of Definition 6.30, and the reusable lemma is Lemma 6.34, which *constructs* (L1) rather than assuming it. Note that the times in  $\Sigma$  must be *strict* stopping times; Remark 6.35 is the trap, and in Lean it is the difference between `IsStoppingTime` for the raw and for the right continuous filtration.

- (e) *Separating classes.* Facts 2.48 and 2.40 should be developed independently of martingale problems; they are of general interest, and within Sections 5 and 6 they are the only place where Polishness of  $E$  is really used. (In Section 8 it is used throughout, and irreducibly: (E3) is what makes Prohorov’s theorem available.)
- (f) *No semigroup theory.* The manuscript rests on Chapters 2 and 3 of [7] alone, as envisaged in the project plan; of Chapter 1 only the notions of a dissipative operator and of the full generator are borrowed, and only for the optional Proposition 4.19 and Remark 4.20. See Remark 2.17. Section 8.2 does not disturb this: the family  $(P_t)$  there is a semigroup of *kernels*, and the balance condition (D1) is assumed in integrated form, so no strong continuity, no generator and no domain theory are needed — which is one reason for stating it that way rather than as [3] do.
- (g) *Both conventions, carried and not chosen.*  $\iota \in \{\text{o}, \text{p}\}$  should be a parameter of the compensator, not a fixed choice. The evidence is in Remark 8.51: the same Markov chain needs  $\text{p}$  on  $\mathbb{T} = \mathbb{N}_0$  and  $\text{o}$  once its grid is embedded in  $\mathbb{R}_+$ . Everything except Section 5 is convention-agnostic, so the parameter costs a hypothesis in one section and nothing anywhere else.
- (h) *The scalar field is RCLike.* Test processes take values in  $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$  (Remark 4.5), and the Mathlib class for this is `RCLike` — `[RCLike  $\mathbb{K}$ ]`, with `re`, `im` and the `star` operation for the conjugation that Fact 2.39 needs. Two points are worth separating. The *martingale* property needs nothing: Mathlib’s `MeasureTheory.Martingale` is already stated for an arbitrary real Banach space, which is more general than `RCLike`, and Mathlib’s conditional expectation API is itself `RCLike`-parametrized. What *does* want a field is everything that *multiplies* test processes: Definition 4.3, where  $Y_t^\circ$  is paired with  $Z_s^\circ$ ; Proposition 4.9, with its products  $\prod_k h_k$ ; and Section 7, where the chain identity is scalar. So `RCLike` is the right class for the *test functions* and a Banach space would do for the *martingales*; formalize the first with `RCLike` and inherit the second. The places needing  $\mathbb{K} = \mathbb{R}$  are listed in Remark 4.5, and the design point is that they are *prerequisites*, not statements: the retreat to  $\mathfrak{R}$  and  $\mathfrak{S}$  happens in exactly one lemma, Corollary 2.23, so in Lean exactly one proof destructs the scalar field and everything above it is uniformly `RCLike`.
- (i) *The state space is fibred.* Take  $E : \mathbb{T} \rightarrow \text{Type}^*$  with  $[\forall \mathfrak{t}, \text{MeasurableSpace } (E \mathfrak{t})]$  and the path space inside  $\prod \mathfrak{t}, E \mathfrak{t}$ , not a fixed  $E$ . The justification is Remark 2.12; the practical points are that the pairs  $(f, g)$  become sections (5), that Sections 5 and 8 should be stated for a *constant* fibre rather than carrying an index they immediately freeze, and that Theorem 6.27 needs the total space  $\Sigma E = \Sigma \mathfrak{t}, E \mathfrak{t}$  because  $X(\tau)$  is a dependent value. Nothing in the abstract layer changes at all, since it never mentions the state space.
- (j) *Augmentation before weak-strong convergence.* Where a functional fails to be continuous at deterministic times — the atoms of the clock — enlarge the path space (Theorem 8.55) rather than weaken the mode of convergence. This keeps Section 8 inside the continuous mapping theorem and Vitali’s theorem, and avoids formalizing Fact 8.61. See Remark 8.58.

## 10 What this manuscript adds to its sources

This section is an audit, not an advertisement. The criterion throughout is narrow: an item is listed only if the statement here is strictly stronger than, or absent from, [7], [2], [10], [16] and [3], and each entry says who uses it. That last column is the discipline that matters. A generalization nobody consumes is not thereby worthless — it may be exactly what a later application needs — but it should not be mistaken for a result, and Section 10.5 lists four that were.

### 10.1 Results that are not in the sources

The three characterization results of [7], Chapter 4 are proved here in a setting with no operator in it, and [2] — who work in that setting — prove none of them. For the càdlàg theorem the omission in [2] is structural rather than accidental: their standing setting presupposes a Polish path space and right continuous test processes, which is what a regularization theorem has to produce.

| Here                              | Nearest in the sources                              | What is new                                                                                                                              |
|-----------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Definition 5.2, Theorem 5.4       | [7] 4.3.6, for an operator                          | the hypothesis is isolated as three conditions on a pair of processes; the compensator may be path dependent and may have atoms          |
| Definition 6.11, Proposition 6.13 | [7] 4.4.2, for an operator                          | uniqueness is proved for an arbitrary family of measures, with no shift, no operator and no Markov structure                             |
| Lemma 6.17, Theorem 6.23          | [7] 4.4.2                                           | four constructions in [7] become one lemma applied four times; the shifted problem is carried as a family, so time inhomogeneity is free |
| Lemma 7.1                         | [7] 4.4.10, on $\mathbb{R}_+$ with Lebesgue measure | the analytic core of duality as an exact telescoping over an arbitrary staircase, on a pre-order, with no probability in it              |
| Lemma 6.26                        | [7] 2.2.13, at bounded times                        | optional sampling at $\tau + s$ for an a.s. finite $\tau$ , from integrability of the increment over the <i>random</i> window alone      |
| Lemma 6.20                        | —                                                   | one-sided determination: nonnegativity on a generating semiring containing $\Omega$                                                      |
| Theorem 8.55, Proposition 8.56    | [2] 3.14, with control variables                    | for discontinuities at <i>deterministic</i> times, enlarging the path space replaces weak-strong convergence entirely                    |
| Lemma 8.10, Corollary 8.18        | [3], Section 5 (outlook)                            | kernel representability yields a <i>restart kernel</i> , hence local uniqueness for a path-dependent problem with no shift system        |

Two of these have a consumer inside the manuscript and the rest are terminal. Lemma 6.26 carries Theorem 6.27; Lemma 6.20 carries Remark 6.19. Theorem 8.55 is used by nothing here and is nevertheless the item this manuscript would most defend, because it removes a whole apparatus from the one place where the apparatus looked unavoidable.

## 10.2 The split that organizes everything: uniqueness is not Markovian

If one thing here is worth carrying elsewhere it is this. [7], Theorem 4.4.2 proves uniqueness *and* the Markov property from one hypothesis, and reads as though the two were the same fact. They are not.

|                                                | needs              | Markovian? |
|------------------------------------------------|--------------------|------------|
| Uniqueness (Proposition 6.13)                  | (32)               | <b>no</b>  |
| Restarting (Lemma 6.5)                         | the tower property | <b>no</b>  |
| (32) from one-dimensional laws<br>(Lemma 6.22) | a shift system     | yes        |
| Markov property (Theorem 6.23(a))              | a shift system     | yes        |

Two consequences that the classical formulation hides. First, *restarting a solution is free* (Lemma 6.5): one may condition on any event observed up to  $r$  and what remains is again a solution. What a shift buys is not the restart but the *re-indexing* of the restarted object to start at time 0, which an unconditional hypothesis needs and a conditional one does not; see Remark 6.6. Second, the conditional form (32) is available in genuinely non-Markovian situations — path-dependent equations, changes of measure, semimartingale problems — where no shift system exists, and Section 8.11 exhibits one.

The negative half is sharper still and is not in the sources at all: Remark 6.25 shows that on a partially ordered index the slogan “uniqueness of the one-dimensional distributions implies uniqueness” is *false*, and locates the obstruction in the induction rather than in the martingale problem.

### 10.3 Generalizations that cost nothing

These are free in the sense that no proof got longer; several are free in the stronger sense that no proof changed at all. The right-hand column is the one to read sceptically.

| Axis                    | Sources                                       | Here                                        | Used by                                                                                                                                           |
|-------------------------|-----------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Time index              | $\mathbb{T} = \mathbb{R}_+$                   | (T0)–(T4),<br>Definition 2.1                | the poset case really survives in Section 4, in Theorem 6.23(a), in Theorem 8.40 and in the kernel layer of Section 8.2                           |
| Compensator             | $\int_0^t g(X_s) ds$                          | a <i>clock</i> , Definition 2.3             | discrete time (Corollary 7.16, Remark 8.27) and fixed times of discontinuity (Section 8.9) become instances, not analogues                        |
| Convention $\iota$      | invisible, the clock being atomless           | both carried                                | Section 5 needs o; Example 8.50 needs o after embedding and p before it (Remark 8.51)                                                             |
| State space             | Polish                                        | (E0)–(E3),<br>Definition 2.7                | martingale problems for $\mathcal{S}'(\mathbb{R}^d)$ -valued processes are covered by Sections 4–7 with no new proof (Remark 2.9)                 |
| State space, fibred     | fixed $E$                                     | $E_t$ varying,<br>Definition 2.10           | forced by the historical process of Section 8.2; buys the space-time martingale problem (5)                                                       |
| Scalar field            | $\mathbb{R}$                                  | $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ | the form the transform literature uses; Theorem 7.12 needs no change at all (Remark 4.5)                                                          |
| Local problems          | a variant                                     | available from Definition 4.2 on            | primary, not optional, for path-dependent rates (Remark 8.72)                                                                                     |
| Path space              | $D_E[0, \infty)$ with $J_1$ , Polish          | any measurable $F$ ;<br>(98)                | Remark 8.41: (C1) and (a) are one hypothesis about finitely many real random variables, so Theorem 8.40 is topology-free and Theorem 8.66 follows |
| Mode of convergence     | $J_1$                                         | also convergence in measure                 | Section 8.10; [12] tightness needs no compact containment                                                                                         |
| Approximating operators | $\ f_n - f\  \rightarrow 0$                   | (97), along the paths                       | Corollary 8.36: locally uniform convergence suffices under (12), which is assumed anyway                                                          |
| Existence from a dual   | [3]: $\mathbb{R}_+$ , Lebesgue, $E_1$ compact | any shift invariant clock, (E1), fibred     | Section 8.2; the compactness of [3] survives only in Proposition 8.7, i.e. in <i>checking</i> (D3)                                                |

Three of these deserve a sentence beyond the table.

*The index bundles pay for themselves by what they exclude.* Separating (T2a) from (T2b) makes visible that Section 5 needs a topology on  $\mathbb{T}$  and Sections 6 and 7 need an algebraic property, which  $\mathbb{R}_+$  happens to supply at once; and it turns up the fact that (T1'), the metric lattice that [7], Section 2.8 builds and that the first draft of this manuscript took as the base assumption, is used *nowhere*.

*The two conventions are not bookkeeping.* Remark 8.51 is the sharpest evidence: the same Markov chain needs  $\iota = p$  intrinsically on  $\mathbb{T} = \mathbb{N}_0$  and  $\iota = o$  once its grid is embedded in  $\mathbb{R}_+$ , because with  $\iota = p$  an atom at  $u$  is compensated *after*  $u$  while  $f(X)$  jumps *at*  $u$ . A manuscript committed to one convention could not state Example 8.50.

*Free is not the same as empty, but it is close to it in one case.* The complex scalar field changed no proof and no hypothesis; its value is entirely that it lets Section 7 be applied in the form the affine and affine Volterra literature actually uses. Whether that counts as a generalization or as a change of font is a matter of taste.

## 10.4 Gaps and slips in the sources

- [7], Chapter 3, Problem 7 is stated as an exercise and used in the last step of their Theorem 4.3.6. Remark 2.49 proves it, and records that the countability the argument needs comes from the state space and not from the separating class.
- [7], Section 2.8 is easy to misread in three ways, and Remark 2.6 sets them out: the lattice is needed for  $\vee$  and not for  $\wedge$  (their Remark 8.3 warns that  $\tau \wedge a$  need not be a stopping time);  $\mathcal{F}_\tau$  needs no infima; and the standing lattice hypothesis is declared before a series of definitions none of which uses it.
- [7], (3.4) quantifies the finite-dimensional criterion over chains  $t_1 \leq \dots \leq t_n \leq s$ . On a preordered index that is too weak, as Remark 4.10 shows, and the proof of Proposition 4.9 needs arbitrary finite subsets of  $\mathbb{T}_{\leq s}$  — the times produced by the Fubini step cannot be chained.
- *Bounded pointwise closure* ([7] 4.3.1) is more than the argument uses. Lemma 4.12 is dominated convergence, applies to unbounded  $f, g$ , and delivers the martingale property for the filtration at hand rather than only for (4); Remark 4.15 isolates what the bp-apparatus is for.
- *Linear Hawkes processes do not explode*, whatever  $\|\phi\|_1$ : the Volterra resolvent exists locally for every locally integrable kernel. What  $\|\phi\|_1 = 1$  separates is stability, not existence (Example 8.74).

## 10.5 Generalizations that turned out to be empty

Four things looked like gains and are not. They are recorded because a blueprint that lists only its successes is not usable as a map.

- (i) “Every process is Markov if the state is the path.” True, and vacuously so:  $\sigma(\hat{X}_t) = \mathcal{F}_t^X$ , an identity of  $\sigma$ -fields (Lemma 6.8). The one-dimensional distributions of the lift are the finite-dimensional distributions of the process, so every hypothesis of the Markov layer becomes its own conclusion. A Markov structure is a *compression* statement, and the lift attains it by deleting the compression; see Remark 6.9.

- (ii) “*The abstract martingale problem is a special case of the Markovian one.*” Formally yes, with  $g = 0$  on the lift, and as empty as the observation that every process solves the martingale problem for the family of its own martingales. The informative lift keeps the generator and lets its integrand be a path functional — which is Setting 8.69, already in the manuscript under another name (Remark 6.10).
- (iii) “*Every clock admits duality.*” Withdrawn. What is true is Corollary 7.4, and it is a deterministic time change of the Lebesgue case rather than a theorem; the atomic case is Proposition 7.8 and needs the atoms below a point to form a finite chain, and the mixed case is open (Remark 7.5).
- (iv) “*Submartingale problems are a further free generalization.*” They are transverse, not wider: formulating a submartingale needs an order and therefore excludes  $\mathbb{K} = \mathbb{C}$ . Over  $\mathbb{R}$  they are a genuine generalization, by (37), but the determining set does not transfer for free (Lemma 6.20) and uniqueness fails outright (Remark 6.21).

Items (i) and (ii) share a moral worth stating once: enlarging the state space never restricts the class of processes, so a construction that only enlarges it cannot prove anything. Item (iii) has a different one, and it is about method rather than mathematics: it began as the repair of a genuine error — the claim that duality *requires* a translation invariant clock — and the repair overshot, turning “translation invariance is unnecessary” into “every clock works” and a substitution into a theorem.

*Remark 10.1* (Where the graduation stops, and why). Definition 2.1 graduates the hypotheses on  $\mathbb{T}$ , and Section 2.9 records which result needs which bundle. What neither says is where the graduation *ends*. It ends in four different places, for four different reasons, and a reader who tries to push past any of them should know in advance which wall is being hit.

| What is being defined                                   | Weakest $\mathbb{T}$ that carries it                                                                                              | What breaks beyond it                                                                                                                                                                                                                                              |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Filtration, adaptedness, the martingale property        | (T0): a preorder with a least element                                                                                             | Nothing: there is no weaker structure in which $E[X_t \mid \mathcal{F}_s] = X_s$ for $s \leq t$ can be written. Mathlib indexes <code>Filtration</code> and <code>Martingale</code> by <code>[Preorder <math>\iota</math>]</code> for the same reason              |
| Stopping times closed under the operations of Fact 2.19 | (T1): a lattice, for $\vee$                                                                                                       | $\max_{k \leq n} \tau_k$ ceases to be an element of $\mathbb{T}$ . It is $\vee$ and not $\wedge$ that is needed, for the reason set out in Remark 2.6(ii)                                                                                                          |
| Càdlàg paths                                            | A <i>tree order</i> : (T2a) relaxed to “ $\{s : s \leq t\}$ is a chain for every $t$ ”, with a countable dense subset as in (T2b) | The left limit stops being a limit. On a merely directed index $\{s : s \leq t\}$ is not linearly ordered, so “the limit from below” is a limit along a directed rather than a linearly ordered filter, and its existence is a hypothesis instead of a consequence |
| The Skorokhod $J_1$ topology                            | (T3), refined: $\mathbb{T}$ a closed subset of $\mathbb{R}_+$ containing 0, with the induced metric                               | The time changes lose control of the paths; see below                                                                                                                                                                                                              |

Rows three and four are the ones that are not obvious, and they fail differently.

*Row three* is the reason (T2a) appears where it does. A tree order — every down-set a chain — is the exact generality in which the càdlàg condition of Definition 2.18 can be stated: left limits are taken along  $\{s : s \leq t\}$ , and that is a chain. This is not idle: the genealogy of a branching process is such an index, and the cluster recursion of Proposition 8.15 builds its process along one. What a tree does *not* have is  $\vee$ , so by row two its stopping times are not closed under max; the branching literature replaces them by stopping lines. The consequence for this manuscript is clean, because it follows the line Section 9 already draws for other reasons: on a tree index, Sections 4, 6.1, 7 and Theorem 8.40 survive verbatim, and everything that invokes Fact 2.25 — Theorem 6.27 and the local theory of Section 6.3 — does not.

*Row four* fails for a reason of its own, and metrizing  $\mathbb{T}$  does not repair it. Billingsley's  $\|\lambda\|^\circ$  is, for a time change fixing the endpoints, exactly

$$\|\lambda\|^\circ = \sup_{s < t} \left| \log \frac{\lambda t - \lambda s}{t - s} \right| = \log \max(\text{Lip}(\lambda), \text{Lip}(\lambda^{-1})),$$

so it is defined on any metric  $\mathbb{T}$ , and the two facts the completeness proof uses —  $\|\lambda^{-1}\|^\circ = \|\lambda\|^\circ$  and  $\|\lambda \circ \mu\|^\circ \leq \|\lambda\|^\circ + \|\mu\|^\circ$  — become the two standard properties of Lipschitz constants. What does *not* come for free is the control  $\sup_t \rho(\lambda t, t) \leq \psi(\|\lambda\|^\circ)$  with  $\psi(0) = 0$ , and without it the norm no longer sees whether  $\lambda$  moves the paths. That control holds when  $\rho$  is additive along the order,  $\rho(s, u) = \rho(s, t) + \rho(t, u)$  for  $s \leq t \leq u$  — and additivity makes  $t \mapsto \rho(0, t)$  an order isomorphism and isometry onto a closed subset of  $\mathbb{R}_+$ . Hence the entry in the table: not an interval, but not more than a closed subset either. The class is larger than it looks — it contains  $h\mathbb{N}_0$ , so Section 8.8 compares objects of one family rather than of two, and it contains the closed range of a subordinator, which is the time set of excursion theory. A tree is not in it: a root-fixing isometry may permute two isomorphic branches, so  $\|\lambda\|^\circ = 0$  while  $\sup_t \rho(\lambda t, t)$  is of the order of the diameter. Restricting to time changes that keep every point on its own branch restores the control, at the price of making the group depend on the tree.

*Manifolds.* The natural question, given rows three and four, is whether a manifold can serve as  $\mathbb{T}$ , and the answer separates the two halves of càdlàg. Existence of one-sided limits is a regularity condition; *choosing the right-hand one* is a choice of a side, and in dimension  $\geq 2$  the discontinuity set of a function that has such limits is generically a hypersurface, which has two sides only if it is co-orientable. A jump across the core curve of a Möbius band has no consistent right-hand side, and the càdlàg condition of Definition 2.18 cannot be stated for it. Where the geometry cooperates the notion does exist and is classical: for  $\mathbb{T} = [0, 1]^d$  with the product order one asks for limits from all  $2^d$  quadrants and continuity from one of them, and that space carries a Skorokhod topology and tightness criteria [21, 19, 22, 23]; on a domain in  $\mathbb{R}^d$  the *BV* and *SBV* theory supplies a  $(d - 1)$ -rectifiable jump set with two traces across an oriented normal [18], which is the same notion carried by geometric measure theory; and for an index ordered by inclusion there is the outer continuity with inner limits of [20]. Each of these fixes the missing datum by hand — a coordinate frame, a co-orientation, a set algebra. A manifold supplies none of them canonically, which is the precise sense in which càdlàg carries a one-dimensional structure that continuity does not.

*And the ceiling that is not there.* None of this constrains Section 8.10. The pseudo-path of Fact 8.64 is the image of a finite measure on  $\mathbb{T}$  under  $u \mapsto (u, w(u))$ ; it needs a measure on  $\mathbb{T}$  and a metric on  $E$ , and no order, no direction and no dimension. Its Borel field is  $\sigma(\pi_u)$ , which is the cylinder field and presupposes nothing. So Theorem 8.40 in the reading

of Remark 8.41, together with Section 8.10, has no ceiling of the kind tabulated above: the clock of Definition 2.3 is the whole of what it uses. That is the sharpest statement of what the abstraction of Section 1.2 bought — not that the results hold under weaker hypotheses on  $\mathbb{T}$ , but that the two that never used  $\mathbb{T}$  can be seen not to.

*The one index in the table that is out of reach for a different reason.* A linearly ordered  $\mathbb{T}$  with no countable dense subset — an uncountable ordinal, the long line — satisfies row three and fails (T2b). There càdlàg is definable and the theory is empty: on  $\omega_1$  every continuous real function is eventually constant. It belongs in this subsection on its own merits.

## References

- [1] R. Degenne, D. Ledvinka, E. Marion and P. Pfaffelhuber. Formalization of Brownian motion in Lean. Preprint, arXiv:2511.20118, 2025.
- [2] D. Criens, P. Pfaffelhuber and T. Schmidt. The martingale problem method revisited. *Electron. J. Probab.* **28** (2023), article no. 19, 1–46.
- [3] A. Depperschmidt, A. Greven and P. Pfaffelhuber. Duality and the well-posedness of a martingale problem. *Theor. Popul. Biol.* **159** (2024), 59–73. Preprint version: arXiv:1904.01564.
- [4] D. J. Daley and D. Vere-Jones. *An Introduction to the Theory of Point Processes*. Vol. I, 2nd edition, Springer, New York, 2003.
- [5] E. Errais, K. Giesecke and L. R. Goldberg. Affine point processes and portfolio credit risk. *SIAM J. Financial Math.* **1** (2010), 642–665.
- [6] O. El Euch and M. Rosenbaum. The characteristic function of rough Heston models. *Math. Finance* **29** (2019), 3–38.
- [7] S. N. Ethier and T. G. Kurtz. *Markov Processes: Characterization and Convergence*. Wiley, New York, 1986.
- [8] A. G. Hawkes and D. Oakes. A cluster process representation of a self-exciting process. *J. Appl. Probab.* **11** (1974), 493–503.
- [9] A. Jakubowski. A non-Skorohod topology on the Skorohod space. *Electron. J. Probab.* **2** (1997), paper no. 4, 21 pp.
- [10] J. Jacod and A. N. Shiryaev. *Limit Theorems for Stochastic Processes*. 2nd edition, Springer, Berlin, 2003.
- [11] T. G. Kurtz. Random time changes and convergence in distribution under the Meyer–Zheng conditions. *Ann. Probab.* **19** (1991), 1010–1034.
- [12] P. A. Meyer and W. A. Zheng. Tightness criteria for laws of semimartingales. *Ann. Inst. H. Poincaré Probab. Statist.* **20** (1984), 353–372.
- [13] T. Jaisson and M. Rosenbaum. Rough fractional diffusions as scaling limits of nearly unstable heavy tailed Hawkes processes. *Ann. Appl. Probab.* **26** (2016), 2860–2882.
- [14] E. Abi Jaber, M. Larsson and S. Pulido. Affine Volterra processes. *Ann. Appl. Probab.* **29** (2019), no. 5.

- [15] C. Cuchiero and J. Teichmann. Generalized Feller processes and Markovian lifts of stochastic Volterra processes: the affine case. *J. Evol. Equ.* **20** (2020), 1301–1348.
- [16] O. Kallenberg. *Foundations of Modern Probability*. 3rd edition, Springer, Cham, 2021.
- [17] D. W. Stroock and S. R. S. Varadhan. *Multidimensional Diffusion Processes*. Springer, Berlin, 1979.
- [18] L. Ambrosio, N. Fusco and D. Pallara. *Functions of Bounded Variation and Free Discontinuity Problems*. Oxford University Press, Oxford, 2000.
- [19] P. J. Bickel and M. J. Wichura. Convergence criteria for multiparameter stochastic processes and some applications. *Ann. Math. Statist.* **42** (1971), 1656–1670.
- [20] B. G. Ivanoff and E. Merzbach. *Set-Indexed Martingales*. Chapman & Hall/CRC, Boca Raton, 2000.
- [21] G. Neuhaus. On weak convergence of stochastic processes with multidimensional time parameter. *Ann. Math. Statist.* **42** (1971), 1285–1295.
- [22] M. L. Straf. Weak convergence of stochastic processes with several parameters. In *Proc. Sixth Berkeley Symp. Math. Statist. Probab.*, Vol. II, University of California Press, 1972, 187–221.
- [23] W. Whitt. *Stochastic-Process Limits*. Springer, New York, 2002.
- [24] J. Kuan. Correlated and uncorrelated long-time asymptotics of type D ASEP. Preprint, arXiv:2607.11376, 2026. Lean formalization: <https://github.com/Jeffrey-Kuan/type-D-asymptotics-lean>.